

Correspondence Analysis and West Mexico Archaeology



Frontispiece. Photograph of Etzatlán taken from the south in 1890 (the partially drained lake is in the background).

CORRESPONDENCE ANALYSIS AND WEST MEXICO ARCHAEOLOGY



CERAMICS FROM THE LONG-GLASSOW COLLECTION

C. Roger Nance

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The book is dedicated to the grandchildren of

CR

Andrew

Charles

David

Finuala

PCW

Alejandro

Helen

Michelle

Sofia

KP

Ryan

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Foreword

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+++The correspondence analysis that Roger Nance, Jan de Leeuw, and their colleagues present in the following pages demonstrates how far ceramic analysis has progressed since my efforts in the mid-1960s to understand the ceramics from the site of Huistla. At that time, seriation analysis was just emerging as something more than a visual process of arranging assemblages, and graduate students such as I did not receive training in quantitative analysis. All this began to change a few years after my analysis of the Huistla ceramics, when archaeologists began to recognize the power of computers and were able to gain access to them. A few pioneering multivariate analyses, such as the Binfords' analysis of French Mousterian assemblages, stimulated this development. Nonetheless, as de Leeuw points out in this volume, correspondence analysis, as one of a variety of multivariate techniques, has a relatively recent history in Americanist archaeology despite its appropriateness for teasing out chronological relationships between categories of artifacts such as pottery types. The fieldwork of the Etzatlán Project occurred during the fall and winter of 1963 and early 1964. Stan Long was the director of the project, and the resulting collections were to be integral to his doctoral dissertation. He asked me to be his field assistant after another graduate

student who was to serve in this capacity had to decline because of other obligations. I had been involved in fieldwork in New Mexico through the summer of 1963, and upon returning to the University of California–Los Angeles (UCLA), I had only a few days to settle my affairs before leaving for Etzatlán. Stan was already there. I took a Greyhound bus to the Mexican border at San Diego, crossed the border into Tijuana, and then took a Tres Estrellas de Oro bus to Guadalajara. In Guadalajara I proceeded to another bus station where I caught a local bus to Etzatlán. I had never been in Mexico aside from a visit to Tijuana with friends several years earlier. I had taken two years of Spanish as an undergraduate, but my speaking ability was still very rudimentary. Needless to say, I experienced a great deal of anxiety during the course of the trip.

I found Stan and his wife, Luty, at a small hotel facing Etzatlán's main plaza. This hotel served as our residence until excavation was completed, and we ate our morning and evening meals there as well. The location of the hotel exposed us to life in a small Mexican town. Occasionally loud reports of skyrockets punctuated the night, apparently in recognition of the death of a town resident. Evenings after supper were a time to watch young adults walking around the plaza, men in one direction and women in the other.

Stan was a well-organized and hardworking field director. Each day we were up early and out to our respective sites. We each supervised two to four hired workers, men who otherwise would be agricultural laborers. Stan put me in charge of excavation at three sites, although he made most of the decisions about placement of test pits. I also had the responsibility of producing a topographic map with a transit of each site I investigated. Huistla was the first, then Anona, and finally Las Cuevas. The sites of Huistla and Anona were on either side of Etzatlán, but Las Cuevas was several kilometers distant. We had a university vehicle, an International Scout, for transporting personnel and equipment. This vehicle ultimately caused us great grief when a rear axle snapped while it was carrying a heavy load up a short but steep grade. As replacement parts for this vehicle were unavailable in Mexico, Stan had the two axle pieces welded back together—not an ideal solution but one that worked so long as we were very careful not to put too much torque on the vehicle's drivetrain.

Excavation at Las Cuevas consumed most of my time. It was a large and complex site, and deposits were generally deeper than at Huistla or

Anona. In addition to excavating pits in habitation deposits on the flatlands at the base of a hill, we also excavated at an obsidian workshop on the hill slope. In one of the pits in the flatlands we encountered a small, shallow double tomb dug partly into the underlying tepetate (volcanic ash) deposits. As I remember, the roofs of the chambers had collapsed, and the chambers were completely filled in. This tomb was the only one we encountered in our excavations, although near the end of the project Stan and his crew removed backfilled deposits from a shaft tomb that had been looted shortly before our arrival.

While working at Las Cuevas, my crew and I were visited each day by an elderly gentleman on his mule. He would spend an hour or so with us, watching the progress of our excavation and discussing local happenings. One day he arrived in a state of obvious agitation, wanting to pass on to me important information he had just learned. Although my Spanish comprehension had improved since my arrival in Etzatlán, I had a great deal of difficulty understanding what he was trying to tell me. I eventually comprehended that President Kennedy had been assassinated, but I was not sure I had understood correctly his extraordinary report until after we returned from the field that day.

After completion of fieldwork at the six sites we investigated, we moved from the hotel to live with a family in the little hamlet of San Sebastián northwest of Etzatlán. Prior to fieldwork, Stan had learned that the entire contents of a shaft tomb in the Etzatlán vicinity had ended up with an antiquities dealer in Los Angeles. Stan ultimately ascertained that the collection had come from a shaft tomb at San Sebastián. He became good friends with the family who owned the property where the tomb was located, and the study of this looted tomb was the principal reason for our move. The family's home was a portion of a derelict hacienda, several rooms of which had been reroofed for their residence, while still-roofless rooms were used to keep farm animals.

Stan had his crew remove the backfill from the tomb so that he could document its shape and structure. One of the family members had been in the tomb when the hollow figurines and other mortuary goods were discovered and removed, and Stan spent a good deal of time interviewing him to determine the placement of interments and mortuary goods within the tomb's chamber. Stan also had his crew go through the backfill of the tomb, which resulted in his finding a number of small items, including jewelry, that the looters had overlooked.

While living at San Sebastián, Stan hired a few young ladies to wash the collections, and one of my jobs was to sort out the body sherds, which were not retained. My other job was to organize my field records and to type up my field notes. I found this task difficult to accomplish, as the onset of cool weather resulted in stiff fingers tapping at the typewriter keys. I also aided in the unsuccessful search for other shaft tombs in the immediate vicinity of the one Stan had re-excavated. This effort entailed use of a geophysical prospecting device known as a seismic hammer, but unfortunately no additional tombs were discovered. Shortly before Christmas, I left Stan and his wife to finish processing the collections. I drove back to California with the two fellows who had come down to do the work with the seismic hammer.

To read Stan's 1966 dissertation, one would never realize that we had carried out test excavations at six sites in addition to the investigation of the San Sebastián shaft tomb. Even Stan's description of the tomb discovered at Las Cuevas only briefly mentions our test excavation at this site. In the end, Stan saw no reason to use the collections from the test excavations, given his focus on the shaft tomb complex in the vicinity of Etzatlán. I suspect he realized that little, if anything, in the collections could be confidently associated with the period during which the shaft tombs were created. The ceramic analysis presented in this volume supports his suspicion.

Upon completion of his dissertation, Stan was appointed to a position (a temporary one, as I recollect) at the Universidad de los Andes in Bogotá, Colombia. In 1967 he met an unfortunate end while engaged in fieldwork there—he drowned while swimming in a river. Had he lived, he eventually would have undertaken the analysis of the collections and other data acquired during the Etzatlán Project. In a sense, therefore, this volume is a partial fulfillment of Stan's intentions.

Before the beginning of the fieldwork, Stan and his faculty supervisors, Clem Meighan and Henry Nicholson, had agreed that I would be able to analyze some small part of the collections as a personal research project. As I remember, Stan proposed that I work with the collection from Huistla. By the beginning of the Etzatlán Project, I had learned some of the basics of ceramic analysis (as it stood in the early 1960s) from Meighan, and while an undergraduate I had been employed as a rough sorter of sherd collections from the Valley of Mexico site of Portezuelo. The analysis of the Huistla ceramics clearly was an important learning

experience for me, and publication of the results helped establish my professional credentials. Interestingly, the analysis presented in this volume reveals that the Huistla ceramics undoubtedly spanned a longer period of time than I realized. As well, one of the types I described and named, Huistla Polychrome, turns out to date later in time than I had determined. This decorated pottery type was well represented not only at the site of Huistla but also at other sites in the Etzatlán vicinity, and it has stood the test of time.

I am pleased that the pottery collections from the Etzatlán Project, after more than forty-five years, finally have received the analytical attention they deserve. The analysis presented in this volume contributes substantially to explicating the diversity of pottery types occurring in the Etzatlán vicinity and their placement in time. We now have a foundation from which to delve into other aspects of prehistoric development in the Etzatlán area and the Magdalena basin as a whole. In this regard, I am intrigued that the Etzatlán Project potsherds appear to be associated mainly with the Postclassic period. Interestingly, they give us only hints about the identities of the creators of the shaft tombs. This apparent anomaly reminds us that the archaeology of the Etzatlán vicinity holds many mysteries yet to be solved.

Acknowledgments

By C. Roger Nance

+++ The archaeology behind this book was a long process that involved many individuals. I am indebted to the Cotsen Institute of Archaeology and the Fowler Museum for permission to study the Long-Glassow pottery. Those facilitating the project at UCLA included Marilyn Beaudry, Richard Leventhal, Wendy Teeter, Julia Sanchez, and Charles Stanish. After considerable time working at UCLA, we lost our laboratory space. At the Westside Pavilion in Westwood, Erica Boatman-Dixon, then marketing manager, agreed to my request for use of a vacant store as an archaeology lab. We used this donated space for over a year and are grateful to Westside Pavilion for not only the space but also the free utilities and benefit of mall security. This donation required a contract between UCLA and Westside Pavilion, a lot to ask by a visiting archaeologist with few ties to the university. A grant by the Cotsen Institute, Charles Stanish, director, covered the expense of painting the store and moving artifacts and furniture between the mall and university. The contract stipulated a professionally rendered sign over the storefront, and this sign was prepared and donated by a company in Culver City. Thanks to Guy Anderson of Designtown, USA.

Others helped out by participating directly in the process of recording potsherds, including several UCLA undergraduate students, most notably Jacob Lee. Thanks go to Helle Girey, also with the Cotsen

Institute, who donated her time to produce the line drawings of potsherds in chapter 3. Also, several archaeologists with substantial experience in the archaeology of West Mexico provided ideas, information, and encouragement at critical junctures in the research: the late Phil Weigand of the Colegio de Michoacán with whom I was to collaborate herein and Christopher Beekman at the University of Colorado–Denver. Later, two other archaeologists read and provided critical comments on a draft of the book: Michael Love, California State University–Northridge, and Marvin Jeter, recently retired from the Arkansas Archaeological Survey and the University of Arkansas at Monticello. Thanks to all of these people as well as to two anonymous reviewers for the University of New Mexico Press who made helpful suggestions.

Closer to home, I would like to thank my son, Charles Nance, for his work in restoring the text in some of the illustrations for chapter 1. My wife, Vally, once again proved indispensable in the writing and completion of this book.

Introduction

By C. Roger Nance, Jan de Leeuw, and Phil C. Weigand

+++Relative to its vast territorial expanse and the complexity of the prehistoric cultures represented there, West Mexico is an area of North America that has received little attention from archaeologists. As a result, portions of the different prehistoric ceramic sequences for the region remain incomplete (see Beekman 1996; Mountjoy 2000; Pollard 2000; Weigand 2000).

Much of the basic fieldwork that has been reported occurred more than twenty-five years ago, when ceramic type distributions were summarized and tabulated by hand. It occurred to us that study of pottery collections using computers and statistics might render some site excavation data more amenable to chronological control and, hence, interpretation. With this question in mind, we approached the Fowler Museum and the Cotsen Institute of Archaeology at UCLA. These institutions gave us permission to study a sample of potsherds from the vicinity of Etzatlán, Jalisco, that had been excavated during the 1960s by UCLA archaeologists and curated since that time at the university.

Over the course of this project, in constructing a ceramic sequence for Etzatlán, we employed correspondence analysis (henceforth, CA), a statistic little used in North American archaeology, including the

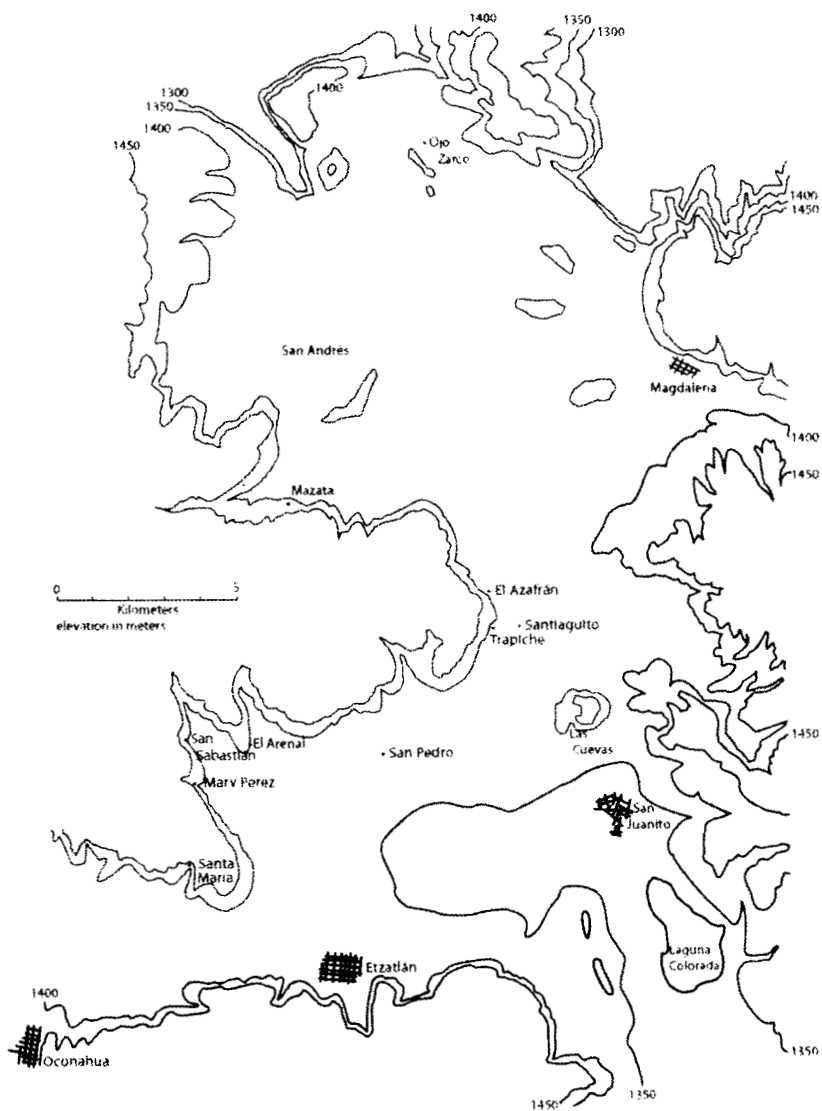


Figure I-1. A map of the Laguna de Magdalena basin, Jalisco. Map by Stanley Long (1966:45); redrawn by Sandra Wong, 2009.

Mesoamerican region (see Nance and Leeuw 2005; Nance et al. 2003). In chapter 2, we discuss the history of CA, both in statistics and archaeology. We then give the basic equations defining CA as either a multivariate exploratory technique or as a statistical model-fitting technique. Similar applications in psychometrics and ecology are discussed as well. As an illustration, we apply CA to one small and one more realistic example. But, of course, the main applications of the technique are given in other chapters of the book, where we analyze typed ceramics from Etzatlán.

The ceramics we selected derived from sites near the town of Etzatlán and in proximity to the dry lake bed of the Laguna de Magdalena in the lake district of northern Jalisco (fig. I-1). In 1963 Stanley Long and his assistant, Michael Glassow, investigated seven sites there through the excavation of randomly spaced pits, usually 1.5 by 1.5 m in dimension and almost always by 20 cm deep arbitrary levels. Apparently, these excavations were to form the basis of Long's dissertation research, but instead he studied elaborate burials from the area (Long 1966), and pottery from the pits in question remained mostly unstudied. Except for an article by Glassow (1967) on one of these sites, Huistla, no one had systematically studied this UCLA collection. This is not to say, however, that archaeological research has been nonexistent in the Laguna de Magdalena basin. On the contrary, Phil Weigand began his own archaeological and ethnohistorical research program there and at nearby localities in north-central Jalisco soon after the Long-Glassow fieldwork. He was active in publishing archaeological fieldwork on the region and conducting pertinent library/archival research from that point up to the completion of this book. In chapter 1, Weigand provides a picture of late prehistoric cultures at Etzatlán and a narrative of key events surrounding the conquest and its socioeconomic and demographic consequences. He also describes archaeological resources in the Etzatlán basin, both as they existed thirty years ago as well as their present, more degraded condition. Chapter 1 also summarizes historical evidence for indigenous languages in this portion of Mexico.

The Collections

The collections and especially the records pertaining to these collections deteriorated to some extent during the thirty-seven years between excavation and the time we began our research. Table I-1 lists extant

Table I-1: Surviving documents for the Etzatlán Project

Site	Feature, Burial Drawings	Artifact Catalog	Discard Sherd Catalog	Site Survey Form	Field Notes	Profile Drawings	Contour Map
Anona	+	+		+	+	+	Fig. I-2
Las Cuevas	+	+	+	+	+	+	Fig. I-4
Santiaguito	+	+	+	+		+	
Tiana		+	+				Fig. I-3

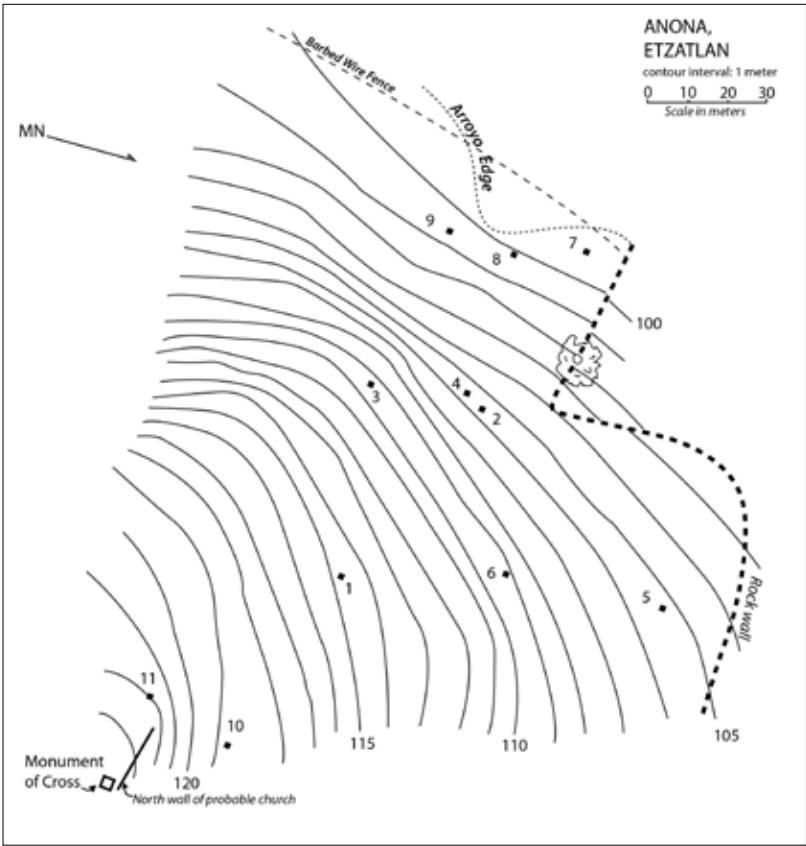


Figure I-2. A contour map of the excavated portion of Anona. Compiled by Michael Glassow, 1963; redrawn by Sandra Wong, 2009.

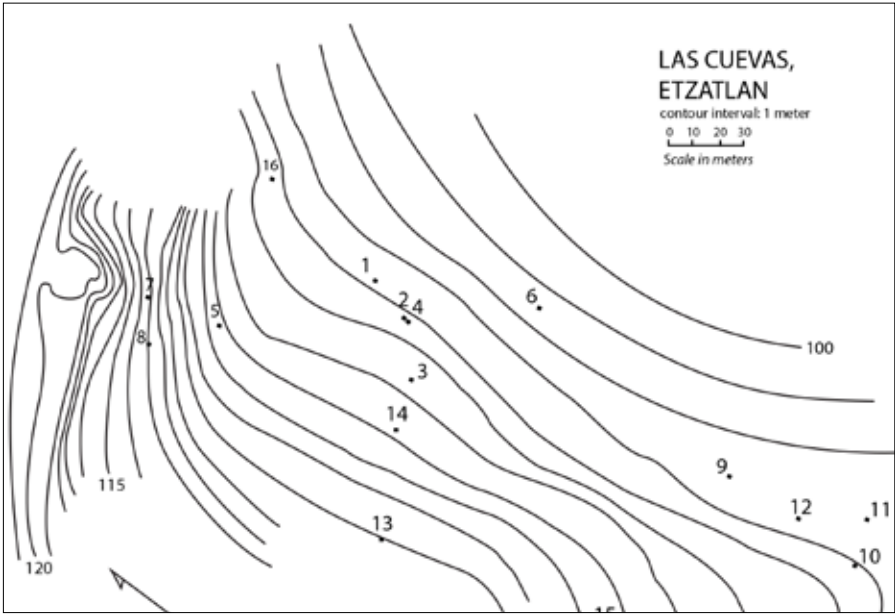


Figure I-4. A contour map of the excavated portion of Las Cuevas. Compiled by Michael Glassow, 1963; redrawn by Sandra Wong, 2009.

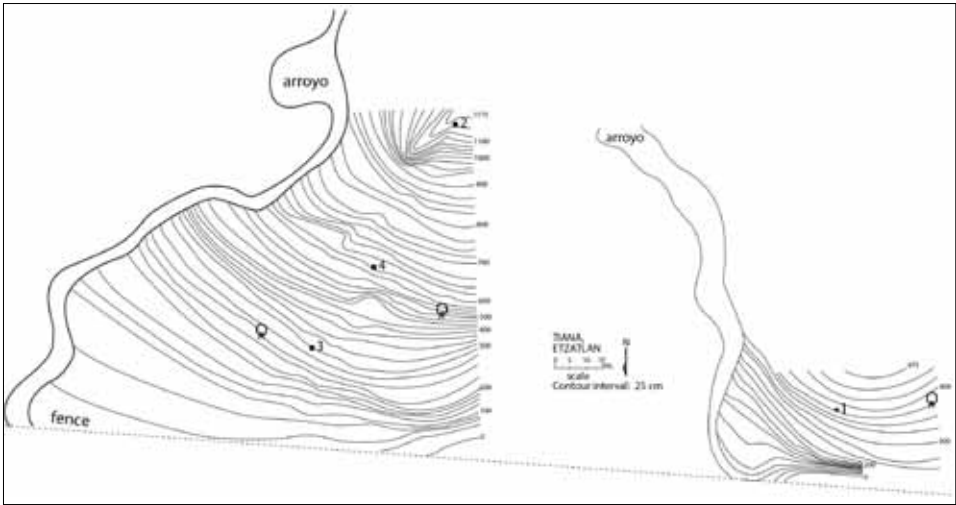


Figure I-3. A contour map of the excavated portion of Tiana. Compiled by Michael Glassow, 1963; redrawn by Sandra Wong, 2009.

records for the four sites investigated in this study. Documentation for Tiana is especially meager: a site survey form and field notes are missing, and we have no map on file showing the site's location. Generally, very few photographs of the fieldwork survive, but contour maps showing pit locations are on file for three of the four sites investigated (figs. I-2–I-4). Some materials were never transported from Jalisco to UCLA, and special “Discard” catalogs enumerate the many hundreds of potsherds discarded in Jalisco by site, level, and square. These sherds are designated as either “red” or “other.” We made no use of this sherd discard data in our research. Finally, some excavated potsherds, reportedly brought to UCLA, could not be located at the time of our laboratory research. For instance, Long (1966:88–89) mentions potsherds from deep levels of Las Cuevas: squares 2 and 4, 120 to 180 cm below the surface. He believed these were contemporary with shaft tomb burials at the site (chapter 7). Small samples of potsherds from these levels are listed in the Las Cuevas catalog, but in our laboratory work, we encountered no samples of potsherds from these pits from levels deeper than 60 to 80 cm.

The collection did have positive aspects, however, that led us to undertake this research. We found potsherds excavated by Long and Glassow to be generally in very good condition. For the site collections we studied, the sherds had been marked with catalog numbers designating site, level, and square. A redundant lot number also had been written on all sherds, which cut down greatly on sherds lost to unreadable catalog numbers. Also, we found these potsherds, for the most part, sealed in their original labeled level bags, as they probably had been packed in the field lab.

While this study deals only with ceramic artifacts, Long and Glassow shipped other excavated materials back to UCLA. In addition to the pottery, these collections include the remains of human burials, animal bones and bone artifacts, hundreds of pounds of obsidian debitage and artifacts, as well as copper artifacts.

Sites Investigated

Las Cuevas

We began our work with the collection from Las Cuevas, a major site on what was once an island in the Laguna de Magdalena, located about 7 km north of the modern-day community of Etzatlán. The lake

was drained through various projects between 1900 and 1960 and the lacustrine deposits turned into farmland (Weigand and Ron Siordia 1987:47–48). Las Cuevas was an important obsidian workshop where raw material from the nearby quarry of La Joya was processed into blades (Weigand and Spence 1982; see also Spence et al. 2002). Long (1966:59–60) describes a tomb and its contents, which he excavated in 1963, and Weigand (1993:figs. 2.3 and 2.8) illustrates several other tombs, one a deep shaft tomb from the same site. (See chapter 1 for more details on Las Cuevas, which is discussed in terms of both that site name and the earlier toponym, *Atitlán*, recorded during a 1525 Spanish *visita* to the then extant community.)

Tiana

While the site of Tiana is located in the vicinity of Etzatlán and probably near the old lakeshore (see Glassow 1967:64), the surviving copy of Long's field map showing site locations is obscure, and the exact location, unknown. Recently, Glassow (personal communication 2009) located a personal journal from his field season in the Etzatlán area and found mention of Tiana as “west of Huistla by somewhat less than a kilometer.” Huistla is within a kilometer of the southwest edge of Etzatlán (Glassow 1967:64). As noted previously, no field notes or other documentation survive, except for the artifact catalog, a contour map, a discarded sherd list, and information written on the artifact level bags.

Anona

According to the Site Survey Record and 1963 field notes, the site of Anona is across a stream from the southeastern edge of the town of Etzatlán, situated on a stream terrace and the sloping hillside behind. Long and Glassow describe the site as covering approximately 1 km². Glassow wrote a brief description of this site the day excavations began there, October 2, 1963:

Began the site of Anona today. This site is located on the southeastern edge of Etzatlán, just on the other side of the first bridge east of the town hall. The site consists of a terrace above a creek and the hillside behind this terrace which flattens out somewhat at a height of ca. 15 meters. On this hillside there is the foundation of a building which is said by the town's inhabitants to be an old church. At this location there is now erected a monument about 3 meters high of a cross. The

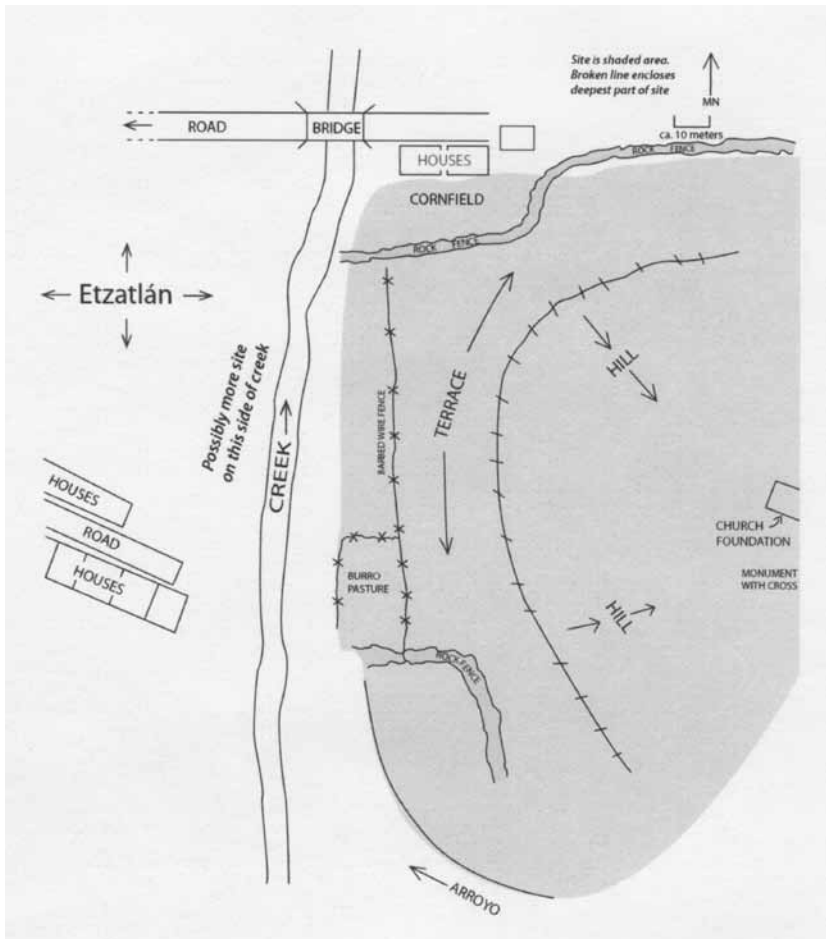


Figure I-5. A field sketch map of Anona. Map by Michael Glassow, 1963; redrawn by Sandra Wong, 2009.

hillside has at least 2 main built up embankments with long terraces behind, running along the contours of the slope. These may be in part fallen down stone fencerows, but they may also be retaining walls for buildings that may have once stood [there]. This is a lot of rock, mostly basaltic, strewn on the surface of the hillside, some of it definitely forming patterns of building foundations. Possibly Etzatlán once extended over this hillside in earlier times, possibly even during the early Spanish

period. At any rate, houses of an earlier date did exist on the site area, as evidenced by the presence of glazed sherds and fired bricks. It would seem logical that a church would have a settlement around it. The water supply for the site must have been the stream which runs along the western margin of the site. Stan [Stanley Long] mentioned reports of an old Spanish graveyard in the vicinity, possibly near the foundations of the problematical church.

Glassow's sketch map of the site, redrawn, is included here as figure I-5.

Weigand amplifies but partially disagrees with this description: Anona is the section of the Etzatlán ruin called Santa Clara (chapter 1). The arroyo/creek Glassow refers to is Santa Clara Arroyo, which was called a *río* in prior times. Today it barely affords a dribble of water even after the heaviest rains. It used to have ponds, many fish and crawfish, reeds, and so forth, as well as clear, clean water. No more. The Santa Clara sector of the Etzatlán ruin is Late Postclassic, and it is from this part of the ruin that we have the AD 1480 ¹⁴C date. It borders on the Chirimoya sector of the ruin, just to the east. Both sectors are characterized by terraces, house walls, pits, platforms (plus a ramp at Chirimoya), burials of both humans and dogs, many Huistla potsherds, obsidian, and copper artifacts (including pits with slag from malachite ores). We have *no* evidence for a chapel, church, or Spanish graveyard here or anywhere nearby, either in the archaeology, the archives, or from the looters who have pocked this area quite thoroughly. As mentioned by Glassow, the area is indeed large (taken with Chirimoya), but it does not cover a square kilometer. Twenty hectares would be more like it. (Also see chapter 1 for sites surrounding Etzatlán.)

Santiaguito

Weigand describes the site as follows: Santiaguito was on an island within the central-west sector of the now-drained Laguna de Magdalena. Its location is noted on the 1579 *Ortelius* map. The location was strategic in that it was near the narrow constriction that divides the northern Magdalena section of the lake from the Etzatlán sector to the south. The island was small, approximately 1 ha in extent, and was separated from the mainland by a shallow section of the lake less than 200 m wide. An ancient causeway may have been constructed over this narrow space to the mainland. Opposite the island, to the west, was a much larger settlement, also now a contemporary village. The two sites were

obviously integrated, as Santiaguito had no farmland, beyond space for small gardens. Very little of either of the pre-Hispanic settlements is currently visible around and beneath contemporary structures. A large platform on the mainland has been completely destroyed by road-building. The summit of the island, marked by a cross, also had a small platform. The terraces utilized by modern Santiaguito houses are most likely pre-Hispanic in origin. The streets are littered with Late Postclassic sherds and obsidian fragments. Looting has produced earlier material, some of which is Late Formative in date. Possible terrace-platform remains are visible within some house lots, and many houses are apparently built on both the pre-Hispanic terraces and platform bases. The morphology of the ancient settlement, however, is now largely unintelligible. The contemporary village of Santiaguito had, before migrations almost emptied it, approximately 150–200 people. The pre-Hispanic island settlement was probably about the same size, though the mainland sector of the site was larger. Before the Magdalena Lake was drained during the 1950s, economic activities included reed-gathering and fishing, accompanied by farming on the mainland. This economic pattern was probably the pre-Hispanic one as well.

An early colonial church or chapel may also have been situated on the island. Such a structure is indicated on one of the two early maps discussed in chapter 1: the *Pintura del nuevo reino de Galicia*. Weigand found no evidence of this structure in the field, but he found the site so altered that he believed a small chapel easily could have disappeared without much of a trace.

In all probability, Santiaguito was on a second island north of Las Cuevas in the Laguna de Magdalena; the town there was recorded as Tenyca by Gonzalo Cerezo and Diego de Coría (1937:559; see chapter 1) during their 1525 visit. They observed that Tenyca had sixty houses and an adult male population of 120.

Ethnohistory

For the Etzatlán region, Weigand and García de Weigand (1996:55) see strong continuity in settlement patterns from the Late Postclassic to colonial periods and persisting into contemporary times. The authors find evidence for this continuity not only in their own field reconnaissance, but also in early maps and documents, as described in chapter 1.

Previous Research

Within the archaeological literature on Jalisco and adjacent areas of West Mexico, relatively few publications contribute to the construction of ceramic sequences. Those we found most useful, which are cited in this publication, include two reports of excavations on the south shore of Lake Chapala, one in far eastern Jalisco (Meighan and Foote 1968), and the other nearby in far western Michoacán (Lister 1949). To the west, the important site of Amapa, as studied by Grosscup (1964, 1976) and Meighan (1976), provided the most detailed comparative data for our purposes. Amapa is located on the coastal plain of Nayarit. Kelly (1945) conducted ceramic research at localities across West Mexico and, most significantly for this study, in the Autlán-Tuxcacuesco area of southwest Jalisco. Other studies providing information on ceramic chronology include Beekman 1996; Beekman and Weigand 2000; Galván Villegas 1991; Gifford 1950; and Mountjoy 1970. Key sites/localities are depicted in figure I-6.

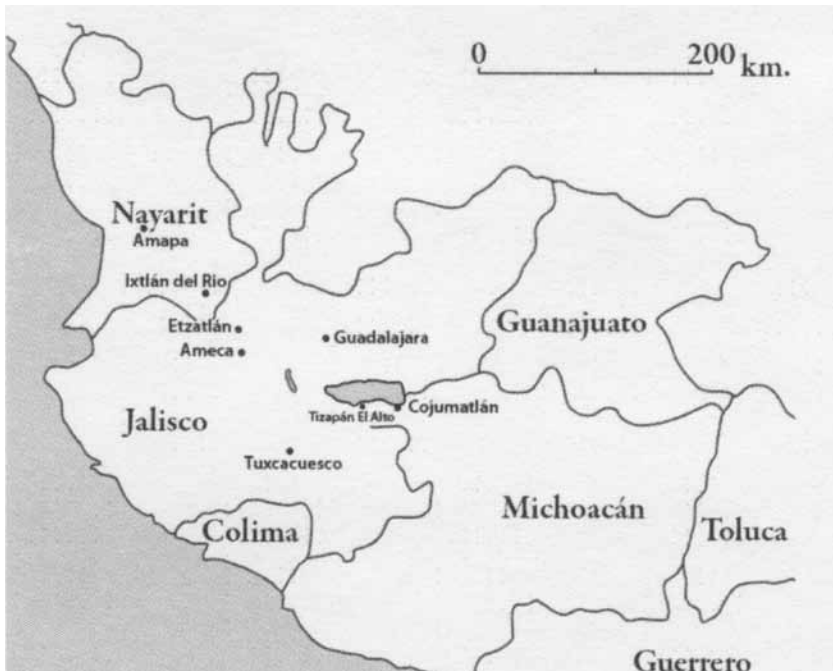


Figure I-6. Key localities in West Mexico. Drawn by Sandra Wong.

Organization of the Ceramic Research

The ceramic study begins in chapter 3 with typological descriptions, the basis for the ceramic classification. Throughout the book, these types are referred to using simple descriptive titles and/or the numerical codes that identify them in the data sets. Chapter 4 presents the analysis, as it developed in two stages. As mentioned, we began our lab work with the largest collection, that from Las Cuevas. We classified over seventy-eight hundred potsherds from this site and entered the data (type and provenience) into the computer with the aim of discovering the site's ceramic sequence. We employed the correspondence analysis (CA) statistic to generate a progression of types, potentially in chronological sequence. Then, in order to find corroborating evidence for this seriation as well as to explore its efficacy as a ceramic chronology, we turned to three more site collections and repeated the process for each: Tiana (approximately eighteen hundred sherds), Anona (approximately twenty-five hundred sherds), and Santiaguito (approximately three thousand sherds).

Through this series of CAs, we found a high degree of consistency between the hypothetical sequence at Las Cuevas and those of the other sites. Tiana appeared to fit in the early portion of the Las Cuevas sequence and Anona to represent the full chronological range. Pottery from Santiaguito was more eroded than that from other sites, but the full sequence did appear there as well, although slightly altered. These results led us to consider seventy-four samples from the three less-impacted sites—fifty-two from Las Cuevas, eleven from Tiana, and eleven from Anona—and to generate a CA for the combined samples. This CA will be referred to in the chapters following as the original three-site CA.

Thus far in the analysis, we had worked only with selected types or groups of related types combined to increase sample sizes. All of these types and type groups had shown some tendency to covary either positively or negatively with one another (sample by sample), which suggested either common or disparate histories in a ceramic sequence. Also, this idea was supported through the CAs mentioned previously. However, most of the types described in chapter 3 had not been included, and the next step was to find a way to expand the tentative sequence. In CA, results are displayed on a two-dimensional grid, and in our case, the types align roughly along the x-axis, or horizontal axis, hypothetically in chronological sequence. Concurrently, though, the CA also projects sample loci onto the same grid. Type loci are arranged so that

those showing similar distributions among samples are situated close together. Similarly, sample loci are projected so that samples with similar type proportions are also close together. And if type loci are aligned across a grid in chronological sequence, then sample loci will be as well.

This fact leads to several consequences favorable to archaeological research. First, one can study to some extent the distribution of types individually through a sequence, not just as a type locus relative to other type loci represented on a CA grid. In this study and by type, we assigned the sample x-axis value to each sherd, according to its sample membership. The distribution of x-axis values by type, then, provides an indication of a type's distribution through time: whether, for example, a type had a relatively long or short history at a site or a normal or skewed distribution. Secondly, the archaeologist can determine the chronological positions of some types not included in the original CA by summarizing x-axis values for each type with a sufficient sample size. Type loci or positions in the sequence, when not produced directly by the CA, are approximated by the median values for these type distributions. Our strategy, then, was to employ in the CA proper high-frequency types with relatively clear typological definitions and/or preliminary evidence of chronological sensitivity. Then, further chronological assessments could be extended to these and other types in the manner described. This approach can also be extended to nonceramic debris, such as stone tool types, debitage classes, or animal bones of a species or category. We summarize ceramic type distributions using this approach with the original three-site CA in chapter 5.

A remaining question had to do with the initial selection of types employed in the CAs. Several decisions were part of the selection process, and it seemed possible that somehow this grouping might have biased the outcome. In other words, selection of another group of types might have led to a different ordering of all types. The issue is explored in chapter 6, where we describe different CAs for which types were selected using different and simpler criteria. Another (related) statistical model was generated using the EDM (exponential distance model) statistic (chapter 2). In all cases, we found generally the same type ordering as in the original three-site CA.

In the concluding chapter of the book, we examine data from other studies, especially pottery type distributions from stratified sites, in order to confirm or reject the sequence suggested by our statistical

study. We have found good support for the proposed Etzatlán sequence in the form of comparable types in parallel sequences, especially from the highly stratified site of Amapa near the coast of Nayarit. Finally, our ceramic research indicates that the indigenous ceramic tradition continued after historic contact and that after the introduction of Spanish Majolica pottery, older prehistoric ceramic forms were modified in this new sociocultural environment. Herein, we explore the implications of these data for our understanding of the prehistoric–early historic interface at Etzatlán.



Roger Nance's interest in West Mexican pottery began over fifty years ago. While a UCLA undergraduate student in anthropology, he was escorted, probably as a class exercise, into a room filled with sherds from an ongoing project. The indelible impression of that ten minutes has remained until the present and probably led, as much as anything, to the writing of this book.

However, other ideas also motivated the Etzatlán research, including our interest in exploring the application of statistics to a collection of Mesoamerican pottery. For de Leeuw and Nance in particular, the emphases were threefold. First, our approach to the search for a ceramic sequence was explorative rather than hypothesis oriented, a mind-set compatible with the use of CA. Second, we put a premium on a large data set as we well understood the difficulties of working with small data samples in archaeology. Third, in the collaboration of a statistician and an archaeologist, we saw potential for the synergism that can follow from cross-disciplinary research.

Another motivation for this research had to do with our interest in the preservation of information through archaeological salvage. Weigand saw an increasing rate of site destruction in Etzatlán over his many years of working there. Etzatlán is no longer the traditional town described in Glassow's foreword. For one thing, the modern city of Guadalajara, 103 km and two hours' driving time from Etzatlán, has grown greatly since the 1960s and now has a population of about 4.3 million. This growth has had its impact. Also, the population of Etzatlán itself has increased by about 50 percent to 18,633, according to the 2010 Mexican census. Phil Weigand's death in 2011 will only compound

the difficulty of trying to study and preserve sites in and around Etzatlán. And what Weigand saw in the field, Nance, to some extent, witnessed in the archaeology lab. The collection has deteriorated at a much slower rate, but the effects have become both noticeable and telling, as we have discussed.

Finally, we did encounter several problems that came close to derailing the project, and their mention might prove beneficial to future students of long-stored collections. The first had to do with lab time. Archaeologists do not often take on such long-term, time-consuming projects, and lab directors might not appreciate that such projects can persist over many years. In these efforts one might face the prospect of moving, with artifacts, to a new locality. In our case, with the cooperation of the Cotsen Institute and Fowler Museum and the generous assistance of the Westside Pavilion Shopping Mall, we were able to move, rent free, to a vacant store in a nearby mall, where we worked for over a year.

Another problem is simply that long-stored collections can outlive the tenure of archaeologists responsible for their excavation. For this project, Nance knew of no locally available expert on West Mexican ceramics to consult during the course of the research, the essential issue being how to relate accurately one's typology of potsherds to published descriptions of types in the literature. As a result of fortuitous events, Christopher Beekman, who works at the University of Colorado-Denver, visited with Nance at the UCLA Fowler Museum laboratory. He examined some typed sherds and was able to clear up a misconception, thus enabling us to move forward.

Archaeology and Ethnohistory of Etzatlán and Its Region

Phil C. Weigand

+++ Obviously, Etzatlán was not founded by the Spanish. The well-developed settlement, and settlement system, that the Europeans and their indigenous auxiliaries (Tlaxcaltecas, Aztecas, and Purépechas) encountered was already at least seven hundred years old and perhaps was founded even before the time of Christ. The huge building complex that the Spanish designed for Etzatlán is a direct indicator of the settlement's importance within the western sections of ancient Mesoamerica. Francisco Cortéz de San Buenaventura, one of the earliest Spanish who systematically explored parts of western Mexico, had heard of Etzatlán's importance during his entrada in Colima (Tello 1968). Thus, Etzatlán's significance within the regional political and economic system was assumed to be a fact by the Spanish, even before they arrived at the site in person.

Reflecting interests quite natural for contemporary towns such as Etzatlán, many people have speculated about what the toponym actually means. The following exercise in etymology is meant to offer, in addition, an analysis of the linguistic information that we currently have on the region. By far the most popular, and certainly the most lyrical, translation for *Etzatlán* is "Lugar de Garzas." This lyrical character of a toponym is seen at quite a number of sites throughout Mexico. Another regional example is *Xuchitepec*, current-day Magdalena

(Jalisco), though this name does not appear in the primary sources that discuss the area at the time of the Spanish conquest. The ancient name for Magdalena included the name or title *Guaxicar*, often rendered as “Guaxacate,” and given in Nahuatl as “Guaxacatlán” (Weigand and García de Weigand 1996, 1997; see figs. 1-1 and 1-2). By Tello’s (1968) time (the mid-seventeenth century), the toponym *Xuchitepec* had appeared. However, toponyms—such as *Xuchitepec* (Mount of Flowers) and *Lugar de Garzas* (which appeared for the first time during the nineteenth century)—more often reflect the romanticism of the past few centuries rather than indigenous realities of the pre-Hispanic period. Certainly there is nothing wrong with lyrical translations and romantic nomenclature for towns such as Etzatlán, as long as these interpretations are understood for what they are.

While certainly no one disputes that in Nahuatl the suffix *-tlán* means “lugar de” (or “place of”), there is far less agreement about what the prefix *etza-* might mean. All suggested meanings for *etza-* are taken from the Nahuatl-Spanish dictionaries, very often with little respect for the phonemic and morphological organization of Nahuatl. Very frequently words are divided and subdivided in ways that do not make sense in Nahuatl and thus are impossible to accept. Using such loose linguistic criteria in interpreting *etza-*, we can arrive at literally dozens of equally acceptable possible meanings. Using Simeon’s (1977) dictionary of the Nahuatl language, for example, we can postulate, with equal probability, as *garzas* such words as *intestines*, *tartamudo*, *pasta de frijoles*, *terreno plantada de frijoles*, *papilla*, *el sexto mes del año*, *costra de sarna*, *jaspe*, *tromba*, *esclavo*, *robalo*, *el 18° mes del año*, *blanco*, *sandalías de los nobles*, and *sangre*, among many other possible candidates.

In considering this exercise in etymology, however, we should make use of the cardinal rule of linguistic studies: the simplest and most economical explanation is also the most likely. And in seeking possible meanings for *etza-*, we must note another very important linguistic and historical fact: that the toponym *Etzatlán* is Nahuatl. Nahuatl, as a language, arrived in this section of Jalisco along with the Spanish. It was not one of the native languages of this region, nor even closely related to them. Nahuatl was more closely related to the Caxcan language (see following) and some dialects in Colima and southernmost Jalisco (Valiñas 1994). Despite the use of the terms *naguatatos* (or *nahualato*) and *lengua de Mexico* in *Guzmán v. Cortéz* (Coría 1937; see following),

the indigenous populations of the Etzatlán area, as well as all their immediate neighbors, were not Nahuatl speakers. Terms like *nagualatos* referred to a general awareness that the speakers of this area were within what we today call the Uto-Aztec language family (of which Nahuatl is also a member), but they did not mean “Nahuatl” (Baus de Czitrom 1982; Valiñas 1994). No historical linguist or ethnohistorian has ever classified the languages of the Etzatlán region as Nahuatl. In addition, *nahualato* is derived from *nahual*, which means an ability to transform oneself or, in the context of language, to be bi- or multilingual. The languages spoken indeed may have included Nahuatl, but the word alone does not refer to that language. The language(s) spoken in Etzatlán and its surrounding area belonged to the Totorame branch of the southern Uto-Aztec family. Cora, Huichol, and Tequal are the three languages that survive within this branch. The Totorame branch is more closely related to Tepiman and Taracahitan than to the eastern Uto-Aztec family, which includes Nahuatl.

As mentioned, the Nahuatl language was introduced to this area by the Tlaxcaltec and Aztec auxiliaries of the Spanish conquerors and colonists. Nahuatl, especially Tlaxcaltec, as a language was already becoming the lingua franca of the Spanish armies and early colonial and evangelical organizations before they arrived in large numbers into this section of western Mexico. Thus, Nahuatl along with Spanish were introduced languages that had nothing to do with the original pre-Hispanic linguistic configuration of this section of the Occidente.

The mythical Aztec migration has also been traced through this area. This migration, according to legend, began along the coast of Nayarit. Aztatlán was the first site suggested as the mythohistorical Aztec origin place. Nowadays, the nearby site of Mexcaltitán, an island in the Marismas Nacionales, is often pointed out as the starting place for this migration. In support of this myth, much is made out of the prefix *mexcalti-*, which is supposedly a variant of *mexica*. The word actually comes from the Totorame language. In contemporary Huichol, the cognate is *metzelli*, meaning “moon.” Aside from this misidentification of the prefix, the simple fact remains that the Aztecs had quite different origins. These are explicitly discussed in their own accounts of their mythological past, such as those discussed by the sixteenth-century Spanish historian Fray Bernardino de Sahagún (1988; see also Boehm de Lameiras 1986). They were self-identified Chichimecas, which gives

them a general point of origin in the northern Bajío and great llanos and deserts of the interior of northern Mexico. The Aztecs never had any pre-Hispanic role at all in the Occidente, until they accompanied the Europeans into this zone as the conquest and colonization proceeded. A further discussion of the Mexcaltitán/Aztatlán mythohistorical problem and controversy can be found in Weigand 1994.

The native languages of this region were numerous but probably all fairly closely related (see Baus de Czittróm 1982; Valiñas 1994). The most important ones in the general Etzatlán area were

1. Totorame/Tecual. This was the one of the major languages of southern Uto-Aztecan and had a very wide distribution over which a great deal of dialectical variation had occurred. The extent of this dialectical variation is a strong indication of the ancient presence of this language in the Occidente. Cora and Huichol are the only survivors of this large group.
2. Coano/Cuano/Cano. This lexical set basically represents the same term. Its original distribution appears to have been spotty and discontinuous in the Occidente, and it was better represented in the southern Sierra Madre just to the north. Tepecano is the last extant dialect from this language, though only a tiny handful of speakers survive. Tepecano is most frequently classified as a language within the extensive Tepiman subfamily of northwestern Uto-Aztecan. Another dialect of Tepiman was found in the Acaponeta area of Nayarit.
3. Tecuexe/Coca. These languages, or dialects, are extinct and only preserved as infrequent personal names and toponyms in the region. While Tecuexe/Coca was also most probably a southern Uto-Aztecan language, and hence related to the better understood Totorame/Tecual, not quite enough vocabulary exists today to say with absolute certainty.

In fact, most of the major settlements of this section of the Occidente appear to have been multilingual, thus incorporating several of the aforementioned dialects, plus others such as Tahue, and languages within the same or close-by communities. The distribution of Tecuexe/Coca was largely centered to the east of the Etzatlán area (Baus de Czittróm 1982), but it was probably present at the site. Totorame/Tequal

was centered in this area just to the north in the area originally called the Nayarita (see figs. 1-1 and 1-2) and west along a large section of the Pacific coast.

The Caxcan language is mentioned as being within the Etzatlán region, too. The clearest reference to it is in the “Relación del pueblo de Ameca” (Acuña 1988), but it is clear that this mention is an artifact of the relocation of Caxcan populations after the Rebelión de Nueva Galicia, a process initiated by Antonio de Mendoza’s conquest of Zacatecas in 1541. Caxcan is often called *mexicano rústico* (see Tello 1968) and therefore was closely related to Nahuatl, probably to the point of some mutual intelligibility. *Mexicano rústico* became the term in the conquest and early colonial periods for languages closely related to Nahuatl, implying some level of common comprehension. This commonality is demonstrated, without room for doubt, by the convergence in the early historical use of this term and the Nahuatl affinities affirmed by later formal linguistic studies. Caxcan was, thus, an intrusive language (Tello 1968; Weigand 1993; Weigand and García de Weigand 1996). However, terms like *naguatatos* for this area are far too vague to have any but the most general value in language assignments. As we have mentioned, *naguatatos* is not a true cognate of the term *Nahuatl*.

The true native toponyms of this area underwent a process of nahuatlization directed by the Tlaxcaltecan and Aztecan allies of the Spanish, who then wrote down what they were told about these place-names and languages. These indigenous allies of the Spanish always vastly outnumbered them: for each single European, there were hundreds of allies, many of them of noble backgrounds. Hence, they were the intermediaries for the Spanish in every contact with the indigenous peoples of the Occidente, and these Nahuatl speakers from central Mexico could call upon a widespread system of bilingual individuals from anywhere within the far-flung frontiers of the ancient Mesoamerican civilization. Only infrequently were the real native toponyms preserved, and Etzatlán’s was not one of them.

Nahuatl and the native languages of this section of the Occidente all belonged to the Uto-Aztecan language family, just as Russian and Spanish both belong to the Indo-European language family. However, just as Russian and Spanish are not mutually intelligible, neither were Nahuatl and the languages of this region (Totorame/Tecual, Tecuexe/Coca, and Coano/Cuano/Cano), except through the use of interpreters.

The original languages spoken in Etzatlán cannot be specifically identified with any real degree of security, although we know that Coano was not one of them. Coano is directly mentioned as the language of Magdalena. The pertinent passage in Tello (1968:128) reads: “Estuvo [Oñate] en esto ocupado cinco días, y supo como el Gobernador Guzmán estaba en Etzatlán, seis leguas de allí, y acuerdo de no irse derecho, sino tomar el río abajo [el Río Grande de Santiago] y ver lo que había, y assi fue por lo alto a *los llanos que entonces se llamaban de Guaxicar* y ahora La Magdalena, *gente de nación coana y distinta de la de Etzatlán*” (emphasis added; also see Baus de Czitrom 1982). The lineage name Tequani is mentioned in the “Relación del pueblo de Ameca” (Acuña 1988) for that pre-Hispanic settlement. In addition, the cognate *Tequal* is seen in many toponyms for this area, including the town of Tequila, the Volcán de Tequila, and the Sierra de Quila, among many others.

In addition, the two most important sixteenth-century maps for this area (the anonymous *Pintura del nuevo reino de Galicia*, ca. 1542 [fig. 1-1], and *Ortelius*, 1579 [fig. 1-2]) have been recently reanalyzed for their content concerning the ethnic organization around Etzatlán (Weigand and García de Weigand 1996). From these maps, we can see how important the Tecualme, Tequalme, and Tezole languages and ethnicities were all along both sides of the barranca of the Río Grande de Santiago in this zone. Certainly, the indigenous peoples of this region had no problem understanding each other, as the rapid spread of the revolutionary words of Tetzcatlipoca, starting with the Caxcanes and accompanying the beginning of the Rebelión de Nueva Galicia, clearly demonstrates.

In none of the languages native to this sector of the Occidente does the suffix *-tlán* appear. Among the equivalents for *-tlán* are possibly the suffixes *-míc* (such as in *Huaximíc*), *-tíc* (as in *Atlitíc* and *Zapotiltíc*), and *-jíc* or *-xíc* (as in *Ajijíc*). Clearly, these are dialectical variants of one another. The first has a western distribution. The second is found in this region and to the south. The last is found in the east. Fortunately, for the Etzatlán area we can actually see the transition from a native toponym to a Nahuatl one and then to a Spanish place-name. The case is San Juanito, Jalisco, the cabecera of the Municipio de San Juanito de Escobedo. The oldest and original name for San Juanito is *Atlitíc*. The Nahuatl name for the same place is *Atitlán*. In this case, the suffix *-tíc* was changed to *-tlán*, and the prefix *atli-* was simplified to *ati-*.

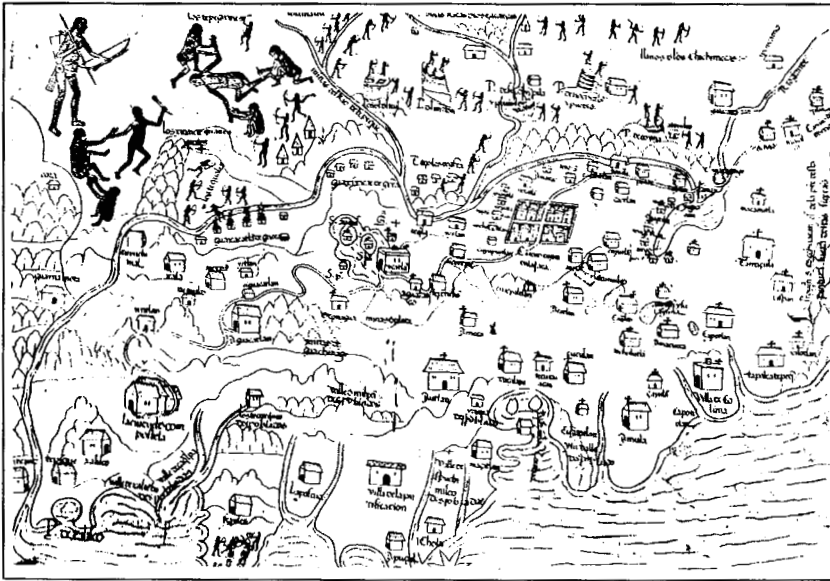


Figure 1-1. The *Pintura del nuevo reino de Galicia*, ca. 1542.

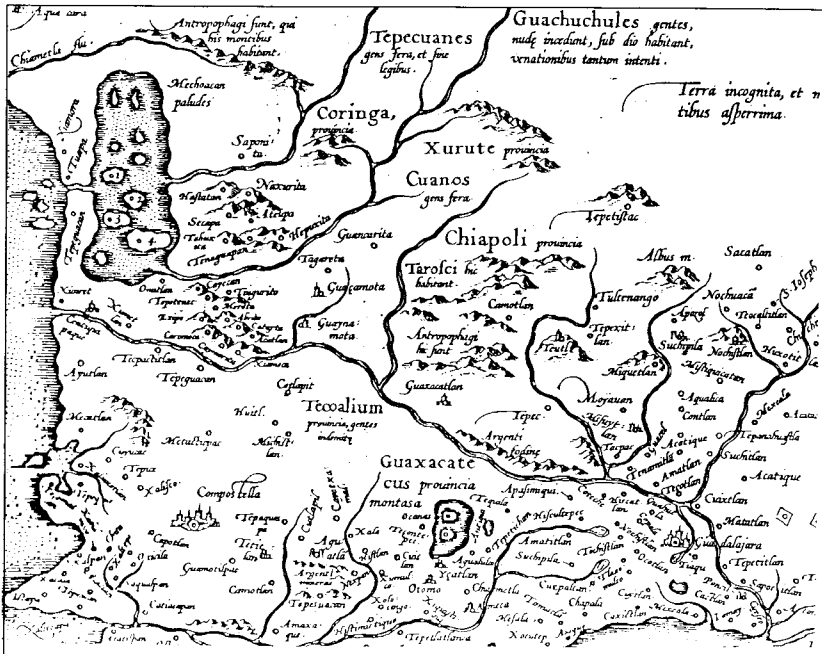


Figure 1-2. Detail of the *Ortelius* map, 1579.

San Marcos was originally Chistic, though the transition to Nahuatl is not recorded. Without a doubt, something very similar took place with Etzatlán. The simplest part is the obvious replacement by the Nahuatl *-tlán* of the native *-tíc*. To consider what the native equivalent of the prefix *etza-* was, however, we must examine how the toponym *Etzatlán* was first recorded by the Spanish.

The use of *etza-* is actually quite late. The first names given to Etzatlán are Icatlá, Icatlán, Yzatlán, Izatlán, and Ezatlán (see the aforementioned maps in figs. 1-1 and 1-2; Coría 1937:558–59; Tello 1968). In other words, *Etzatlán* as a toponym is later and obviously derives from some other prefix/suffix set. And *Etzatlán* is preceded by still another variant: *Itzatlán* or *Ytzatlán*. Thus, using the rules that we observed for the *Atlitíc–Atitlán* transition, we can postulate that the original native name for Etzatlán was probably Yzatic/Izatic (which are really one in the same) or Ycatic/Icatic.

The prefixes *iza-* or *ica-* may be intelligible through an examination of Nahuatl cognates—one of the few options left to us since we have no lexica for this area. Such an examination suggests the terms *iztatl* (salt), *itztli* (“obsidian,” also *navaja de obsidiana*), or *Iza/Itza* (a lineage name, such as seen in the hybrid Maya/Nahuatl toponym *Chichén Itzá*). The Itzá were apparently a great noble family of merchants and warriors who first entered the historical record during the Late Epiclassic and Early Postclassic periods (ca. AD 800–1000). Their purported origins were within the Toltec world of central Mexico and the Bajío. At first, the term *Itzá* was a lineage name, and only later did it become a toponym. This transformation, which is documented among the Maya, may also have transpired here in the Occidente, resulting in the Izatíc toponym. Thus, *Etzatlán* may mean “Lugar de los Itzá.” We should note that the phonetic elements /t/ and /z/ are somewhat redundant in Nahuatl, and one or the other is easily omitted in practical speech. Therefore, one or the other could have been discarded when translating from a western Uto-Aztecan language to Nahuatl. The phonetic element /c/ is extremely close, too, though representing a softer sound.

The most promising alternatives are related to the terms for obsidian (hence, “Lugar de Obsidiana” or “Navaja de Obsidiana”) or salt (hence, “Lugar de Sal”). While there are minor deposits of salt near Etzatlán (the closest are nearby El Amarillo, on the municipal boundary between Ahualulco and Teuchitlán, and El Salitre, in the Municipio

de San Martín Hidalgo), none were large enough or of adequate quality to have served as an inspiration for such a major toponym as Etzatlán, despite the phonetic similarities. However, salt is mentioned as an economic product of the Etzatlán region in the 1525 Neogallego census (Coría 1937:558–59). The salt consumed and distributed through trade by the Etzatlán polity probably came from the major saltworks in the Zacualco and Atoyác-Sayula basins (see Liot 2000) or, alternatively, from the Pacific coast, where saltworks are very numerous. Etzatlán was quite possibly a major transporter of salt within the pre-Hispanic Occidente, perhaps enough so as to have foreshadowed or influenced the formation of its toponym.

Obsidian and fine prismatic obsidian blades were major items for export from this area (Weigand 1993; Weigand and García de Weigand 1994). Atlitíc/Atitlán was a settlement that was tributary to Etzatlán when the Spanish arrived in the region. At this site, one of the greatest prismatic blade workshops of all Mesoamerica can be seen, and millions of blade fragments are still visible on the site's surface. The obsidian came from the mines near La Joya (Municipio de Magdalena), where 1,265 mine/quarry entrances exist. This obsidian has been found in archaeological strata dating from the Postclassic period (ca. AD 900–1520) from Guasave (Sinaloa) and the Durango area in the north all along the Pacific coast. Obsidian prismatic blades constituted the major economic export from this area within the overall Mesoamerican trade structure, attracting in return turquoise and elaborate polychrome ceramics, among many other commodities. The names for obsidian (which is possibly at the root of the prefix *atli-*) and salt along with the lineage name Itzá remain as the three most probable native terms that inspired the Nahuatl/Spanish variant *Etzatlán*, which we use today. These three possibilities conform more closely with the known pattern for toponyms throughout the rest of Mesoamerica, too, in which economically important regional products or attributes (such as salt and obsidian), prominent lineage names, and gods and calendrical dates are frequently featured.

The etymology of the toponym for Oconahua is equally fascinating. We know from both interviews and the *Ortelius* map (see fig. 1-2) that a very old name for Oconahua was Ocomo. Our interviews were conducted almost thirty years ago, when fragments of indigenous vocabulary could still be recalled among the pueblo's oldest inhabitants. In

addition to that term, local folklore has preserved a version of how the pueblo received the name it has nowadays. According to the story, when the eagle of the Aztecs approached a group of women washing clothing in an arroyo, they became frightened. Calling out among themselves, they decided to scare the eagle away, questioning only whether to do it “con piedras o con agua,” hence the toponym *Oconahua*. But the story’s mixture of the Spanish and indigenous languages makes it an unlikely historical explanation, however ingenious and charming.

The earliest Nahuatl toponym for Oconahua is *Ocotitlán* (Coría 1937:559). This toponym obviously derives from the Nahuatl word *ocoti* (resinous pine) as the prefix (*ocoti-*) and the suffix *-tlán*. Hence, the first Nahuatl toponym for Oconahua meant “Lugar de Pinos Aromáticos” or “Lugar de Astilla de Pino.” Clearly, the prefix *oco-* is what survives from these early toponyms (*Ocomo* and *Ocotitlán*), which are most likely cognate terms in different (but related) languages or dialects. What about the current suffix *-nahua*? In the testimony of *Guzmán v. Cortéz* (Coría 1937:559), the inhabitants of Oconahua were called “naguatatos.” This term, as we have seen, is not an accurate indication of language or ethnicity, except in an extremely general sense. In addition, it was not always used in an objective fashion and at times was akin to terms like *naco* or *indito*, though there is no overt indication of this type of attitude in the testimony. However, a simplification of *naguatatos* leaves us with the suffix in question: *-nahua*. Thus, *Oconahua* could have resulted from a combination of an ethnic term and either *Ocomo* or *Ocoti-*. *Oconahua*, as a Nahuatl toponym, would mean “Nahuas de los Pinos.” The original name was perhaps *Ocomotíc* or *Ocotíc*, if not *Ocomo* as the Ortelius map preserves, and thus a possible proper name.

Oconahua’s toponym, in all its variants, describes what must have been one of its major resources in the pre-Hispanic period: the products of pine trees per se (resins, wood, charcoal for paint and dye) and the pine forests in general. Of all the major Postclassic settlements of the Etzatlán region, Oconahua’s physical location is the most unusual. It is located higher off the valley floor, and hence further from the lake shores, than any other settlement. This special location may have allowed for the development of an economic specialization in pine forest products.

Etymology, however, is not the only data set at our disposal for researching the toponyms of this area. Iconographic analysis of

archaeological materials offers another independent approach. As is often the case in other areas, scholars do not agree on the etymology and iconography surrounding ancient Etzatlán. Of course, the iconographic data set is very incomplete due to a variety of factors, the most important being our lack of a large corpus of Postclassic motifs for this area. This situation is in itself due to the destruction of the database due to urbanization and massive looting in and around the ancient site of Etzatlán.

Despite these limitations, however, several important iconographic observations have been made. A set of ceramic stamp seals that has been recovered from the site over the past twenty years forms the most important material data set. In other areas of Mesoamerica, these seals often mark property with the name(s) of owners, lineages, or ceremonial sites by reference to the deities important at those sites. Figure 1-3 shows the impression of one such seal found in the Santa Clara component of the ancient settlement of Etzatlán (see fig. 1-4). Two skulls, or defleshed faces, flank and look outward from a clear glyphic element in the center of the design panel. The bottommost sector of this glyph reads “lugar de . . .” or “cerro de . . .” The three dots in the center of the glyph represent the number 3. The top of the glyph is a rendition of either the agave or of tules. Both plants are prominent in the area, though tules are nowadays fairly rare due to the desiccation of the Laguna de Magdalena. The glyph could thus be read as “Lugar de Tres Agaves/Tules.” The former seems the more likely because in the tradition of Mesoamerican iconography, tules are most often shown in bundles rather than individually.



Figure 1-3. An impression from a ceramic seal from the Santa Clara section of the ancient site of Etzatlán showing possible toponymic information.

The skulls on this ceramic stamp, if they represent defleshed faces, could reference a regional variant of the pan-Mesoamerican god Xipe Totec. If they represent skulls per se, then they might be the regional variant of Tetzcatlipoca (also almost pan-Mesoamerican in distribution, though, like Xipe Totec, with different names in different regions). The former is a grim symbol of fertility: the skin from human sacrificial victims was worn by priests until it rotted away, thus symbolizing the renewal of life after death and the dry season (Nicholson 1971). The obvious parallel in nature is the shedding of skin by snakes, especially the rattlesnake. The latter is also a symbol of fertility, but with a different and more cosmic implication: Tetzcatlipoca, as the counterpart to Quetzalcoatl, symbolized death, warfare, dryness, night, and the afterworld, thus representing the other half of nature's fertile and green realm of growth, life, and rainfall (Nicholson 1971).

The Spanish identified a regional variant of Tetzcatlipoca in this area during the crisis set off by the *Rebelión de Nueva Galicia*, though they most frequently just called him the *demonio* (Weigand and García de Weigand 1996). The symbolism surrounding a regional variant of Xipe Totec can be seen as early as the Late Formative period (El Arenal phase: 300 BC to AD 200). Ceramic figurines of priests from this time period occasionally show the symbolic wearing of flayed skins, with the priests' eyes and teeth showing through the cut-outs in the skin of the sacrificial victim.

Of course, the glyphic material on this ceramic stamp (as well as others) may refer only to a barrio of ancient Etzatlán, hence representing a lineage or corporate group rather than the entire settlement or polity. In this sense, the extant iconographic data set, while very incomplete and fragmentary, is not in conflict with the etymological material. Instead, the different data sets complement one another and thus expand our understanding of the social complexities of ancient Etzatlán. Within the larger settlements of ancient Mesoamerica, each barrio had its own ceremonial and economic life. It was in these social units that most people recognized their primary membership within the overall society (see Carrasco 1971).

Of course, many settlements have numerous toponyms, and the one chosen for formal histories is often a matter of chance or the prestige of the individual or group sponsoring a particular name or term. We need only look at San Juanito's interesting toponymic history to see

this process in action. The settlement has been known by many variants, some of which obviously only refer to part of the site, including Atlitíc, Atitlán, San Juan Atitlán, San Juan, San Juanito, La Otra Banda, Las Cuevas, and Antonio Escobedo.

Even though we cannot resolve with absolute certainty the etymology and linguistic heritage of ancient Etzatlán, we can describe the bare outlines of how the pre-Hispanic settlement appeared to the first Spanish who explored, and later conquered, this region. Let us return to an earlier question: why was such an important and monumental *templo/convento* constructed here in Etzatlán?

While the religious fervor of the Franciscans has to be considered, we should look to the physical, economic, and political character of the pre-Hispanic settlement to understand why such a structure would have been built. We should first consider the fact that the templo/convento in Etzatlán was designed to serve as a fortification in addition to its religious functions. These needs dictated, in great part, the type of architectural design and style that could be employed in such a multipurpose edifice. As it turned out, this choice was simplified by the Franciscan experience with the late Romanesque style for churches and the straightforward, but sophisticated, megaron style for palaestra *conventos*. Stylewise, the templo/convento complex in Etzatlán is certainly competent but rather uninspired, though later modifications and additions certainly improved its aesthetic qualities and appeal, as Arq. Gónzales and colleagues' (2000) study carefully documents.

The distinguishing quality of the templo/convento complex lies elsewhere than its style. The monumentality and scale of the complex, as originally conceived, set it apart from almost every other building project undertaken in early sixteenth-century Jalisco. While today this complex does not look that grandiose, it is the measure of the 1530s that counts. What it lacked in grace and beauty, it more than made up for with its massive walls and fortresslike revetments. It was one of the most impressive structures in western Mexico during much of the first century of Spanish rule. Such edifices actually were (and are) quite rare for the Occidente, though the structure of the first archbishopric at Compostela is another good example. Most of the churches and conventos dating from this period have been remodeled and reconstructed to such a degree that the original project is difficult to understand. This is not the case in Etzatlán, though of course many subsequent building

projects have indeed altered the original structure. However, its original format is still visible and intelligible.

Cortéz de San Buenaventura's expedition of 1525 was the first documented direct Spanish contact with Etzatlán. It is possible that even before his visit, but most certainly before that of Nuño de Guzmán (1530–1531), the European epidemic diseases had already made their presence felt in the region (Weigand 1993; Weigand and García de Weigand 1996; cf. Morín Tamayo 1992). The first Spanish in this area either accompanied or followed the epidemic that within the century had reduced the indigenous population thereabouts by almost 90 percent. Thus, the Spanish did not encounter pristine Mesoamerican societies in the Occidente. In addition, the pan-Mesoamerican trade structure had broken down, which had a major impact on this region's economy. When Tenochtitlan and Tzintzuntzan were conquered and destroyed, the carefully balanced commercial networks of ancient Mesoamerica completely collapsed (Weigand 1993; Weigand and García de Weigand 1996).

Given the social, cultural, and epidemiological catastrophe that was under way, we cannot regard even the earliest of the Spanish accounts of this region as accurately describing an entirely indigenous world. And unfortunately, even with that reservation, the accounts we do have are woefully brief and superficial. We do not have a *Florentine Codex* or a *Relación de Michoacán* for this region. One of the best early colonial documents for the Occidente, the *Lienzo de Tlaxcala* (1964), does not even discuss Etzatlán because it was not considered part of Guzmán's conquest, even though Guzmán spent part of a year there.

However, along with the two early maps that we have already briefly discussed we have another important document concerning the first Spanish observations in this area: the various testimonies collected into the *Guzmán v. Cortéz* litigation of the 1530s, which fortunately includes the 1525 Neogallego census material (Coría 1937, see especially pages 558–60 of the Cerezo/Coría section). This invaluable material cannot stand by itself, however, and therefore the archaeological data set is critical. From the outset, we should remember that the historical materials and the archaeological data set offer quite different kinds of information. While at times it seems as if these materials and data sets are contradictory, in reality they are complementary.

Within the materials collected for *Guzmán v. Cortéz* are brief descriptions of the three major centers that constituted the pre-Hispanic

polity of Etzatlán: Ezatlán (Etzatlán or Izatíc [?]), Atitlán (Atlitic), and Ocotitlán (Oconahua or Ocomo). The descriptions of these three centers are highly interesting but only partial and extremely brief. One important point to note is that the *visitador*, Gonzalo Cerezo, and his *escribano*, Diego de Coría, described these settlements using “a barrios,” rather than “de barrios.” The distinction is important because within a settlement organized “a barrios,” the barrios are not continuous with one another but rather more dispersed. A settlement described as “de barrios” is one wherein the barrios are continuous with one another. In addition, the officials only described the major barrio of each of these settlements, thus leaving us with a superficial impression of rather small sites. As we will see, this is not the same impression that one gets from examining the archaeological record or, for that matter, from other descriptions of these same settlements, such as those summarized by Tello (1968).

We should also note that Cerezo and Coría spent only three or four days visiting the entire Etzatlán region, thus averaging several settlements during each day of the visitation. While invaluable, the allotted time for their research could only allow for very cursory and sketchy observations. However, these do constitute the first descriptions of the settlements within the Etzatlán polity.

The following narratives are taken from the *Guzmán v. Cortéz* litigation:

1. Etzatlán: “El lunes VI de hebrero del dicho año [1525], antel señor Gonzalo Cerezo, vesitador, y en presencia de mi, el Escribano [Diego de Coría], vesito la *provincia e* pueblo de Ezatlán, donde hay una gran laguna dulce, de ques *señor Coyulan*, el cual dijo que tiene la cabecera que se llama Ezatlán C [100] casas, y visto e moderado por el dicho vesitador, le parecio que tiene CCC [300] *casas y seiscientos hombres*, el cual esta en la costa de la dicha laguna [the barrio where Coyulan resided], y parte de lo poblado en una ladera de unos cerros [other barrios] junto a la dicha laguna, *a barrios* [dispersed barrios], y esta entre mucha arboleda de frutas, las casas son muchas dellas las *paredes de piedra* y la cobertura de paja; *hay tiangués*. Fuele prequntado al dicho señor de que viven, dijo: que de *sal* y de maiz y pescado de la dicha laguna, y de algun

algodón; los mas dellos [los elites] son de la lengua de México [see previous discussion—not a reference to Nahuatl, but much more likely to Uto-Aztecan]. Hay en esta laguna muchas *canoas* muy bien hechas, y son de cañas y de enca; confinan con los *teules chichimecas* [Caxcanes—i.e., north of the Río Grande] por un cabo y por el otro, con la tierra de *Michuacán* [Tarascans or Purpechas—i.e., somewhere in the lake zones to the south], y *tenía guerra*, es gente muy pobre; puede tener esta laguna diez leguas de boxo y es hondable, esta esta cabecera seis leguas de Cocula” (Coría 1937:558, emphasis added).

1. San Juanito: “Vesito el dicho señor esta día un peñol [Isla de Las Cuevas, just northwest of the current site of San Juanito] que tiene en la dicha laguna, *muy poblado*, que es dice *Atitlán*, una legua de la cabecera que le cerca el aqua; es de media legua de boxo, el cual dijo que tiene LXX [70] casas, e visto por el dicho vesitador le parecio que tiene *doscientas y cincuenta casas e quinientos hombres*, y esta gente que esta en este peñol es de la cabecera que por miedo de las guerras se metieron dentro, y tienen *sus labranzas fuera en la tierra*; las casas deste peñol son las *paredes de piedra* y la cobertura de paja, hay ques a manera de los de Calna y las *pedras labradas*; esta este peñol dos tiros de ballesta [ca. 200–300 m] de la tierra firme, tratan de mucho pescado, los mas destos son *naguatatos* [not a reference to Nahuatl, but probably to Uto-Aztecan]” (Coría 1937:558–59, emphasis added).
1. Tezontepeque (while mentioned separately from Atitlán, nonetheless it is so close as to constitute a barrio of that larger settlement; it is most probably the site of Cerro Colorado, just across from the narrowest portion of the Laguna de Magdalena and opposite from Las Cuevas): “Vesito el dicho señor, que tiene un *pueblo pequeño* en la costa de la dicha laguna que se dice *Tezontepeque*, y tiene puesto *dos calpixques* [barrio leaders] que se dicen el uno Huichichillo y el otro Zule, los cuales dijeron que tienen XXX [30] casas, y visto y moderado por el dicho vesitador le parecio que tiene *sesenta casas y CXX [120] hombres*, esta una legua de la cabecera de Ezatlán, por la dicha laguna, y dos tiros de ballesta [200–300 m] del peñol de

Atitlán; tratan de pescado y maiz y algodón, son otomies [usually a euphemism for a completely unknown language—the presence of Otomi in this area of the Occidente is questioned in all recent historical linguistic studies (see Valiñas 1994)]” (Coría 1937:559, emphasis added).

1. Oconahua: “El siete de hebrero del dicho año, vesito el dicho vesitador el pueblo de Ocotitlán ques por si de la provincia de Ezatlán; llamase el señor Coyul[an], el cual dijo que tiene CLX [160] *casas* y CCLXXX [280] hombres; esta este pueblo en un valle de arboleda y esta poblado *a barrios* cerca de una laguna pequeña [Laguna Palo Verde], y esta una legua de la cabecera de Ezatlán; tiene *tiangués*. Fuele preguntado al dicho señor de que tratan, dijo: que de maíz e frixoles e *sal*; son los mas destos, *naguatatos* [probably a reference to Uto-Aztecán], visten de ropa de algodón y maguey” (Coría 1937:560, emphasis added).

The Cerezo-Coría account mentions even more briefly their visits to five other settlements, all smaller than the aforementioned ones, but all tributary directly to either Etzatlán (Tenyca and Tiazantleyco) or Oconahua (Atlexicayan, Atletotone, and Coyntequepaque). Recalling that Oconahua itself was tributary to Etzatlán, we see that six settlements were thus directly tributary to the rulers of Etzatlán. Three additional settlements paid tribute through Oconahua, for a total of nine.

The testimony contains much important information. First, Etzatlán is called a “provincia” and headed by a “señor.” This nomenclature means, without any doubt, that the area was territorially organized and directed by a political hierarchy. The polity was fairly large, as its neighbors were the Caxcanes to the north and the Purpéchas to the south, though mention of this frontier undoubtedly is a reference to the general limits of the zones being systematically raided from Michoacán, rather than Michoacán *per se*. Hence, this frontier was probably somewhere north of the Zacoalco lake zone. This territorial description conforms closely to the zone included within the early parochial entity dominated by Etzatlán (Weigand 1993), which reinforces the reliability of political information that the local leader, Coyulán, gave to Cerezo and Coría.

Frontiers in ancient Mesoamerica were not like those of today, as they were far more pervious and malleable. Therefore, this fairly large

territory may have been simply the maximum political and military expression that Etzatlán was capable of attaining, rather the daily state of affairs. This situation is especially probable given the fact that no major settlements outside the Etzatlán Valley are mentioned as tributary to that site in the Cerezo-Coría statement. The polity was probably organized at the level of an advanced *cacicazgo* or segmentary state. But in either case, it is unlikely that its direct and day-to-day control extended beyond the Ahualulco Valley to the east or into the Ameca Valley to the south. The Ameca Valley had a quite viable political organization of its own and was clearly not subject to Etzatlán or anyone else, despite being periodically raided by the Purépechas (“Relación del pueblo de Ameca,” Acuña 1988; Weigand and García de Weigand 1996).

The reference to Etzatlán’s northern frontier is quite reasonable given that the Caxcanes had already arrived by the time of the conquest to the nearby barranca of the Río Grande de Santiago (Tello 1968; Weigand and García de Weigand 1996). The western frontier must have been located somewhere in the mountains beyond San Marcos, as Etzatlán is mentioned by Cerezo and Coría as directly bordering upon the important Ahuacatlán polity.

As mentioned, Etzatlán was organized *a barrios*, which means that the component segments of the settlement had green expanses between the concentrations of buildings. This configuration is exactly what we have documented during the course of our field studies (fig. 1-4). By taking advantage of work on water lines, sewer excavations, and, more recently, the installation of telephone cables, we have examined profiles of ancient structures on almost every street in the contemporary town at least once. In addition, we have examined the open lots within the settlement and the fields surrounding Etzatlán. The materials that date from before the Postclassic period are largely, but not exclusively, confined to the Puerta de Veracruz zone within the site, where a small circular ceremonial circle was located and from where a shaft tomb was entered and looted (fig. 1-5). Other observations concerning the Formative and Classic period sites and settlement systems can be found in Weigand 1993 and Weigand and García de Weigand 1996. The results of these surveys have supported the observations made by Cerezo and Coría, as well as considerably amplified them, concerning the Late Postclassic settlement.

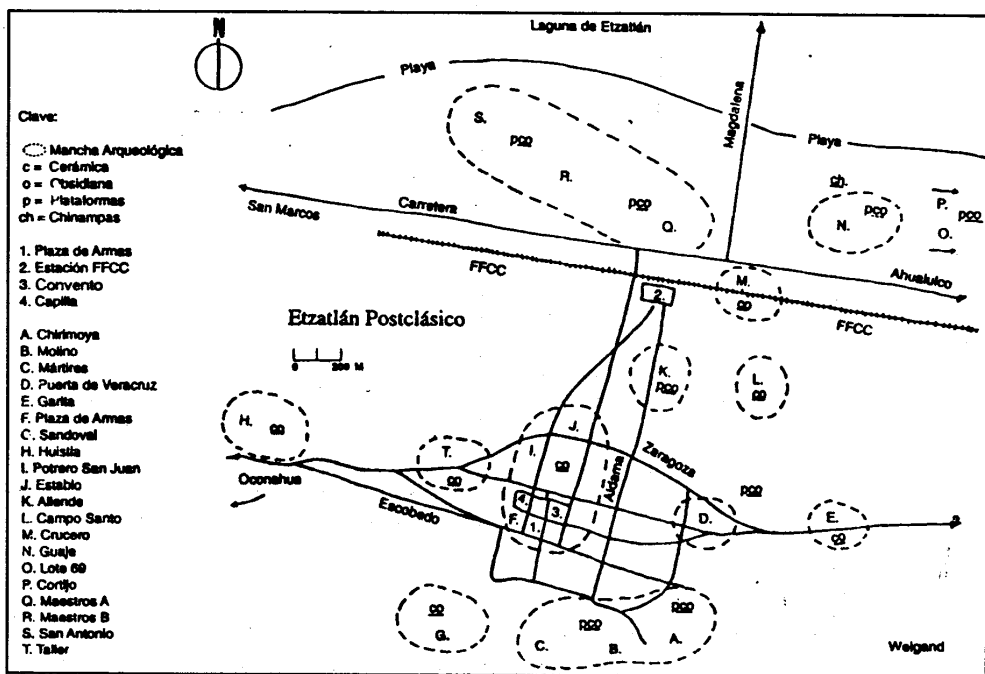


Figure 1-4. The Late Postclassic settlement of Etzatlán showing its organization "a barrios."

Keeping in mind that the Laguna de Magdalena's playa was just north of the current Ahualulco-San Marcos highway, we can provisionally locate some of the barrios that Cerezo and Coría mention. The hillside barrios obviously correspond to the Santa Clara, Chirimoya, and Cruz de Sandoval areas but may also include La Garita and Huistla. The main barrio, which had three hundred houses, some of them made from stone, is either the Plaza de Armas area, where excavations have always turned up both architecture and artifacts, or the Rancho San Antonio, Colonia de los Maestros, and Guaje zone. All of these areas have been heavily impacted by later construction activity, erosion, and systematic looting, so very little architecture is preserved. The best preserved areas are the terrace systems still visible at Chirimoya and within the Arroyo de Santa Clara. Low mounds near Rancho San Antonio used to be visible though today have all but disappeared.

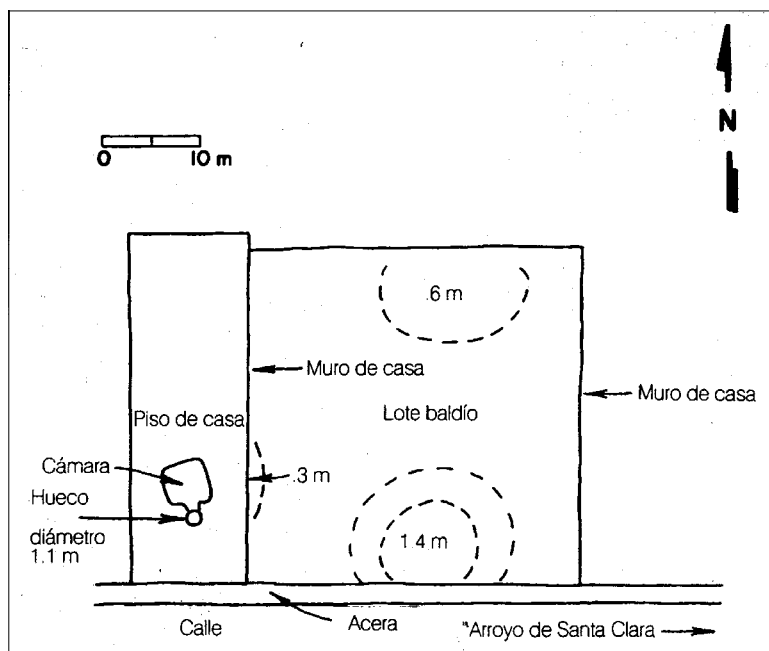


Figure 1-5. A circular structure and shaft tomb from the Puerto de Veracruz section of Etzatlán.

The contemporary town of Etzatlán covers approximately 400 ha. The ancient settlement covered about 600 ha, though within that area were many green spaces. The settlement, as the term *a barrios* implies, was only seminucleated (see fig. 1-4) and contained numerous open spaces. If the largest barrio had three hundred houses and six hundred men, then, by extrapolation, the total settlement may have had over fifteen hundred houses and three thousand men. This extrapolation is not just an abstract exercise in mathematics because, using the best preserved sections of the ancient site, we can calculate an average of three residential compounds per hectare. Some compounds had more than one household, or platform, in addition. Multiplying those figures against the archaeological deposits' extent gives us roughly the same number of houses.

The rough figure of three thousand men helps us calculate the population from still another perspective. In other areas of Mesoamerica, we know that each household contained, on average, two adult men.

A “man” in these figures can mean (1) an individual of sufficient age to be a warrior who may not have his own family and/or (2) an individual within a larger household who is responsible for tribute payment and who does have his own family. Either way, composite, or extended, families are also indicated by this type of demographic figure. Splitting the difference between the possibilities and using five as the average size of each family, the total population may have been around ten thousand inhabitants. Tello (1968) gives a higher figure for Etzatlán, though one must use several sections of his text to arrive at his estimate (for more detail concerning this argument, see Weigand 1993; Weigand and García de Weigand 1996).

In the preserved areas of the ancient settlement, we can see that the residential compounds were arranged as one, two, or, less frequently, three modest platforms facing a common open patio. In the adobe pits at the edge of the Arroyo de Santa Clara, several of these platforms have been profiled (fig. 1-6 shows an example). About twenty others have been seen in profile within the aforementioned excavations inside the current town. The platforms average 10–14 by 6–10 m to a side and are seldom more than 60–80 cm high. The terraces still visible at Chirimoya and Santa Clara average around 15–20 m wide and 10–15 m deep. They are seldom straight and more often form lunates protruding from the hillside. This configuration, of course, may represent erosion more than original design, as some of terraces preserved at Atitlán are rectangular. The terrace edges are, most frequently, all that are preserved of the patio/platform complexes. Slightly larger platforms were found in the Rancho San Antonio and Colonia de los Maestros area, and some of these may represent small ceremonial structures.

No ceremonial or administrative structures of any kind were mentioned in the Cerezo-Coría account, but this does not mean that they did not exist. In fact, we can see them all over the landscape in the general Etzatlán zone. Some are truly monumental, such as the Palacio de Ocomo, or El Mirador, near San Juanito (see following). A settlement of the size and importance of ancient Etzatlán would have had both types of compounds. The best archaeological indications for Etzatlán’s ceremonial center have been found in the Plaza de Armas area. The artificial platform and terrace upon which the Templo/Convento de la Concepción is situated most likely represents the remnants of a

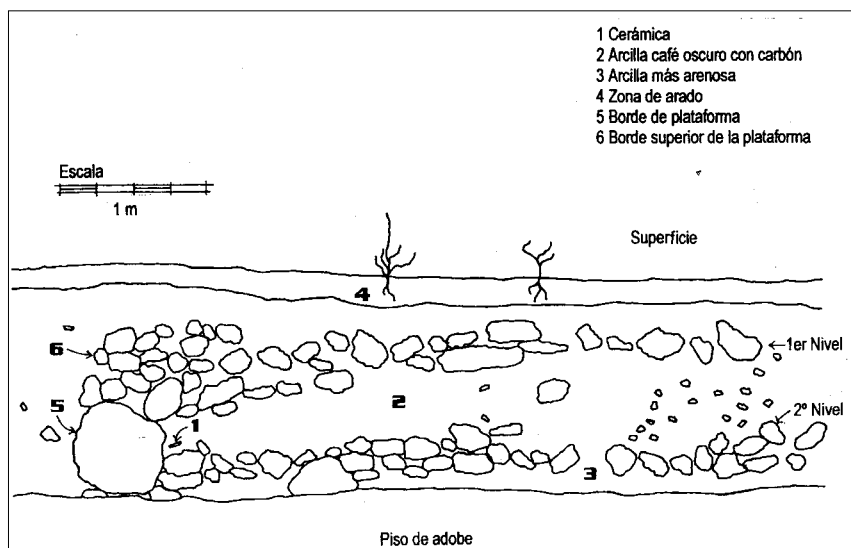


Figure 1-6. A profile of a habitation platform from the Arroyo de Santa Clara section of Etzatlán.

pre-Hispanic platform/pyramid complex, and the pattern of placing a European religious building on top of a pre-Hispanic structure is seen all over Mexico. The various excavations around the edges of the Plaza de Armas (the most recent by Telmex) have also produced numerous artifacts and clear architectural remains. Aside from figurines, ceramic vessels, pieces of obsidian, stone beads, shell objects, and ground stone implements, excavations in the overall Plaza de Armas area have turned up segments of walls over many years. These walls are very poorly preserved and are difficult to interpret. Obviously, this general zone of the site had been churned up on many occasions prior to becoming the landscape we see there today; for example, a system of brick-lined cisterns and underground aqueducts now passes through the central sections of the contemporary settlement. In addition, the Plaza de Armas and templo/convento complex is at the center of one of the largest concentrations of pre-Hispanic materials noted to date for Etzatlán.

The most probable pattern for the major ceremonial complex at Etzatlán can be seen at two nearby sites: Portezuelo (Municipio de Ameca; figs. 1-7 and 1-8; Weigand and García de Weigand 1996) and

Techaluta (Municipio de Techaluta; figs. 1-9 and 1-10), where well-preserved compounds still can be examined. At all four of these precincts, as is indicated at Etzatlán, a pyramid faces across a sunken patio, most frequently containing an altar, to a platform usually on the patio's exact opposite side. At Techaluta, these pyramids were stone faced and terraced, as were the patios and platforms around the banquette. Adjacent patios and structures usually accompany buildings like those found at Portezuelo and Techaluta. The stone facing at Techaluta was very finely worked and quite elegant and very similar to that which originally covered the Palacio de Ocomó, within Oconahua, and the buildings on the citadel of Las Cuevas/Atitlán. The stone facing at Portezuelo is no longer visible, though local informants have mentioned its existence. We have noted, in addition, the presence of fine tabular *canteras* (worked stones) at four sections of the ancient Etzatlán settlement: Santa Clara, Chirimoya, Crucero, and Calle Allende.

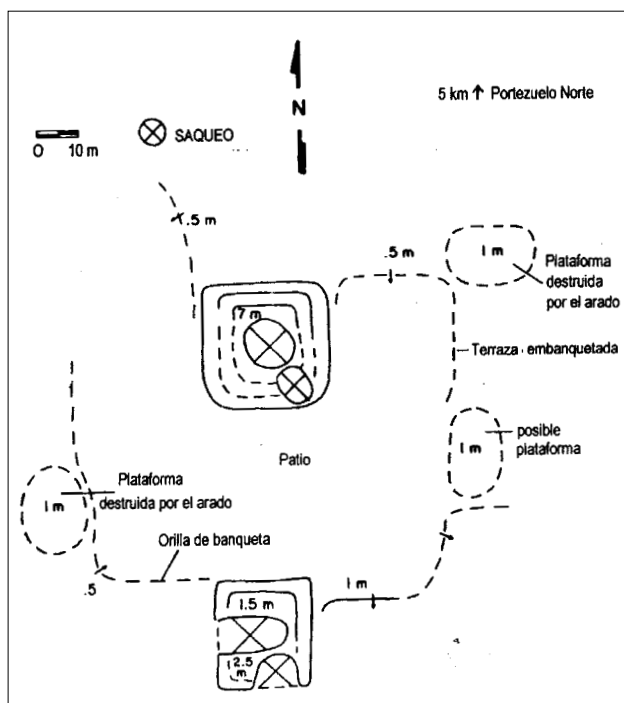


Figure 1-7. The Portezuelo Sur complex near Ameca.

Figure 1-8. The Portezuelo Norte complex near Ameca.

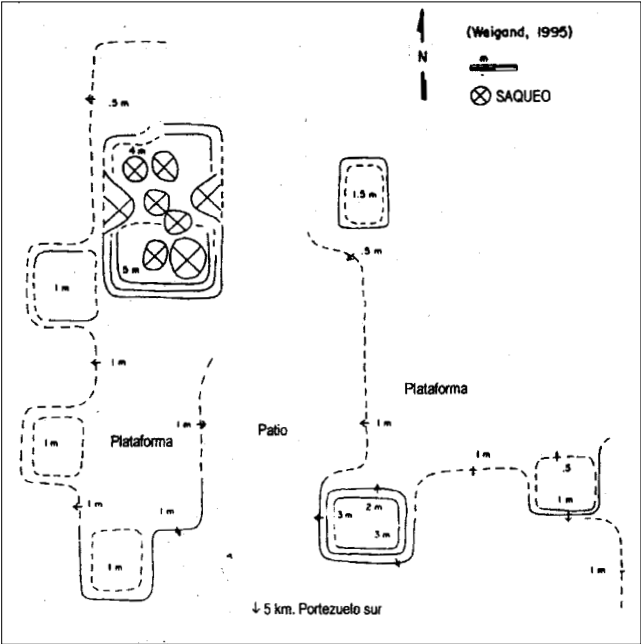
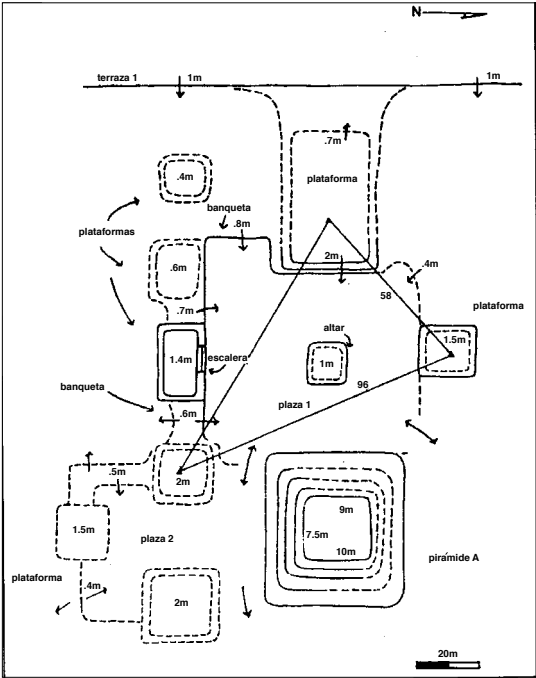


Figure 1-9. The Techaluta "A" complex, Municipio de Techaluta.



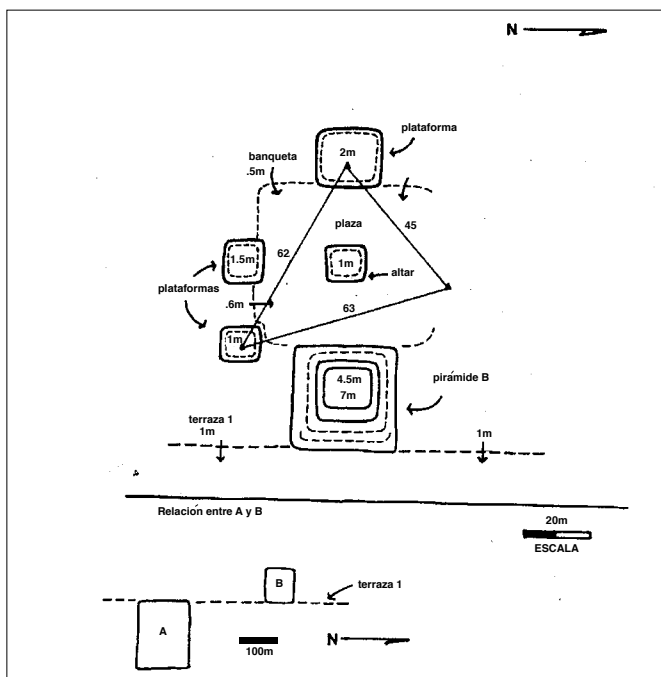


Figure 1-10. The Techaluta "B" complex, Municipio de Techaluta.

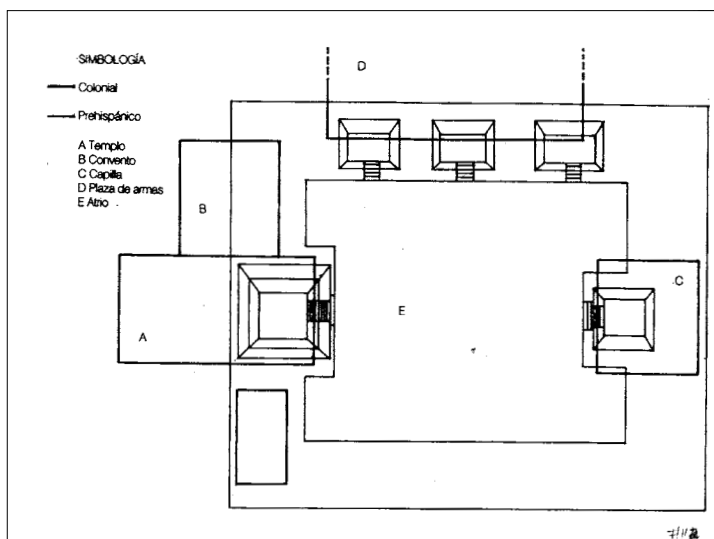


Figure 1-11. An interpretative reconstruction of Etzatlán's pre-Hispanic ceremonial area near the Convento and Plaza de Armas.

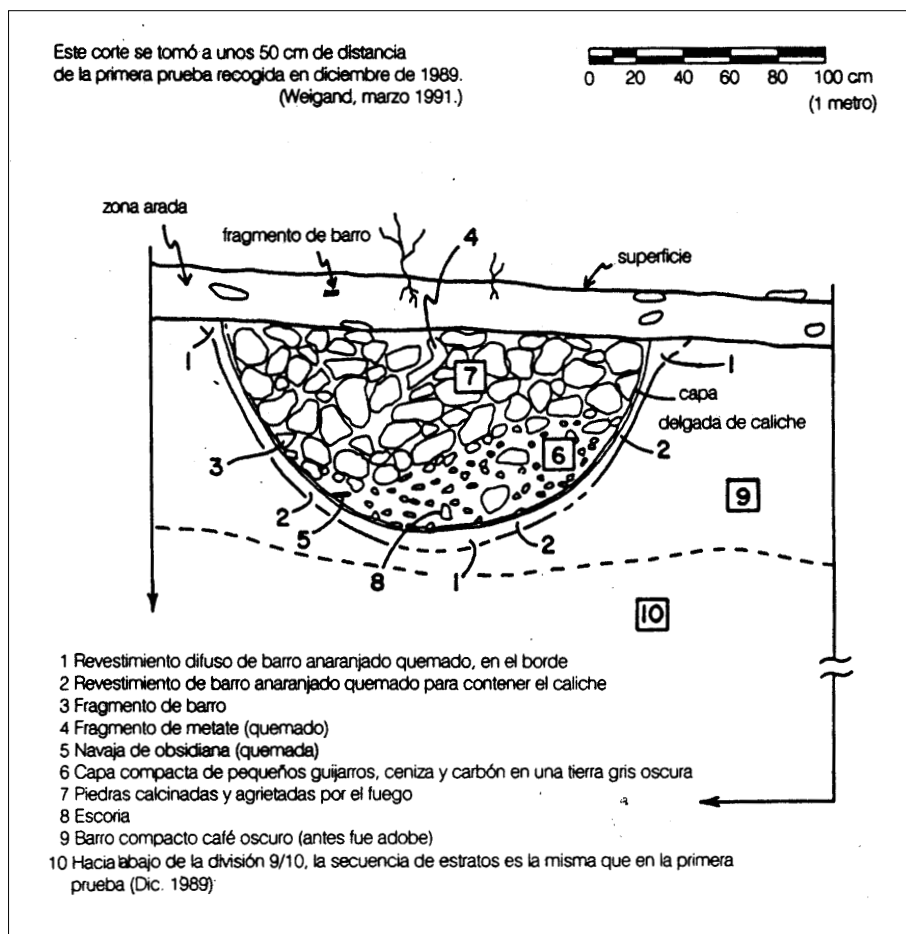
While arguing from analogy has its limitations, Etzatlán's importance within the region was certainly marked by architecture of some sort. The Spanish made very little detailed mention of any type of architecture for this entire region and did not even note the extremely monumental buildings, such as the Palacio de Ocomo. Therefore, Etzatlán is not the only site suffering from an incomplete description. Figure 1-11 offers a possible reconstruction of the ancient ceremonial center of Etzatlán, as the Spanish may have found it, based upon the excavations that have occurred over many years in the Plaza de Armas area of the town and the analogies with nearby architectural pyramid/patio complexes that have been preserved.

Large cooking and storage pits have also been seen in profile all around Etzatlán. Figures 1-12 to 1-15 show examples from the Arroyo de Santa Clara. Most of the pits in this area were clay lined and filled with fire-cracked rock, carbon, and, at times, burnt bone, along with occasional ceramic and obsidian artifacts. Burnt wood from the pit was ¹⁴C dated at around AD 1465 (see fig. 1-12). The clay linings were heavily burned, thus indicating that these pits were utilized as cooking ovens. The pits throughout the town are most frequently associated directly with the platforms but are found to the sides of patios rather than in the patios themselves. Some of these pits were reused for human burials. Clearly, these burials were not prestigious, as they were seldom accompanied by offerings. The remains are always in a flexed position, in order to take advantage of the pit's prior design. One pit contained the burial, or disposal, of a small dog.

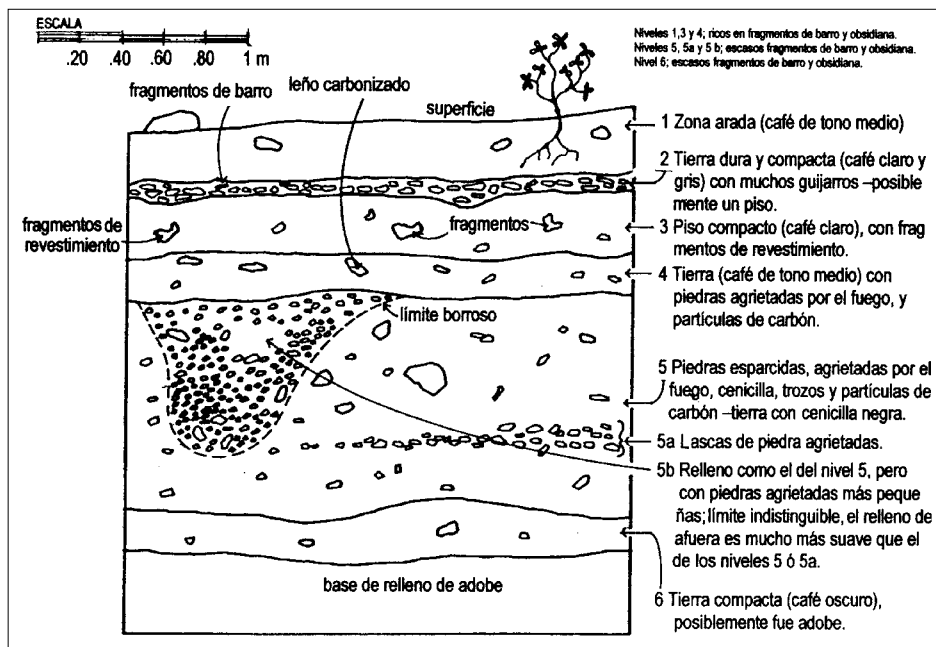
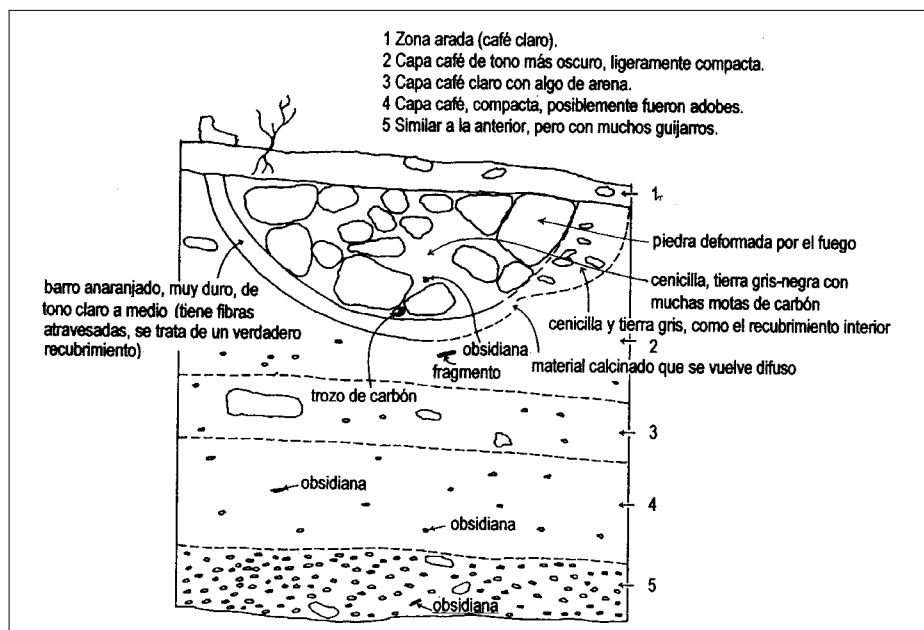
The ceramics from all the Postclassic areas of the Etzatlán settlement conform very closely with the published description from the Huistla section of the settlement researched by Glassow (1967). Aside from the elegant imported polychrome vessels (especially from the coast of Nayarit, the Chapala basin, and the Autlan area), the Huistla Polychrome series, composed of three subtypes, is well represented over the entire extent of the ancient settlement. The most elegant of the subtypes has three colors (black, white, dark red) in fairly unelaborated lineal designs (which occasionally are outlined by incised lines) over a lighter red or orange slip. These vessels are tripods, and the feet are often molded. The molded faces look like eagles or serpents. In fact, many of the Huistla Polychrome vessel bodies were mold made. Very often, the tripod feet are hollow and were made into rattles. The interiors of the

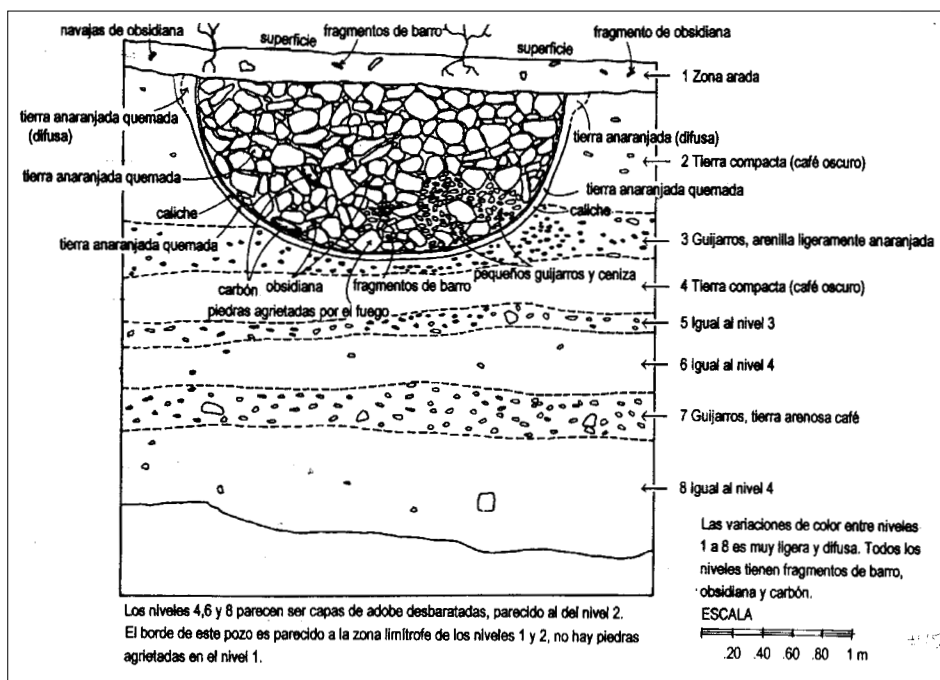
tripods have roughly drawn *molcajete* crosshatches. This design represents the ceramic style that characterized the Etzatlán area, and obviously much of it was made within or very near the settlement.

The most common ceramic forms, however, are ollas and *cantaros* of all sizes, that is, cooking and storage vessels. These are usually orange, though some have rough red slips and and/or smudges of red paint. Polished black and red vessels, some with incised designs, also are found in quantity.



Figures 1-12-15. Profiles of cooking, storage, and trash pits from the Santa Clara area of Etzatlán.





The so-called Tula-Mazapan figurines are found in the Early Post-classic sequence at and near Etzatlán, though they were no longer being made by the time that the Europeans arrived. They, too, are mold made and come in three basic sizes. These figurines are flat and tabular in shape. They wear elaborate headdresses that frame faces superficially similar to the visages on the monumental statues found at Tula (Hidalgo). Despite the name, the Tula-Mazapan figurines were locally made and indeed are found with far more frequency in this area than anywhere else.

It is interesting to note that obsidian is not mentioned in the Cerezo-Coría document, nor are the copper, silver, and quartz crystal mines of the Sierra de Ameca. Obsidian and quartz crystals were apparently of little interest to the Spanish and barely receive any mention at all in any subsequent documents either. But this disregard undoubtedly reflects a European prejudice concerning stone versus metal and therefore should not be used to downplay the importance of obsidian for

the region's ancient economy. The area's metal mines, however, were recorded by about 1542 on the *Pintura del nuevo reino de Galicia* (see fig. 1-1). Ancient quarries and mines can still be seen in the Sierra de Ameca and its foothills, though most of these have been reworked by colonial and contemporary miners in the ongoing quest for mineral wealth. Quartz crystals were also mined in this area, and several of these quarries are impressive.

Fish, salt, and cotton are mentioned as important economic products of the area or as at least passing through the zone. The existence of a *tianguis*, or market, in both Etzatlán and Oconahua is a significant marker of the importance of these sites in the regional settlement and economic system. The *tianguis* may also reflect the importance of long-distance trade in which Etzatlán was a participant.

Elegant polychrome ceramics from the Nayarit coast, worked shell artifacts, shell trumpets, turquoise artifacts, other semiprecious stones from both the region and farther away, copper artifacts, probably feathers, most certainly dried or powdered fish and fowl meats, textiles, plus other commodities and exotica were all present at the site, belying somewhat the Cerezo-Coría comment about Etzatlán's poverty. By the time they had reached Etzatlán, the Spanish reputation for looting was already quite well known. It would only be natural for Coyulán to plead poverty in hopes of avoiding a sacking. While the settlement was not sacked by the Cortéz de San Buenaventura group, it was by Guzmán's. The settlement, judging from the artifact record, was wealthy and well connected within the overall Mesoamerican trade structure. Its size; excellent resource profiles in arable land, water, and mineral and lithic resources; and its territorial extent gave the settlement a relative cultural wealth that made it a leader in the non-Tarascan and non-Caxcan Occidente.

Other features of interest in our consideration of the character of the ancient settlement include poorly preserved fragments of small irrigation canals, which appear to date from the Postclassic period, near the Rancho el Guaje, visible in the unplowed forest cover there and at the edge of the beach of the Laguna de Magdalena. The configuration is roughly rectilinear. Huistla Polychrome ceramics, along with other later pottery types, are found within this area as well. Within this possible chinampa zone are the fragments of one of Etzatlán's ancient canoe ports, though the stone from the ramps has been quarried in recent times. Long stone-surfaced ramps extended from the higher part of the

playa to the deeper parts, thus offering a way to launch and board without becoming stuck in the muddy edge. The ramps that used to be visible at El Guaje were morphologically the same as those from El Relíz, near San Juanito (see following), and hence very likely pre-Hispanic, at least in origin. The ramps at El Guaje, though, were clearly reused and remodeled during the colonial and contemporary periods, while those at El Relíz never were.

The fact that Oconahua had three of its own tributary settlements without doubt indicates that it was a polity in its own right prior to its incorporation into Etzatlán's. Even after its incorporation, Oconahua retained its social and political identity. As we have seen, the presence at Oconahua of one of the most monumental *tecpan*/palace structures for all Postclassic Mesoamerica strongly supports this postulate (fig. 1-16). This huge structure is one of many archaeological indications of a sizable and very important pre-Hispanic settlement at Oconahua. Excavations at the *tecpan* are currently under way and showing finely built stone walls, a magnificent stairway, slag, and ceramics that largely date from the Epiclassic through the Early Postclassic periods. The building measures approximately 125 m to a side, enclosing a plazuela that measures 60 by 70 m. The highest preserved point measures to 6 m. An external plaza to the north measures 50 by 180 m and contained a series of stelae. These stelae had glyphs, as seen on the fragment that survived the Franciscan destruction around one hundred years ago. The settlement history at Oconahua extends back into the Formative and Classic periods as well. A poorly preserved circular ceremonial complex from those periods is located in the adjacent Potrero Grande.

Our examination of water and sewer lines in Oconahua has not been systematic, but those from the central part of the pueblo and the northern sector contain ample indications of a large archaeological deposit. Aside from these indications, the Cerro de la Campana, on a hilltop above Oconahua to the north, is covered by a dispersed habitation area called Cacalotlán. Again, this Nahuatl term, meaning "Lugar de Ruidos/Resonados," obviously is the base for the current Spanish name. This barrio of ancient Oconahua is characterized by freestanding platforms and terraces (fig. 1-17). The largest platform is near the Piedra Campana, measures about 50 by 40 m, and stands 2 m tall at the highest point. Most platforms are smaller than this one. In the most concentrated area, the distance between them averages 30–40 m. Several

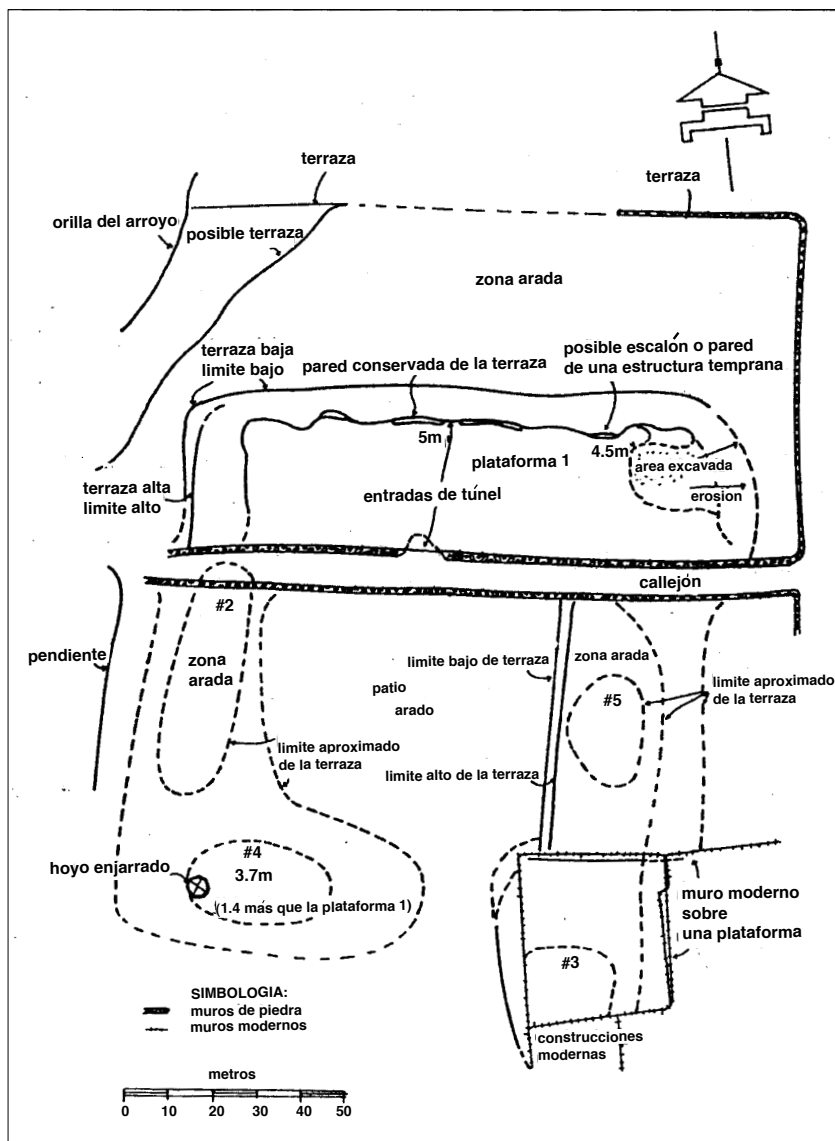


Figure 1-16. The Palacio de Ocomo, Oconahua.

possible plazas and patios are visible, but in general there appears to be little regularity in the placement of the platforms and terraces, except in compound clusters. No pyramids were built, except the structure covered by the colonial Templo de San Miguel Arcangel.

The major structure of the settlement is the aforementioned Palacio de Ocomo (see fig. 1-16). Recently protected by a *patronato* signed by a very large group of Oconahua's citizens and registered with both the Instituto Nacional de Antropología e Historia and the Secretaría de Cultura del Estado de Jalisco, this structure is one of the jewels of Mesoamerican Postclassic archaeology.

Buildings of this style are rare in the Occidente, but the Ocomo structure is unique in its monumentality. Several recent house constructions on the structure's eastern side, plus an old tunnel that had been dug through the northern platform, have allowed us to describe at least two major building phases for this structure. The earliest phase produced orange annular-base vessels plus other ceramic forms, such as "Puebla-Mixteca" red, orange, black, and white polychromes from the Autlan and Chapala areas, which are apparently indicative of the Epiclassic or Early Postclassic periods. The first construction level consists of rock terraces over an adobe mud and rock rubble fill, though the fill does not contain much rock. A possible floor is visible atop this first level, which is 1.5 to 2 m beneath the current surface. Above this level are a number of other minor construction phases that are marked as levels in the abode-like fill but are not rock covered. Some of these soil levels appear to have been burned. Several are capped with a rough caliche cement. While a total of two major building phases are suggested, the last phase had at least four subphases.

Originally, this structure was footed, around both its outside perimeter and its interior face, by large well-carved *cantera* blocks, some measuring 40 to 50 cm to a side, though most are smaller. Many of these have been quarried and used for more modern buildings. Some of these blocks were used for a bridge abutment on the east edge of the pueblo as well as for some of the foundation courses for the primary school located across the street from the Templo de San Miguel Arcangel.

The exterior surfaces of the platforms that compose the tecpan were terraced and covered with a fine *cantera* veneer. Some of the larger pieces of this veneer were carved with geometric motifs and/or possible glyphs, and fragments of these still exist in the area. The veneer,

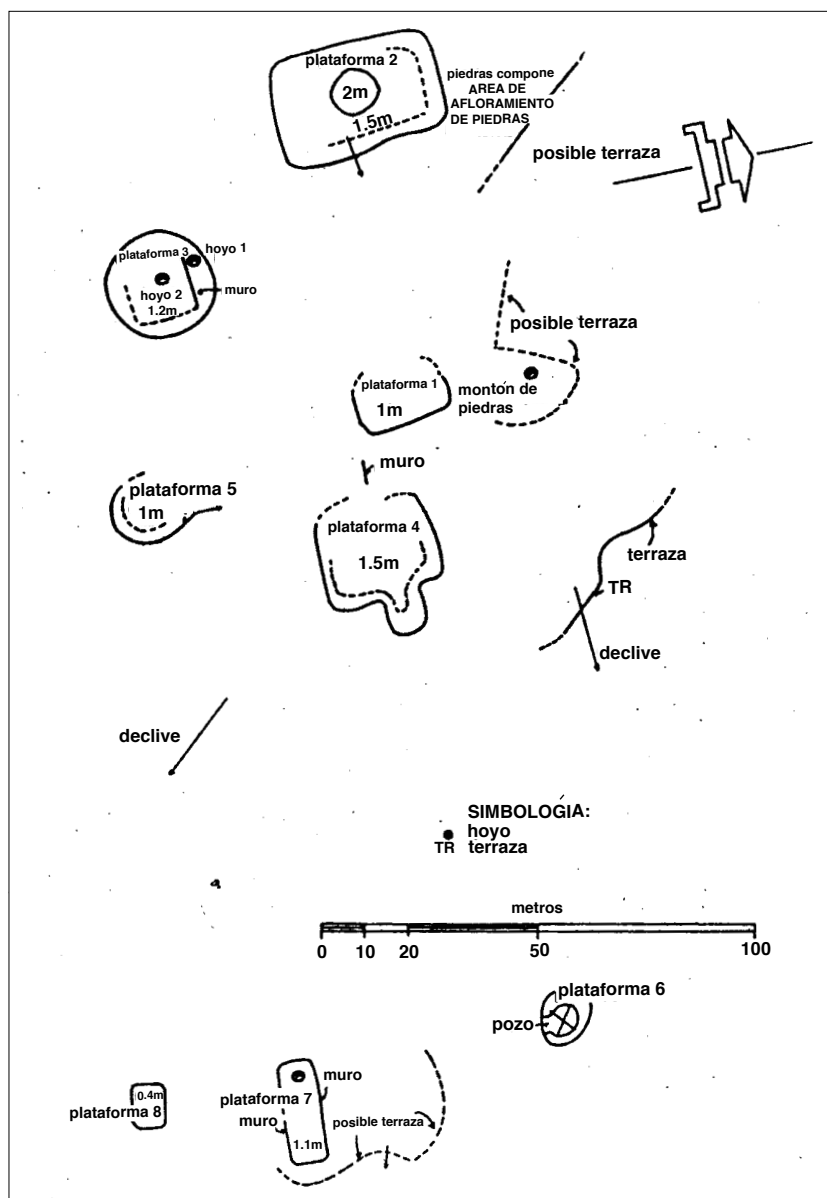


Figure 1-17. A map of a section of the Cacalotlán area of the Postclassic Oconahua settlement.

however, has largely disappeared, though some of it can be seen on part of the aforementioned church, along with many other contemporary structures within the pueblo. Given its size, plus the elaborate and elegant finish, this structure must have been a most impressive and important monument within the pre-Hispanic social order of the region.

The tecpan was constructed on a sloping surface, so its northern side is the highest. The southern side is very eroded and has suffered from the presence of some contemporary housing. The large gap along this side clearly constituted the entrance to the interior plazuela of the building. Both the flanking platforms (to the east and west) have also been impacted, though from different postoccupational processes: plowing to the west and house-building to the east. The house-building actually has helped us define the western platform, as the back walls of the lots have preserved virtually intact the interior/plazuela platform face. These lateral platforms were rather narrow, probably measuring about 25 m in width. The northern platform is separated nowadays from the rest of the complex by an alley (*callejón*) that was excavated into the northwest and northeast corners of the plazuela a number of decades ago.

The northern platform is the largest component feature of the entire structure, measuring 40 m wide. It appears to have a ramplike structure on its exterior northwestern corner as well. This platform alone contains approximately 20,000 m³ of construction fill, composed largely, as mentioned, of adobe-like prepared earth and some rock. The other platforms, all together, add around another 12,000–15,000 m³ of fill, for a total of around 30,000–35,000 m³.

The exterior, northern edge of the structure was also built up. As mentioned, this area is about 180 by 50 m, about 2 m tall at the highest preserved point, and appears to have constituted an exterior plaza. The oldest inhabitants of the pueblo, interviewed around thirty years ago, remembered being told by their antecedents that four or five great stone statues once stood in an east–west line in this area. Four modest elevations can still be seen, and small carved stone fragments from the structure are kept in local collections. The exterior plaza perhaps adds another 6,000 m³ of construction fill to the complex, though its northwestern corner has eroded away.

On a number of occasions, we (Weigand 1993, 1996; Weigand and García de Weigand 1996) have drawn parallels between this archaeological complex and the Palacio de Quinatzín. The latter palace is

illustrated in plan in the *Códice de Quinatzín* and originally was a Late Postclassic structure located in or near Texcoco (fig. 1-18). The Spanish overwrote the Nahuatl drawings and glyphs, thus giving us a fairly exact interpretation of the functions carried on within that palace. The first thing that we should note, however, is the complete congruence of morphology between the structure at Oconahua and that in the *códice*. Both are semiclosed U-shaped buildings with interior plazuelas and the largest and most important platform located directly opposite the narrow and relatively modest entranceway. This congruence between the two structures cannot be coincidence, based on what else we know about the Mesoamerican styles of architecture during the Postclassic.

Extrapolating from the *Códice de Quinatzín*, we propose that the northern platform would have been the actual office of the highest dignitary of the building, who would have been flanked by his most important aides. The lateral platforms constituted offices, each one closely identified by function in the *códice*. The entranceway was guarded by soldiers, and the plazuela was filled with people engaged in a wide variety of administrative duties. The building was thus an administrative, not a religious, one. The fact that Oconahua and the Etzatlán area could have had at least one of these structures is another indicator of the region's general importance.

The other major architectural complex within Etzatlán's pre-Hispanic polity (as defined in the Cerezo-Coría testimony) is located within and near San Juanito. While the aforementioned testimony did not characterize Atlitíc/Atitlán as "a barrios," this community settlement pattern is clearly what we see on the ground. The most poorly preserved section of this ancient settlement is found within the current pueblo, between the plaza and the hillock to its south. Within the solars of two houses in this area are the poorly preserved remnants of a pyramid of modest proportions. Around half of it has been excavated away in order to level the terrain within the lot. Thus, its exact measurements are very difficult to reliably calculate. We estimate that the structure was probably at least 25–30 m² and 3–4 m high. Subsoil excavations in this section of the pueblo always produce a wide variety of artifacts, but we have never seen a collection from this area. Given the square or rectangular configuration of the badly damaged pyramid, however, it undoubtedly dates to the Postclassic period. No map has ever been made of this structure.

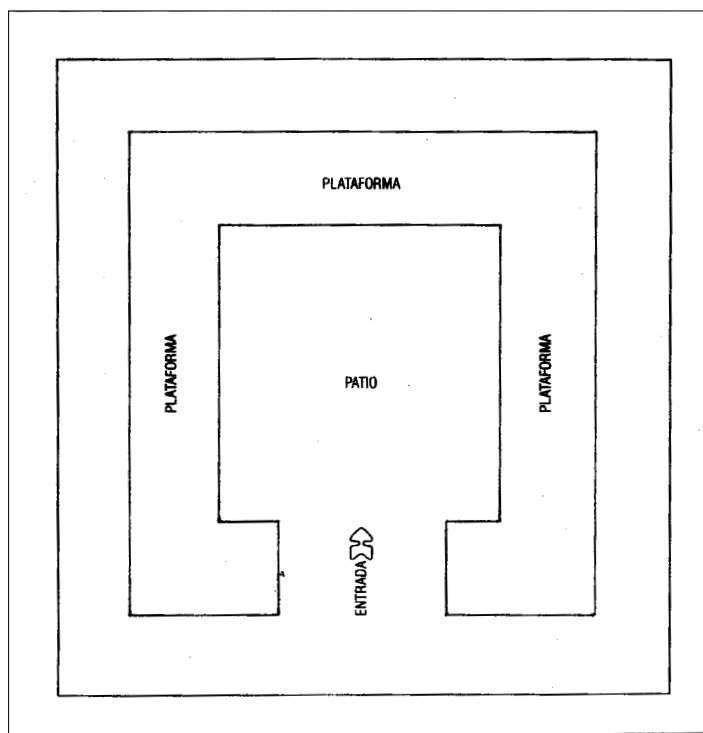


Figure 1-18. The Palacio de Quinatzín from the *Quinatzín Codex*, Texcoco.

Better preserved sections of the ancient settlement are found at Las Cuevas (fig. 1-19), El Miradór (fig. 1-20), and El Relíz (fig. 1-21). The barrio of Las Cuevas was obviously the most important to the settlement. It is the only one mentioned in the Cerezo-Coría account (Coría 1937:558–59), though the village called Tezontepeque may have formed another barrio of this settlement located on the nearby mainland. The Las Cuevas barrio was located on the island *peñol* (hill) northwest of the contemporary pueblo and was judged important enough to merit the construction of a Catholic *capilla* on its summit. This capilla, along with the pre-Hispanic structures of the citadel, is in ruins and has suffered from an enormous amount of looting. The last surviving large stone block from the capilla's altar was carried off by looters during the mid-1990s.



Figure 1-19. The citadel of the Las Cuevas/Atitlán section of the San Juanito ruin.

The original citadel straddled a natural saddle between the two uppermost points of the volcanic cinder cone that constitutes the island. The saddle was terraced and elevated in order to construct a plaza. Several modest platforms, the terraced plaza, and a complex of rooms are the most prominent features remaining from the pre-Hispanic component, along with a possible ball court on a lower terrace just to the south. The buildings atop the cinder cone were covered with a very fine tabular veneer that, for the most part, has been quarried and carried away. Several of the veneer fragments show signs of carving. A larger slab obviously represents the top of a fragmented stela. Still visible on this fragment is a solar glyph and a section of an elaborate plumed headdress (figs. 1-22a and 1-22b). This fragment is the most complete piece of a stela that we have encountered during almost thirty years of field research.

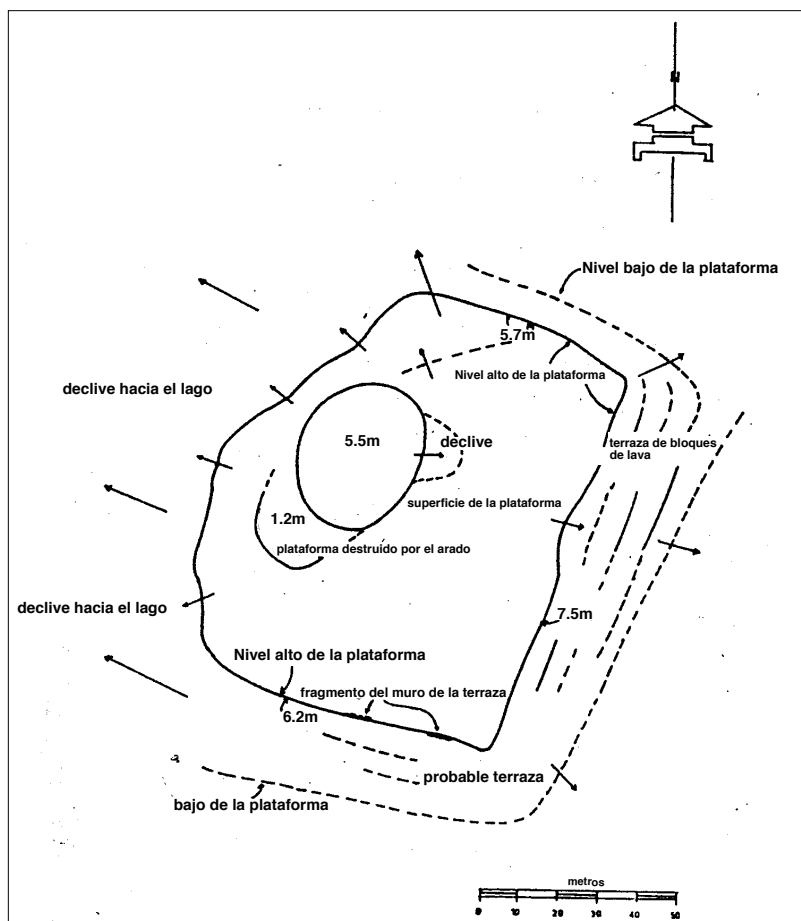


Figure 1-20. The monumental platform/fortification at El Miradór section of the San Juanito ruin.

The most prominent feature of this barrio, however, is the terraced habitation area and the aforementioned huge obsidian workshop located on the southern, and most gradual, face of the peñol. No other obsidian workshop of this size has ever been located or described in the Occidente. Its specialized product was fine, prismatic blades, and tons of this material—millions of blade fragments and flakes—have been found here. Obsidian-working at this scale can be characterized as “industrial.” The wealth that the workshop was able to attract was

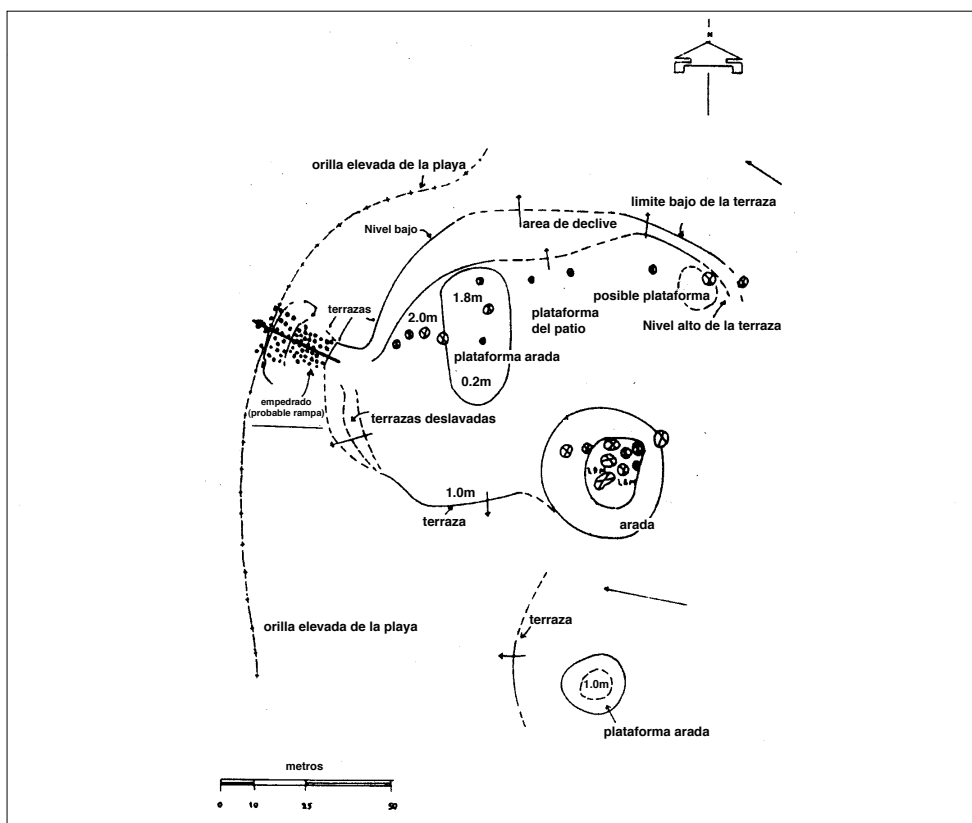


Figure 1-21. The canoe ramp and platform complex at El Reliz section of the San Juanito ruin.

considerable, and the two most visible materials were polychrome ceramics (mostly from the Nayarit coast and southern Jalisco) and turquoise from the far north.

The regional historian and writer Antonio Domínguez Ocampo has documented in a series of talks the important roles that tules played in the economy of San Juanito over the ages. Indeed, this industry only conclusively collapsed when the Laguna de Magdalena was finally completely drained during the 1950s. Tules were the raw material for a wide variety of artifacts during the pre-Hispanic and colonial periods, and bundles of them were used to form canoes (the numbers of which were commented upon by Cerezo and Coría), petates, basketry, construction



Figure 1-22a and b. A solar glyph and plumed headdress on a stela fragment, Las Cuevas.

materials, and so forth. Tules must have been very important economically among the ancient inhabitants of this peñol, along with obsidian blade manufacturing, fishing, waterfowl hunting, and reptile and insect harvesting. Few sites in this region show as much natural potential for such a wide variety of economic activities. The inhabitants of Las Cuevas had everything except agricultural land. For that, they had to go ashore.

Another major pre-Hispanic architectural feature of the peñol is the artificial cave system that is found along the beach of the island's western side. The habitation and ceremonial areas of the peñol are virtually inaccessible from this beach; therefore, the shrines dug into the cinder facing the beach probably did not constitute part of the ancient canoe port for the peñol's habitation and ceremonial areas. That port more likely was on the southern face, where the cinder cone's topography is very gentle. The artificial caves have been badly impacted by

modern quarries for construction fill, and only a few of them survive intact. Formerly, as many as a dozen existed. The three best preserved show two basic morphological forms. One cave really looks like a natural structure, though we find pick and hammerstone marks on the interior walls. It forms a U in the hill and has two entrances.

The other caves are quite different: they form large, and at times even spacious, rooms excavated into the cinder and opened directly onto the playa. One has an banquette/altar carved into the back wall, and a tiny spring once existed at the interface of the altar and the back wall within this shrine. Before the Laguna de Magdalena was drained, these shrines were venerated by the Huicholes as the residence of Grandmother Growth, and the overall peñol was thought to be the remnant of Noah's ark (Weigand 1992). The Huicholes no longer visit these shrines, having transferred them to the shores of the Lago de Chapala. They did this for two reasons: the desiccation of the Laguna de Magdalena destroyed its fundamental ceremonial value for growth and renewal, and the shrines were consistently being vandalized and looted of their modest contents. These shrines, however, must have played an important role in the region's ceremonial life during the pre-Hispanic period.

The structure at El Miradór (see fig. 1-20) is not well preserved. It is one of the three small peaks called the Cerro de los Tres Reyes and still overlooks the entire ancient, as well as contemporary, settlement. The structure is a large platform built on the uppermost slopes of the hillock in order to obtain a flat surface that measured at least 180 by 140 m. Apparently, much of the western edge of the terrace has collapsed, so the platform was somewhat larger. The edge of this great terrace averages between 5 and 7 m high. The lower edge of the platform measured about 230 by 180 m, though it is difficult to follow because of natural rock outcrops.

The natural peak of the hill protrudes through the terrace in the middle of the structure and was perhaps modeled to resemble a pyramid. Fragments of the terraced walls holding up the great platform are still preserved. Obviously, a great deal of the natural outcrop of rock was incorporated into the lower sections of the terraces. Subtracting the volume of the natural peak within this structure, we estimate that perhaps 30,000–40,000 m³ of fill and rock for terracing had to be carried to the hillock's top. We have no evidence for any veneer finishing on this structure.

The structure clearly represents a fortification, though other activities—those meant to be witnessed by the entire settlement—certainly would have been visible from below. Very few artifacts were found associated with this structure, though sherds and obsidian artifacts are fairly frequent on this hillock's lower slopes. The Huistla Polychrome series is well represented among these sherds. The fortified structure's placement above the Postclassic pueblo without doubt dates it to that period. Volumetrically, it is one of the largest Postclassic structures so far recorded for the region. In terms of design, this structure is unique.

The areas north and east of the hillock (opposite the island of Las Cuevas and grading into the current settlement of San Juanito, respectively) have extensive sherd and obsidian covers. These areas clearly represent portions of the ancient habitation zone of Atlitíc/Atitlán, though very few terraces or other traces of architecture are preserved.

One of the best preserved stone ramps in the entire region is at El Relíz (see fig. 1-21). Its placement at the edge of the ancient playa leaves little doubt as to its original function as a feature of a canoe port. Just above the ramp is a large basal platform (about 50 by 60 m in area and 2 m at the highest elevation) that grades into the hillock rising at its eastern edge. Two very altered platforms sit atop this base. Plowing and looting have damaged these features severely. Another platform is located around 20 m to the south. The general area has a number of other terraces and platforms. This area may have functioned as one of the settlement's many embarkation and disembarkation points for the extensive lake traffic mentioned by Cerezo and Coría.

As mentioned, the first Spanish who explored this region in detail had heard about Etzatlán from as far away as Colima. The polities of this area, from Sayula to Tonalá to Etzatlán, were caught in an uncomfortable "double frontier" of better organized, expansive states located in both the north (the Caxcanes) and the south (the Purépechas). Military raids into the Sayula-Atoyác, Tlajomulco-Tonalá, and Tala-Ameca-Etzatlán zones were frequent and destructive enough to be recounted in some detail to the Spanish. The best-preserved such statements describe Tala and Ameca (Weigand and García de Weigand 1996). While Ameca was able to resist the most destructive Purépecha raids (Acuña 1988), Tala was completely destroyed by the one that occurred around 1480 or 1490 and was still in ruins when the Guzmán party reached the site in 1530 (Tello 1968). The reference to warfare in the Cerezo-Coría testimony

concerning Etzatlán is not completely clear, though it appears to be a reference to warfare against both the Caxcanes and the Purépechas (see previous; Coría 1937:558).

Whether or not these raids included actual plans for conquest, in addition to looting, is unknown. The Purépechas had conquered several non-Tarascan polities in southern Jalisco and parts of Colima. The Caxcanes had expanded into the Tecuexe-speaking areas of Los Altos de Jalisco and were raiding into the general Tonalá and Magdalena areas in addition. Thus, conquest may have been an ultimate goal, though not one evident at the point of Spanish conquest. Clearly, the appearance of the Europeans completely altered the geopolitical situation, though ironically the Purépechas, as Spanish auxiliaries, did indeed finally participate in the conquest of the overall Etzatlán region. In addition, Caxcanes were used as auxiliaries within this area after they were defeated by Antonio de Mendoza in 1541 (Weigand and García de Weigand 1996).

Whether or not this area was a target for actual conquest during the pre-Hispanic period, its riches were certainly well known and appreciated. These riches included excellent, well-watered farmlands; obsidian outcrops of the best quality; minerals, such as copper, silver, lead, ochre, and cinnabar; crystals and semiprecious stones such as quartz, malachite, and azurite; easy access to the routes of communication and trade with the coastal regions of the Vallarta areas and the Nayarit littoral; and a relatively high demographic profile.

The area's wealth, therefore, certainly served as an inducement for raiding. The Purépechas' original imperial plans for expansion seem to have focused on areas that had mineral wealth, especially copper and silver (Pollard 1987 and 1993). Since Etzatlán had such resources, the possibility remains open as to what the ultimate Purépecha intentions may have been.

Except for Tala, the general Etzatlán area seemed to have been just out of reach for the Caxcanes and Purépechas. Warfare was apparently endemic in the overall region, from Colima and Sayula in the south to Tonalá and Etzatlán in the north. Certainly, some of this warfare was oriented toward local expansion, improved access to and control over resources, and prestige. The arrival, by conquest, of the Tequani lineage in Ameca, probably around 1440–1450, plus its subsequent expansion testify to that type of activity (“Relación del pueblo de Ameca” [1579],

in Acuña 1988). By far most references to warfare, however, mention the Purépechas and/or the Caxcanes, as in the Cerezo-Coría example, and we must conclude that the area was being heavily impacted by the expansionist policies of its better organized northern and southern neighbors. Thus, the situation into which the Spanish erupted was one of constant and continuous military and sociopolitical flux.

Etzatlán was one of several areas contested by the two groups of Spanish conquerors who entered the Occidente between 1525 and 1530. The forces of Cortéz (representing the interests of the Audencia de Nueva España) and the forces of Guzmán (ultimately representing Nueva Galicia) contested for control over Etzatlán. Guzmán's forces spent the winter of 1530–1531 in Etzatlán, thus effectively taking possession of the site from the Cortéz faction for a period (Coría 1937; Martínez 1991; Morín Tamayo 1992; Tello 1968; Weigand and García de Weigand 1996). The justification that Guzmán gave for his action was that the Etzatlán area had been effectively abandoned by the Cortéz faction and was in a state of chaos. This assertion was without a doubt true. The controversy over who controlled what became extremely bitter and polemic and directly affects the reliability of the historical record itself. However, for Etzatlán the territorial dispute was finally decided in favor of Nueva España. Thus, in a very real sense Etzatlán never was part of the Neo Gallegan orb until much later. Another reason for Etzatlán's continued incorporation into Nueva España was the presence of the silver mines. Almost all such localities were administered directly from Mexico City during the early colonial period.

This decision to divide authority in the heart of the Occidente wound up having serious repercussions as the Rebelión de Nueva Galicia gained momentum. While the monumental convento in Etzatlán was begun in 1534, just before the rebellion swept over this area, the authorities in Guadalajara had committed few resources to defending a region that was outside their jurisdiction. Soldiers were stationed in the relatively unimportant settlement of Tequila, but they only occasionally visited the Etzatlán zone. When they did so, it was in unimpressive numbers. Hence, the monumental convento at first could not well serve in one of its primary architectural functions, that of a fortification.

Guzmán had understood the need for a comprehensive strategy for conquest and control in the Occidente, though when and how he came to these realizations are unknown. The conquest would have to first

neutralize the Caxcan area before the rich lake region, in which Etzatlán was located, could be secured. Leaving such a powerfully organized flank open would be to invite disaster. Only after an attempt to secure that flank had been made could the extremely rich west-central lake districts of Jalisco and further on into Nayarit be systematically conquered. Then, a series of key settlements would have to be used as control points, along an east–west axis, to keep the entire area secure. The key points that Guzmán picked were Nochistlán (to secure the Caxcan area), Etzatlán (to secure the central-west lake valleys), and Compostela (to secure the Nayarit zone and the approaches to the Pacific coast). Guzmán's strategy seems sound but failed concerning all three of the key points. Nochistlán had expelled the Spanish (thus ending the first Guadalajara) in 1533. While Guzmán had originally planned this Guadalajara as an access point for his realm in Panuco, this idea eventually became untenable. However, it is also clear that the Caxcanes were instrumental in pressuring the Spanish from that position. The second failure was to secure direct control over Etzatlán. After the illegal Spanish slave raid on Magdalena in 1535, this area gradually spun out of control, with the indigenous leader Guaxicar assuming a prominent role in the revolutionary events that followed. Rebellion started even earlier in the Compostela area. As early as 1532, in the Xalisco-Tepic area, that zone was becoming more and more unstable.

Divided authority in the heart of one of the two theaters of the *Rebelión de Nueva Galicia* was compounded by the exit of Francisco Vázquez de Coronado and his army in 1540 on their quest for gold in Cíbola. This commander took with him the only military force that could have slowed the rebellion or stopped it from spinning out of control. As a result of this blunder, the entire area was shortly thereafter conclusively out of the Spanish realm. Except for an occasional Franciscan or military patrol, Etzatlán was independent again until Antonio de Mendoza arrived in 1542 and dumped thousands of his indigenous veterans in Etzatlán, adding more chaos, grief, and suffering to the already heavily impacted community (Weigand and García de Weigand 1996).

After that date, epidemic diseases accomplished the rest of the conquest process. While the epidemics had started as early as the 1520s, the years after the collapse of the *Rebelión de Nueva Galicia* were the worst. Only the policy of *reducciones*, plus the importation of other indigenous groups and, later, large numbers of African slaves, kept Etzatlán alive

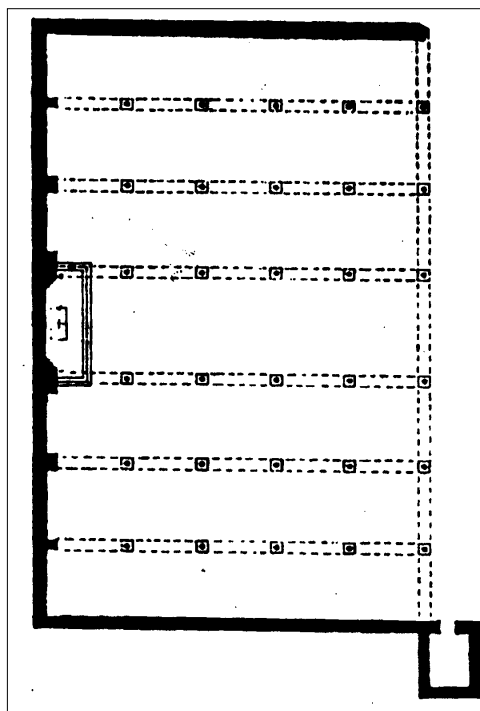
during the dark epoch of the seventeenth century. The beginning of the end of the indigenous world spanned the years 1525 to 1542. By the end of the century, the demographic collapse had taken most of what was left (Borah and Cook 1974). Only isolated communities, such as Oconahua, or the absolutely impoverished and composite barrios, such as the hillside Barrio de los Yndios in Etzatlán, were able to maintain remnants of their former cultural life beyond that century of depression.

The Templo/Convento de la Concepción was not the only important public colonial monument in Etzatlán. Two other building complexes deserve mention in this context: the Capilla de la Virgen and the Casa de la Moneda. At the northwestern edge of the contemporary Plaza de Armas, apparently constructed over a pre-Hispanic platform (see fig. 1-11), an open capilla was designed and built sometime during the early colonial period (fig. 1-23).

While today this religious structure is dedicated to the Virgen de Guadalupe, it is possible that the first appearance of this Virgin within the region occurred at Oconahua rather than at Etzatlán. The first capilla in Oconahua was located across the street from the Templo de San Miguel Arcangel, beneath the site of a contemporary primary school. The original capilla was destroyed, probably during or shortly after the Rebelión de Nueva Galicia. Subsequent religious building activity took place across the street, as mentioned. When this school was first built, the excavations for its foundation courses penetrated the early colonial archaeological deposit. A rough cantera statue of the Virgen de Guadalupe was uncovered and later taken away from Oconahua by a Franciscan priest as part of his personal property. The statue was carved in an indigenous style, though it was already clearly affected by many Spanish artistic conventions. It stood about 1.3 m high. Despite the possibility of this very early appearance of the Virgen de Guadalupe in the region, it was in Etzatlán where a freestanding capilla was dedicated to her.

Architecturally speaking, this was no ordinary capilla. It was one of the less than twenty freestanding open capillas built in Mexico during the colonial period and one of only a very few built in the Occidente. Open capillas were usually inspired by Mozarabic Christian traditions of southern Spain and had originated in the Islamic religious architectural styles of northern Africa. This style for religious structures was probably introduced there as early as the eighth century, when the Arab/Berber conquest of the Iberian Peninsula was under way. In Mexico, one

Figure 1-23. An interpretative drawing of the open *capilla* at Etzatlán.



of the very best examples of this style is well preserved at Actopan, State of Hidalgo. Figure 1-23 is a sketch map of how the open *capilla* might have appeared in Etzatlán prior to the several reconstruction projects that covered over its open patio and moved its retablo to the street. The original layout of this structure is still intelligible, though the Oficina de Telégrafos also occupies a part of the site. This Arab architecture in Etzatlán undoubtedly means that a number of Etzatlán's earliest Spanish inhabitants were from southern Spain, where the Mozarabic tradition was largely concentrated.

With the completion of the open *capilla*, the hispanization of the ancient ceremonial center of Etzatlán was finished. Along with the Campo Santo de los Españoles, to the north of the templo/convento, a great *atrio* spanned the area between the two Catholic compounds and became the focal point for early colonial religious life. The *tianguis*, dancers of the early *cofradías*, *matachines*, Penitentes, and processions of the saints at various times all swarmed over this large sacred space.

The Plaza de Armas, itself atop more pre-Hispanic buildings, became the focus for secular life, as it is today.

Later in the colonial period, the Casa de la Moneda was constructed. This structure was put into its final form during the early eighteenth century, though it was probably begun a century earlier. During the early eighteenth century, silver mining was revived throughout all Mexico, including the Intendencia de Guadalajara. With the change of dynasties from the Hapsburgs to the Bourbons and after the War of Spanish Succession was concluded, Mexico revived economically and culturally from its long economic and social depression (Brading 1971). The audiencia system was revised, as well. Nueva Galicia became the Intendencia de Guadalajara, and Etzatlán became one of its important *jurisdicciones*. With the renewed activity in the silver mines of the Sierra de Ameca came the need for processing centers for the increased quantities of ore being produced. Hence, the Casa de la Moneda's construction and refitting.

This building is located at the corner of Calle Colón and Calle Abasolo, an area that was at the very edge of the settlement then. Befitting a building of that status, it was built in an elaborate baroque style still visible today on the massive corner column supporting the portal. While that portal is now filled in by walls and windows, the arches are plainly discernible. The portal section of the Casa de la Moneda was dedicated to administrative functions and the storage of bullion. The western half of the complex constituted the patio where the actual final processing of the silver ore took place. The *azogue* pits were located in this patio, where mercury, saltpeter, and powdered silver ore were mixed together in order to obtain the bullion. The powdering of the ore was accomplished at the water-powered mill site in the Arroyo de Santa Clara. The arroyo in those times was called a *río*, and water was brought to the site by a series of aqueducts and canals. This mill remained in sporadic production after the colonial period was over, until the massive deforestation of the Etzatlán section of the Sierra de Ameca dried the river to an insignificant trickle.

The azogue pits in this complex were placed at the edge of the settlement for an extremely good reason: mercuric intoxication represented a serious health hazard for the nearby inhabitants. While no coins were cast or stamped at this Casa de la Moneda, silver ingots were. We have no reliable depictions of what the Etzatlán ingot stamp(s) and seals may

have looked like, except that all such stamps and seals carried the same basic information. These stamped ingots were the final product of the Casa de la Moneda in Etzatlán.

The fact that Etzatlán had such a large number of nearby silver mines, and later a Casa de la Moneda, meant that it was considered a Real de Minas. Whether or not this designation became completely formal is still a matter for historical research. Its formal designation as a settlement type during the colonial epoch was *villa*, and at one time it probably was granted an *escudo*, or coat of arms. No rendition of this escudo has been preserved locally, nor have any archival studies to this date encountered a sketch of it. The escudo, if indeed it ever existed, was probably associated somehow with the Casa de la Moneda, though that does not mean that it formed part of the Casa de la Moneda's ingot seal. The emblem that the Villa de Etzatlán currently displays is not an escudo but a *blasón*. The difference between an escudo and a blasón is fundamental: an escudo is assigned from above, in Etzatlán's case as a *real* and/or villa from the Crown, while a blasón is a creation of the pueblo, marking in symbols what the inhabitants themselves think best represents the historical and cultural spirit of their settlement. The Blasón de Etzatlán covers many bases: from a latinized saying attributed to Guaxicar, the designative "Villa y Real," to depictions of the natural environment featuring the Laguna de Magdalena, a pre-Hispanic pyramid, and the Franciscans and Spanish.

Correspondence Analysis of Archaeological Abundance Matrices

By Jan de Leeuw

Introduction

+++Correspondence analysis (CA) is a technique used to analyze data matrices of non-negative numbers. CA is related to *principal component analysis* (PCA) and *multidimensional scaling* (MDS), that is, it is a form of *proximity analysis*. CA is most frequently applied to rectangular tables of frequencies, also known as *cross tables* or *contingency tables*, although applications to binary incidence or presence-absence matrices are also quite common.

The most often used statistical technique for analyzing cross tables computes and tests some measure of *independence* or *homogeneity*, such as chi-square. In the analysis of independence we investigate whether the body of the table is the product of the marginals. Or, if one prefers an asymmetric formulation, if the rows of the table differ only because they have different row totals (and the columns only differ because they have different column totals).

Pearson's chi-square and related measures quantify how different an observed table is from an expected table, based on the row and column totals. Pearson residuals are used to investigate deviations from independence. CA supplements this classical chi-square analysis because it makes both a *decomposition* and a *graphical representation* of the deviations from independence.

History

CA has a complicated history, both in statistics and in archaeology. The prehistory of CA, starting with work by Pearson around 1900 and ending with the reinvention of the technique by Fisher and Guttman around 1940, is discussed in de Leeuw 1983. The technique has been re-reinvented under many different names, in many different countries, and in many scientific disciplines. New reincarnations still continue to appear, although at a slower pace than before, in the data mining and data analysis literature. Beh 2004 is a recent comprehensive bibliographic review.

The history of CA in archaeology is discussed by Baxter (1994:133–39). Although the literature contains some earlier applications of CA to archaeological examples, the credit for the introduction of the technique to archaeologists usually goes to Bølviken and others (1982). Early applications almost without exception came from archaeologists in Continental Europe, under the influence, no doubt, of the French *analyse des données* school and the leadership of Benzécri (1973a, 1973b). A good overview of these various Continental archaeological applications of CA is found in, for example, Müller and Zimmerman 1997.

It is clear from Baxter's discussion that archaeologists in Continental Europe were ahead of archaeologists in Great Britain, who came on board around 1990. Orton (1999:32), one of the deans of quantitative archaeology in Britain, argues that CA was the most important technique introduced into archaeology in the 1980s. From Britain archaeological CA migrated to the United States, where it arrived shortly before 2000. Duff (1996:90) indicates in an influential article from the mid-1990s that CA was "not well established in Americanist literature." And very recently, Smith and Neiman (2007:55) have concurred: "CA has a long history of use by archeologists in continental Europe but its use by Americanist archeologists is both more recent and rare."

There are several possible reasons why CA did not rapidly become popular in archaeology in Britain and the United States. Most importantly, perhaps, archaeological methodologists tend to look to statisticians for guidance, and in statistics CA was not really known until about 1980, despite the work of Hill (1974). Except in France, of course, but French statistics was relatively isolated from that of the mainstream. The dominant multivariate techniques applied in archaeology were MDS and PCA (sometimes in the disguise of factor analysis). The

most influential work in the area in the seventies was Hodson et al. 1971, which concentrated on the MDS techniques of Boneva, Kendall, and Kruskal. These are all forms of proximity analysis, but they differ from CA in various ways.

In a pioneering article, LeBlanc (1975:22) predicted, "Proximity analysis seems to hold a great deal of promise and will in all probability supplant all other seriation methods." If we interpret this prediction narrowly, in terms of the methods that were available in 1975, it turned out to be incorrect, for reasons that are quite obvious in hindsight. Data, in archaeology and elsewhere, come in many different forms. Sometimes we deal with cross tables, sometimes with incidence matrices, and sometimes with multivariate data that describe archaeological objects in terms of a number of qualitative or quantitative variables. There is no reason to expect that a technique designed for one particular type of data will also work, or even be appropriate, for another type of data. A data analysis technique must obviously take the nature of the data into account, and forcing all data into a common "proximity" format may not be an optimal strategy. But the basic advantages of proximity analysis mentioned by LeBlanc (1975:22) are still very much on target: "In the past, the basic goal of seriation has been to order a series of cultural units on the basis of an assumed single underlying variable, usually time. It is now possible to seriate units according to two or more variables by using a form of proximity analysis or MDS. This increases the power of seriation greatly, and among other advantages, it gives a much better idea of the fit of data to one variable (e.g. time alone) than have previous methods."

Because CA was rediscovered and reintroduced in different countries at different times, most authors in the field of archaeology feel obliged to give some sort of introduction to the technique, even in such recent articles such as Poblome and Groenen 2003 and Smith and Neiman 2007. Our discussion of CA differs in some respects from the ones traditionally encountered in archaeology. In other respects it is quite standard. First, and this is actually quite common, we do not present the technique exclusively as a seriation method. Archaeological sites may be similar or dissimilar for many different reasons, and, to quote Kruskal (1971), "time is not the only dimension." Most CA plots are, of course, two-dimensional maps in the plane, which already suggests that more than one dimension may be relevant. Second, we discuss CA

both as an exploratory technique and as a method of fitting a particular statistical model. And finally, we relate the least squares fitting of the CA model to the maximum likelihood fitting of the exponential distance model (EDM). EDM can be considered to be an alternative, and closely related, form of correspondence analysis.

Types and Attributes

LeBlanc (1975) compares *type seriation* and *attribute seriation* (see also Duff 1996). We can discuss this comparison by distinguishing the different types of data that CA can be applied to. In a CA context, attribute seriation corresponds to multiple correspondence analysis (MCA), which is treated in Gifi 1990:chap. 3, and type seriation corresponds to simple CA, treated in Gifi 1990:chap. 8. Or, to translate this into software, attribute seriation corresponds with the R package *homals* (de Leeuw and Mair 2009a), while type seriation corresponds with the package *anacor* (de Leeuw and Mair 2009b).

LeBlanc (1975:24) carefully distinguishes the terms *attribute*, *type*, *variable*, and *dimension*. Actually, he uses *variable* and *dimension* interchangeably, but *dimension* is probably best reserved for the axes in multidimensional representations of data. A “variable” is then a formally defined aspect of the group of objects in the study. Each variable is measured in terms of a scale, and the mutually exclusive characteristics of the scale are called “attributes.” In the book by Gifi (1990), a variable is defined similarly as a mapping of the objects in a study into the categories of a variable. Defining a number of variables on a set of objects creates, in the terminology of the R software system (R Development Core Team 2007), a “data frame.” More specific to archaeology is the notion of a “type,” which Leblanc (1975:24) defines as “the existence of a non-random association between the attributes of two or more dimensions.” Thus, types are aggregations of attributes over different variables, and consequently, they can be counted more easily and are more susceptible to be treated with frequency-based techniques.

This discussion also makes it possible to compare CA with MDS and PCA. In MDS the first step is usually to derive some symmetric matrix of *similarities* between the sites, assemblages, proveniences, or cultural units. Similarities can be defined in many ways, and often the choice of a particular similarity measure is somewhat arbitrary. Moreover, instead of computing similarities between sites, we could also

decide to compute similarities between the variables describing the artifacts found in sites. A commonly used similarity measure between variables is the correlation coefficient. However, it is unclear how the MDS analysis of the sites and the MDS analysis of the variables are related. In PCA we usually start with a correlation matrix between variables and then derive component loadings to describe the variables and component scores to describe the sites. This means PCA can be used to make a joint plot, also known as a biplot (Gower and Hand 1996). Biplots are compelling ways to visualize multidimensional information, and as such they go beyond simple seriation.

One oft-mentioned disadvantage of PCA is that it assumes linear relations between the variables. This disadvantage, however, no longer applies to modern nonlinear versions of PCA, such as those reviewed in de Leeuw 2006. Moreover, nonlinear PCA and MCA are closely related, so closely, in fact, that nonlinear PCA can be carried out with the MCA package *homals* (de Leeuw and Mair 2009a).

The CA framework of Gifi 1990 gives one single class of techniques to analyze attribute matrices of artifacts by variables, frequency matrices of types by sites, and incidence matrices of types by sites. It is basically, to use a term from Benzécri's *analyse des données*, all a matter of "codage." One can code both types and sites as attributes of artifacts, and then the type by site frequency table is just the bivariate cross table of those two variables.

One important advantage of CA and MCA over MDS and PCA is that they stay as close as possible to the original data, no matter if the data are frequencies or incidences or variables with attributes. There is no need to first choose a measure of similarity or correlation, and there is no need to aggregate data into correlation or product matrices. It is true that CA can be presented in terms of a particular measure of dissimilarity, the chi-square distance, and we will give such a presentation in this chapter. But it is only one interpretation of the technique, and the chi-square distances have close connections with the familiar chi-squares that can be computed from the frequencies.

Typical Archaeological Applications

We will discuss some of the typical applications of CA to archaeology in more detail to illustrate where the technique may be appropriate and what archaeologists look at.

Bølviken et al. 1982 uses three data sets from the Stone Age in northern Norway. The first one, from Iversfjord, uses thirty-seven lithic types found in fourteen house site assemblages. Because of interpretational difficulties the analysis was repeated after grouping the thirty-seven types into nine tool categories. The joint plot in two dimensions of the house sites and tool categories is interpreted in terms of economic orientation and settlement permanence. The second example is from the Early Stone Age in the Varanger Fjord area—data counts frequencies of sixteen functional tool types in forty-three sites. Two-dimensional plots give a refinement interpreted in terms of earlier qualitative archaeological hypotheses. The analysis was repeated after the tools were grouped into seven classes, which yielded less informative results. In the third example CA was used to establish a chronology. Data came from a farm mound on the island of Helgøy in Troms. Nineteen classes of artifacts were distributed over fifteen excavation layers, carbon dated from the fourteenth to the nineteenth centuries AD. The analysis shows the layers mapped on a two-dimensional, or *arch*, curve. Projections on the curve can be used to reorder the rows and columns of the data matrix, producing a seriation closely corresponding with the one based on carbon dating.

The article by Duff (1996) on micro-seriation compares attribute and type seriation, following LeBlanc (1975). But whereas LeBlanc used multidimensional scaling for the type seriation, Duff used CA. The data are counts of six ceramic types from forty proveniences in Pueblo de las Muertas, in the Zuni (Cibola) region of New Mexico, from the thirteenth to the fourteenth century AD. The two-dimensional CA solution exhibits a weak arch, with lots of scatter around it, but produces essentially the same ordering of the units as the MDS analysis of Leblanc.

Early on, Clouse (1999) applied CA to Americanist materials and used it to analyze artifacts found in excavations at the military settlement in Fort Snelling, Minnesota. Sites include eight defense buildings, eleven support buildings, and eight habitation buildings. At all sites artifacts were counted and classified into fourteen groups, such as culinary, armament, commerce, and furniture. Separate abundance matrices are given for defense, support, and habitation buildings, and separate CAs are computed. Both joint plots, showing units and artifact groups in two dimensions, and unit plots, which only show the units, are presented. Groupings of the units conform to what is expected on the basis of the military site model but provide more detailed information. Clouse

(1999:105) argues that CA makes expected and unusual features more clearly visible than the numerical summary given by the table.

The excellent paper by Smith and Neiman (2007) aims to compare frequency seriation, in the tradition of Ford (1952), with CA. They use two case studies. The first case study is from the Gulf Coast area, near the Chattahoochee and Apalachicola Rivers in Alabama, Georgia, and Florida. Data are from the Middle and Late Woodland periods (100 BC to AD 900). Ceramic data were collected at many sites, of which twenty-nine were selected because they had more than eighty painted sherds. The twenty-nine sites were subdivided into eighty-four assemblages, and the sherds were classified into eighteen pottery types. Obviously, the way in which artifacts and proveniences are grouped into the rows and columns of the table is important for the eventual outcome of the technique, and the CA of the eighty-four assemblages shows a very clear arch pattern, with a clear grouping of sites along the curve. “The CA results confirm what the clean seriation solution suggests: there is no significant source of variation in type frequencies other than time” (Smith and Neiman 2007:61). The analysis was repeated after removing some of the later assemblages. This smaller CA was validated (as a seriation method) by plotting CA scores against radiocarbon dates for selected sites.

The second case study in the Smith and Neiman article is from Kolomoki, a well-researched multimound site in southwestern Georgia, and is an intrasite analysis, not an analysis with multiple sites. The CA uses twenty assemblages and nine pottery types. Separate two-dimensional plots for assemblages and types show no arch, but a significant and interpretable second dimension. The CA solution shows effects, for example spatial ones, not detectable by the inherently one-dimensional frequency seriation. The first CA dimension is again validated as time, using radiocarbon data. We will use the same Kolomoki data set as one of our illustrative examples in this chapter.

Seriation

There is an interesting parallel historical development of what could broadly be called “seriation methods” in psychometrics, ecology, and archaeology. The main steps in this development occurred in the same order in each field but at different moments in time, not unlike archaeological artifacts in different sites. Let us look at psychometrics first.

Psychometrics

In the 1940s at the war department, Guttman (1944) discovered scalogram analysis, a method to simultaneously order attitude or achievement items (columns) and respondents (rows) with data in a binary data matrix. Initially scales were constructed by trial-and-error methods, in which row and columns of the binary data matrix were permuted to create the “consecutive ones” property. Normally, we look to order rows and columns in such a way that all ones are next to each other. This result was achieved manually with various ingenious devices. At the same time, the theory for principal components–based computations was already available (Guttman 1941, 1950). In fact, Guttman’s (1941) paper was the very first to rigorously define MCA, and he proves that the first MCA dimension provides the consecutive-one ordering for error-free data (1950). The monumental book by Coombs (1964) gives a systematic presentation of these heuristic pencil-and-paper techniques applied to the various data matrices in proximity analysis. And although Coombs’s conceptual framework is still relevant, the techniques had already been superseded by computerized methods at the time the book appeared.

Archaeology

Guttman’s methods were published around 1950, almost simultaneously with Robinson 1951. To discuss this work, we borrow some terminology from Kendall 1969. An incidence matrix of, say, sites by types is a *Petrie matrix* or *P-matrix* if in each column all ones occur consecutively. A non-negative symmetric matrix is a *Robinson matrix* or *R-matrix* if rows and columns are unimodal and attain their maximal values on the diagonal. By *unimodal* we mean that entries increase to a maximum and then decrease again. Similarities between sites whose incidence matrix is a P-matrix often form an R-matrix. Again, there is an interesting connection with psychometrics here. In the original definition of the Spearman model for general intelligence, dating back to 1904, a battery of tests satisfied the model if their correlation matrix was an R-matrix.

The notion of a P-matrix can be generalized to abundance matrices, that is, to any matrix with non-negative entries. An abundance matrix is a *Q-matrix* if its columns are unimodal. That is the same as saying that the columns of the abundance matrix can be represented as a series of *battleship plots*, as defined in Ford 1952 or Smith and Neiman

2007. Many of the original archaeological seriation techniques take an incidence or abundance matrix and permute the sites in such a way that that it becomes a P-matrix or a Q-matrix. The permutation that is found then orders the sites in time, that is, it is a seriation. Ultimately, however, finding optimal permutations, especially for large matrices, is what is known in computer science as NP-hard, which basically means that the optimization problem, although finite, cannot be solved in a practical amount of time, even by the fastest computers (Arlif 1995).

One way around the impractical computations involved with permutations is to use other related definitions of optimality. As we noted, Guttman already proved in 1950 that CA can be used to find the optimal permutation to a P-matrix in the error-free case (for abundance matrices, see also Gifi 1990:chap. 9 or Schriever 1983). In fact, these publications prove more. They also show that in the error-free case, the second dimension of the CA will be a quadratic function of the first—that plotting the sites in the plane will show a quadratic curve.

Kendall (1971) and others later developed the well-known HORSHU program, which applies MDS to similarities derived from abundance matrices and then derives the order from the projection of the sites on the horseshoe or arch. “We view the arch as a relatively benign indicator that the underlying data do, in fact, contain battleship-shaped curves,” write Smith and Neiman (2007:60).

Ecology

In ecology the key concept is that of a “gradient.” The emphasis in the data analysis is not on time, as in archaeology, but on environmental characteristics. What is called “seriation” in archaeology is called “ordination” in ecology (Gauch 1982). Plant and animal species do well under certain circumstances and do best at some optimum level of, for example, humidity or altitude. Different species need different altitudes and/or different degrees of humidity. The major advantage of ecology, of course, is that environmental gradients such as altitude can be directly measured, unlike in psychometrics, where aptitude and attitude are theoretical constructs, and archaeology, where direct information about the origin in time of an artifact is usually missing. Ecologists use direct gradient analysis and plot frequencies of species as a function of the gradient. In many cases the result is unimodal distributions, that is, the abundance matrix is a Q-matrix.

Initially, as in psychometrics and archaeology, ordination techniques in ecology required pencil-and-paper methods to reorder the rows and columns of the abundance matrix or of derived similarity matrices with a Robinson structure (Whittaker 1978). These methods changed with the advent of the computer, and, like archaeologists and psychometricians, ecologists turned to PCA and MDS for ordination and to a host of measures of resemblance or similarity.

CA was introduced in ecology by Hill (1974) as “reciprocal averaging.” Ter Braak (1985) showed how CA was related to the unimodal response model without going into precise mathematical detail. Ecologists initially were worried about the arch because they considered it an artifact without any empirical significance. We now know more precisely where the arched structures come from, and we know that they indicate strong unidimensional effects (see in particular Schriever 1985 or Van Rijkkeversel 1987). We consequently tend to be pleased if we see a clear arch, especially in archaeology, where we have more reason perhaps to expect unidimensionality. (We will discuss the relationship between unimodal response models, in particular the Gaussian model of Ihm and Van Groenewoud [1975], in more detail when we discuss the exponential distance model.)

Abundance Matrices

We now formalize some of the concepts we mentioned in the introduction. Consider an $r \times c$ table N with *counts*. Rows correspond with r *sites*, columns with c *types*. Frequency n_{ij} indicates how often type j was found in site i . Such a matrix with counts N is called an *abundance matrix*. We also define the row sums $n_{i\bullet}$ and column sums $n_{\bullet j}$ of the table. The *grand total* is the sum of all the counts in the table, which we will also abbreviate simply as $n_{\bullet\bullet}$.

It should perhaps be mentioned that *presence-absence matrices* or *incidence matrices* are a special case of abundance matrices in which all entries of the table are either zero or one. An entry merely indicates if a type is present in a site or not, which means our discussion of abundance matrices also covers presence-absence matrices.

A more general type of data matrix is also quite common in archaeology. Suppose the observation unit is an artifact such as a pottery sherd,

a piece of obsidian, or maybe a fish bone. The units can be described in terms of a number of variables that can be either qualitative (categorical) or quantitative (numerical). The abundance matrix is a very special case in which only two categorical variables are used to describe the units, namely *site* and *type*.

The abundance data N can be coded as an $n \times 2$ matrix, where n is the grand total of the table, the first column is *site*, and the second is *type*. The table N is then the *cross table*, or the *contingency table*, of the two variables. But clearly, in a more general case, variables such as size, color, weight, or composition could be used as well. For these more general multivariate data we need a technique such as MCA, also known as *homogeneity analysis* (Gifi 1990; Greenacre and Blasius 2006). Since the data analyzed in this book are all of the simpler bivariate contingency table format, we shall not discuss MCA any further. As we mentioned in the introduction, MCA is the perfect technique for attribute-based seriation in the sense of LeBlanc 1975, in which data are not aggregated to types and assemblages or to counts in a cross table.

Examples

Throughout the chapter we shall use two examples to illustrate the concepts of CA. The first example of an abundance matrix comes from a much larger matrix of sherd counts for sites by pottery types. All samples are from surface collections made around 1940 in Jalisco, Mexico, by Kelly (1945).

This example is not a realistic application of CA because it is too small and too simple. The results of CA do not really add anything to what we can easily see by just looking at the table, but this very fact makes the example useful as an illustration of the basic concepts and calculations (table 2-1).

The second example is pottery data from the Kolomoki burial mounds in Georgia (Pluckhahn 2003; Sears 1956), analyzed previously with CA by Smith and Neiman (2007). We have already discussed these data in the introduction; they include twenty assemblages and nine pottery types.

Associated Matrices

With the abundance matrix we can associate several other matrices. First is the matrix P of *proportions*, whose elements are defined by

Table 2-1: Abundance matrix from Kelly data

	Site	TYPE			
		AutPol	MiReBr	AuWhRe	AltRed
21	8	14	0	0	22
34	19	35	0	0	54
23	138	6	0	1	145
37	299	11	0	2	312
9	102	12	22	271	407
7	34	14	59	246	520
	600	92	81	520	1293

Source: Data come from Kelly 1945.

Note: The codes for the types, used as column headers, are AutPol for Autlan Polychrome, MiReBr for Miscellaneous Red on Brown/Buf, AuWhRe for Autlan White on Red, and AltRed for Altillos Red Ware. Site 21 (Cofradía No. 1) and Site 34 (Hacienda Nueva) are included in the Cofradía Complex (early); Site 23 (Cofradía No. 3) and Site 37 (Amilpa) are included in the Mylpa Complex (intermediate); and Site 9 (Altillos) and Site 7 (Mezquitlan) are included in the Autlan Complex (late).

$$p_{ij} = \frac{n_{ij}}{n_{\bullet\bullet}}$$

The matrix with proportions shows more clearly how the counts are distributed over the cells. Again, the row marginals are $p_{i\bullet}$, and the column marginals are $p_{\bullet j}$ (table 2-2).

Independence

We say that the row variable (site) and the column variable (type) are *independent* if $p_{ii} = p_{i\bullet}p_{\bullet i}$. Independence can be interpreted to mean that the body of the table does not give additional information, that in fact all the information is contained in the marginals. If we know the relative frequencies of the sites and the types, then we can predict perfectly how many of each type will be in each site.

We measure independence by what is called *inertia* in CA, borrowing a term from physics, and define the table Z of *Pearson residuals* with

Table 2-2: Proportions matrix from Kelly data

		TYPE			
	Site	AutPol	MiReBr	AuWhRe	AltRed
21	0.006	0.011	0.000	0.000	0.017
34	0.015	0.027	0.000	0.000	0.041
23	0.107	0.005	0.000	0.001	0.112
37	0.231	0.009	0.000	0.002	0.241
9	0.079	0.009	0.017	0.210	0.315
7	0.026	0.011	0.046	0.190	0.273
	0.464	0.071	0.063	0.402	1.000

Source: Data come from Kelly 1945.

Note: The codes for the types, used as column headers, are AutPol for Autlan Polychrome, MiReBr for Miscellaneous Red on Brown/Buf, AuWhRe for Autlan White on Red, and AltRed for Altillos Red Ware. Site 21 (Cofradía No. 1) and Site 34 (Hacienda Nueva) are included in the Cofradía Complex (early); Site 23 (Cofradía No. 3) and Site 37 (Amilpa) are included in the Mylpa Complex (intermediate); and Site 9 (Altillos) and Site 7 (Mezquitlan) are included in the Autlan Complex (late).

$$z_{ij} = \frac{p_{ij} - p_{i\bullet}p_{\bullet j}}{p_{i\bullet}p_{\bullet j}}$$

The elements of Z show the deviation between the observed proportion and the expected proportion on the hypothesis of independence (corrected for the standard error of the proportion). Positive elements indicate that we see more in the corresponding cell than we expect, and negative elements mean that we see less. The *inertia* is defined simply as

$$X^2 = \sum_{i=1}^r \sum_{j=1}^c z_{ij}^2$$

In the Kelly (1945) example the inertia is 0.9338, and the Pearson residuals are shown in table 2-3.

Table 2-3: Pearson residuals from Kelly data

Site	TYPE			
	AutPol	MiReBr	AuWhRe	AltRed
21	-0.02	+0.28	-0.03	-0.08
34	-0.03	+0.44	-0.05	-0.13
23	+0.24	-0.04	-0.08	-0.21
37	+0.35	-0.07	-0.12	-0.31
9	-0.18	-0.09	-0.02	+0.23
7	-0.28	-0.06	+0.22	+0.24

Source: Data come from Kelly 1945.

Note: The codes for the types, used as column headers, are AutPol for Autlan Polychrome, MiReBr for Miscellaneous Red on Brown/Buff, AuWhRe for Autlan White on Red, and AltRed for Altillos Red Ware. Site 21 (Cofradía No. 1) and Site 34 (Hacienda Nueva) are included in the Cofradía Complex (early); Site 23 (Cofradía No. 3) and Site 37 (Amilpa) are included in the Mylpa Complex (intermediate); and Site 9 (Altillos) and Site 7 (Mezquitlan) are included in the Autlan Complex (late).

Table 2-4: Z-scores from Kelly data

Site	TYPE			
	AutPol	MiReBr	AuWhRe	AltRed
21	-0.69	+9.94	-1.17	-2.97
34	-1.21	+15.90	-1.83	-4.66
23	+8.62	-1.34	-3.01	-7.50
37	+12.81	-2.38	-4.42	-11.02
9	-6.32	-3.15	-0.69	+8.39
7	-10.14	-2.22	+7.84	+8.73

Source: Data come from Kelly 1945.

Note: The codes for the types, used as column headers, are AutPol for Autlan Polychrome, MiReBr for Miscellaneous Red on Brown/Buff, AuWhRe for Autlan White on Red, and AltRed for Altillos Red Ware. Site 21 (Cofradía No. 1) and Site 34 (Hacienda Nueva) are included in the Cofradía Complex (early); Site 23 (Cofradía No. 3) and Site 37 (Amilpa) are included in the Mylpa Complex (intermediate); and Site 9 (Altillos) and Site 7 (Mezquitlan) are included in the Autlan Complex (late).

If the data are a random sample, and if types and sites are independent, then nX^2 is distributed as a chi-square random variable with $(r - 1)(c - 1) = 15$ degrees of freedom. In our example, nX^2 equals 1207.508. Moreover, each $\sqrt{n}z_{ij}$ is approximately standard normal; that is, it is what is commonly known as a z-score, and it can be tested for significance in the usual way. The z-scores are listed in table 2-4.

In the Kelly example the total inertia is clearly far too big, the z-scores are mostly hugely significant, and the two variables *site* and *type* are very far from being independent. Of course, in most archaeological applications data are very far from being a random sample because we generally enumerate and classify all the artifacts found in the site. Nevertheless, we can still take inertia as a guideline to indicate how much structure there is in the data or, more precisely, how much structure there is in the data that cannot be predicted from the marginals.

Conditioning on Rows and Columns

In archaeological studies the hypothesis of independence is not the most natural way to look at abundance matrices. Independence is the appropriate concept if the contingency table results from a random sample from a discrete bivariate distribution, that is, if we sample both sites and types. Usually, however, sites are not sampled. They are fixed either by design or by geographical circumstances.

What really interests us is a comparison of the distribution of types in the different sites that we have selected. Thus, we are mainly interested in comparing the rows of the abundance matrix because each row defines a distribution over types. Fortunately, the hypothesis of homogeneity of rows is mathematically equivalent to the hypothesis of independence. We can most easily see this equivalence by normalizing the rows—dividing each row by its row sum.

To keep our treatment symmetrical, we also consider the case (less common in archaeology) in which it may be interesting or appropriate to also compare the columns. Using the row and column sums, we can normalize the frequency table (or, equivalently, the table with proportions) by dividing the entries of the table by their row or column marginals. This process defines two new tables, the first one conditioned by rows, the second conditioned by columns. The elements are defined by

$$p_{j|i} = \frac{n_{ij}}{n_{i\bullet}} = \frac{p_{ij}}{p_{i\bullet}} \quad \text{and}$$

$$p_{i|j} = \frac{n_{ij}}{n_{\bullet j}} = \frac{p_{ij}}{p_{\bullet j}}$$

The hypothesis of independence can now be written in the two equivalent forms

$$p_{ij} = p_{i\bullet}p_{\bullet j} \quad \text{and}$$

$$p_{j|i} = p_{\bullet j}$$

which we can call *homogeneity of rows* and *homogeneity of columns*. Homogeneity of rows says that the probability distribution of types is the same for all sites. Homogeneity of columns says that the probability distribution of sites is the same for all types, which in our context seems a less natural way of expressing the same basic mathematical fact.

Table 2-5 shows the distribution of types over each of the sites and within the last row the distribution of types over all sites, that is, the $p_{\bullet j}$. We have homogeneity if and only if all rows of the table, including the last row, are the same. Table 2-6 shows the distribution of sites over each of the types and within the last column the distribution of sites over all types, that is, the $p_{i\bullet}$. We have homogeneity if and only if all columns of the table, including the last column, are the same.

We can define appropriate measures of homogeneity of the rows and columns. These are again called *inertias* in CA, and one inertia exists for each row and each column. They are defined by

$$X_{i\bullet}^2 = \sum_{j=1}^c \frac{(p_{j|i} - p_{\bullet j})^2}{p_{\bullet j}} \quad \text{and}$$

$$X_{\bullet j}^2 = \sum_{i=1}^r \frac{(p_{i|j} - p_{i\bullet})^2}{p_{i\bullet}}$$

Rows with a large inertia differ from the average row, that is, the

vector of column marginal proportions. And columns with a large inertia differ from the average column.

Previously, we have defined the *total inertia*. Because of the simple relationship

$$X^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(p_{ij} - p_{i\bullet}p_{\bullet j})^2}{p_{i\bullet}p_{\bullet j}} = \sum_{i=1}^r p_{i\bullet} X_{i\bullet}^2 = \sum_{j=1}^c p_{\bullet j} X_{\bullet j}^2$$

the total inertia is the weighted sum of the row and column inertias.

Under the hypothesis of random sampling from sites and homogeneity of rows, the $nX_{i\bullet}^2$ are distributed as chi-squares with $c - 1$ degrees of freedom. If we have random sampling and homogeneity of columns, the $nX_{\bullet j}^2$ are distributed as chi-squares with $r - 1$ degrees of freedom.

Table 2-5: Conditioning on the rows in Kelly data

	TYPE				
Site	AutPol	MiReBr	AuWhRe	AltRed	$X_{i\bullet}^2$
21	0.36	0.64	0.00	0.00	4.98
34	0.35	0.65	0.00	0.00	0.04
23	0.95	0.04	0.00	0.01	0.11
37	0.96	0.03	0.00	0.01	0.24
9	0.25	0.03	0.05	0.52	0.31
7	0.10	0.04	0.17	0.47	0.27
$p_{\bullet j}$	0.46	0.07	0.06	0.40	0.93

Source: Data come from Kelly 1945.

Note: The codes for the types, used as column headers, are AutPol for Autlan Polychrome, MiReBr for Miscellaneous Red on Brown/Buf, AuWhRe for Autlan White on Red, and AltRed for Altillos Red Ware. Site 21 (Cofradía No. 1) and Site 34 (Hacienda Nueva) are included in the Cofradía Complex (early); Site 23 (Cofradía No. 3) and Site 37 (Amilpa) are included in the Mylpa Complex (intermediate); and Site 9 (Altillos) and Site 7 (Mezquitlan) are included in the Autlan Complex (late).

Table 2-6: Conditioning on the columns in Kelly data

TYPE					
Site	AutPol	MiReBr	AuWhRe	AltRed	$p_{i\bullet}$
21	0.01	0.15	0.00	0.00	0.02
34	0.03	0.38	0.00	0.00	0.04
23	0.23	0.07	0.00	0.00	0.11
37	0.50	0.12	0.00	0.00	0.24
9	0.17	0.13	0.27	0.52	0.31
7	0.06	0.15	0.73	0.47	0.27
$X_{\bullet j}^2$	0.64	4.06	1.18	0.68	0.93

Source: Data come from Kelly 1945.

Note: The codes for the types, used as column headers, are AutPol for Autlan Polychrome, MiReBr for Miscellaneous Red on Brown/Buf, AuWhRe for Autlan White on Red, and AltRed for Altillos Red Ware. Site 21 (Cofradía No. 1) and Site 34 (Hacienda Nueva) are included in the Cofradía Complex (early); Site 23 (Cofradía No. 3) and Site 37 (Amilpa) are included in the Mylpa Complex (intermediate); and Site 9 (Altillos) and Site 7 (Mezquitlan) are included in the Autlan Complex (late).

Exploratory Correspondence Analysis

The basic purpose of exploratory CA is to make a *map of the types* and a *map of the sites*. By a “map” we mean a low-dimensional geometric representation. If we choose dimensionality equal to two, for instance, a map of the types consists of c points in the plane, with one point corresponding to each type. If we choose dimensionality three, then a map of the sites consists of r points in three-dimensional space. Sometimes even a one-dimensional map, which puts all sites on a straight line, is already enough to present the essential information in the table.

The location of the points in the map is not arbitrary, of course. If we make a two-dimensional map of the types, for example, we want the distances between the c points in the plane to be approximately equal to the distances between the c columns of the abundance matrix N . And similarly for the map of the sites and the rows of N .

Distance on the map is defined in the usual way, “as the crow flies.” In other words, it is ordinary Euclidean distance. But distance between

columns of the abundance matrix depends on weights that take into account the statistical stability of the cell counts. Specifically, in CA we use *chi-square distances* (Gifi 1990; in de Leeuw and Mair 2009a we use *Benzécri distances* instead). The squared chi-square distance between row i and row k of table N is given by

$$\delta_{ik}^2 = \sum_{j=1}^c \frac{(p_{j|i} - p_{j|k})^2}{p_{\bullet j}}$$

and the squared chi-square distance between column j and column l of table N is

$$\delta_{j\ell}^2 = \sum_{i=1}^r \frac{(p_{i|j} - p_{i|\ell})^2}{p_{i\bullet}}$$

We give the squared chi-square distances for the rows and columns in the Kelly example in tables 2-7 and 2-8.

If we look more closely at table 2-7 we can already predict what CA will do. If we want a geometric representation in which the distances approximate the chi-square distances, then it is pretty clear how such a representation would look. The chi-square distances between sites 21 and 34 and between sites 23 and 37 are almost zero. Thus, in a map sites 21 and 34 will coincide, and sites 23 and 37 will also coincide. Sites 9 and 7 are close as well, and 21/34 is about equally distant from the two groups 7/9 and 23/37. A two-dimensional map will thus look like an isosceles triangle with the three groups of sites at the edges. The shorter side is somewhere around $\sqrt{2}$ or $\sqrt{3}$, and the two longer sides are around $\sqrt{6}$. We also see that it will in general be impossible to map the distance information on a straight line because in that case we would have to let 7/9 coincide with 23/37. In this small example we can easily see what a map would look like, but in a larger example, such as the Kolomoki one, this map becomes much more complicated. That is why we have CA, which approximates the chi-square distances to the Euclidean distances in a precise way on the map.

In CA we approximate chi-square distances *from below*. Let me explain this concept. In any CA map of the sites, for instance, we will

Table 2-7: Squared Benzécri distances, rows (sites)

	21	34	23	37	9	7
21	0.000					
34	0.002	0.000				
23	5.721	5.950	0.000			
37	5.841	6.072	0.001	0.000		
9	6.353	6.550	2.188	2.208	0.000	
7	6.812	6.999	3.207	3.233	0.259	0.000

Source: Data come from Kelly 1945.

Note: Site 21 (Cofradía No. 1) and Site 34 (Hacienda Nueva) are included in the Cofradía Complex (early); Site 23 (Cofradía No. 3) and Site 37 (Amilpa) are included in the Mylpa Complex (intermediate); and Site 9 (Altillos) and Site 7 (Mezquitlan) are included in the Autlan Complex (late).

Table 2-8: Squared Benzécri distances, columns (types)

	AutPol	MiReBr	AuWhRe	AltRed
AutPol	0.000			
MiReBr	4.921	0.000		
AuWhRe	3.221	6.203	0.000	
AltRed	2.539	5.780	0.436	0.000

Source: Data come from Kelly 1945.

Note: The codes for the types, used as column and row headers, are AutPol for Autlan Polychrome, MiReBr for Miscellaneous Red on Brown/Buf, AuWhRe for Autlan White on Red, and AltRed for Altillos Red Ware.

always have $d_{ik} \leq \delta_i$, where d_{ik} is Euclidean distance between points i and k on the map. More precisely, CA constructs a sequence of maps: the first one has only one dimension, the second has two, and so on. The final map has $t = \min(r - 1, c - 1)$ dimensions, that is, three in the Kelly example and eight in the Kolomoki example. The maps are *nested* in the sense that the projection on the first dimension of all the maps is identical to the one-dimensional map, the projection on the plane of the

first two dimensions of all maps with at least two dimensions is equal to the two-dimensional map, and so on. If $d_{ik}^{(s)}$ represents the distances on the s -dimensional map, with $1 \leq s \leq t$, then

$$d_{ik}^{(1)} \leq d_{ik}^{(2)} \leq \dots \leq d_{ik}^{(t)} = \delta_{ik}$$

Thus, the t -dimensional map has distances exactly equal to the chi-square distances. Maps in fewer dimensions approximate the distances, and the approximation becomes better for each of the distances when the dimensionality increases. Approximation is from below because map distances are always smaller than chi-square distances, no matter what the dimensionality of the map is. Of course the same reasoning applies to chi-square distances between columns and the CA map for types.

The map does not only approximate chi-square distances between sites or types, it also approximates the inertias of the sites and the types. In the sites map, for instance, the inertia is approximated (from below, as usual) by the distance of the site to the origin of the map. Or, equivalently, by the length of the vector corresponding with the site. This means that a site that differs little from the average site, and thus has a small inertia, will be close to the origin of the map. And sites that are different from the others will tend to be in the periphery of the map. As a consequence, the center of the map, the area near the origin, can quite easily be cluttered with sites that are similar to the average site.

A CA program (we use *anacor* by de Leeuw and Mair [2009b]) typically takes the abundance matrix and the desired dimensionality of the map as its arguments. It then outputs coordinates for the maps of the row objects (sites) and the column objects (types). In addition it can provide a variety of plots, and it provides a *decomposition of the inertia*. This type of decomposition is familiar from PCA. Consider the weighted squared length of the projections of the site points on the first dimension, on the second dimension, and so on. This decomposes the total inertia of the vectors into a component due to the first dimension, the second dimension, and so on. By dividing the components by the total, we can say that a certain percentage of the inertia is “explained” by the first dimension, another smaller percentage by the second dimension, and so on. Ultimately, there are $t = \min(r - 1, c - 1)$ dimensions, and each of them takes care of a certain decreasing percentage of the total inertia.

CA can also make *joint maps* or *biplots* in which we basically take the site plot and the type plot and put them on top of each other. We then have a plot in which types will tend to be close to sites in which they occur more frequently than one would expect on the basis of the marginals. We say “tend to” because there is no chi-square distance defined between a site and a type, and thus there is no approximation in some well-defined mathematical sense. The CA program *anacor* basically lets the user make four choices for the joint plot. The first one is to put the two Benzécri plots on top of each other. Distances between sites and distances between types approximate chi-square distances, but distances between sites and types have no simple relation to the data. The second option, which is called Goodman scaling in the program, is to adjust the length of the site and type vectors in such a way that their inner product approximates the Pearson residual. Unfortunately, this result invalidates the interpretation of site and type distances as approximations of chi-square distances. The last two options use the *centroid principle*. We can take the Benzécri map for the sites and then plot the types by taking weighted averages (centroids) of the sites using the frequencies of the types in those sites as weights. This process produces a joint plot in which site distances approximate chi-square distances. The locations of the types in the plot again only differ in vector length from the locations in the Benzécri type plot. Type distances cannot be interpreted as approximating chi-square distances between types anymore, but they do have a clear geometric interpretation as weighted averages of site points. By symmetry, there is a second centroid principle in which we use the Benzécri type plot and then plot the sites as weighted averages of types.

The centroid principle can also be used to fit passive sites or types into the plots. Suppose an additional site, not used in the analysis, is excavated, and the objects are classified using the same typology as the one used in the analysis. The type scores from the analysis can be used to compute the score for this new additional site just by calculating the average CA score of the site on each of the dimensions. In the same way we could use the site scores to add additional types to the analysis, for example if we decided to split one original type into two new types. Of course, the alternative is to repeat the CA with the additional sites and types, which then would actively determine the overall CA solution.

The Kelly Example

Let us illustrate exploratory CA with the small Kelly example. The two-dimensional maps for sites and types from CA are shown in figure 2-1.

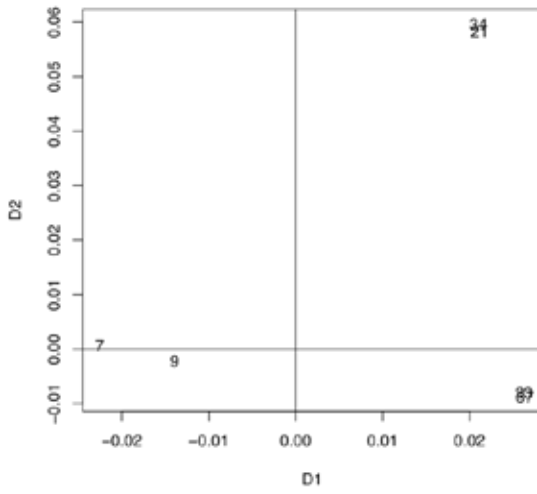
As expected, in the sites map we see the three clusters of points at the vertices of a triangle, and as we know, the one-dimensional map is simply the projection of all points on the horizontal axis.

In figure 2-2a we see the approximation of the chi-square distances between sites in one dimension and in Figure 2-2b in two dimensions. Chi-square distances are on the horizontal axis, Euclidean map distances on the vertical axis. Approximation from below means that all points are below the 45-degree line of perfect fit. But as we can see, fit in two dimensions is already almost perfect. In one dimension some of the larger chi-square distances, in particular those between 21/34 and 23/37 are seriously underestimated.

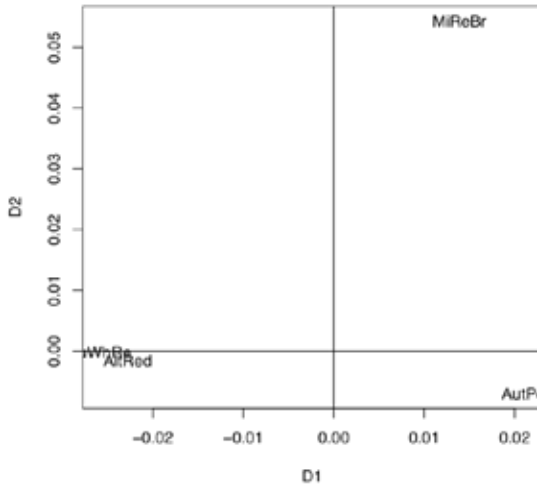
We finally show the chi-square decomposition for the Kelly example (table 2-9). Not surprisingly, the two first dimensions account for 97 percent of the total inertia, and the third dimension is of very little importance.

The Kolomoki Example

We now apply CA to the Kolomoki data, our more realistic example. The chi-square decomposition is given in table 2-10. Two dimensions account for 80 percent of the inertia, three dimensions for almost 90 percent. The CA maps for the types in two and three dimensions are given in figures 2-3 and 2-4. Again, the two-dimensional map is just the projection of the three-dimensional map onto the horizontal plane (except for a possible rotation). Note that the points in the two-dimensional maps are the centers of ellipses of varying sizes. These ellipses represent 95 percent confidence regions for the points. Confidence region computations, which are shown in de Leeuw and Mair 2009b, are based on the assumption that the abundances are a large random sample from a population. As with chi-squares, this assumption may not be appropriate in archaeological examples, but also as with chi-squares, the sizes of the ellipses do give a useful representation of variability. Without the random sampling assumption they measure how the location of the points in the plot changes with small perturbations of the data. We see larger ellipses for outlying points, which generally



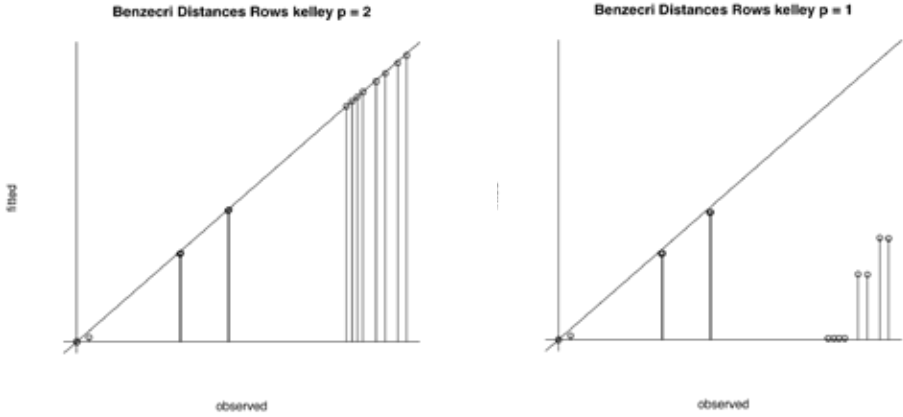
(a) Sites



(b) Types

Figure 2-1. A two-dimensional CA map of the Kelly data.

(a) One Dimension



(b) Two Dimensions

Figure 2-2. An approximation of the Benzécri distances for the Kelly data.

Table 2-9: Chi-square decomposition of Kelly data

	χ^2	%	Cum %
1	787.9	0.65	0.65
2	390.0	0.32	0.97
3	29.6	0.03	1.00
Total	1207.5		

Source: Data come from Kelly 1945.

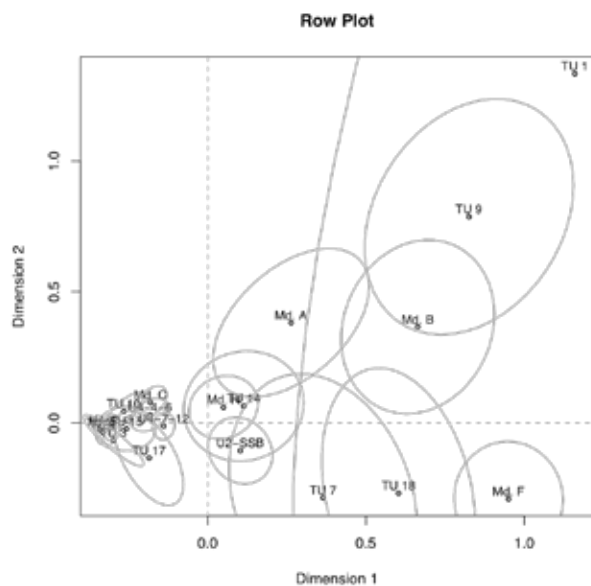
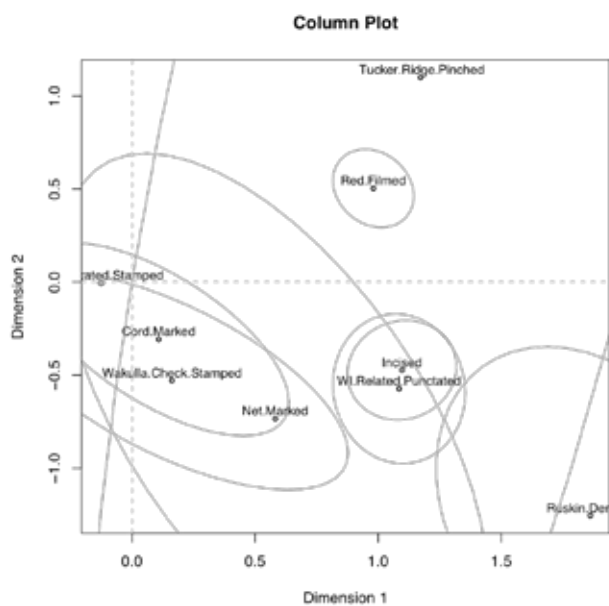


Figure 2-3. CA maps of the Kolomoki data.



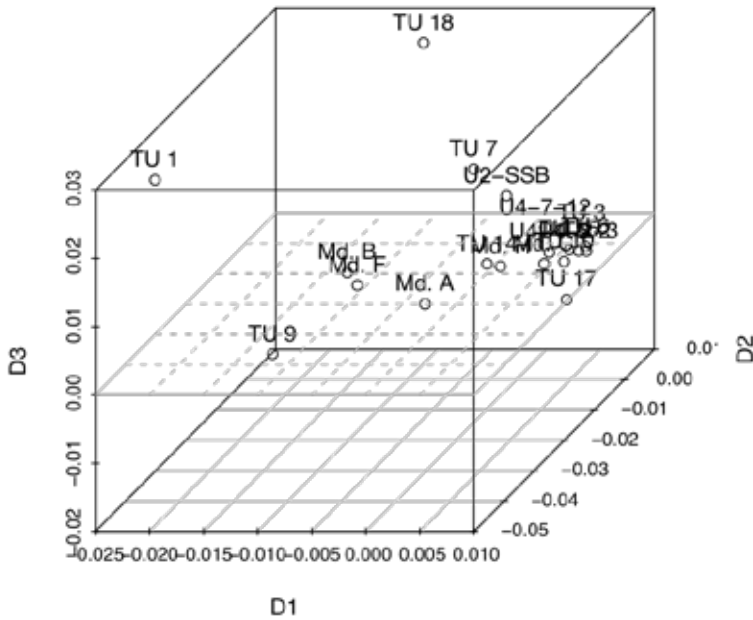
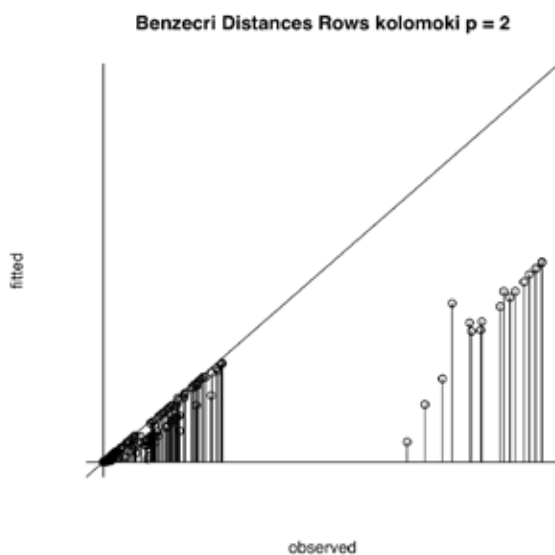


Figure 2-4. A three-dimensional map of the Kolomoki data.

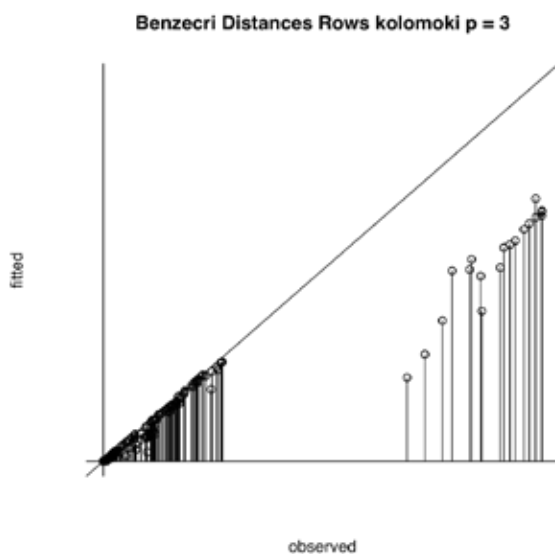
correspond with smaller abundances, and we see examples of overlapping ellipses for sites or types that cannot really be distinguished.

For the interpretation of the two-dimensional Kolomoki results, we refer to the experts Smith and Neiman (2007). The third dimension does not add much (only 9 percent of the total inertia), but it does allow us to better approximate some of the larger chi-square distances. In particular, the third dimension emphasizes the differences between the outliers T9 and T1/T18.

If we continue to add dimensions, we will probably see each new dimension take care of a group of the large chi-square distances, which are still seriously underestimated in three dimensions. See the Benzécri plots in figure 2-5.



(a) Two Dimensions



(b) Three Dimensions

Figure 2-5. An approximation of the Benzécri distances for the Kolomoki data.

Table 2-10: Chi-square decomposition of Kolomoki data

	χ^2	%	Cum %
1	1018.8	0.63	0.63
2	261.6	0.16	0.79
3	144.7	0.09	0.88
4	128.0	0.08	0.96
5	38.6	0.02	0.98
6	17.9	0.01	0.99
7	9.0	0.01	1.00
8	3.8	0.00	1.00
Total	1622.5		

Source: Data come from Kelly 1945.

Variations of Correspondence Analysis

Several variations of CA are also used. We have not applied them in our example, but we briefly mention them for completeness. One may wonder, for example, if approximation from below is such a good idea. It seems obvious that a better approximation of the chi-square distances is possible if we allow some of the map distances to overestimate and others to underestimate. This idea is exploited in de Leeuw and Meulman 1986. The idea, basically, is to compute chi-square distances first and then apply multidimensional scaling to these distances.

A second question is whether there are suitable alternatives to the chi-square distances. Remember that chi-square distances are used because we correct the proportions for their standard errors, on the assumption of independence. Chi-square distances have a natural connection to chi-square, to the weighted sum of squares, and thus to Euclidean distance. Alternative methods for weighting the proportions are indeed possible, as in the spherical CA of Domingues and Volle (1980), but generally the connection with Euclidean geometry becomes less transparent.

And finally, we can get away from the interpretation of abundance matrices in terms of relative frequencies. Instead, we can think of them

as *compositional data*. Each row is a vector of proportions adding up to one, but the proportions may come from a chemical analysis of samples and may not come from counts. Compositional data are very common in chemometrics and the earth sciences and are also quite common in archaeology. Variations of principal component analysis for compositional data similar to, but not identical with, CA are discussed in the monograph by Aitchison (2003).

Exponential Distance Models

In ecology (Ihm and Van Groenewoud 1975; Ter Braak 1985), and to some extent in archaeology, much attention has been paid to the Gaussian ordination model (GOM). The model says that for site i and species j the expected value of the abundance is

$$\mathbf{E}(f_{ij}) = \alpha_i \beta_j \exp\left(-\frac{1}{2} \left(\frac{(x_i - y_j)^2}{s_j}\right)^2\right)$$

Thus, sites and types can be scaled on a common one-dimensional scale. Abundance f_{ij} is, except for the marginal row and column effects α_i and β_j , related to the distance between the scale value of site i and the scale value of type j . More precisely, a type will be abundant in sites whose scale value is close to the type's scale value, and it will be largest if type and site coincide on the scale. Rows of the abundance matrix will be unimodal: they have a single peak and then level off in both directions. Or, to use Kendall's terminology, they are Q-matrices. Again, except for the marginal effects, the same thing is true for the columns. Thus, if the model fits we can reorder the sites and types in such a way that both rows and columns of the abundance matrix are unimodal.

The GOM can be generalized easily to more than one dimension.

$$\mathbf{E}(f_{ij}) = \alpha_i \beta_j \exp\left(-\frac{1}{2} \sum_{s=1}^p (x_{is} - y_{js})^2\right)$$

For obvious reasons we call this the exponential distance model (EDM). The EDM is unimodal in a more general geometrical sense.

The response curves in the plane, if $p = 2$, have a single peak and level off in all directions. There are many ways in which the EDM can be fitted to abundance matrices. Most of them are based on multinomial maximum likelihood, and thus they naturally come with large-sample significance tests and confidence regions. Not surprisingly, contributions have been made by both psychometricians and ecologists. For a recently proposed technique, and a good overview of earlier work, see De Rooij and Heiser 2005.

We can simplify the EDM, by expanding the square and collecting terms, to the equivalent form

$$\mathbf{E}(f_{ij}) = \alpha_i \beta_j \exp\left(\sum_{s=1}^p x_{is} y_{js}\right)$$

This equation shows how we expand the abundances into the product of marginal effects and an interaction term, which is an inner product of row and column effects and is actually quite close to CA. In the social sciences this is often referred to as the row-column or RC-model. For small arguments we have $\exp(x) \approx 1 + x$ and consequently

$$\mathbf{E}(f_{ij}) = \alpha_i \beta_j \left(1 + \sum_{s=1}^p x_{is} y_{js}\right)$$

This model is fitted by CA, using weighted least squares. Thus, we see that CA can be interpreted as a convenient and inexpensive approximation to EDM but also as a model in its own right in which the multiplicative (exponential) interactions are replaced by additive ones. Besides this relationship, of course, both EDM and CA can be discussed as data reduction and data representation methods, without necessarily referring to a statistical model.

The two-dimensional Kolomoki EDM solution is given in figure 2-6. We will not give an interpretation of the result but merely point out that there are some differences from the CA solution. The grouping of sites and types is approximately the same, but the EDM solution displays less of an arch, and this is typical.

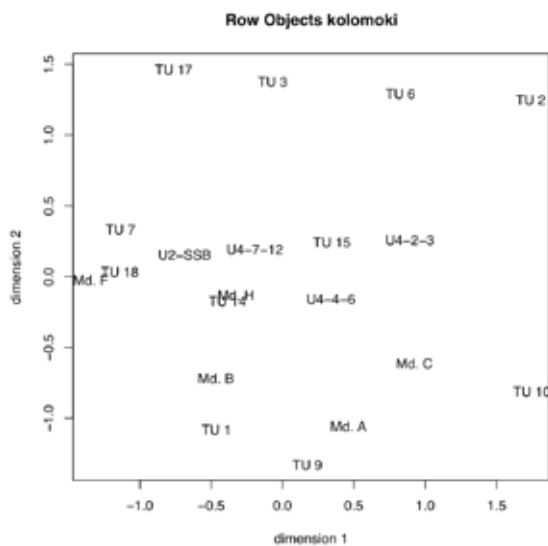
Discussion

This chapter could be called “The Many Faces of Correspondence Analysis,” and in it we have tried to provide various interpretational frameworks to look at CA plots in terms of distances, centroids, association models, and chi-square. It also shows how the same models and techniques appear in many different disciplines, often under different names, and that combining ideas from these disciplines gives us additional possibilities for interpretation.

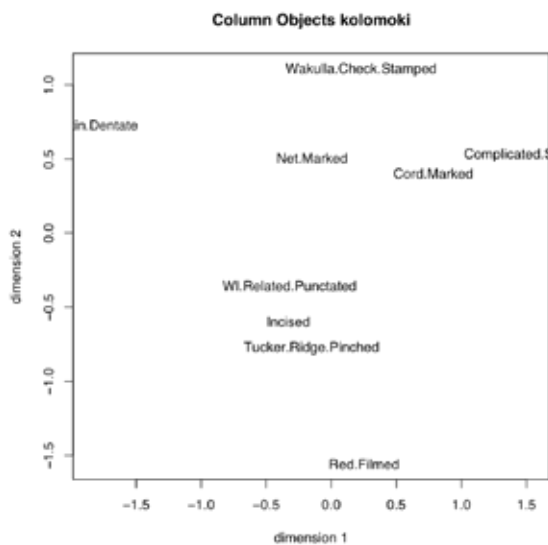
We have also discussed the EDM model in its various disguises as the GOM or the RC-model. It can be used to embed a form of CA into a maximum likelihood framework and to shift the emphasis from multivariate exploration to model testing.

Archaeologists not familiar with CA can use this chapter to look at previous examples in their discipline and to think in a different way about abundance and incidence matrices. We have tried to emphasize the continuity between CA and previous seriation methods used in archaeology.

As we have indicated, convenient, free R packages are available for CA. We mentioned *homals* and *anacor*, but in de Leeuw and Mair 2009b other available packages are discussed as well. All standard statistical systems, such as SAS, SPSS, and Stata, also have CA methods as either built-ins or add-ons.



(a) Rows



(b) Columns

Figure 2-6. EDM maps of the Kolomoki data.

Ceramic Type Descriptions

C. Roger Nance

Structuring the Typology

+++ We began our work with pottery from the site of Las Cuevas. Here, we encountered thousands of sherds, all from relatively shallow deposits. Preliminary observations showed, as far as we could tell, no tendency for different ceramic forms to cluster either in lower or upper 20 cm levels or, for that matter, in different localities of the area excavated. The purpose of the typology thus became the devising of categories that would prove to be time sensitive and represent different portions of the site sequence. Various considerations and influences came into play as the different types first took the form of piles of morphologically similar potsherds.

The author has studied collections of prehistoric potsherds elsewhere: in Alabama (e.g., Nance 1976, 1988) and in eastern Mesoamerica (Nance 1992; Nance et al. 2003). In his experience, traditional categorizations based on surface finish, color and design technique, and, if painted, design color have often proved to be reliable chronological indicators. These variables figure in many type definitions in the present study. At the same time, certain design motifs or other distinctive decorative elements were found to appear on Las Cuevas sherds where the sherds varied in surface color and finish. In some cases, such sherd groupings were given type status because they were perceived as potentially representing a limited portion of the sequence.

In pursuing the goal of discovering or developing time-sensitive types, we thought it reasonable to explore as much as possible the inherent variability in ceramic vessel form and decoration represented in the collection. The strategy was to define as many types as we could conveniently handle in sorting, with each type representing a limited manifestation of the overall variability. Type designations, in the form of a numeric code, were to be entered into the computer for every potsherd studied, along with a sherd's site affiliation, a unique identification number, and the sherd's provenience within the site. The fact that sherd data would be computerized had an impact on the form of the typology. The strategy of fashioning many types would be advantageous only if types at times could be grouped together into encompassing classes. Because of this requirement, we structured the study so that types were linked taxonomically where possible. Large classes of sherds were divided into smaller groupings (types) with distinguishing characteristics. At times, we would want to work with these larger type groupings in order to avoid problems of sampling error in the analysis. At other times, our interest would turn to the narrowly defined types that might have more delimited time ranges. With computerized data, it is relatively easy to move from one level to another within such a type/type-group hierarchy.

Another concern centered on the problem of bias that could compromise typological assignments and limit the effectiveness of the study. One potential source of bias, for example, would be different individuals making type assignments. Another might involve changing type perceptions as sorting progressed. A third might be entertaining hypotheses regarding type age during sorting. Every sherd in the study collection was inscribed with its provenience, including its 20 cm level. Also, pottery from a few stratified sites in the greater region had been published along with assessments of type age. Where Las Cuevas sherds resembled those from published types, and especially for those cases where type assignment was difficult, there might be a tendency to begin to see a sequence in the Las Cuevas material and unwittingly to make a few type assignments based on sherd depth in the deposits. This problem is discussed by Adams and Adams (1991:51–52) under the heading "Foreknowledge." We countered these potential biases as much as possible. Sherd lots from Las Cuevas and other sites were processed in no particular order, although we did proceed on a site-by-site basis. Nance,

although joined by others in the effort of recording potsherds from Las Cuevas and other sites investigated, made all final assignments of type affiliation and continuously made visual comparisons with previously typed sherds. Finally, no detailed local ceramic sequence was extant, and Nance did not attempt to incorporate previously published types from farther afield into the typology being formulated. We remained uninformed, to some extent, about published types for the greater region, and comparative research was mostly postponed until we had completed the classification and recorded and entered data into the computer for all sherds in the study.

During the sorting process, we became aware that our perceptions of what constituted accurate type assignments changed to a small but noticeable extent, in spite of the fact that we continued to use the typed Las Cuevas collection as a reference throughout the sorting of potsherds from the remaining sites. That is, type definitions, once established, seemed to remain constant, but decisions as to which sherds ought to be included in a given type changed slightly as we became more experienced with the collection and as the body of typed sherds expanded. At the same time, new types occasionally were being discovered and added to the typology. Adams and Adams (1991:60–61) describe the continual flux or learning that accompanies typological sorting as “the classificatory dialectic.” For this reason, each sherd in the collection was examined at least twice, once during initial sorting and secondly when provenience data and sherd identification numbers were transcribed from sherds to “sherds by type and site” lists. Later, we entered data from these lists into the computer. All initial sorting was done by Nance, except for the site of Tiana, where sherds were classified by Prado but checked by Nance. Re-examination during the recording process was undertaken by Nance, Prado, or Verity. If by Prado or Verity, problematic sherds were called to Nance’s attention for a second appraisal.

One other concern in the initial stages of type formation had to do with ease of classification. We knew there were thousands of sherds to process and that we had no guarantee that any sequence would emerge from our efforts. Therefore, we wanted to move through the collection with some speed. For this reason, all types are based on easily observed characteristics, and we did not examine the paste of each sherd microscopically. Another feature of this classification is that we decided to type all sherds in lots investigated, ignoring only very small potsherds

that lacked two clear surfaces. This meant that eroded sherds or possibly eroded sherds had to be placed in residual categories. Two of these “types” (1 and 4) are included in the data set but not incorporated in any of the CAs that produced the seriated type sequences for these sites.

Grater Bowls

Grater Bowls, Black Band—Type 10

These sherds are from well-fired *molcajetes* decorated by a black band below the rim on interior surfaces. The band is of thick paint, polished, and about 2.0 to 2.5 cm wide. At times the band extends onto the everted and flattened rim. Exterior surfaces are not well finished. A definite ridge is present on most vessel exteriors, which encircles them about 2.0 cm below the rim (fig. 3-1a). In form the vessels were shallow bowls, apparently supported on solid conical feet. The bowls' flat bottoms are covered with closely spaced incised lines, and these grater surfaces sometimes show pitting from use. Nonpainted surfaces are gray in color. Temper consists of crushed white rock, and fresh break surfaces reveal black cores, which suggest incomplete oxidation. The sherds average about 8.6 mm in maximum thickness.

Grater Bowls, Red and Black Stripes on Buff—Type 8

These *molcajetes* were decorated in a rigidly stylized way: around interior rims and above the incised-line grater surfaces, a broad painted red band is flanked by two narrow black stripes, both below and above. Above the latter, a parallel red stripe occupies the rim surface proper (fig. 3-1b, c). These vessels were shallow bowls with solid conical feet; rims tend to be rounded and slightly expanded or thickened. Exterior surfaces were smoothed, and most rim sherds exhibit an exterior ridge about 2.0 to 2.5 cm below and parallel to the rim. On several sherds, grater incisions are partially obliterated through use wear. Temper is very fine crushed rock, and in cross section, sherds appear well oxidized with pronounced dark cores infrequent. Surface colors vary from pinkish gray (7.5YR7/2) or pink (5YR7/3) to light reddish brown (2.5YR6/4).

Grater Bowls, Red and Black on Buff, Complex—Type 7

Sherds of this type are very similar to those of type 8, the key difference being that the basic design of type 8 has been embellished. In the upper pair of black stripes, the lines have been moved slightly farther apart,

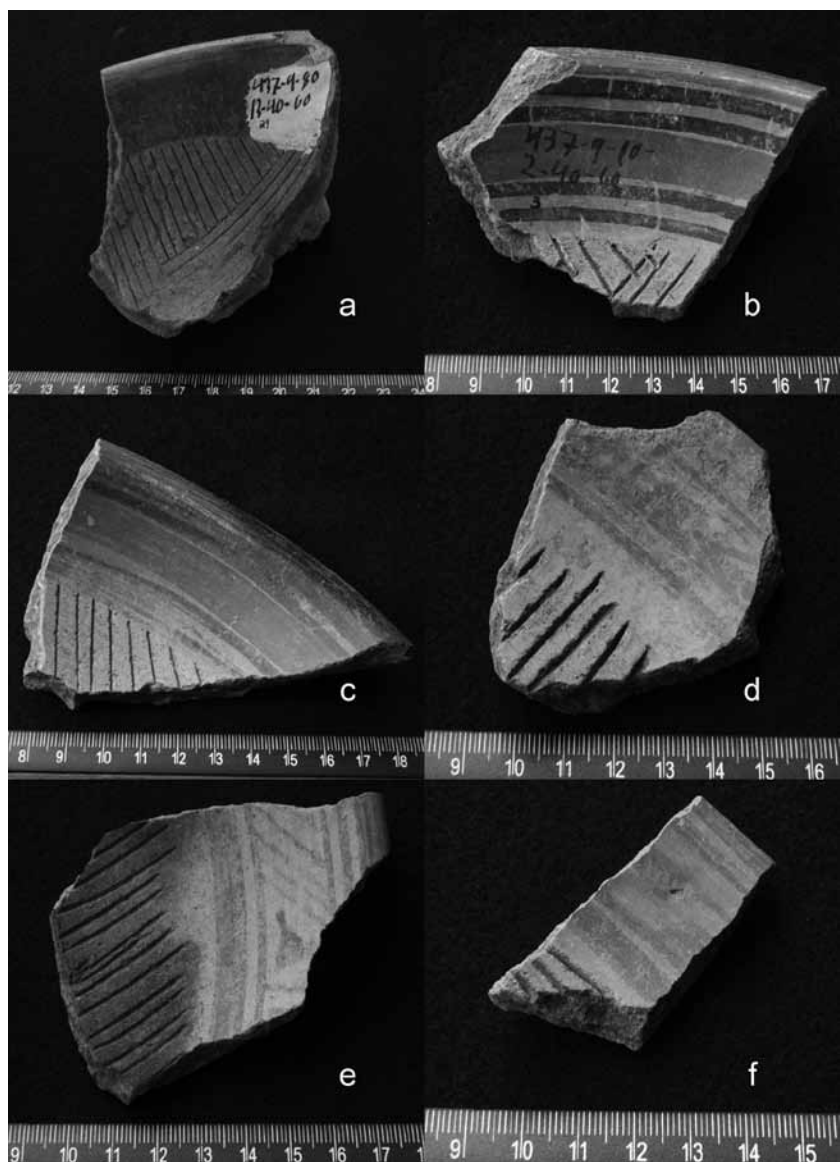


Figure 3-1. Potsherds: (a) Grater Bowls, Black Band (type 10); (b-c) Grater Bowls, Red and Black Stripes on Buff (type 8); (d-f) Grater Bowls, Red on Buff (type 11).

and a wavy or zigzag black painted line has been added, connecting one of the parallel lines to the other (fig. 3-2a, b; see fig. 3-9 for the drawing color codes). In a minority of sherds in this category, one can see other variations on this theme.

*Grater Bowls, Red and Black on Buff, Complex or Parallel Lines—
Type 9*

Sherds of this category come from grater bowls, either with the simple banded decorations of type 8 or the more complex design of type 9. On these sherds, the design is incomplete and assignment to either type impossible.

Glassow (1967:68–80) describes pottery of the three Black and Red on Buff grater bowl types (7, 8, 9) as Huistla Polychrome, based on the collection he studied from the site of Huistla in the vicinity of Etzatlán.

Grater Bowls, Red on Buff—Type 11

Again, designs around the interior rim distinguish this class of grater bowls from others. Here, the painted designs above the grater incising

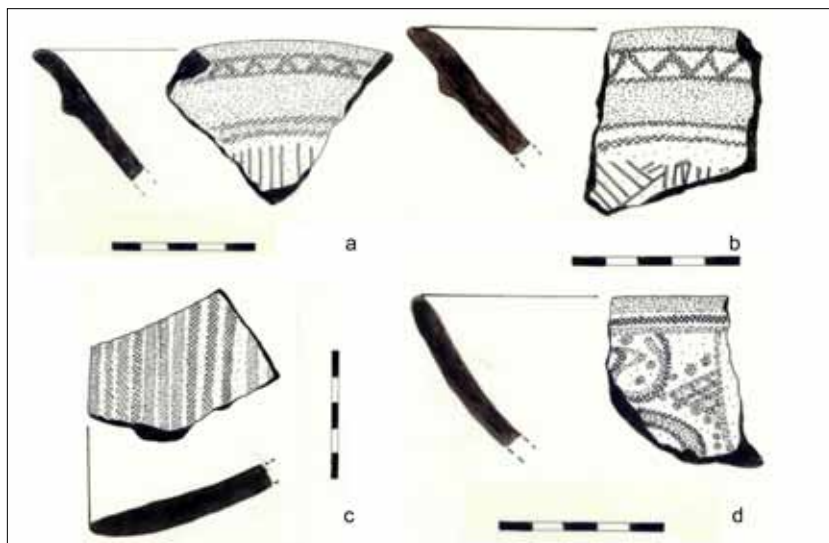


Figure 3-2. Potsherds: (a–b) Grater Bowls, Red and Black on Buff, Complex (type 7); (c) Red and Black on Buff, Striped (type 21); (d) Red and Black on Buff, Complex (Unique) (type 20). Drawings by Helle Girey.

are red only and mostly in elaborate patterns (see fig. 3-1d, e, f), although sherds with simple stripes parallel to the rim occur as well. One sherd shows evidence of a foot support, and grater surfaces are well worn on some examples. The sherds show further differences in comparison with other types of decorated grater bowls. Exterior surfaces are better finished here and lack ridges below the rim. Also, rims do not expand but are straight with rounded lips. Finally, these sherds are thinner, averaging about 7.7 mm in thickness. Surface color ranges from reddish brown (2.5YR5/4) to pinkish gray (5YR7/2). Temper varies from very fine to unsorted crushed rock of fine to moderate size.

Grater Bowls, General Form—Type 3

The great majority of molcajete sherds fall in this residual category. Most are body sherds with the typical multiple parallel-line incising on interior surfaces. Some are rim sherds with decorations that are indistinct or eroded away.

Grater Bowls, Fine Ware—Type 13

These molcajete fragments are separable from others on several accounts. First, incised lines on the interior of bowl bases tend to be more carefully executed, deeper, and spaced further apart (fig. 3-3a, b, c). Also, patterns are more complex and have designs included within one or several incised lines encircling interiors at wall/base junctures. Some, if not most, of these vessels had solid conical feet, and several examples show stamped decorations, one of an animal effigy (see fig. 3-2of). One sherd, indicative of vessel form, is from a shallow bowl with a straight upturned wall with rounded lip. Paste tends to be finer than that of other molcajetes. Surfaces outside the grating areas can be black, slipped and polished, or polished gray. Most sherds have dark cores, suggesting incomplete oxidation. Temper is fine crushed rock with occasional evidence of very fine ground-up shell. Maximum potsherd thickness averages about 6.8 mm. Surface colors vary from pink (5YR7/3) to gray (5YR5/1).

Red and Black on Buff

Red and Black on Buff, Striped—Type 21

These sherds all are from well-finished, large bowls, decorated near the rim on interior surfaces. Red and black stripes are painted parallel to

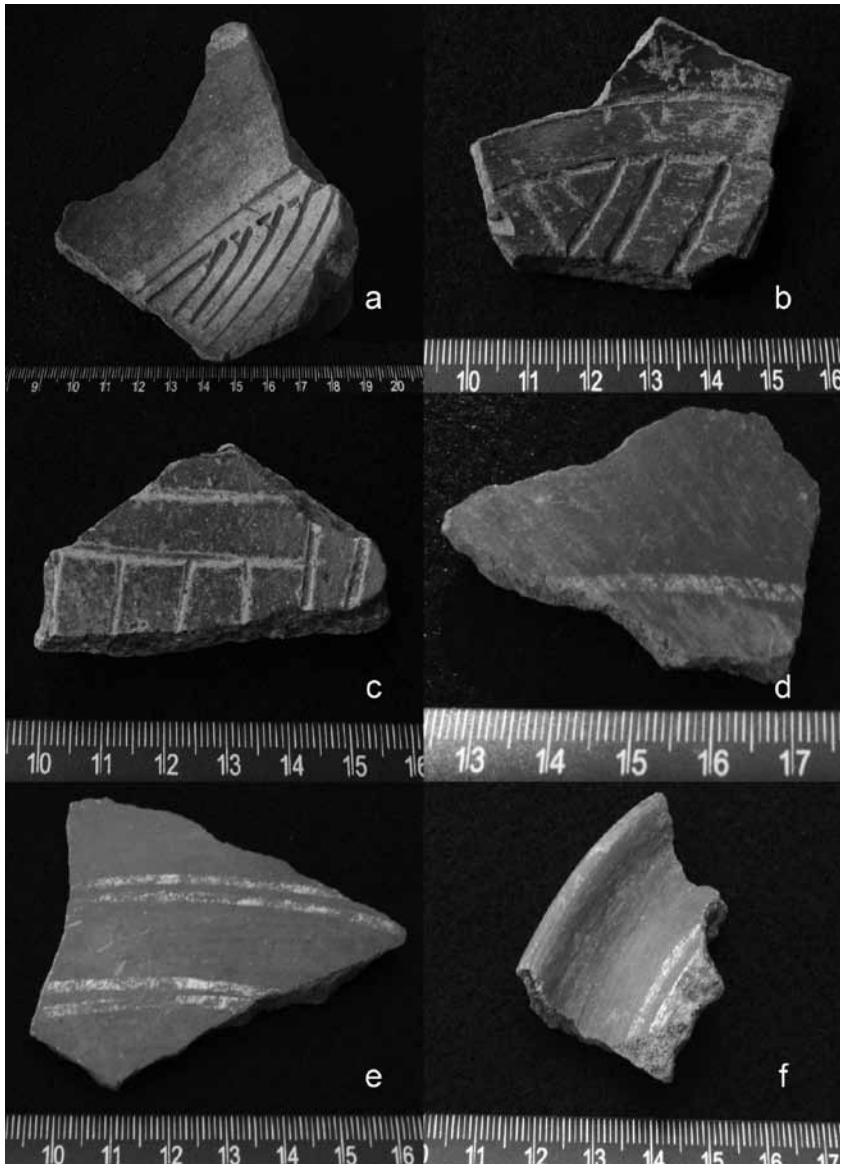


Figure 3-3. Potsherds: (a–c) Grater Bowls, Fine Ware (type 13); (d) White on Red, Parallel Lines Outline Red Stripes (type 18); (e–f) White on Red, Simple (type 16).

the rim, the background color is buff, and buff strips alternate with the painted stripes. Some sherds are probably from molcajetes and show the same motif of a red band bordered by double black stripes described for type 8. Other sherds show different color sequences (see fig. 3-2c). Striped areas are polished. Surface colors vary from reddish yellow (5YR5/6) to pinkish gray (7.5YR7/2). Sherd thicknesses average about 6.8 mm.

Red and Black on Buff, Complex (Zigzag)—Type 19

Sherds of this type are similar to red and black molcajetes with complex designs, except they lack evidence of molcajete incising and for the most part lack the standard array of black and red stripes described for type 7. One of the sherds that does display this pattern is a small rim sherd fragment and may well be from a grater bowl. But on one large vessel fragment, a flat-bottomed footed bowl, decorations continue toward the bowl center. In this instance, basal decorations are curvilinear and complex. Generally, rims tend to be straight and terminate in rounded lips. The pots are well finished on both surfaces; decorated areas are

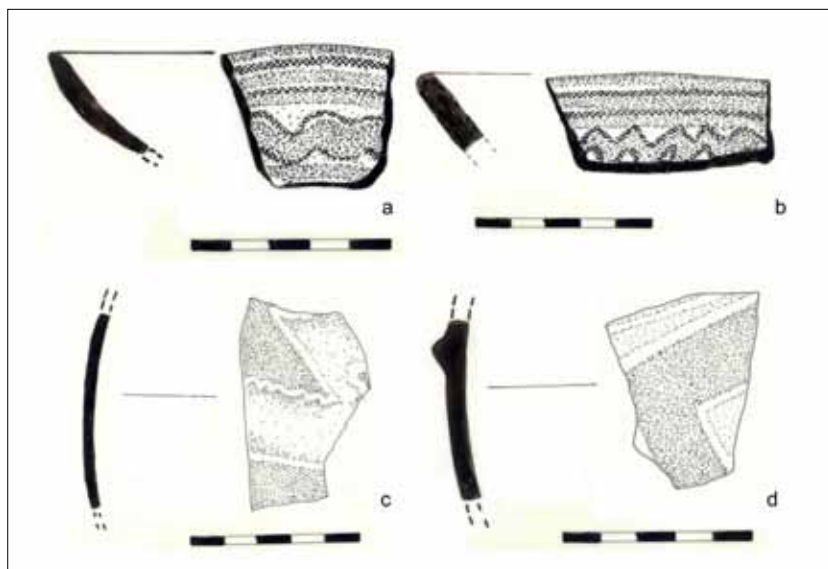


Figure 3-4. Potsherds: (a-b) Red and Black on Buff, Complex (Unique) (type 20); (c-d) White on Red, Parallel Lines Outline Red Stripes (type 18). Drawings by Helle Girey.

polished. Surface colors range from light brown (7.5YR6/4) to reddish gray (5YR5/2), and sherd cross sections tend to be light gray (5YR7/1) in color, which suggests complete firing and relatively thorough oxidation. Temper is finely crushed rock, and mean thickness for the sherds is about 7.4 mm.

Red and Black on Buff, Complex (Unique)—Type 20

Here, designs are highly variable and boldly executed in a casual or sloppy manner. One sherd is from a shallow, rounded bowl, but otherwise vessel form is unknown. Painted decorations are mostly on interior surfaces, but not always. At times, crude designs in red paint are bordered with black lines (see fig. 3-2d). On one sherd, black and red parallel lines set off areas of black or red hatching. On other sherds, zigzag or oscillating lines of red and black alternate and/or tend to be separated by multiple stripes of red and black, all on a buff background (fig. 3-4a, b). All decorations on interior surfaces appear to be in the vicinity of the rim. Sherd cross sections are light gray in color or else evidence black cores. Surface colors range from pinkish gray to reddish yellow (5YR6/6). Also, sherd thickness averages 7.2 mm, and temper, sparsely distributed, is finely crushed rock.

The previous three Red and Black on Buff types (19, 20, 21) are the non-grater bowl variants of the published type Huistla Polychrome (Glassow 1967:68–70).

White on Red and Black and White on Red

White on Red, Parallel Lines Outline Red Stripes—Type 18

One of three closely related types, sherds in this grouping display broad, roughly executed red stripes, which appear to result from negative painting (a resist material applied to lighter areas with a red background). Their borders were then painted with carefully applied white lines (see figs. 3-3d, 3-4c, d). Decorations are on exterior surfaces only. While one sherd appears to be a necked jar fragment, it and all others are body sherds, and no other vessel form is suggested. Exterior surfaces appear to be slipped and polished. The background color of the red (10R4/6) stripes is pale red (10R6/3) to red (2.5YR5/6). Interior surfaces are well finished and light (5YR7/1) to dark gray (5YR3/1). Compared to other types, these sherds, although well finished, have more abundant and

coarser crushed-rock temper. Also, these sherds are relatively thin, averaging about 5.7 mm in thickness. In cross section, sherds tend to be light gray and appear well oxidized, with relatively few displaying dark cores.

These sherds closely resemble the type Iago Polychrome of the Ceritos phase, as defined by Grosscup (1976:229) for the site of Amapa, Nayarit.

White on Red, Simple—Type 16

These sherds resemble those of the previous type, except no red stripe is discernible. Exterior surfaces have light red (10R6/6) to red (10R5/8) backgrounds, and single or at times parallel painted white lines extend across sherds from one edge to another (see fig. 3-3e). Most sherds are body fragments and not indicative of vessel form, although one appears to be the rim fragment of a necked jar (see fig. 3-3f). Sherds are well finished on both surfaces; pot exteriors were probably slipped then polished. Crushed rock or sand temper is abundant and of moderate size. Viewed on fresh break surfaces, cross sections tend to be light gray and

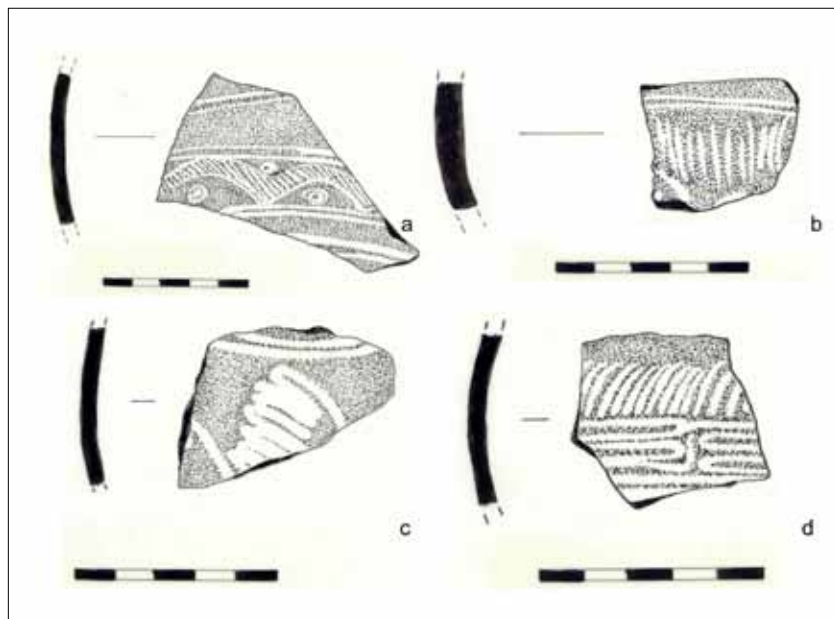


Figure 3-5. Potsherds: (a-d) White on Red, Complex (type 17). Drawings by Helle Girey.

to lack dark cores. Interior sherd surfaces vary in color from pinkish gray (5YR7/2) to gray (5YR5/1). Sherds are also relatively thin, averaging about 5.6 mm in thickness.

White on Red, Complex—Type 17

Quite similar to sherds in the other two White on Red categories, these specimens have complex white-painted designs on possibly red-slipped exterior surfaces. Designs consist of rectilinear or curvilinear motifs and/or hatching (fig. 3-5). In several cases, white designs are confined to carelessly painted darker red areas, like type 18. The only vessel form represented is the necked jar. These sherds, as for the previous two types, evidence the liberal use of sand or crushed-rock temper. Surface colors show the same range as well, and average sherd thickness is about 5.2 mm.

The type compares closely to Santiago White on Red defined for the Ixcuintla and Santiago phases at Amapa (Grosscup 1976:235).

Black and White on Red—Type 14

Here, on elaborately decorated vessels, black and white designs were painted on red-painted surfaces. On one sherd, this red area also evidenced negative painting, where a resist material had been applied in long narrow parallel stripes (fig. 3-6a). Typical designs involve broad black stripes bordered with thin white lines (fig. 3-6b, c). Additionally, a single row of white dots sometimes runs along the center of the black stripe. Overall, design elements are curvilinear and/or rectilinear. Exterior surfaces are well finished but tend not to be polished; interiors are best termed smoothed. Almost all sherds are body fragments, and vessel shapes remain unknown. Average sherd thickness is about 5.9 mm. Interior surface colors run from pinkish gray (5YR7/2) to gray (5YR5/1). The abundant temper is probably sand, and paste colors, viewed on fresh break surfaces, are light.

Red on Cream

Red on Cream, Thin Parallel Lines—Type 22

Sherds of this type are decorated with multiple red-painted stripes that are parallel and straight. These are relatively thin, in most cases, 2 to

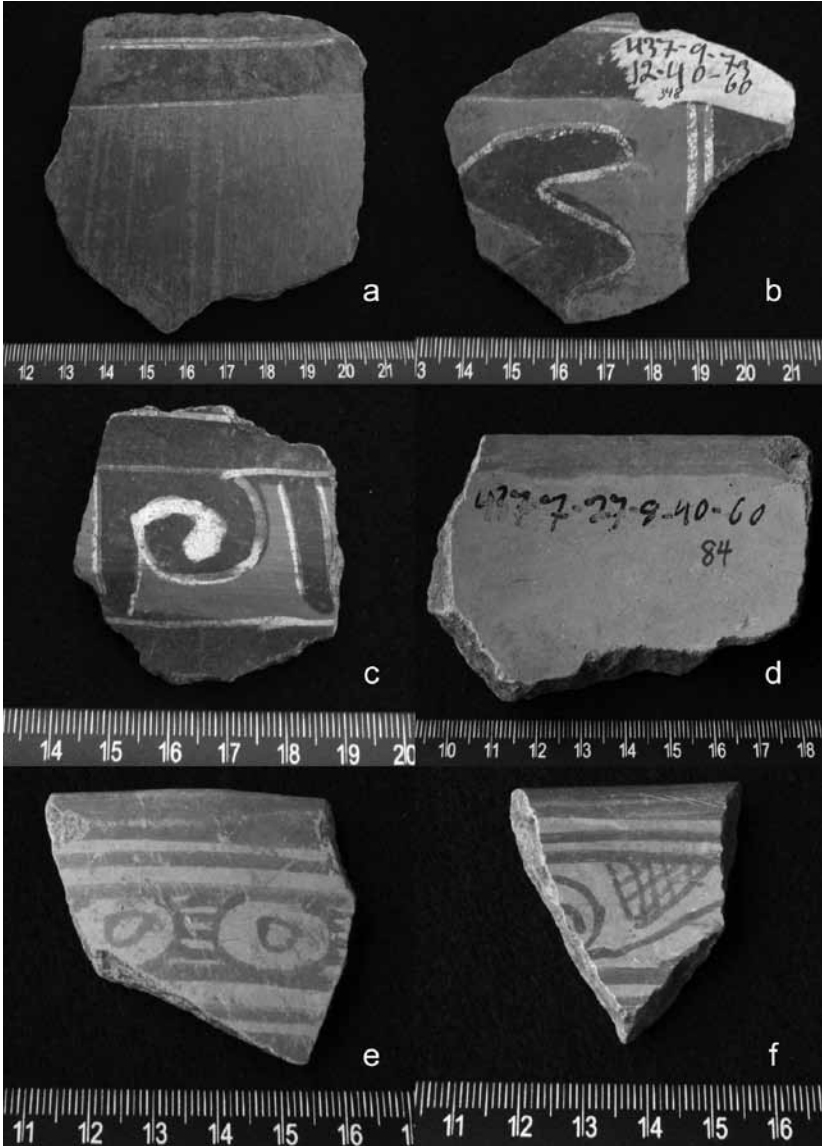


Figure 3-6. Potsherds: (a-c) Black and White on Red (type 14); (d) Red on Cream, Broad Parallel Lines (type 23); (e-f) Red on Cream, Complex (type 24).

4 mm wide. Decorations are usually confined to interior surfaces, and on rim sherds, the stripes are parallel to the rim, with the uppermost stripe on the rim proper (fig. 3-7a). Decorated areas tend to be well polished, and both surfaces are well finished. Viewing the paste on freshly broken cross sections, we see that most specimens lack dark cores and have light paste colors. Temper varies from fine and sparsely distributed crushed rock to sand of moderate to large grain size in abundant amounts. Nonpainted surface colors vary from pinkish gray (5YR6/2) to reddish brown (5YR5/4). Average sherd thickness is about 7.4 mm.

Red on Cream, Broad Parallel Lines—Type 23

Here, sherds were grouped together on the basis of broad parallel red-painted bands, with band widths varying from 5 to 15 mm. On some bowl fragments (see fig. 3-6d), a single band is indicated, encircling the rim on and just interior to it. On other sherds, these bands are multiple and parallel. Several specimens are from large necked jars with streaky red stripes applied on both sides of the rims. The sherds tend to have

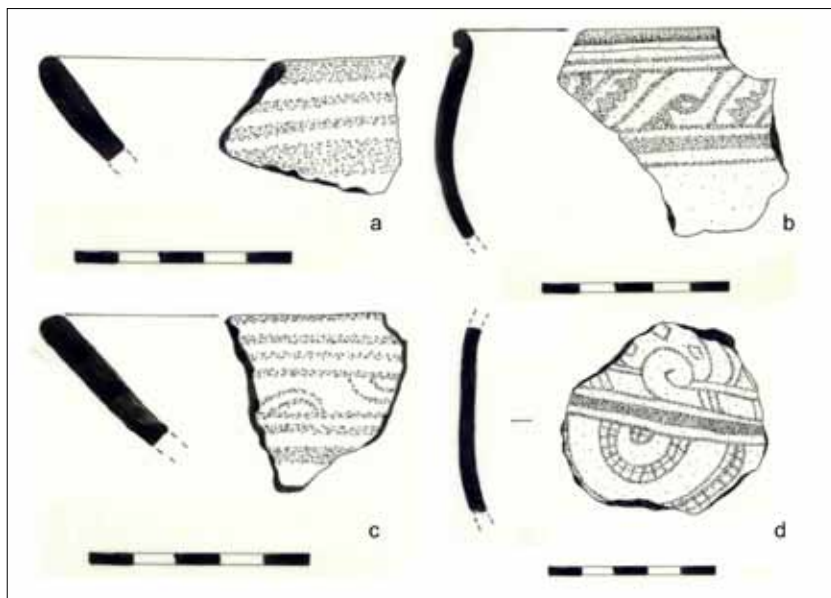


Figure 3-7. Potsherds: (a) Red on Cream, Thin Parallel Lines (type 22); (b–d) Red on Cream, Complex (type 24). Drawings by Helle Girey.

light paste colors and abundant sand temper. Average sherd thickness is about 7.8 mm.

Sherds covered by this type are probably identical to those described by Glassow (1967:68) as Acueducto Red on Buff for Huistla, a site in close proximity to modern-day Etzatlán.

Red on Cream, Complex—Type 24

A variety of design motifs is present on these well-finished sherds, most of which appear to be from large open bowls; some may be rim fragments of molcajetes. One other form represented is the restricted-mouth jar or *tecomate*. All designs include painted red lines on a buff or gray background, with decorated areas being polished. Most designs appear to have been on bowl interiors, although others were rendered on vessel wall exteriors. Motifs consist mostly of lines parallel to the rim, but elaborations occur between these lines in the form of series of concentric circles (see fig. 3-6e), hanging triangles filled with cross-hatching (see fig. 3-6f), or other design elements (fig. 3-7b, c, d). Of the cross sections examined, half contained dark cores, while the remainder evidenced a very light paste color. Temper in all cases is both fine and sparsely distributed through the clay matrix. Mean sherd thickness is about 6.6 mm.

Glassow (1967:70) mentions elaborately decorated red on buff pottery at Huistla and a great diversity of such pottery, described as red on cream or red on buff, occurs in archaeological sites in north-central Jalisco (see Beekman and Weigand 2000).

Grosscup (1976:222, plate 2) describes and illustrates similar design motifs, with small repeated elements enclosed between painted parallel lines for the type Amapa Red on Orange.

Major Polychrome Types

Incised Polychrome—Type 15

For this distinct class of sherds, a single vessel form is indicated: a shallow flat-bottomed bowl with out-flaring sides and probably, in some cases, vessel supports. Decorations apparently extended across the entirety of vessel interiors and around exterior walls as well. On several sherds with support remnants, the legs are hollow. Decorations consist of carefully executed incised patterns with the zones so demarcated

filled with paint (fig. 3-8). Often the result was polychrome bowls with filled zones of different colors, such as red zones alternating with white. At times, several colors are present in a single zone, for example, white on black or orange on black. Overall, these vessels took the general form of molcajetes, but incising in the central flat portions of the interiors probably would not have served a grating function satisfactorily and seems entirely decorative. Incised lines there formed curvilinear patterns and defined zones for polychrome decoration. These bowls may have served as incense burners. Painted portions of these vessels were well polished. Viewed in cross section, sherds manifest very fine temper and mostly a light, oxidized paste color. Sherd average thickness is about 7.7 mm. (Fig. 3-9 shows the drawing color codes for figs. 3-2, 3-4, 3-5, 3-7, and 3-8.)

The type compares to Cerritos Polychrome (Grosscup 1976:228) and Cojumatlan Polychrome (Meighan and Foote 1968:96-111).

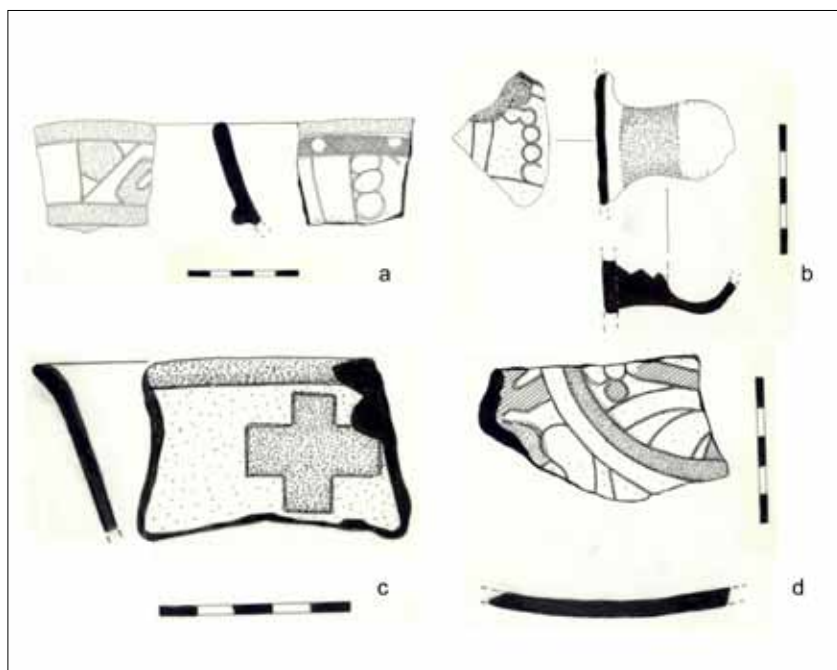


Figure 3-8. Potsherds: (a-d) Incised Polychrome (type 15).
Drawings by Helle Girey.

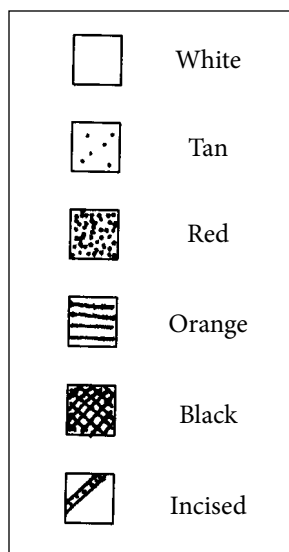


Figure 3-9. Color codes for potsherd drawings.

Polychrome, White Dots—Type 26

These polychrome sherds show great variability in design motif, with elaborate decorations often covering both surfaces. The defining characteristic, though, is at least one alignment of painted white dots, usually running along the center of a painted black band (fig. 3-10a, b). Painted decorations are often applied to red-slipped and polished surfaces and at times occur in conjunction with incising to form zone-incised polychrome. In these cases, without aligned white dots, the specimens would have been

included in type 15, Incised Polychrome. In form, most vessels appear to have been bowls, either small and hemispherical or shallow and flat bottomed with out-flaring sides. Hollow feet supported one small bowl (fig. 3-10c). Polychrome decorations were executed in white, black, red, and orange. These sherds are well finished and polished on at least one surface. Temper is generally very fine and present in moderate to abundant amounts. Paste color varies from reddish brown (5YR5/4) to pink (5YR8/3) with black cores. Apart from surfaces covered with red slip, nondecorated surface color varies from light gray (5YR7/1) to reddish yellow (5YR7/6). Average sherd thickness is about 6.1 mm.

With distinctive aligned white dots on black stripes, this type compares closely to Gavilán Polychrome, assigned to the Gavilán phase of the Amapa sequence (Grosscup 1976:223–24).

Well-Finished Monochrome

Gray/Buff Slipped and Polished—Type 5

These are all plain fine ware sherds with both surfaces well polished. Evidence of vessel shape indicates moderately sized open bowls with curved sides, although two specimens suggest restricted-mouth jars

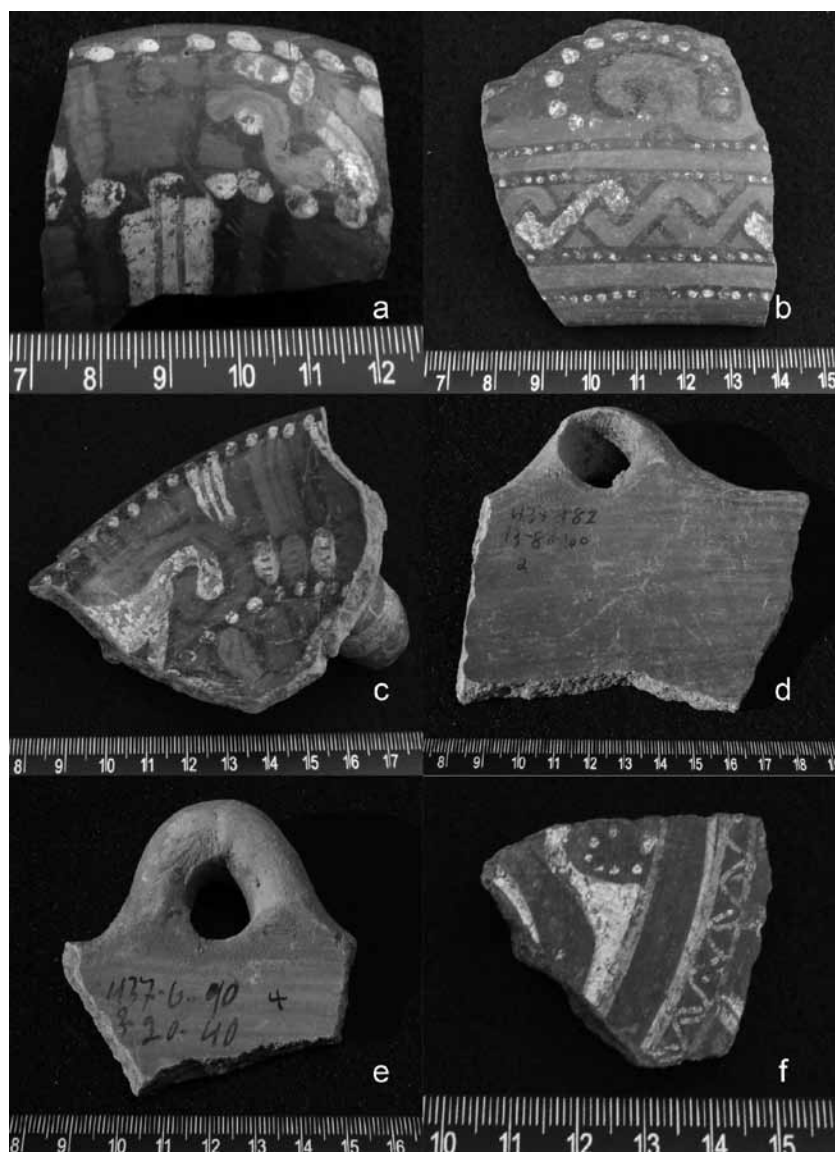


Figure 3-10. Potshrds: (a-c) Polychrome, White Dots (type 26); (d-e) Comales (type 2); (f) White on Black (type 31).

(*tecomates*). Some sherds seem to have been slipped as well as polished; rims tend to be straight and lips, rounded. Viewed in cross section, temper is highly varied both in the size of the crushed rock grains as well as in abundance. More than sherds of most types, these tend to have dark cores or a dark paste color. Sherds average about 6.4 mm in thickness. Surface colors vary from light reddish brown (5YR6/4) and pinkish gray (5YR6/2) to reddish brown (5YR4/3) and dark gray (5YR4/1).

Black/Brown Slipped and Polished—Type 6

In this type, plain, finely finished sherds resemble those of the previous type (5) in terms of vessel form. However, the well-polished and possibly slipped surfaces are darker in color, ranging from dark reddish gray (5YR4/2) or gray (5YR5/1) to black (5YR2.5/1). Surface color tends to be variegated within both types, so much of the color range for each can be manifested on a single sherd. Viewed in cross section, there is a tendency for paste color to be dark or for sherds to have dark cores. Crushed rock temper tends to be fine and abundant. Finally, average sherd thickness is about 6.5 mm.

Residual Types

Gray/Buf Utility—Type 1

These sherds could not be assigned to more specific types. Most are from thick-walled utility pots, but some show well-finished surfaces; others are eroded, some with eroded or burned traces of decoration.

Eroded Fine Ware—Type 4

Here, the sherds are from what were originally well-finished, relatively thin-walled pots; sometimes the bowl form can be discerned. They were distinguished as a class generally on the basis of their fine paste (temper), but due to their eroded surfaces, further distinctions were not attempted.

Comales—Type 2

Clearly identifiable as comales, the sherds are from flat ceramic disks. Viewed in cross section, these platters are thick at the rim but become progressively thinner toward their centers. Upper surfaces tend to be polished and at times coated with a red slip or painted with a broad red stripe around the rim circumference. Top surface color varies from

pinkish gray (5YR7/2) to red (2.5YR5/6) or dark reddish brown (5YR3/2). Bottom surfaces are poorly finished and often have a brushed appearance. Many are blackened from use. Many if not all of these comales had thick loop handles affixed to their rims (fig. 3-10d, e). Cooks probably moved comales on and off the fire by inserting sticks into the handles. Rims are generally squared off at the edge and average about 12.0 mm thick. Away from the rim, these sherds have an average thickness of about 8.2 mm. Viewed on a break surface, paste color tends to be light in color and temper of fine to moderate size is usually abundant.

Other Paint Decorated

Negative Painted—Type 29

Here, artisans produced decorations through negative painting, so that designs are in relief against a red- or brown-painted background. The sherds are too small to reveal overall decorative patterns. Several specimens appear to be from bowls and are decorated on the interior; most sherds are polished on both surfaces. Temper is abundant, moderate-sized grains of sand, and cores tend to be black. Average thickness is about 6.4 mm.

White on Black—Type 31

All body sherds but one, these sherds have a black paint or slip covering their surfaces, upon which exist finely executed designs in white paint (figs. 3-10f and 3-11a). Most of these decorated surfaces appear to be unpolished. The single rim sherd is from a necked jar with out-turned rim. If these sherds had been larger and encompassed more of the vessel, they might have been assigned to type 14, Black and White on Red. Decoration is confined to exterior surfaces, which, along with roughly finished interiors, suggests that most vessels of this type in fact were necked jars. Temper consists of abundant moderate to coarse sand, and paste color is variable with black cores seemingly infrequent. Sherd average thickness is about 5.7 mm.

Red on Black—Type 33

It is difficult to understand the decorative technique(s) represented in these sherds. Several definitely have red paint applied to a black background. Others seem to be negative painted, with the resist material

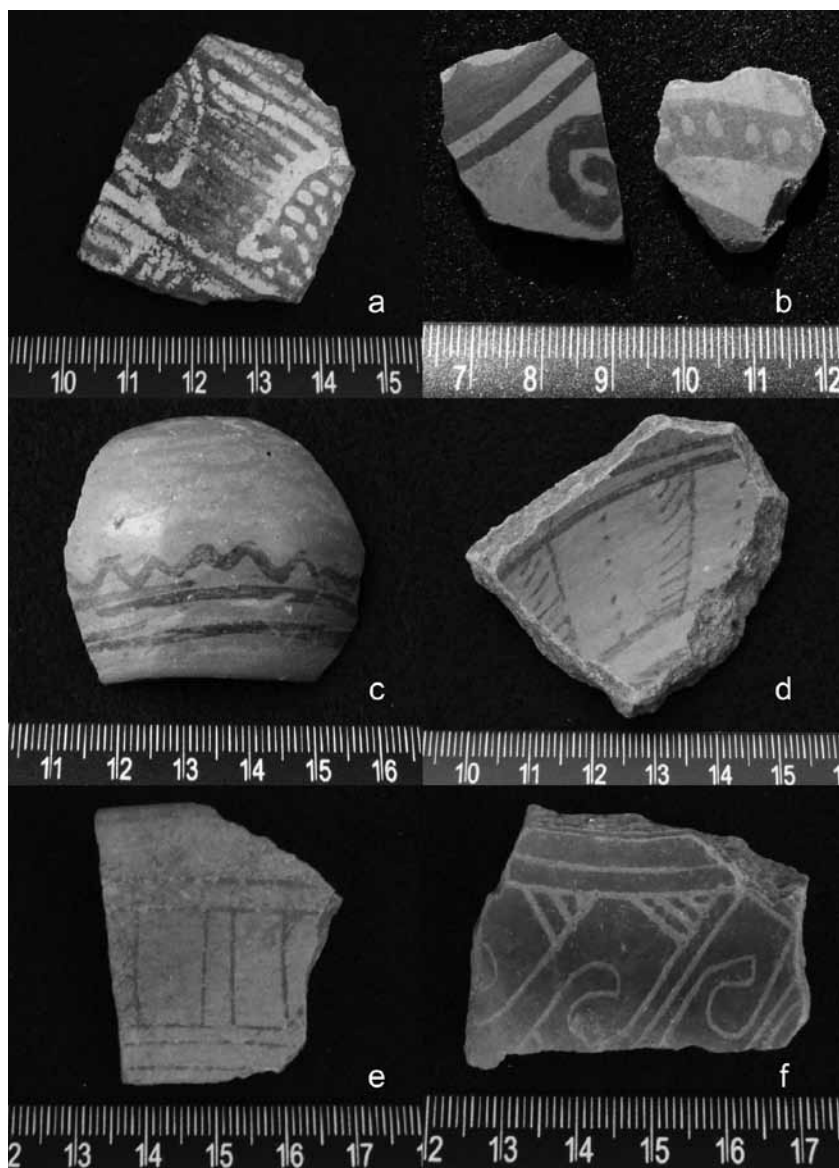


Figure 3-11. Potsherds: (a) White on Black (type 31); (b) Red on Black (type 33); (c) Black on Red Fine-Line, Complex or Red on Black Base, Negative Painted (type 37); (d) Black on Red (type 38); (e-f) Miscellaneous Monochrome Fine Engraved or Incised (type 39).

applied to a black background before application of red paint (fig. 3-11b). Designs appear as rectilinear or curvilinear stripes, in one case with a line of red dots running along a back band. Nondecorated surface colors are reddish brown (5YR5/3) to pinkish gray (5YR6/2). Viewed in cross section, sherd paste tends to be light in color (pinkish gray [5YR7/2] to yellowish red [5YR5/6]), and temper is generally abundant fine sand. Average sherd thickness is about 5.8 mm.

Red on Black (?)—Type 34

These sherds are more problematic than those of the previous type. They appear to be red on black, that is, red stripes on a black background, but actually their surfaces may have been painted almost entirely black, while an underlying red surface was left exposed as one or several stripes. Both surfaces of these sherds tend to be dark—gray (5YR4/1) or black—and polished. Temper is abundant fine sand, and paste color is more often than not dark (dark gray or black). Average sherd thickness is about 6.8 mm.

Black on White, Black on Gray, Brown on White—Type 36

These sherds have black or brown decorations painted on backgrounds ranging from gray (5YR5/1) to reddish yellow (5YR6/6) to pink (5YR7/4). Several sherds are probably from bowl rims and decorated on interior surfaces. Decorations tend to be parallel lines in rectilinear to curvilinear motifs. Paste colors are mostly light (white to light gray [5YR7/1]), and temper varies from moderate-sized crushed rock to very fine sand or crushed rock in moderate to sparse amounts. Average sherd thickness is about 5.7 mm.

Black on Red Fine-Line, Complex or Red on Black Base, Negative Painted—Type 37

This small grouping of sherds is like those of types 33 and 34, where the exact decorative technique is usually unclear. In several of these cases, though, it is obvious that negative painting was used. One of these sherds is from a very small red, probably slipped and polished, bowl. The design, in underlying black, consists of three parallel lines, all parallel to the rim, one wavy, the others straight (fig. 3-11c). Nondecorated surfaces tend to be light reddish brown (5YR6/6). These sherds

have light-colored paste coupled with thin, dark cores, and temper is very fine and sparse. Average sherd thickness is about 6.6 mm.

Black on Red—Type 38

This heterogeneous grouping of sherds has in common black painted lines on a red surface, either on the interior or exterior. Some of these sherds appear to be from slipped and polished vessels, while on others, black lines cross surfaces that appear only to be well smoothed. Decorations vary from straight parallel lines parallel to the rim to haphazardly crossing black lines to more intricate motifs (fig. 3-11d), in one case on the interior of a footed bowl. Nondecorated surfaces vary in color from red (10R4/6) to light gray (5YR7/1). Temper and paste color are also diverse, the former varying from abundant and coarse sand to very fine, sparsely distributed crushed rock. Average sherd thickness is about 6.2 mm.

Engraved and Incised

Miscellaneous Monochrome Fine Engraved or Incised—Type 39

This residual category of sherds is unassignable to other incised or engraved types. For some of these sherds, decorations were applied postfiring and hence are termed *engraved*; incised sherds just as clearly were decorated before firing. For the latter, the grooves are even, while for engraved sherds, lines have a scratchy appearance, although they still evidence considerable skill. We could not make this distinction for some sherds due to their small size or slightly eroded surfaces (fig. 3-11e, f). All of these potsherds are from well-finished vessels, and both surfaces on uneroded specimens are polished. Some rim sherds appear to be from straight-sided bowls; several parallel lines encircle the rim of these, and more complex designs appear beneath, all on exterior surfaces. Two other sherds appear to be from flat-bottomed bowls with out-flared sides of the general molcajete style. Both of these are engraved on their interior surfaces. Surface colors range from red (2.5YR5/6) and dark gray (5YR4/1) to pinkish gray (5YR6/2). In terms of paste, about half the sherds have dark cores, suggesting incomplete oxidation. Temper generally consists of moderate-sized sand grains present in moderate amounts. Average sherd thickness is about 6.3 mm.

Fine Engraved, Arcaded—Type 40

These sherds were grouped together on the basis of a single design motif applied to the polished, nonpainted exterior surfaces of what apparently were hemispherical bowls. The design, depicted in various ways, consists of multiple engraved arches, contiguous to one another and encircling the vessel. Above the arches, depicted through one or two closely spaced parallel lines, are one or two parallel lines encircling the rim (fig. 3-12b). Often, these arches terminated in long lines that extended down and converged toward the base of the pot (fig. 3-12a, c). At times, these designs were executed through incising, not engraving. Another variation can be seen in figure 3-12d. In cross section, the specimens tend to have fine temper present in moderate to abundant amounts. Core color for about half the sherds is dark, and surface colors range from light reddish brown (5YR6/3) or light gray (5YR7/1) to dark reddish brown (5YR2.5/2). Rims are straight and lips, rounded. Finally, sherd average maximum thickness is about 6.3 mm.

This type, in its distinctive and consistent decorative design, is very like the type Cerritos Engraved, described by Grosscup (1976:228) for Amapa.

Washboard Exterior—Type 42

This grouping contains no rim sherds, but interior surfaces are poorly finished, suggesting that the specimens derive from jars. Well-polished exterior surfaces are covered partially or completely with parallel grooves to create a “washboard” appearance (fig. 3-12f). We have indications that some of these pots were supported by small nubbin feet. Exterior surfaces tend to be dark (dark reddish gray [5YR4/2] or dark reddish brown [5YR2.5/2]). Also, most sherd cross sections show a distinctive pattern, having light paste near the exterior surface but blackened paste next to the interior wall. The pots may have been fired upside down, producing this distinctive pattern of incomplete oxidation. Temper is of moderate grain size and abundant. Average sherd thickness is about 7.7 mm.

Brushed Plaques—Type 43

These ceramic artifacts appear to be fragments of thick rectangular or square plaques. Edges are straight, and one surface has been brushed, mostly producing striations in line with or perpendicular to the edge

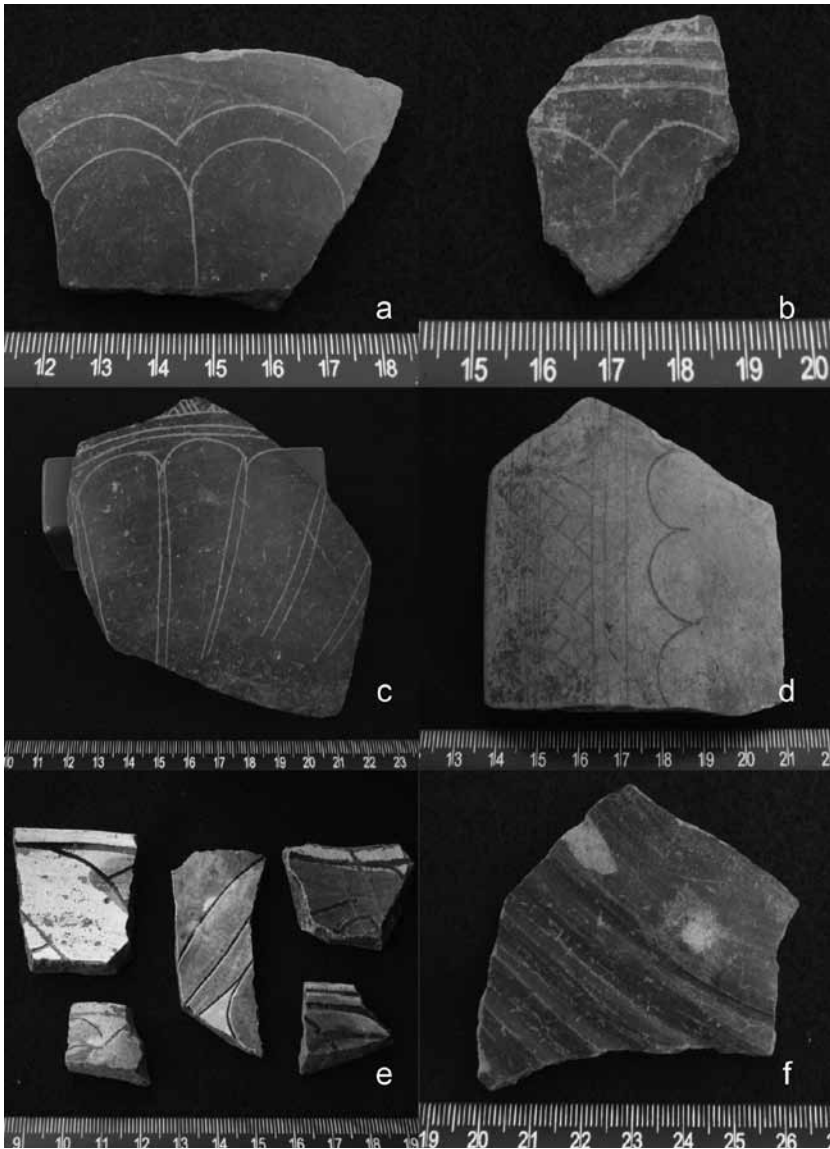


Figure 3-12. Potsherds: (a-d) Fine Engraved, Arcaded (type 40); (e) Historic, Glazed Majolica (type 41); (f) Washboard Exterior (type 42).

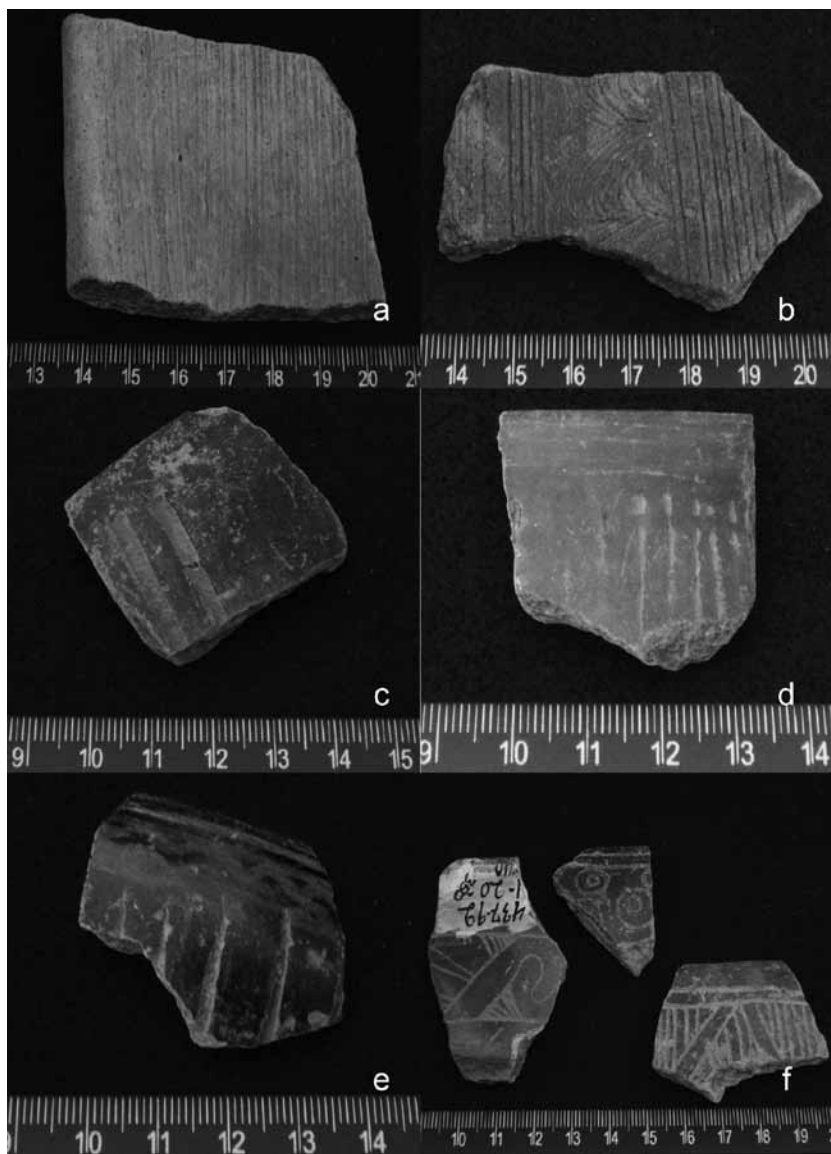


Figure 3-13. Potsherds: (a-b) Brushed Plaques (type 43); (c-e) Hatched from Rim Exterior, Incised (type 44); (f) Black Crude Engraved, Complex (type 45).

(fig. 3-13a). Occasionally, these striations form curvilinear patterns (fig. 3-13b). Smooth border areas, about 1 cm wide, occur next to the rounded edges. Reverse surfaces are smooth but unpolished. The size and function of these artifacts is unknown. They have abundant amounts of moderate to fine sand temper and vary in paste color. Surface color tends to be pinkish gray (5YR6/2) or light reddish brown (5YR6/3), and mean thickness is 9.2 mm.

Meighan (1976:84–87) describes ceramic plaques for Amapa, but those differ in having elaborate incised decorations.

Hatched from Rim Exterior, Incised—Type 44

The distinguishing feature of these potsherds is the presence of deeply incised parallel lines perpendicular to the rim. On some, the lines were deeply grooved into the wet clay with a comb, often with square-ended teeth (fig. 3-13c, d). Both surfaces of these sherds are well finished, most polished, and some probably slipped as well. The vessels represented were probably open bowls with rounded sides; rims were straight and lips, rounded. Paste color is variable, and temper is mostly abundant very fine crushed rock or sand. Surface colors vary, too. About half appear to be black slipped and polished on both surfaces. Several others are red slipped and polished on the exterior, and one of these has a black band below the rim interrupted by negative-painted designs (fig. 3-13e). Mean sherd thickness is about 5.9 mm.

Black Crude Engraved, Complex—Type 45

These sherds appear to have dark-slipped and polished surfaces altered by complex engraved decorations. Designs were crudely rendered, and sometimes a relatively large area of surface was scratched away to leave the design in relief (fig. 3-13f). Several sherds appear to bear the multiple-arcade motif and could have been assigned to type 40, although again, the designs were crudely executed. Other motifs include nested diamonds, zoned hatching, concentric circles, and interlocking scrolls. Most decorations occur on exterior surfaces of round-sided bowls, one sherd may be from a necked jar, and several are decorated on interior surfaces of what apparently are fragments of shallow bowls with flanged rims. Viewed on fresh break surfaces, these sherds show evidence of having been coated with an organic substance or having been fired in close proximity to organic debris. Such firing might have produced

smudged surfaces, which were then polished black. This seems likely, because the paste (in sectional view) is often blackened near black, polished surfaces. Cross sections often appear multilayered: black cores, blackened edge sections, and intervening light-colored bands. Temper is finely crushed rock distributed in sparse to moderate amounts. Both surfaces of these sherds tend to be black (5YR2.5/1) to very dark gray (5YR3/1), although several interiors are polished gray (5YR5/1). Rims tend to be straight and lips, rounded. Average sherd thickness is about 5.6 mm.

Similarities exist with the Amapa type Tuxpan Engraved (Grossman 1976:236–37), especially in the scratchy appearance of designs, where sometimes the engraving technique was utilized to remove small areas around a vessel's fired and polished surface (see Grossman 1976:plates 175–76).

Black Crude Engraved, Suspended Triangles—Type 46

Sherds of this category would have been merged with those of the previous type except that these could be classed together on the basis of a single design motif: crudely rendered, contiguous triangles suspended beneath several parallel lines encircling the rim. The triangles are filled with hatching (fig. 3-14a, b, c), and all decorations appear to be on the exterior surfaces of open, round-sided bowls. One almost complete specimen had a flat bottom and probably was supported by three solid, conical feet. Paste and surface color characteristics are the same as those of the previous type. Average sherd thickness is about 6.3 mm.

Similar hanging triangles can be seen on several photographed bowls of the Tuxpan Engraved type (Grossman 1976:plates 175 D, 176 B), although on Etzatlán specimens, designs are simpler and more crudely rendered.

Dark Red to Black Engraved, Complex—Type 47

The engraving here is more precise, and surface color is more variable, from black to dark red. These characteristics distinguish this grouping of potsherds from those of the previously described engraved types. Also, designs can be more complex, and some of the motifs described previously are absent. Rim sherds predominate, and apparently, decorations were confined to areas near rims (fig. 3-14d, e). Several tecomates are represented along with straight-sided bowls, all decorated on

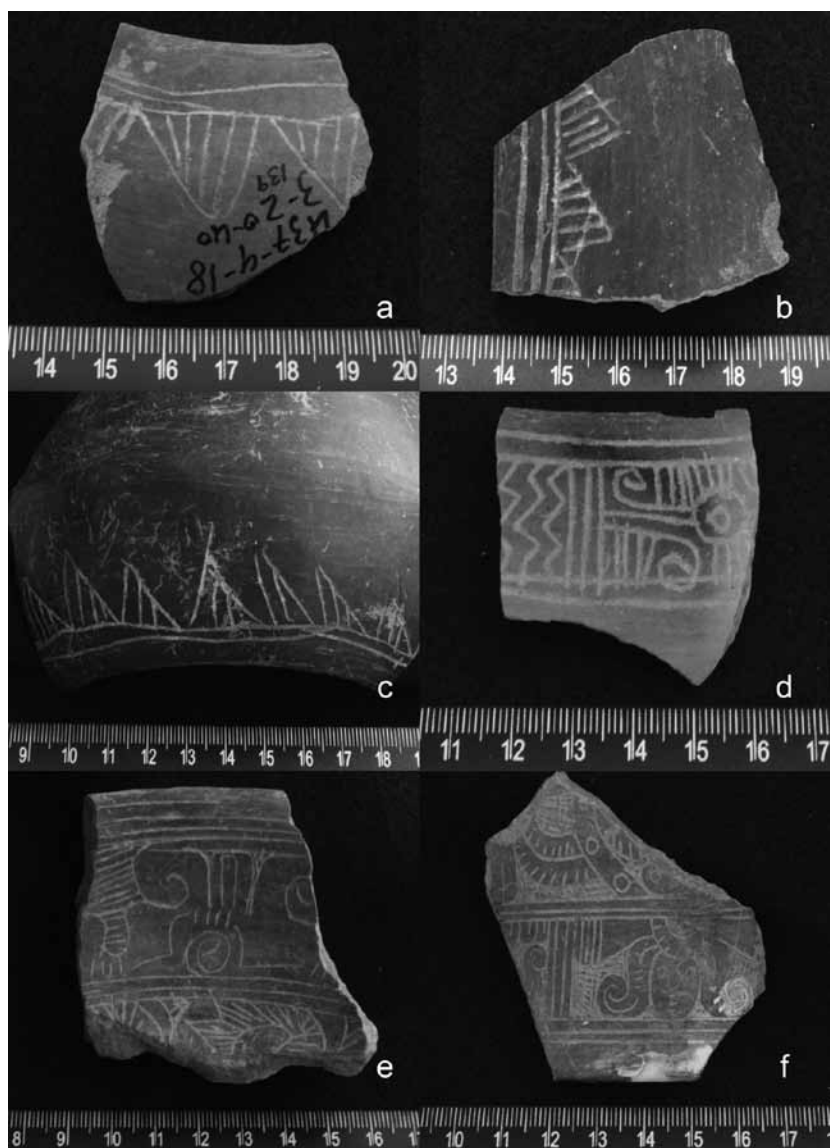


Figure 3-14. Potsherds: (a-c) Black Crude Engraved, Suspended Triangles (type 46); (d-f) Dark Red to Black Engraved, Complex (type 47).

the exterior. One sherd is from a molcajete-shaped bowl with engraving on the interior, and another sherd bears the crude depiction of a human face (fig. 3-14f). Surface colors range from yellowish red (5YR4/6) to very dusky red (2.5YR2.5/2) or very dark gray (5YR3/1). Regarding the paste, most sherds appear to be virtually temperless under low-power microscopy, others show moderate amounts of fine temper, and a few, abundant sand temper. About half the inspected sample had dark cores or exhibited a dark paste color, for example, very dark gray (5YR3/1). Average sherd thickness is about 5.8 mm.

Reed Punctate—Type 48

Here, included sherds have one consistent attribute: the presence of multiple hollow tube punctuations, probably produced with small reeds or bird bones. One included specimen is from a well-finished red bowl with a turned-out or flanged rim (fig. 3-15a).

Exterior Incised, Deep Parallel Lines—Type 49

The specimens examined are all body sherds, so it is impossible to know the orientation of multiple straight incised lines parallel to the rim. Incisions were carefully executed on exterior surfaces, with lines being both broad and deep (fig. 3-15b, c). Both surfaces tend to be well finished and polished. Exterior surface colors range from reddish brown (5YR4/3) to very dark gray (5YR3/1). Paste was tempered with moderate to fine sand present in moderate quantities, and most sherds examined have dark gray cores. Average sherd thickness is about 5.7 mm.

Red Fine Incised—Type 50

These sherds were grouped together on the basis of their red surface color and the delicate and precise nature of their incised decorations. The lines are so thin that we first believed them to be engraved. Later microscopic examination, however, revealed no evidence to support this interpretation. The sherds do not appear to be slipped, but they are highly polished on both surfaces. Elaborate motifs are contained within parallel-incised lines encircling the rim, occurring on what seem to be mostly open bowl exteriors (fig. 3-15d, e). Temper is fine, though it tends to be unsorted, and is probably crushed rock, present in sparse to moderate quantities; paste color tends to be gray or lighter in value. Exterior surface color varies from weak red (10R4/6) to dark reddish brown

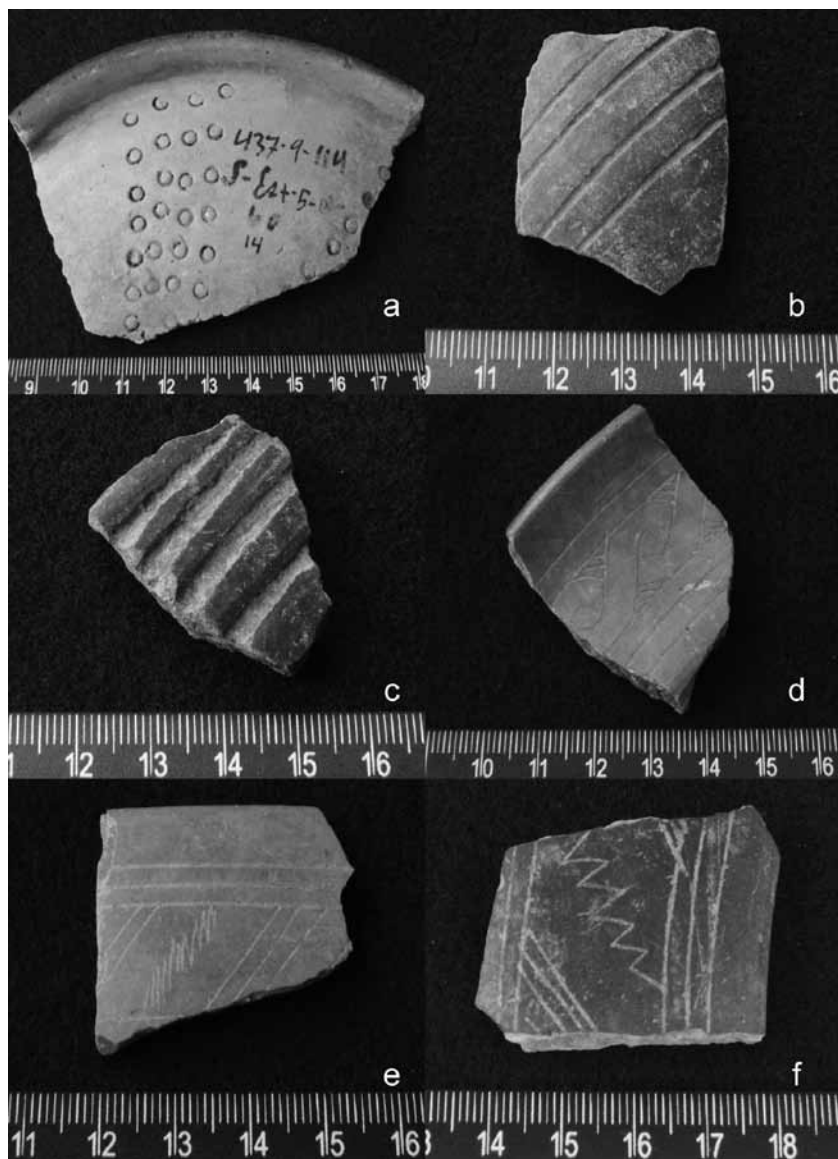


Figure 3-15. Potsherds: (a) Reed Punctate (type 48); (b-c) Exterior Incised, Deep Parallel Lines (type 49); (d-e) Red Fine Incised (type 50); (f) Black Fine Engraved (type 51).

(2.5YR3/4). Several of the rims curve outward, but most are straight, and lips are rounded. Average sherd thickness is about 6.3 mm.

Black Fine Engraved—Type 51

Compared to the ambiguities of the previous type, designs here were more clearly rendered through engraving. Lines are even thinner, and designs are less accurately depicted, indicating the difficulties of the engraving technique. However, the work is much finer than that of the other two engraved types (45 and 46) described previously. Again, designs appear to be confined to zones encircling the rim (fig. 3-15f). At least one sherd likely is from a tecomate. Several interior surfaces are poorly finished, while all other sherd surfaces seem to be both slipped and polished. Surface colors vary from dark gray (5YR4/1) to black (5YR2.5/1), and the paste is also variable, both in temper and color. Sherd thickness averages about 5.6 mm.

Miscellaneous Broad-Line Incised—Type 52

This residual category includes small potsherds with several parallel, straight lines incised on their exteriors. Surfaces tend to be polished and to range from yellowish red (5YR4/6) to very dark red (5YR3/1) in color. Paste color tends to be dark (very dark gray [5YR3/1]), and temper appears to be poorly sorted and mostly fine crushed rock in abundant amounts. Average sherd thickness is about 6.4 mm.

Complex Broad-Line Incised—Type 53

This distinctive grouping of sherds is set off by deep broad-line incising and complex design motifs (fig. 3-16c), which are either rectilinear or curvilinear. One characteristic design is a spiral, either curvilinear or squared off, accompanied by a single zigzag line (fig. 3-16a, b). Most sherds appear to be from thick-walled, open bowls with straight rims and rounded lips. Surfaces are well finished, although few are highly polished. Most decorations are on the exterior surface near the rim, but there are exceptions. As for paste, dark cores are relatively infrequent, the color tending to be reddish brown (5YR4/4). Crushed rock temper is fine to moderate in grain size and generally sparse. Surface colors range from light reddish brown (6YR6/3) to dark gray (4YR4/1), but one sherd we examined has a red exterior (10R4/3). Finally, mean sherd thickness is about 6.9 mm.

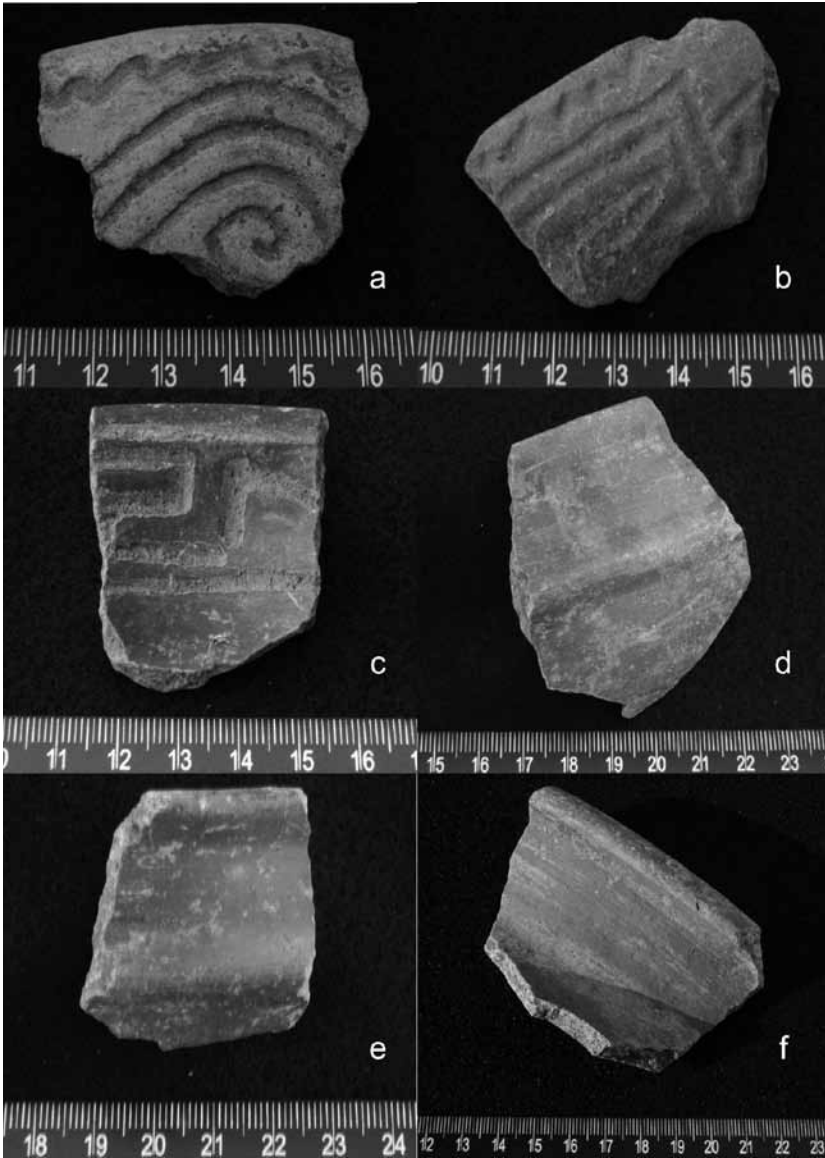


Figure 3-16. Potsherds: (a–c) Complex Broad-Line Incised (type 53); (d–e) Fine Red (type 61); (f) Red Utility (type 62).

Historic

Glazed Majolica—Type 41

Majolica sherds for the most part are small and lack defining characteristics. However, a few sherds, larger than others, have green and yellow painted areas outlined by dark brown painted lines, all over a white background. These sherds (see fig. 3-12e) resemble illustrated specimens of the type Santa María Polychrome, defined by Lister and Lister (1982:28, fig. 3.41) for the Valley of Mexico.

Red Ware

Fine Red—Type 61

Here, potsherds were grouped on the basis of their fine red finish. Some sherds, viewed microscopically in cross section, appear slipped on their exterior surfaces. Temper—probably crushed rock for the most part, otherwise sand—is usually of moderate size and tends to be plentiful. Only one of the examined sherds contains a dark core, and paste color is generally light. Apparently, the vessels became well oxidized during firing. Several sherds are indicative of a composite silhouette vessel form (fig. 3-16d, e), and others are from simple bowls with curving sides. Also, a minority of sherds are thin walled and well finished on one surface only. These may be from moderate-sized necked jars. None of these sherds is decorated, and most are red (5YR5/6 to 10R4/6) on both surfaces. Mean sherd thickness is about 5.8 mm.

Red Utility—Type 62

Many of these sherds are from the rims of large necked utility jars with everted rims. A red slip or paint is often confined to two broad bands around the rim, one on each side of the lip (figs. 3-16f and 3-17a). Other sherds are from the rims of large deep bowls, some of which may have been slipped red and polished on both surfaces. Viewed under low-power microscopy and on fresh break surfaces, temper is large grained and moderate to high in concentration. At times, this temper is clearly sand; in other sherds, it has the appearance of crushed rock. About half the sherds have dark and apparently incompletely oxidized cores. Average sherd thickness is about 9.3 mm.

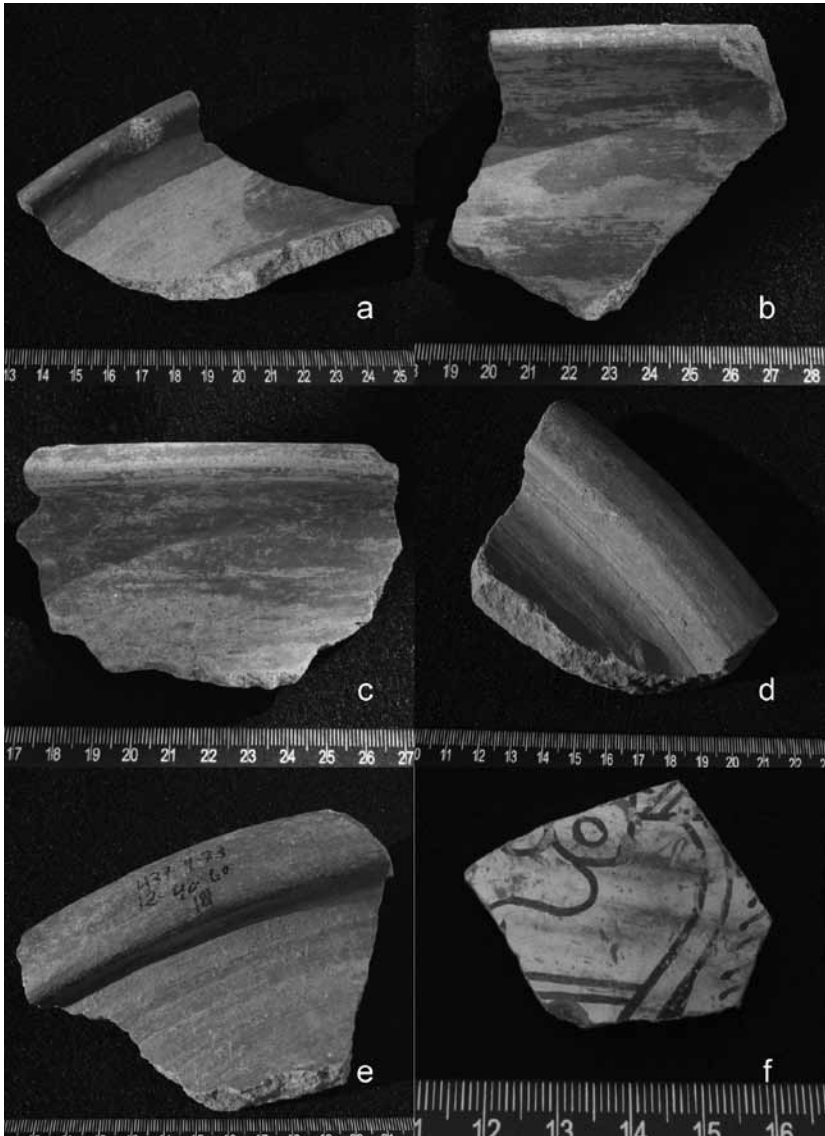


Figure 3-17. Potsherds: (a) Red Utility (type 62); (b–c) Red-Painted Jars, Streaky Polish (type 63); (d–e) Red Utility, Very Thick (type 64); (f) Red on White (type 27).

Red-Painted Jars, Streaky Polish—Type 63

We distinguished sherds of this type on the basis of the necked jar form, red exterior surface, and the presence of streaky polishing. Almost all sherds in this grouping are rim fragments, and on the interior surface, the red paint of the exterior extends into the mouth as far as the base of the vessel neck. Jar rims were thickened and sometimes partially flattened at about a 60-degree angle (fig. 3-17b, c). These were large utility pots, but body walls were relatively thin: at the base of the neck, these sherds measure about 5.8 mm in average thickness. Several specimens are burned black on their interiors, and some vessels may have been used in cooking. Viewed in cross section, most sherds have abundant sand temper, with grain size being moderate to large. One sherd we examined has coarse, crushed rock temper. Finally, most sherds have light-colored paste, so apparently the vessels were well fired.

Red Utility, Very Thick—Type 64

Here, sherds were grouped together because of their massive thickness and red surfaces. All are rim sherds, and the only vessel form suggested is that of a very large globular pot with a thickened and everted rim. Rims were folded out in many cases to create a definitely protruding lip (fig. 3-17d, e). These pots probably functioned as stationary storage vessels, and these reinforced and out-turned rims would have facilitated the use of tied-down fabric covers. Surface finish varies, but it appears that some vessels were covered with an exterior red slip and then polished. Paste examination of some specimens revealed uniformly light colors suggestive of thorough firing. Sherd thickness is variable, but rims can be up to 26 mm thick. Body thickness ranges from 9 to 22 mm.

Other Minority Types

Red on White—Type 27

Sherds in this category have one or both surfaces covered with a white slip, which served as a background for red-painted decorations. Designs range from simple parallel bands to complex curvilinear or rectilinear motifs (fig. 3-17f). In the study sample, one sherd is from a necked jar with straight rim; others are from open bowls mostly decorated on interior surfaces. Some decorated surfaces are polished. Inspection of the paste revealed a few sherds that are virtually temperless, while for most



Figure 3-18. Potsherds: (a) Red Wash over Black Painted Design (type 35); (b) Black and White on Red, Slipped and Polished (type 25); (c) Crude, Punctated and Incised (type 55); (d-e) Polished Utility (type 60); (f) Unique Sherd A (type 57).

others, temper is fine to moderate-sized crushed rock. It tends to be sparsely distributed. In most cases, the paste is light gray in color. Average sherd thickness is about 7.2 mm.

Red Wash over Black Painted Design—Type 35

The actual design technique employed on these potsherds is unclear but might have involved black bands painted on possibly a white or other light-colored background with the entire surface then coated over with a red wash. However, the designs seem in part the product of negative painting, with the original banding produced in this way (fig. 3-18a). The only vessel form in evidence is the curving-sided, open bowl. Decorations occur on one or both surfaces, and both surfaces tend to be polished. Also, paste color is most often light gray; temper consists of moderate amounts of sand. Average sherd thickness is about 6.9 mm.

Black and White on Red, Slipped and Polished—Type 25

Not a well-defined grouping, some of these sherds would have been assigned to type 14, except that the sherds are well finished, both slipped and polished on their exterior decorated surfaces. Others are from well-finished bowls with unique decorations around the rim interior (fig. 3-18b). Crushed rock temper varies from abundant and of moderate size to sparse and very fine.

Thick White Paint—Type 28

These sherds have white paint or stucco covering their exterior surfaces. Several specimens appear to be from a ring-based bowl. Temper in one case is abundant sand and in others, crushed rock varying in grain size and abundance. Paste color tends to be pinkish gray (5YR7/2) or red (2.5YR5/6). The sherds tend to be relatively thick, with an average mean thickness of 7.6 mm.

Crude, Punctated and Incised—Type 55

For this small grouping of three sherds, decorations consist of punctations on one, zoned punctations on a second (fig. 3-18c), and punctations aligned between incised lines on the third. The sherds are smoothed but neither slipped nor polished. Temper seems to be fine crushed rock, some of it quartz.

There is a resemblance here to incised and punctated sherds of the Formative Capacha phase defined by Kelly (1980:fig. 18) for Colima.

Polished Utility—Type 60

This type represents an intergrade between Gray/Buff Utility (type 1) and Gray/Buff Slipped and Polished (type 5) in which the sherds are from larger vessels than indicated for the latter type but are better finished than those of the former. Some of these sherds are from necked jars with out-flaring rims (fig. 3-18d), while others derive from large open bowls (fig. 3-18e). Surface colors range from reddish yellow (5YR5/3) or light reddish brown (5YR5/4) to pinkish gray (5YR6/2). Most sherds we examined have dark cores, suggestive of incomplete oxidation; temper in most cases appears to be crushed rock of moderate grain size present in moderate amounts. Average thickness is about 8.6 mm.

Unique Sherds

Unique Sherd A—Type 57

This body sherd probably is from a well-finished bowl. On the interior surface an elaborate design of contiguous concentric circles (fig. 3-18f) was executed in fine incising. This surface is slipped red and polished, and line incisions seemingly were filled with white paint. The exterior surface, light reddish brown (5YR6/4), is polished. Paste color is light gray (5YR7/1), and the temper, poorly sorted, is fine to very fine black crushed rock.

The sherd appears to conform to the type Santiago Engraved of Amapa (see Grossman 1976:plate 160 F).

Unique Sherd B—Type 58

Here, the specimen may be from a shallow bowl. Crude hatched designs cover the exterior surface, the interior is blackened, and neither surface is well finished. It appears that a flanged rim once surrounded the periphery (fig. 3-19a), and the sherd may be a ceramic lid fragment.

Unique Sherd C—Type 70

The remains of this vessel show that a single black band was apparently painted just below the exterior of the rim (fig. 3-19b). The sherd is unique

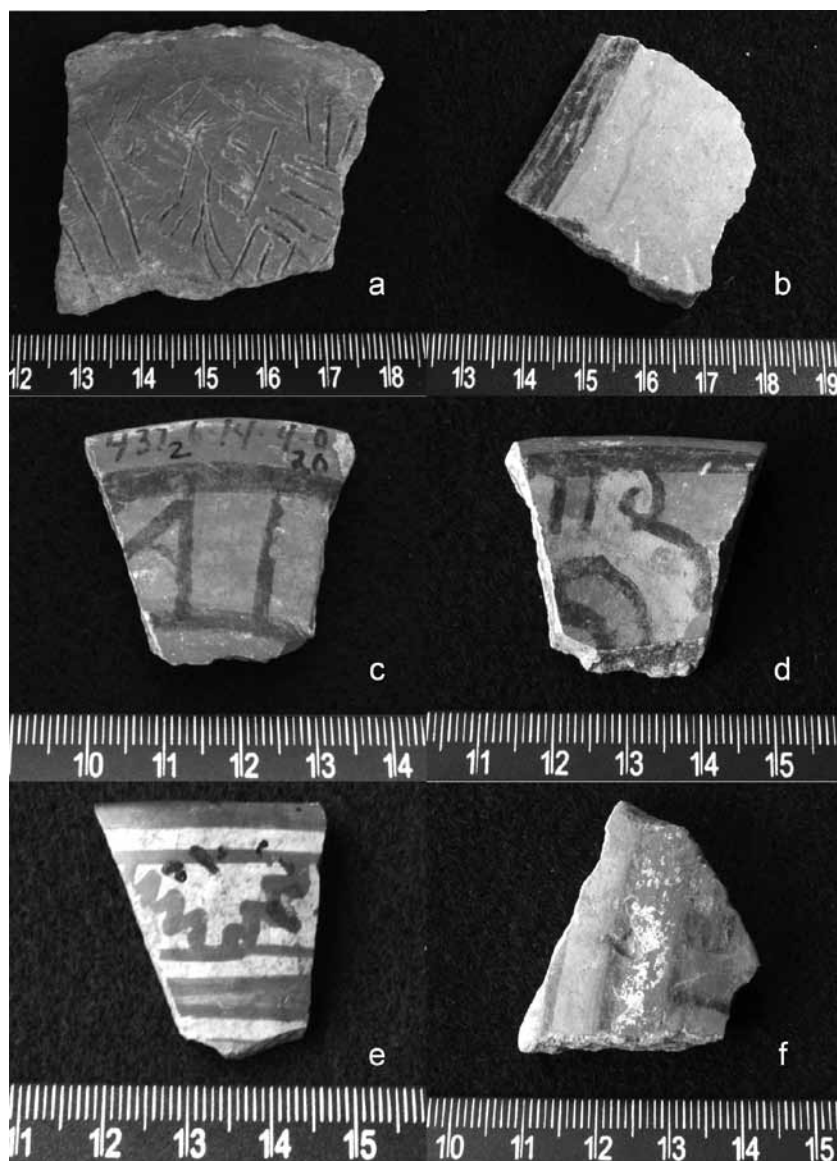


Figure 3-19. Potsherds: (a) Unique Sherd B (type 58); (b) Unique Sherd C (type 70); (c-d) Unique Sherd D (type 71); (e) Unique Sherd E (type 32); (f) Unique Sherd F (type 32).

in this regard and also due to the orange color (yellowish red [5YR6/6]) of its exterior slip. The interior is unslipped but highly polished and pinkish gray (5YR6/2). The specimen probably originated from a thick-walled tecomate. Temper consists of a moderate amount of fine sand, and the paste is generally light gray. The sherd is 8 mm thick.

Unique Sherd D—Type 71

This sherd could have been assigned to the type Red and Black on Buff, Complex (Unique) but has been singled out because its decoration is so unusual compared to sherds in that group. The sherd is from a thin-walled (5.5 mm thick) plate or shallow bowl, decorated on both surfaces (fig. 3-19c, d). The paste is light gray, and temper consists of very fine sand present in a moderate amount.

Unique Sherd E—Type 32

This red and black on white polychrome sherd, decorated on both surfaces, seems foreign relative to other sherds in the collection and may be from a trade vessel. It is notable for its thinness (6 mm thick) and the finely executed design painted on the exterior (fig. 3-19e). It probably comes from a relatively small tecomate. The paste is dark gray (5YR4/1); temper is fine and sparse.

Unique Sherd F—Type 32

Again, this polychrome sherd is not easily assigned to another type. Both surfaces are covered with red slip. Rectilinear designs are rendered in orange, black, and white paint on the exterior surface (fig. 3-19f). Additionally, both surfaces are polished. Temper is a soft, light-colored substance, and paste color is reddish yellow (5YR4/6); thickness measures 7 mm.

Unique Sherd G—Type 32

This potsherd, slipped on both surfaces, has elaborate polychrome designs painted on one surface. The painted lines are more delicate and the designs more intricate than found on other polychrome pottery in the Etzatlán collection. The sherd is thought to be from a vessel traded into the region (fig. 3-20a). Paste color is gray; temper is poorly sorted gray crushed rock in moderate abundance.



Figure 3-20. Potsherds: (a) Unique Sherd G (type 32); (b) Unique Sherd H (type 32). Other ceramic artifacts: (c-d) Mazapan figurines; (e) spindle whorl; (f) effigy vessel foot.

Unique Sherd H—Type 32

Like the previous type, this polychrome sherd is also isolated by the polychrome design painted on its exterior surface (fig. 3-20b). Both surfaces appear slipped and polished. This sherd is likely from an imported vessel as well. Temper is fine crushed rock and the paste, variegated in color.

Other Ceramic Artifacts

Included here are a few ceramic artifacts not contained in the ceramic data sets. Most of these were excavated from Las Cuevas. Included are so-called cookie-cutter or Mazapan figurines (fig. 3-20c, d), considered “a standard late horizon time marker for the region” (Meighan 1976:69). Excavators also recovered small ceramic spindle whorls, which are



Figure 3-21. Other ceramic artifacts: (a–b) ceramic stamps; (c) pipe stem; (d) colonial figurine.

characteristic of regional Classic to Postclassic sites (Lister 1949:63–67; Meighan 1976:79–82; Meighan and Foote 1968:126–32). Etzatlán spindle whorls vary in form from biconical to a truncated cone to disk shaped. Most bear incised decorations (fig. 3-20e). These artifacts average about 2.25 cm in diameter and 1.41 cm in thickness, as measured through their drilled centers. Another artifact category to consider consists of hollow vessel feet, each molded in the form of an animal's head (fig. 3-20f). Very similar specimens are reported and illustrated for Cojumatlán (Lister 1949:21–22, 26), a site near Tizapán El Alto on Lake Chapala and in part contemporary with it (cf. Meighan and Foot 1968). Other ceramic artifacts to be mentioned are ceramic stamps with elaborate geometric designs (fig. 3-21a, b), similar to those from Amapa (Meighan 1976:81–82, plates 71–72); a whistle fragment (cf. Meighan 1976:76–77, plates 55–57); several modeled figurine fragments; clay pellets, probably from hollow vessel-foot rattles; and a possible pipe stem (fig. 3-21c; cf. Meighan 1976:81–83). Finally, there is one small figurine of particular interest. The specimen is a mold-made rendition of a human head, definitely of Spanish colonial inspiration, but seemingly produced through the indigenous ceramic technology. The item lacks glaze and appears to have been fired at low temperature (fig. 3-21d). The chronology of these artifacts will be discussed in chapter 5.

Ceramic Analysis

By C. Roger Nance, Jan de Leeuw, Kathleen Prado, and David S. Verity

Las Cuevas

+++Our quest to determine the ceramic sequence for the Etzatlán region began with research on the relatively large collection from the site of Las Cuevas. Here, Long and Glassow had excavated sixteen pits scattered over a relatively flat promontory of land that extended into the lake on the eastern side of the island. We processed about seventy-eight hundred potsherds from these sixteen units, classifying and physically sorting the sherds into sixty-five different types. We also added a unique specimen number to each sherd. Next, we recorded the data in notebooks, listing for each sherd its type, unique ID number, and provenience (square and depth). Finally, we entered the data into a computer and then compared computer printouts to handwritten data, line by line, to eliminate data entry errors.

When we attempted to study type distributions in a traditional way, looking for stratigraphic changes through time, level by level, the data appeared confusing—trends in some units did not appear in or were even contradicted by data from others. Part of the problem was that we were dealing with part of a large open site that had been intensively occupied. Deposits with cultural debris tended to be shallow with 94.9 percent of all potsherds from Las Cuevas excavated from deposits less than 1.0 m deep. Also, we found ample evidence of disturbance:

field notes for the Las Cuevas excavation record fragmentary burials, a rock oven (?), rock concentrations, intrusive pits, and a rock wall foundation. As a result, we took a statistical approach and concentrated at Las Cuevas on data from fifty-two lots of potsherds (samples from individual square levels) with more than fifty sherds each. We also concentrated on those types present in large frequencies in order to reduce the problem of sampling error. Our approach was to look for covariation of types among the samples with the idea being that types, which covaried closely, would tend to be contemporaneous. Initially, we used the correlation coefficient in an explorative way to discover this covariation among important or high frequency types. At this stage of the Las Cuevas analysis, we were able to discern two type complexes. Each group was defined by positive correlations among some of these types, while negative correlations separated these same two complexes on a type-to-type basis. Also, we had some reason to believe that one complex was earlier than the other, given slightly different average levels for the two groups of types in the deposits. The types in each of the suggested groups are listed in table 4-1. In order to increase sample sizes, in several cases we combined sherd data from related types for the analysis. Specifically, sherd data for Huistla Polychrome grater bowl types (7, 8, and 9) were grouped, as were data for Huistla Polychrome non-grater bowl types (19, 20, and 21) and White on Red types (16, 17, and 18). Morphological differences among these individual types are described in chapter 3.

Unfortunately, we had been forced to use type (or combined type) percentages by lot in constructing the correlation matrix in order to obtain any patterning at all. This practice of employing the correlation coefficient with percentages instead of frequencies is considered

Table 4-1: Two provisional ceramic complexes

Early Complex	Late Complex
Incised Polychrome	Huistla Polychrome
Gray Slipped and Polished	Huistla Polychrome Grater Bowl
White on Red	Brown Slipped and Polished
	Comal

to be statistically unreliable. Our initial effort, however, did show some potential for a seriated sequence of ceramic types at Las Cuevas and indicated how those types might figure in such a sequence.

We turned next to a different statistical treatment of these data, correspondence analysis (CA). This procedure uses the chi-square statistic to measure the degree of covariation among both types and samples and maps the results on a two-dimensional grid. CA is described and its archaeological potential evaluated in chapter 2. Evidence for the two complexes (table 4-1) can be seen in the results of an initial CA (fig. 4-1). The three types provisionally designated as early are to the left: they have lower x-axis values than the four types assigned to the late complex. At this point, though, we still did not have enough information to determine if these two complexes represented different portions of a ceramic sequence. The two type groupings, for example, could represent functional rather than chronological differences. In order to explore this issue, we turned to the study of potsherds from three additional sites.

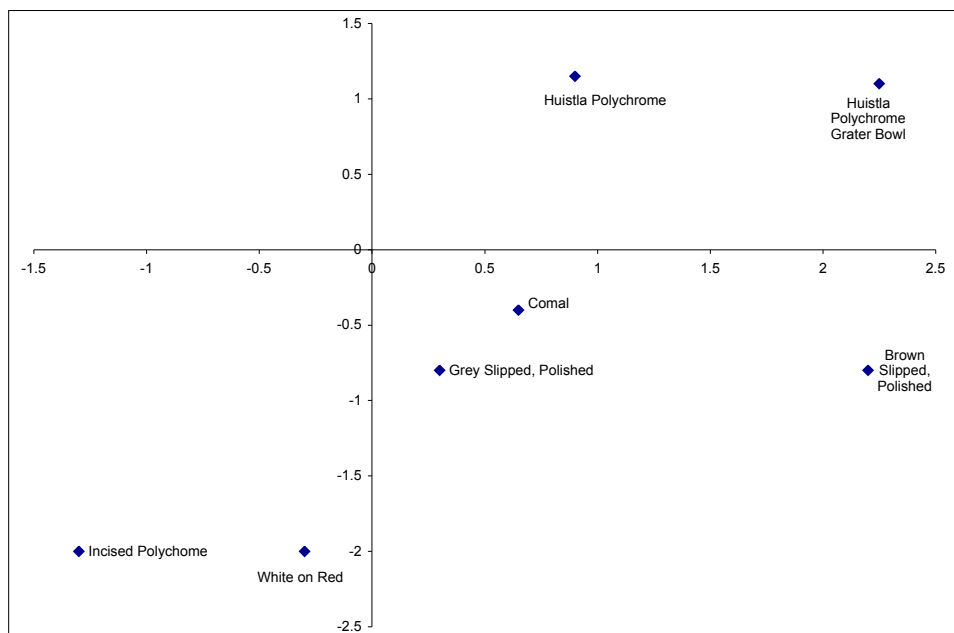


Figure 4-1. Type plot, original Las Cuevas CA.

Tiana

Long excavated few squares at the site of Tiana, and we obtained only eleven lots with the minimum fifty sherds, distributed over four pits. However, when the data were summarized in terms of the original two complexes (table 4-1) and 20 cm level samples, we could see stratigraphic evidence (table 4-2) that the two complexes we proposed for the site of Las Cuevas do represent to some degree different portions of the local ceramic sequence. They align stratigraphically in the suggested chronological order. One should keep in mind here that the four excavation units at Tiana were not contiguous but scattered over the site.

In the course of the Tiana analysis, however, we identified an inconsistency with findings from Las Cuevas. Sherds of the late complex types (see table 4-1) from Tiana did not include those of Huistla Polychrome. Only three Huistla Polychrome sherds were found among approximately eighteen hundred sherds excavated. Instead, the late designation for levels depicted in table 4-2 was based mainly on the presence of sherds of the Comales type. Paint-decorated sherds from these same levels tended to be from three types ignored so far in the analysis: those with red-painted decorations on a cream or buff background. We had not included these types in the Las Cuevas CA because they were not found to correlate significantly with other types when first viewed in the Pearson correlation matrix. Now these Red on Cream types took on significance. In the CA for Las Cuevas (fig. 4-1), Comales can be seen occupying an intermediate position between the Huistla Polychrome types on the one hand and Incised Polychrome and White on Red on

Table 4-2: Distribution of provisional ceramic complexes at Tiana by level and square

Depth of Excavated Unit (cm)	Unit 1	Unit 2	Unit 3	Unit 4
0–20	Late	—	Mixed	—
20–40	Mixed	Late	Mixed	Late
40–60	Mixed	Late	Early	—
60–80	Early	Mixed	—	—
80–100	—	—	—	—

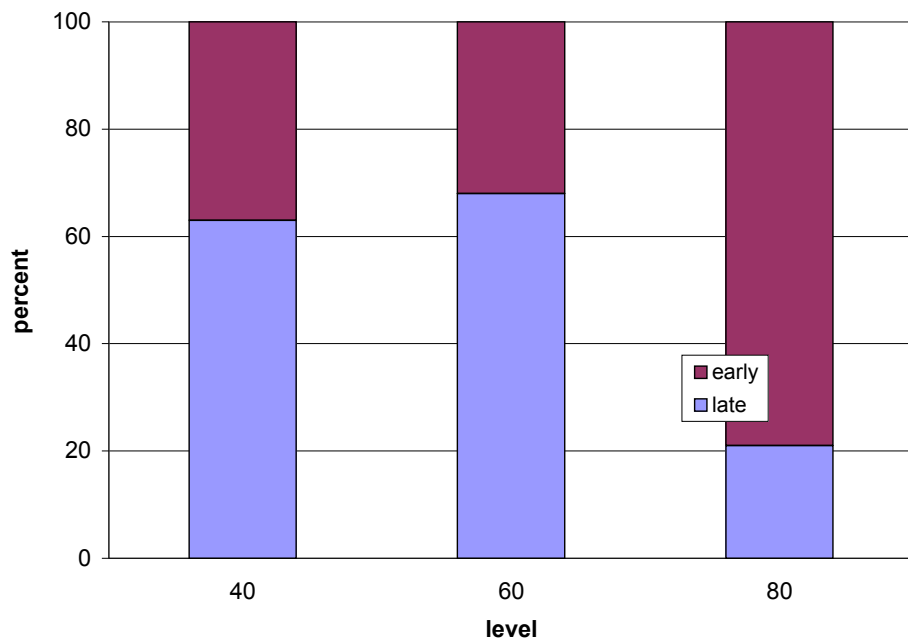
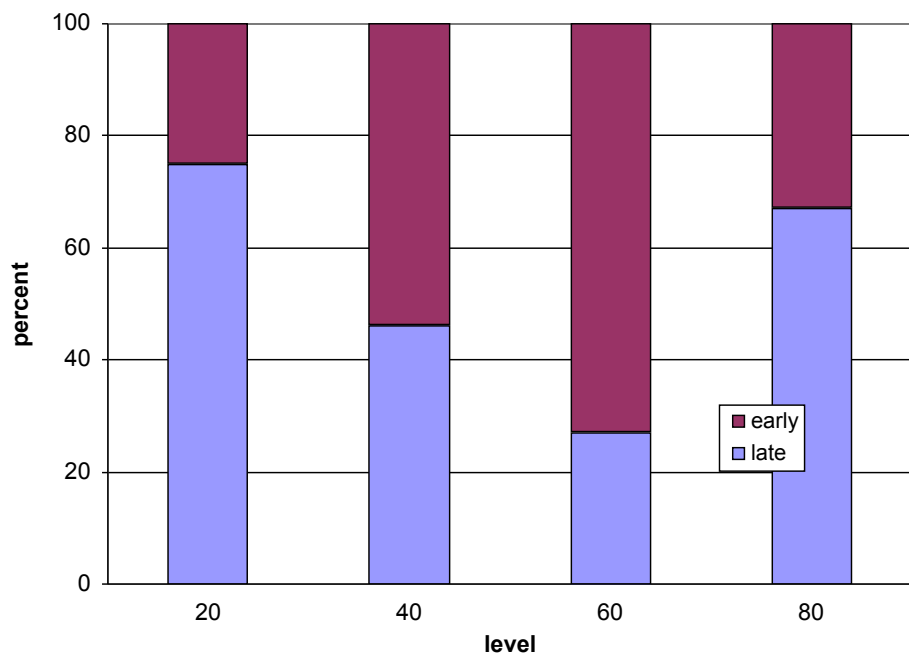
the other. If figure 4-1 represents to some extent a ceramic sequence for the Etzatlán area, then it seemed possible that Red on Cream pottery (i.e., the three types lumped together) might also be diagnostic of an intermediate stage or interval as well. That is, this pottery might not just occur in the upper or later levels at Tiana, it might concentrate there.

We examined the stratigraphic distribution of Red on Cream pottery at Tiana, but such a simple picture failed to materialize. Red on Cream sherds occurred throughout the deposits at Tiana and were also associated in early levels with the Incised Polychrome and White on Red. At the outset, we had constructed a fine-grained typology (chapter 3) and had defined three Red on Cream types (22, 23, and 24). In terms of these individual types, we did find at Tiana vertical separations, which potentially could help to define a portion of the Etzatlán regional sequence. Two of the types (22 and 23) could be identified by the presence of red bands present either singly or as two or more parallel bands, often parallel to the rim. Sherds of these two types tended to be in the upper levels of Tiana. For the third type (24), the red-painted designs were highly varied and more complex, being either rectilinear or curvilinear, with design quality being highly varied as well. Sherds of this type tended to occur in deeper levels of the site.

Figure 4-2 (data in table 4-3) shows the distribution of what we are calling Early versus Late Red on Cream pottery at Tiana. Looking at ratios of one group to the other, we see that the two late banded types combined increase by level and presumably through time at the expense of the third, based on the three excavation units with multiple superposed fifty-sherd samples.

Other evidence for culture change at Tiana can be seen in figure 4-3, which shows proportions of late Red on Cream relative to the contrasting bichrome, White on Red.

At this point, we carried out a CA for the site of Tiana, employing sherd data from the eleven lots in question but now adding data for two additional categories, Early and Late Red on Cream. As discussed in chapter 2, CA also deals with chi-square distances among samples and generates sample loci as well as those for variables (i.e., types). In figure 4-4, both sample (sherd lot) and type loci are depicted. Sample loci are labeled according to the original assessment of type content included in table 4-2. As can be seen, the samples are aligned roughly in the form of a curve, commonly referred to as an arch in CA (chapter 2),



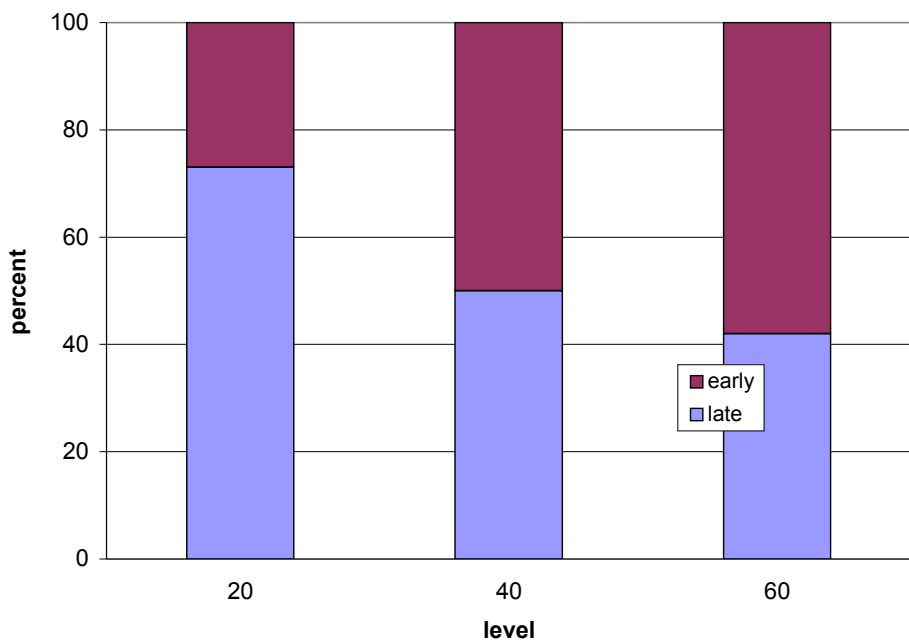


Figure 4-2. Early versus late Red on Cream by level and square, Tiana.

Table 4-3: Type frequencies by sample for the Tiana CA

Lotno	Un/Lv	Comal	GrySP	BrnSP	Inc. P	W/R	LR/C	ER/C	Total
1	1/20	9	0	0	3	1	6	2	21
2	1/40	24	2	1	1	28	12	14	82
3	1/60	14	4	2	1	17	6	16	60
4	1/80	4	6	1	2	13	2	1	29
6	2/40	34	1	1	1	1	5	3	46
7	2/60	57	7	2	5	2	13	6	92
8	2/80	6	1	2	4	1	3	11	28
9	3/20	12	0	1	7	5	8	3	36
10	3/40	41	2	3	10	20	25	25	126
11	3/60	14	4	1	5	35	5	7	71
15	4/40	8	0	0	0	2	1	2	13

from early (E in fig. 4-4) to mixed (M) to late (L). Types line up roughly along the same curve in a generally predictable order from early types, Gray Slipped and Polished, White on Red, Early Red on Cream, and Incised Polychrome to the later Brown Slipped and Polished, Late Red on Cream, and Comales. Keep in mind that Huistla Polychrome is all but absent from the Tiana collection. Type frequencies by sherd lot and type for the Tiana analysis are included in table 4-3.

Anona

Data from Anona are summarized in table 4-4, again in terms of the original two complexes (see table 4-1). These complexes segregate spatially at Anona as well, but here the separation is horizontal and not vertical. Missing excavation units and cells with missing data reflect level samples with overall sherd totals of less than fifty. Figure 4-5 depicts the CA for this site with two-dimensional plots for both types and lots. Again, we see evidence of an arch. The difference from the CA for the site of Tiana (see fig. 4-4) is that Huistla Polychrome ceramics

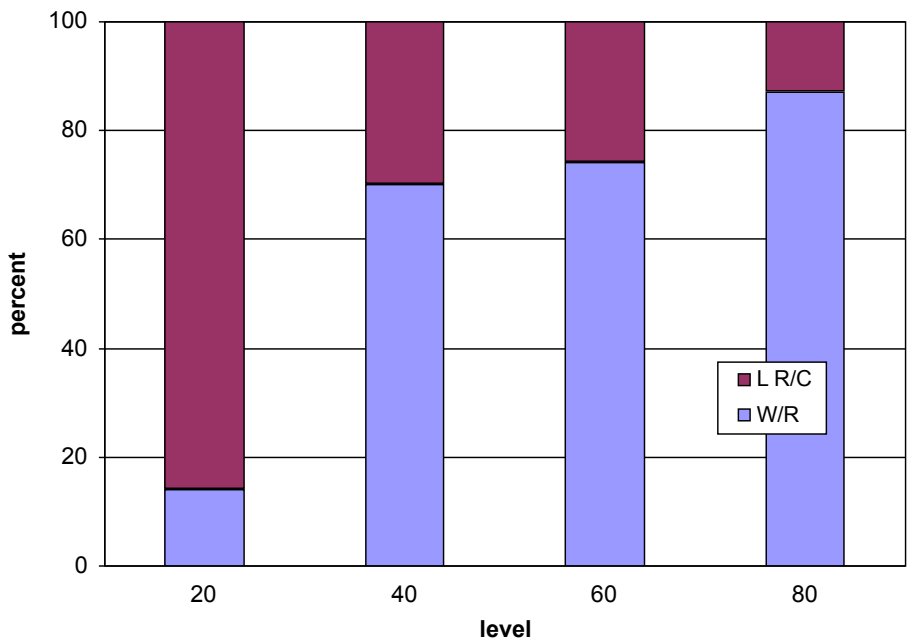
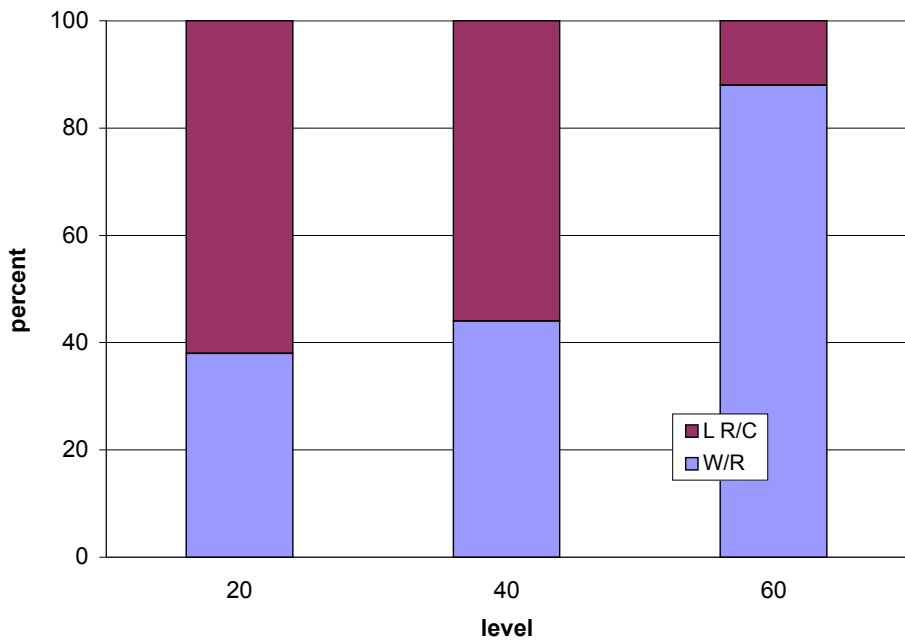
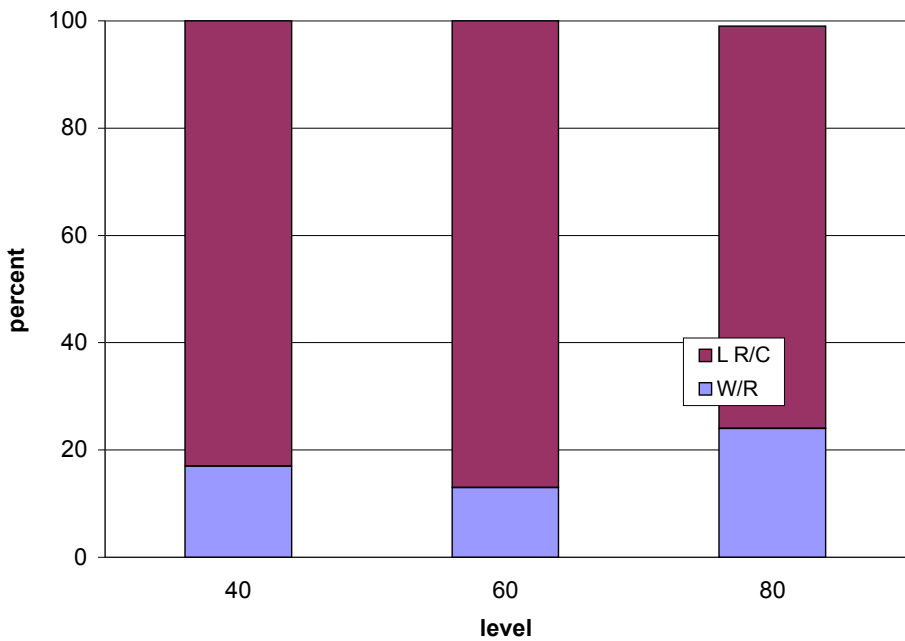


Figure 4-3. Late Red on Cream versus White on Red by level and square, Tiana.



are abundant in the Anona collection. Along the curve, samples align from early (E) to late (L; designations from table 4-4), and type-wise the progression begins with the early types/type group Incised Polychrome, Early Red on Cream, and White on Red. Late Red on Cream and the Comales type occupy an intermediate position, and the two categories of Huistla Polychrome follow at the late end of the sequence.

Las Cuevas Reconsidered

Next, we reconsider the site with which we began, Las Cuevas. Figure 4-6 portrays a second CA for the site, now including the two categories Early and Late Red on Cream. In this graphic, only types are included (i.e., sample loci are excluded), and the hypothetical time line is straight across the chart. There is no arch, probably due to the fact that we are dealing here with many more samples: fifty-two versus eleven for both the sites of Tiana and Anona. As for Anona, we see Late Red on Cream

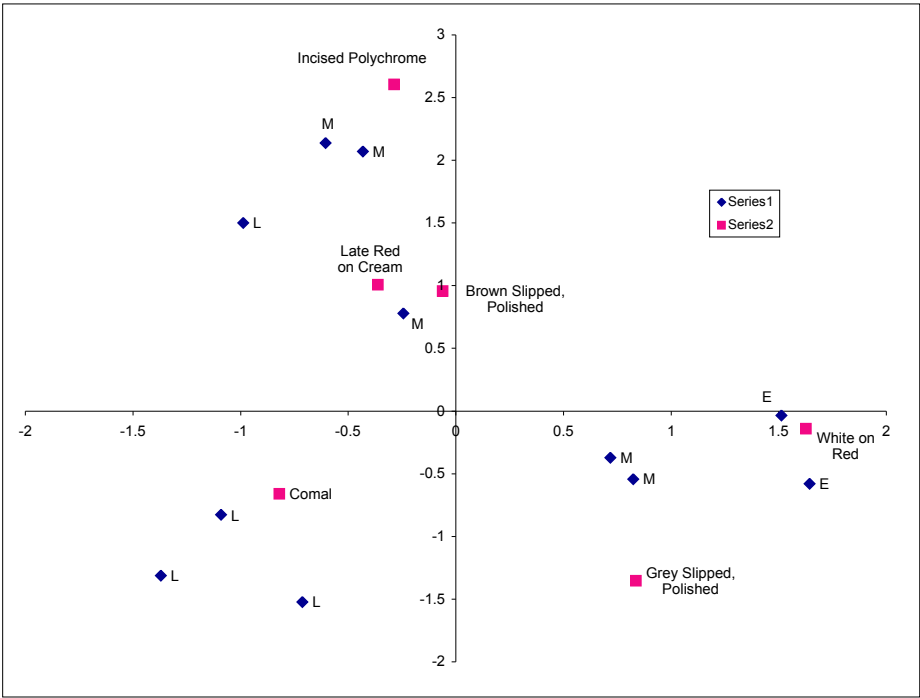


Figure 4-4. Type and sample plot, Tiana CA.

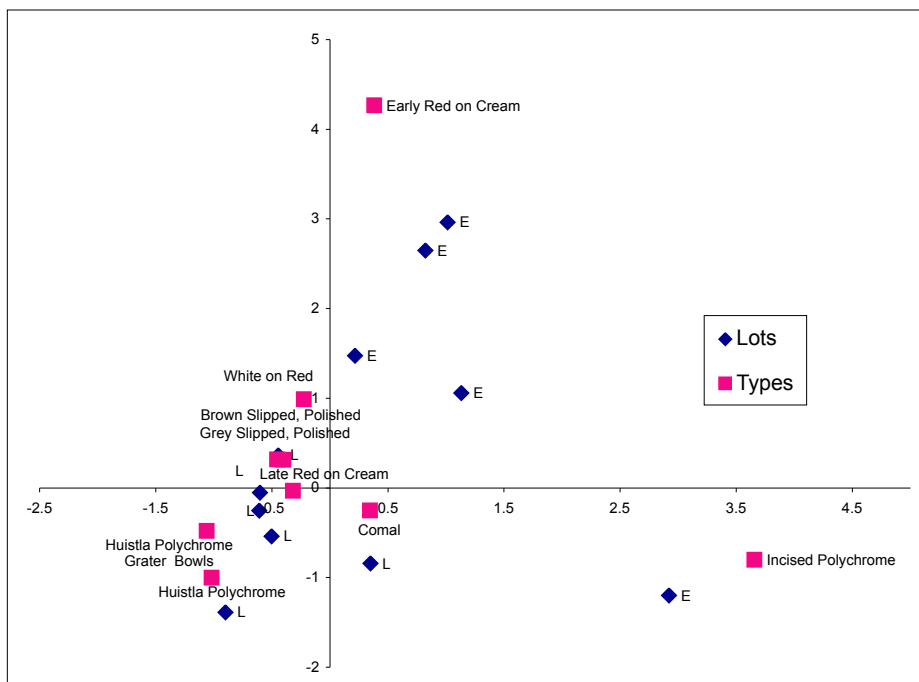


Figure 4-5. Type and sample plot, Anona CA.

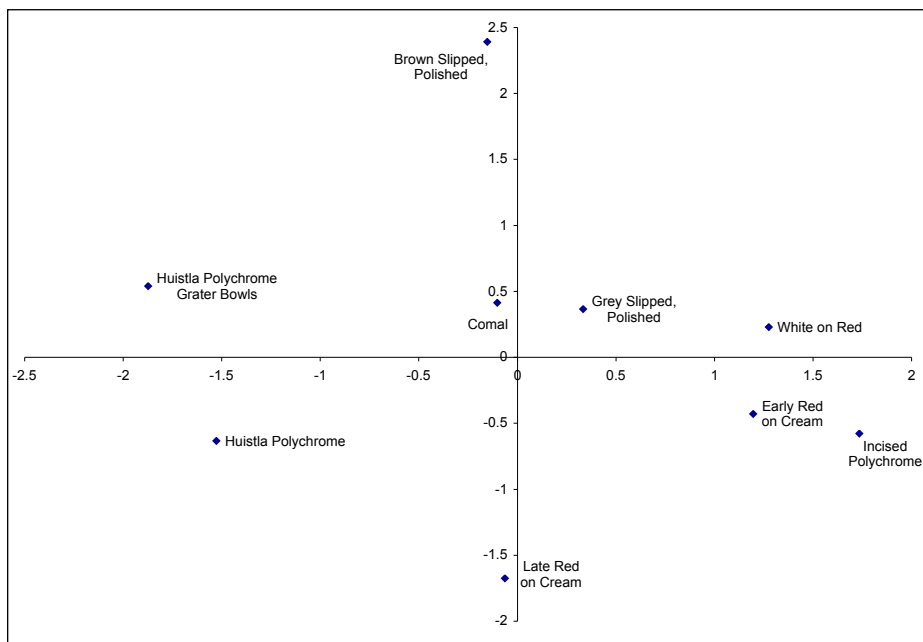


Figure 4-6. Type plot, expanded Las Cuevas CA.

along with Comales in an intermediate position between earlier types on the one hand and Huistla Polychrome and Huistla Polychrome Grater Bowls on the other.

Three-Site Correspondence Analysis

Finally, figure 4-7 depicts the type plot for a CA based on data from all three sites; that is, all seventy-four samples are included. Again, we see a straight-line progression of types, although along the proposed time line, the type progression across the graph is reversed. This fact is not surprising since in any seriation, the actual order is what counts, and in CA the signs for the plot values could all be reversed and the results would be considered identical. Again, we see roughly the same alignment of types, with Late Red on Cream and Comales in an intermediate position.

Santiaguito

Long and Glassow dug twenty pits at Santiaguito and recovered a relatively large sample of pottery, but unfortunately, we found that many of these sherds have eroded surfaces. Sherd erosion at Santiaguito was much more extensive than in other site collections. Because of this erosion, we were able to classify potsherds from certain lots only, and even for those cases, sherd preservation was only marginally satisfactory. Of 103 pottery lots excavated, we classified potsherds from only 40. When we charted the distribution of the original two complexes defined for

Table 4-4: Distribution of provisional ceramic complexes at Anona by level and square

Depth of Excavated Unit (cm)	Unit 6	Unit 7	Unit 8	Unit 9	Unit 12
0-20	Early	—	—	—	—
20-40	—	—	Late	—	—
40-60	—	—	Late	Early	Early
60-80	—	Late	Late	Early	—
80-100	—	—	Late	Early	—
100-120	—	—	Late	—	—

Las Cuevas within these Santiaguito lots, we found a lack of either horizontal or vertical segregation (table 4-5). As a result, and given our lack of confidence in the collection stemming from varying degrees of erosion, we put further study of the Santiaguito data set on hold. Upon completing the research that led to the three-site CA, however, as well as the alternative analyses discussed in chapter 6, we thought it worthwhile to reconsider the Santiaguito data. How would a CA perform on data compromised in this way? Could anything worthwhile be gleaned from this collection, given the time and effort expended in the field and, more recently, in the laboratory?

For the Santiaguito CA, we selected the twenty-two lots with at least fifty classified sherds and employed the nine types/type groupings already utilized in the three-site CA described previously. Figure 4-8 depicts the distribution of both types and lots in the CA two-dimensional plot. Here, one can see types generally in the same progression described for the three-site CA, except that early and intermediate

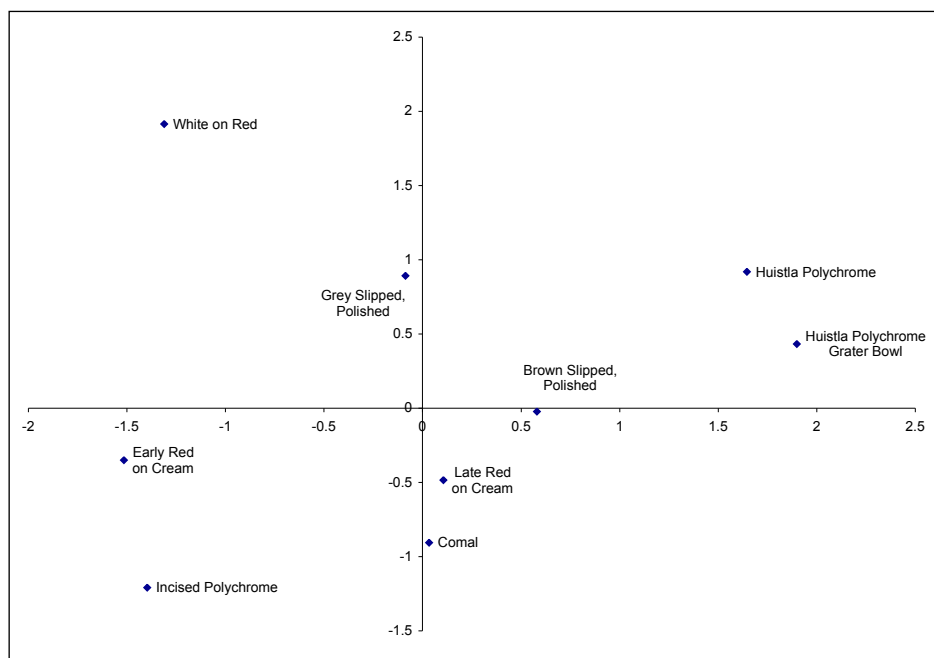


Figure 4-7. Type plot, three-site CA.

Table 4-5: Distribution of provisional ceramic complexes at Santiaguito by level and square

Depth of Excavated Unit (cm)	2	3	5	9	10	13	14	15	16	18	18ext	19	20
0-20	—	—	—	—	—	—	—	—	—	—	—	—	—
20-40	—	—	—	—	—	—	—	—	L	L	L*	—	E
40-60	—	L	—	L	E	—	E	E†	E	E	—	E	E
60-80	E	—	L	—	—	L	L	—	E	L	—	—	—
80-100	—	—	—	—	—	—	E	—	—	L	—	—	—

*The sample is from a pit extension excavated in a single 40 cm level (0-40).

†Two lots (nos. 74, 75), both early, were assigned to this same provenience. Both are included independently in the CA.

types are somewhat compressed and separated from the two Huistla Polychrome categories. Within the early and intermediate type distributions, the one discrepancy of note is the earlier positioning of the types Gray/Buff Slipped and Polished (5) and Black/Brown Slipped and Polished (6). Since identification of these types depends on surface finish, it could be that later deposits had been subjected to more erosion, tending to mask the presence of these types in later lots.

Among lots, we should note the single outlier located beyond and above the Huistla Polychrome type groupings (fig. 4-8). It is distinguished from other lots by having a much higher x-axis value. In figure 4-9, only lot loci are included, and the outlying lot has now been excluded. Lots are labeled as to their closest affinity, either the early or the late ceramic type complex as defined provisionally through use of correlation matrixes (see table 4-1). As we can observe, early and late lots, as defined, cluster in separate localities along the x-axis with little overlap. They separate cleanly when viewed along a diagonal extending from the upper left to the lower right quadrant.

We have good evidence, then, that the ceramic sequence identified for the other three sites under investigation is also present at Santiaguito. The outlier lot is of interest because of its very late position in the sequence. As a matter of fact, the historic type Glazed Majolica (41) occurs at Santiaguito. We identified fourteen sherds of this type in our research, all, it so happens, coming from lots that figured in the CA.

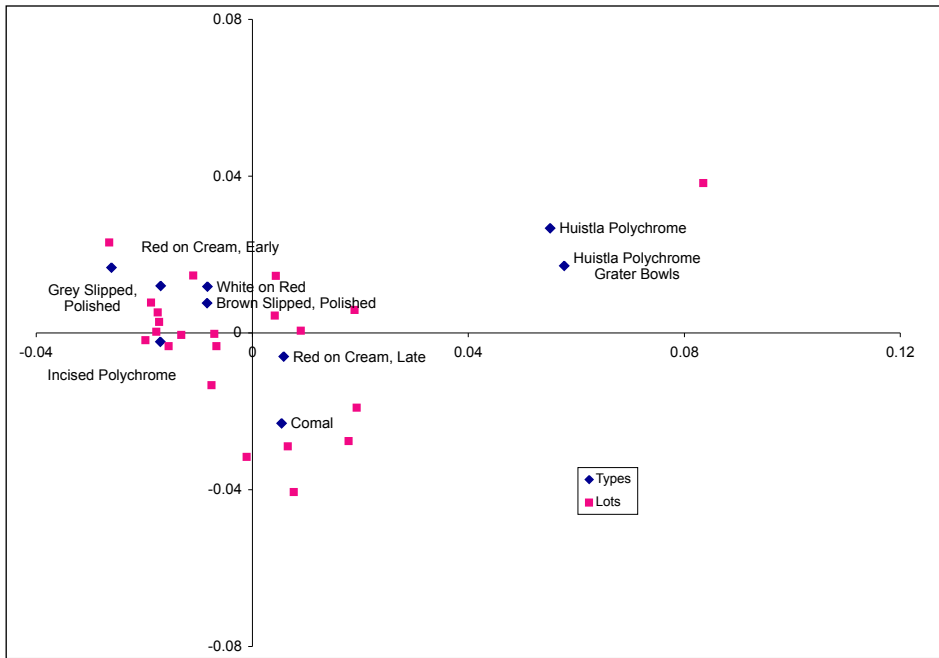
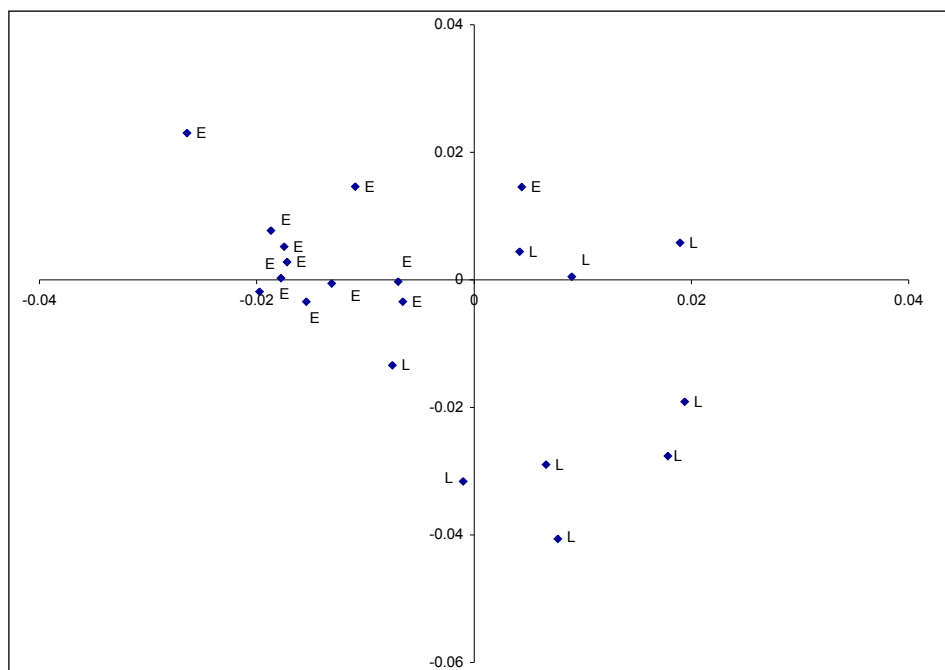


Figure 4-8. Type and sample plot, Santiaguito CA.

Eight of these sherds are from this single outlier lot. Furthermore, this lot was not a normal 20 cm level sample, but a 0–40 cm sample taken from an extension to pit 18 (table 4-5). No clear reason for this extension survives in the record, since field notes from the Santiaguito excavation are missing. Since types included in the Santiaguito CA are all of indigenous origins, this finding supports the idea that, in the vicinity of Etzatlán, indigenous pottery continued to be made into the historic era.

Extending the Sequence

As indicated in figures 4-4 and 4-5, CA plots for the sites of Tiana and Anona graphically describe relationships (i.e., covariation) among types and samples as aligned along arches. We have discussed evidence that these alignments of types and samples are chronological. For the composite or three-site CA, generally the same alignment of types extends roughly parallel to the horizontal axis. If this alignment represents a chronological progression, then it is reasonable to assume that the seventy-four samples from this three-site analysis, as viewed



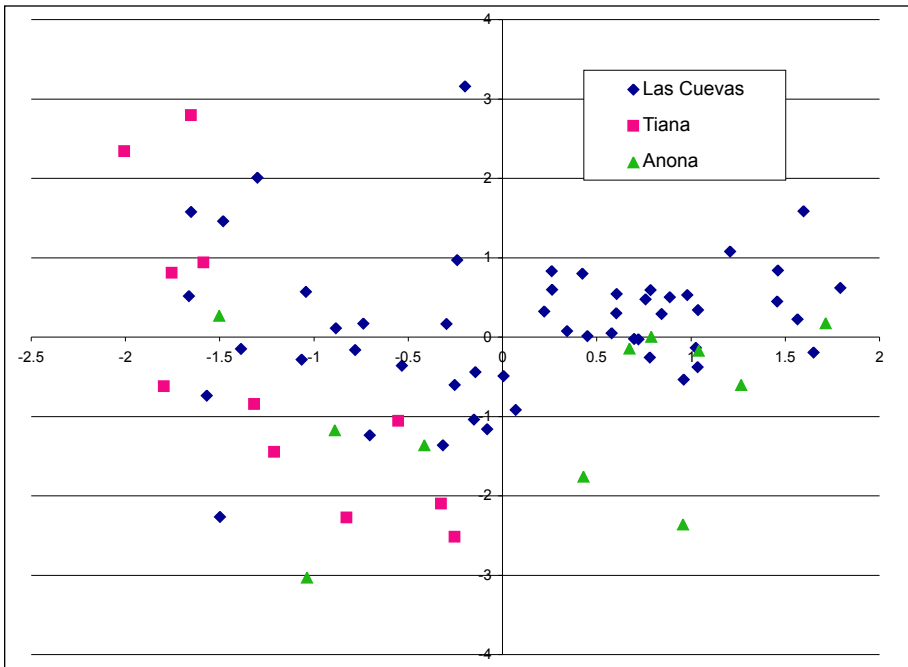


Figure 4-10. Sample plot, three-site CA.

fit into—the relative chronology developed so far? Altogether, 10,594 classified potsherds were included in the seventy-four samples, and of these, we utilized 4,369 (41 percent) in the three-site CA. Of the remaining, 2,461 sherds had been classified into thirty-three potentially well-defined types, with type frequencies varying from 9 to 680 sherds. We found we could obtain an estimate of relative age in the sequence for each of these types by applying the lot or sample horizontal-axis (x) value from the three-site CA to each sherd according to sample membership and then summarizing the distribution.

For example, the type Hatched from Rim Exterior, Incised (44) is represented by forty-three sherds, distributed through twenty-two of the seventy-four samples. Summarizing the sample x- or horizontal-axis values (CAvalues) for these sherds results in the distribution shown in figure 4-11. By contrast, the type Dark Red, Complex (47) had forty-one sherds distributed through twenty-three samples, but the summary by CAvalue produced a much different result (fig. 4-12). While these

trends should be treated cautiously, they may well represent the beginnings of a detailed ceramic sequence for the Etzatlán region. In chapter 5, we make use of this approach to provide information on the chronologies of individual types.

CA for Animal Bones

Elsewhere, Porcasi (2012) deals with animal bones recovered during the course of excavations at these sites. In a search for changes or continuity in the procurement of animal resources, an attempt has been made to place bone samples in relative sequence, based on associated ceramics. In other words, the sequence was extended to animal bones in the same way it was to pottery types that did not figure in the CAs, as just described. Initially, this process was accomplished by analyzing bones from the seventy-four level/square samples employed in the three-site CA described previously. X-axis CAvalues were assigned to associated bones according to provenience. Porcasi found, however, that only 39 percent of 1,676 identified specimens from these sites derived from the

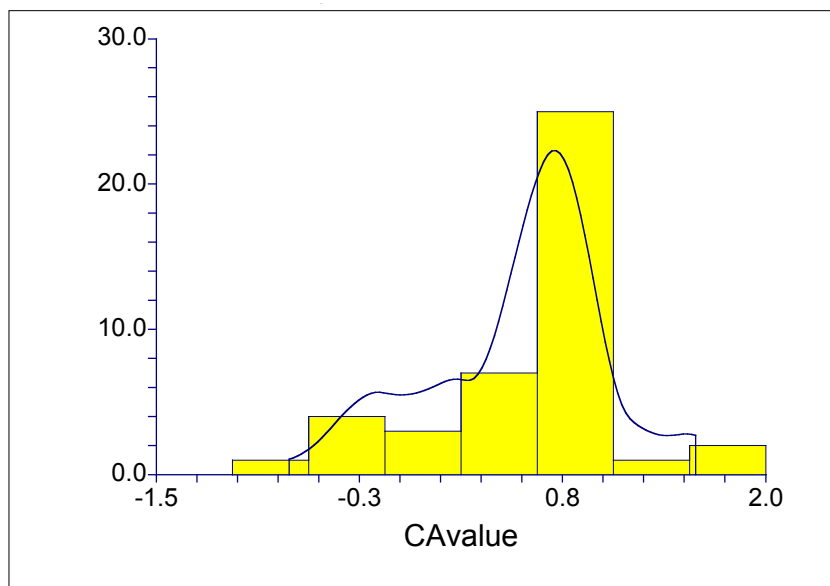


Figure 4-11. CAvalue distribution for type 44, Hatched from Rim Exterior.

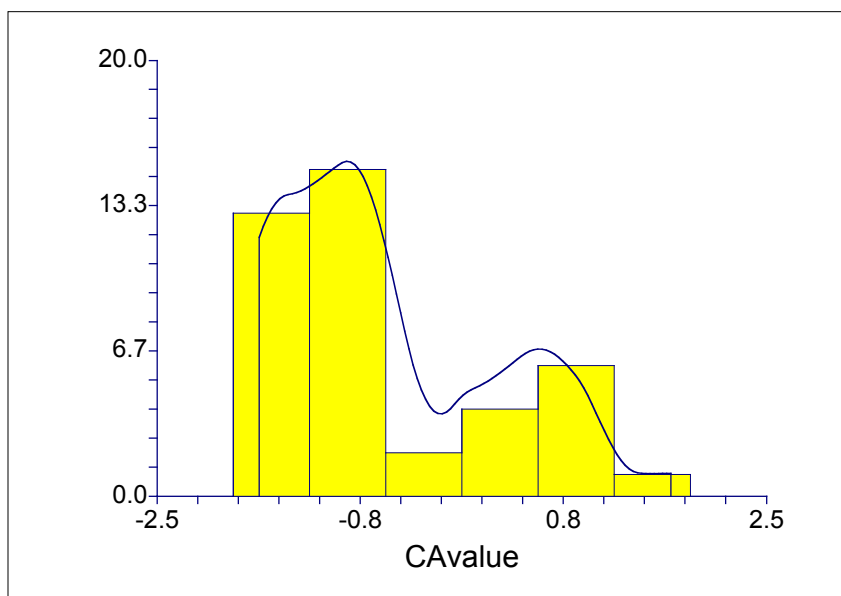


Figure 4-12. CAvalue distribution for type 47, Dark Red, Complex.

seventy-four samples because many bones came from samples with less than fifty sherds and from samples including pottery that remained unclassified for other reasons. Also excluded were bone samples from Santiaguito because no Santiaguito ceramic samples were included in the three-site CA. In order to increase the fraction of bones in relative sequence, we generated a CA for animal bones. This CA included all classified ceramic samples from levels/squares that also contained animal bones, with the minimum number of required sherds/samples dropped from fifty to six. This CA includes sherd samples from all four sites, including Santiaguito. The CA for animal bones is based on the same types/type groupings employed in the three-site CA and all other CAs described previously, except the original Las Cuevas CA. Figure 4-13 displays the two-dimensional CA plot for types, and virtually the same progression of types is manifested, consistent with the CAs we have described. The progression of samples for both ceramic types and, by extension, animal bones may not be as reliable as that generated through the three-site CA due to the inclusion of smaller sherd samples

(< 50 sherds) and samples from the site of Santiaguito. However, it still encompasses considerable data: seventy-six samples of pottery altogether. Of these, fifteen samples (20 percent) contain fewer than 50 sherds, for an average of 19.7 sherds/sample.

Based on this CA, the fraction of animal bones now included in the sequence is 1,196 of 1,617 specimens (74 percent). Porcasi's (2012) findings will be discussed in the concluding chapter (7) of this book.

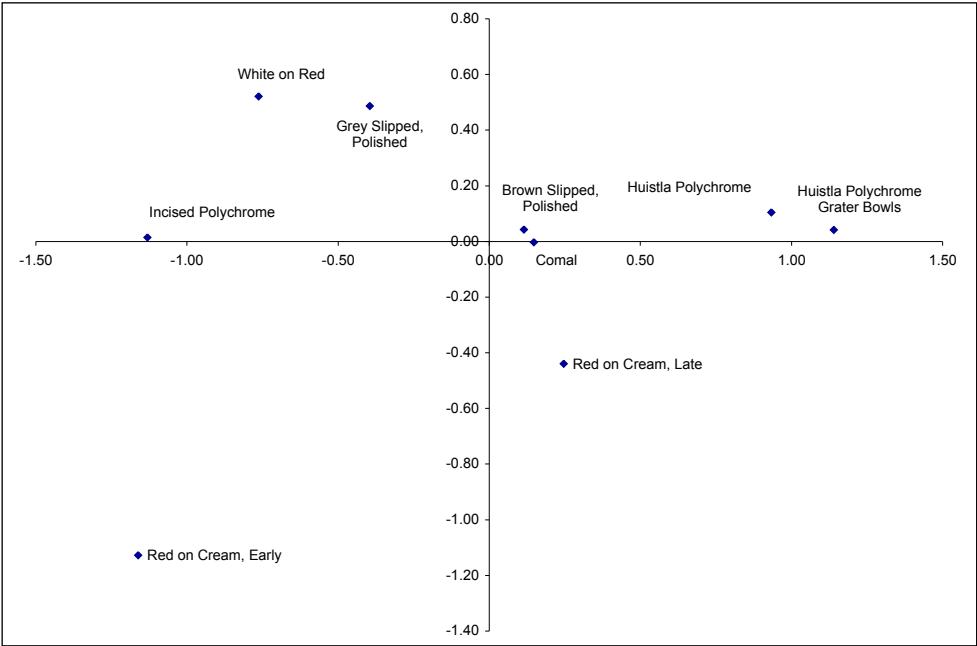


Figure 4-13. Type plot, CA for animal bones.

Chronological Considerations

By C. Roger Nance, Jan de Leeuw, Katheen Prado, and David S. Verity

Introduction

†††As indicated in chapter 4, the chronological position of each ceramic type can be estimated through the three-site correspondence analysis. We will deal here only with sherds from level/square samples that figured in that analysis and assign to each sherd a sample horizontal-axis CAvalue according to its sample provenience. This procedure will be followed whether or not the sherd is from a type actually included in the CA. This sample horizontal- or x-axis value (CAvalue) serves as an estimate of relative age. The mean and/or median CAvalue by type, then, should, at least ideally, indicate the age of a type relative to other types for which these same CAvalue statistics have also been computed.

Grater Bowls

Table 5-1 lists grater bowl types, their totaled sample frequencies, their mean and median CAvalues, standard deviations, and skewness statistics.

The bulk of included sherds are of the General Form type (3), and for this type the mean CAvalue (table 5-1) is close to that for all grater bowl sherds (.305). This figure is consistent with the type's high frequency and the probability that nondiagnostic sherds from both early and late types in a sense are included. Setting aside this General Form type, the

Table 5-1: Grater bowl types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CValue	Median CValue	Skewness
3	General Form	1,254	.847	.288	.429	-.448
7	Red and Black on Buff, Complex	86	.454	.982	.993	-.151
8	Red and Black on Buff, Stripes	79	.450	.752	.759	-1.709
9	Red and Black on Buff, General	79	.612	.744	.788	-.915
10	Black Bands	20	.825	.173	.343	-1.175
11	Red on Buff	32	.803	-.132	-.152	-.027
13	Fine Ware	83	.765	-.789	-1.038	.894

statistical significance of these findings can be assessed through several chi-square tests. Table 5-2 compares the three types (7, 8, 9) with the highest mean CValues to the three (10, 11, 13) with the lowest. The mean CValue for all 379 sherds in these six types combined is approximately .36, and Table 5-2 also groups these sherds according to whether each sherd's CValue falls above or below that mean. Viewed in this way, the data are significant; it is highly unlikely that these differences in CValue by type grouping are due to random fluctuation.

Strictly speaking, significance levels are only theoretically appropriate for large samples of independent observations. Both assumptions are always only approximately true. To remind the reader of this limitation, we report significance levels in this book using asterisks, replacing traditional p values as follows: probability level = * for $p < .05$, probability level = ** for $p < .01$, and probability level = *** for $p < .001$.

Types 7, 8, and 9 encompass the Huistla Polychrome (i.e., Red and Black on Buff) grater bowl component of the CA. The sherds of these

Table 5-2: Grater bowl types by CValue

CAvalue	Types 7, 8, 9	Types 10, 11, 13	Total
> .36	206 84.4%	28 20.7%	234
< .36	38 15.6%	107 79.3%	145
Total	244	135	379

Table chi-square = 149.231
Probability level = ***

Table 5-3: Red and Black on Buff Grater Bowl types by CValue

CAvalue	Type 7	Type 8	Total
> .87	50 58.1%	25 31.6%	75
< .87	36 41.9%	54 68.4%	90
Total	86	79	165

Table chi-square = 11.657
Probability level = ***

three types were lumped together for the CA, but the types can now be considered individually. Clearly, they are closely related, with type 9 containing the more fragmentary sherds that could not be assigned specifically to either type 7 or 8. Table 5-3 compares CAvalue distributions for type 7 (complex zigzag elements) to those of type 8 (simple red and black stripes). The mean CAvalue (.87) for the two types was used to divide sherds into potentially early and late groupings. And again, the difference is highly significant statistically.

Finally, there are the apparently early types (10, 11, 13) included in table 5-4. These also display differing CAvalue distributions by type, and the mean for the three types is -.49. Type 13, Fine Ware Grater Bowls, has more sherds from early contexts than the types Black Banded Grater Bowls (10) and Red on Buff Grater Bowls (11). Differences are highly significant, though we should keep in mind the small sample size for the Black Banded type.

We can hypothesize a sequence of grater bowl types as follows. Fine Ware Grater Bowls are the earliest identified or at least seem to peak during a relatively early period in the sequence. The types Black

Table 5-4: Early grater bowl types by CAvalue

CAvalue	Type 10	Type 11	Type 13	Total
> -.49	16 80.0%	20 62.5%	26 31.3%	62
< -.49	4 20.0%	12 37.5%	57 68.7%	73
Total	20	32	83	135

Table chi-square = 20.014
Probability level = ***

Banded and Red on Buff Grater Bowls follow. Then, during a relatively late period, the Huistla Polychrome types become important, with type 7 showing the latest peak in popularity.

In figure 5-1, a box plot displays the distribution of sherd CAvalues as percentiles by grater bowl type. In these box plots, horizontal bars represent the median CAvalue by type (fiftieth percentile), and each ellipse encompasses the twenty-fifth to the seventy-fifth percentiles of the type observations—what is referred to as the interquartile

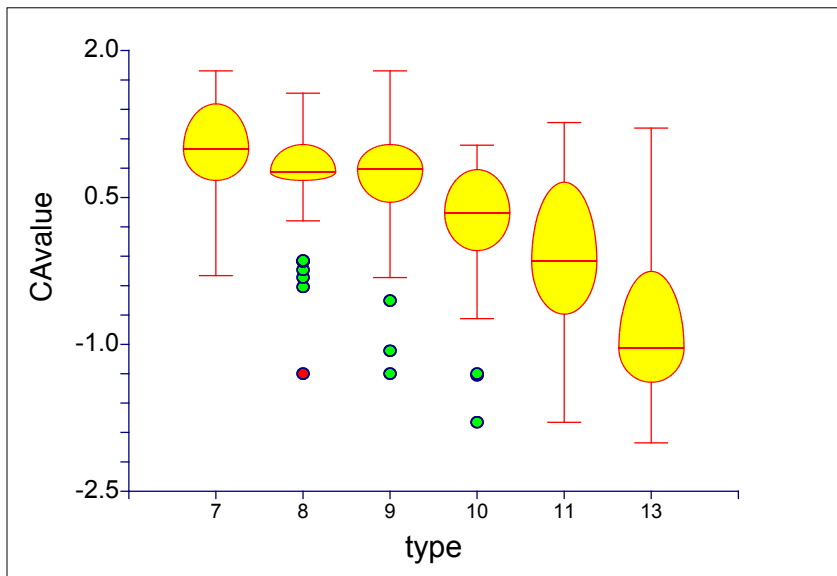


Figure 5-1. Grater bowls, CAvalue distributions by type.

range (IQR). Adjacent values are encompassed by T-shaped drawings above and below the ellipse, extending from the twenty-fifth percentile minus 1.5 times the IQR to the seventy-fifth percentile plus 1.5 times the IQR. Values outside this range are regarded as outliers. This plot provides some idea of the progression and possible overlap of these types through time. (For further details on box plots, see McGill et al. 1978.)

One statistic shown in table 5-1 but not yet discussed is the skewness value included for each type. Skewness is one measure of a distribution's deviation from the normal curve. A negative skewness value indicates a more pronounced tail to the left (i.e., toward negative values) and a positive value, a pronounced tail to the right. The differences can be seen in figure 5-2 in which the distribution for type 7 shows a slight negative skew, and figure 5-3, which depicts a positively skewed distribution for type 13 (cf., figs. 4-11 and 4-12). Grater bowl type skewness values and medians were compared through the correlation coefficient, and the result was a negative correlation not quite significant statistically. That is, skewness values decrease generally as median CA values increase.

These findings would seem most compatible with an archaeological sequence involving essentially two components. Early types, introduced suddenly at the onset of occupation, gradually decreased proportionately through time as a result of culture change or a tendency to mix into later deposits containing sherds of the later component. Later types, on the other hand, either were either introduced gradually into the prehistoric community and slowly replaced earlier types, or else they were introduced suddenly but tended to mix through time into deposits containing sherds of the earlier component.

Other Huistla Polychrome Types (Non-Grater Bowl)

The three Huistla Polychrome (i.e., Red and Black on Buff) types listed in table 5-5 were combined in the CA but are examined here individually in terms of their mean CA values. All three types are in a late position (table 5-5) with little difference among them. The Red and Black on Buff, Stripes type (21) has a slightly lower mean than found for the others, paralleling the finding for Red and Black on Buff, Complex (type 7) and Red and Black on Buff, Stripes (type 8; see table 5-3) grater bowls, but here the difference is not statistically significant. It can be seen, however, in the elongated ellipses for types 19 and 20 in figure 5-4.

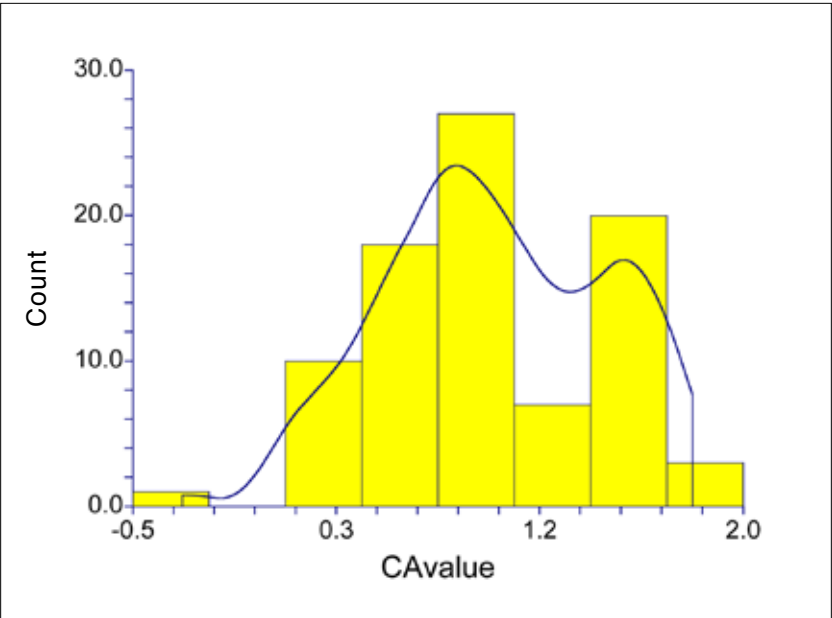


Figure 5-2. CAvalues for type 7, Red and Black on Buff Grater Bowl, Complex.

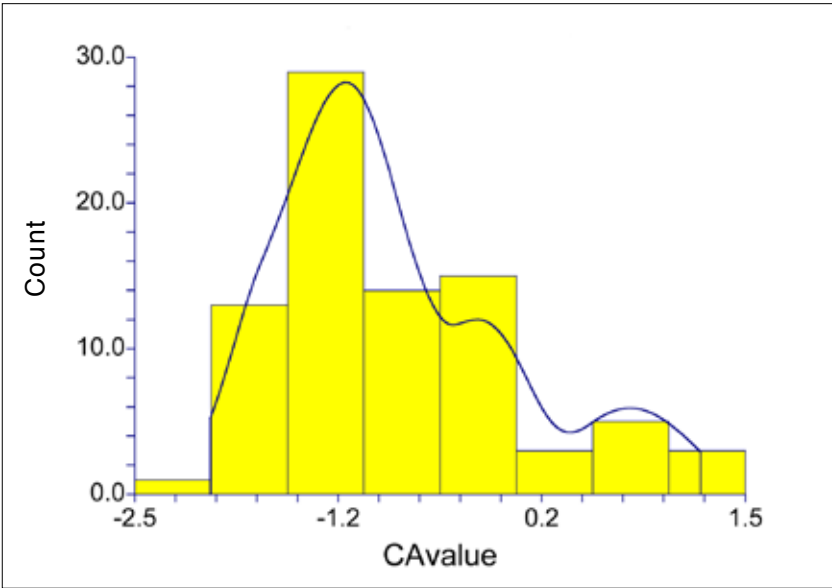


Figure 5-3. CAvalues for type 13, Fine Ware Grater Bowl.

Among the three types, the one with the highest median CAvalue (type 20) also has the lowest skewness value (table 5-5).

White on Red, Black and White on Red Types

Three of these types, all White on Red, were combined for the CA, while the Black and White on Red types were not included. One of these, Black and White on Red, Slipped and Polished (25), has a high standard deviation combined with a small sample frequency (table 5-6) and probably should be disregarded. This grouping of sherds was hardly homogeneous in the first place (see type descriptions in chapter 3).

The three White on Red types, combined, have a low mean CAvalue (-.573) and as a general class of pottery, count early in the sequence. Yet their distributions, when compared one to another, proved to be dissimilar. Type 16 with simple white lines and type 17 with more complex designs tend to be spread out timewise, as indicated by their high standard deviations (table 5-6), but the latter type became popular later in

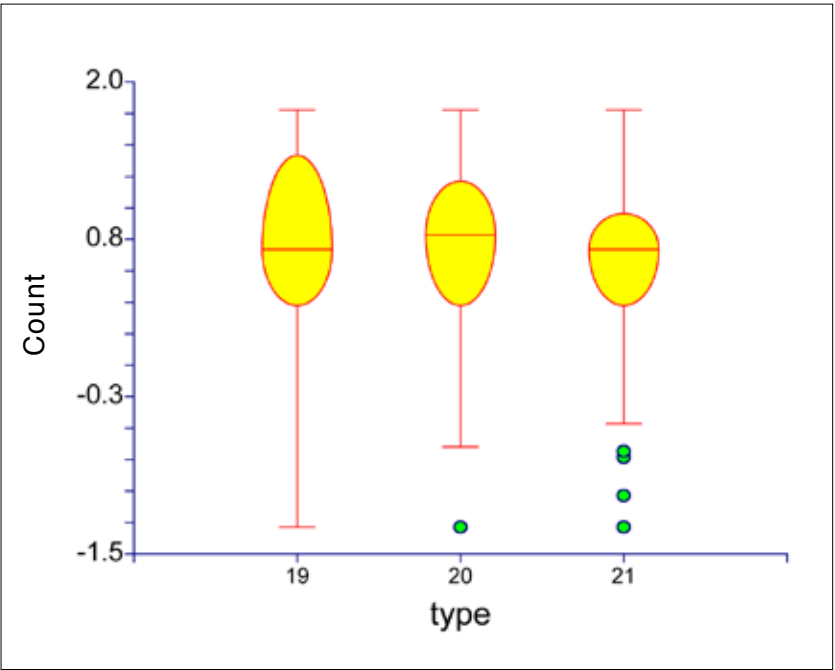


Figure 5-4. Red and Black on Buff, CAvalue distributions by type.

Table 5-5: Red and Black on Buff types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
19	Red and Black on Buff, Zigzag	148	.677	.797	.759	-.664
20	Red and Black on Buff, Unique	78	.662	.792	.865	-.851
21	Red and Black on Buff, Stripes	245	.600	.649	.759	-.789
Total		471	.638	.719	.759	-.704

Table 5-6: White or Black and White on Red types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
14	Black and White on Red	249	.705	.556	.720	-1.269
16	White on Red, Simple	324	1.042	-.601	-1.043	.388
17	White on Red, Complex	257	1.014	-.321	-.240	.074
18	White on Red, Parallel	79	.775	-.959	-1.300	1.275
25	Black and White on Red, SP	14	.929	-.698	-1.054	.560

Table 5-7: White on Red types by CAvalue

CAvalue	Type 16	Type 17	Type 18	Total
> -.537	136 42.0%	88 56.1%	17 21.5%	241 43.0%
< -.537	188 58.0%	69 43.9%	62 78.5%	319 57.0%
Total	324	157	79	560

Table chi-square = 25.92
Probability level = ***

Table 5-8: Black and White on Red, White on Red by type and CAvalue

CAvalue	Types 16–18	Type 14	Total
> -.226	206 36.8%	224 90.0%	430
< -.226	354 63.2%	25 10.0%	379
Total	560	249	809

Table chi-square = 195.72
Probability level = ***

the sequence (table 5-7). Type 18, with its very distinctive design, clusters earlier than either of the others, and these differing distributions among the three types are highly significant statistically (table 5-7).

The last type considered here, Black and White on Red (14), is similar to the White on Red types in its lack of polished surfaces, relatively thin vessel walls, and abundant sand temper. Yet the type peaks later in the sequence than any one of the White on Red types or all of them combined, as can be seen in table 5-8, which also is highly significant. The distribution of all these types is displayed graphically in figure 5-5.

Skewness and median values for these five types were compared through a correlation coefficient, as was also done for grater bowl types. Again, the correlation is negative ($r = -.960$) and in this case statistically significant (probability level = **).

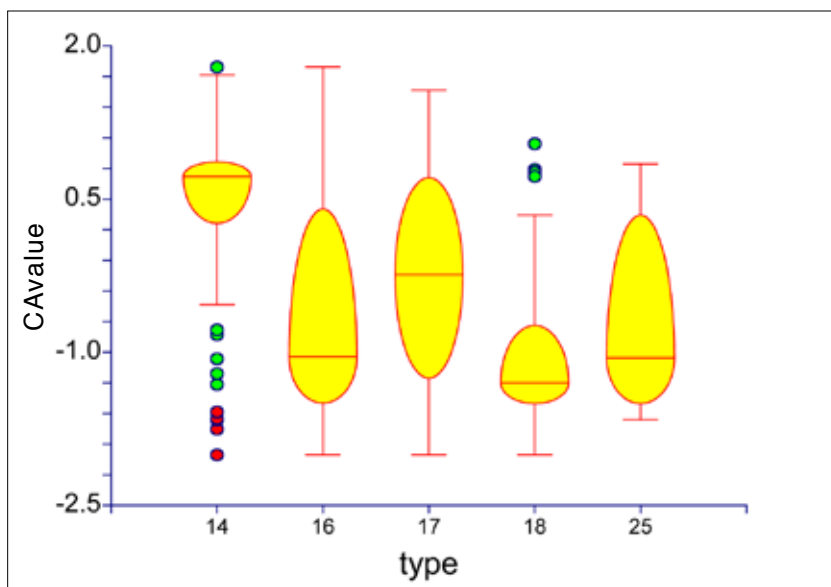


Figure 5-5. White or Black and White on Red, CAvalue distributions by type.

Red on Cream

At the site of Tiana, two Red on Cream types with single or parallel straight lines (22, 23) were found to be later stratigraphically than those with complex designs (24), as described in chapter 4. In the CA, then, types 22 and 23 were combined, and type 24 functioned separately. In terms of CAvalue distributions by type, this contrasting distribution shows up clearly in figure 5-6. By themselves, however, types 22 and 23 have significantly different distributions when CAvalues are compared (table 5-9). Red on Cream sherds with broad stripes (type 23) tend to be more frequent later in the sequence than the narrow-striped sherds of type 22. (See also table 5-10 for summary statistics.) Sherds of the former type are from larger, thicker-walled vessels compared to the latter, so the shift may have significance beyond discussion of simple ceramic chronology. Altogether and consistent with other type groupings, the three types produced a negative correlation when median and skewness values were compared, although it was not significant statistically.

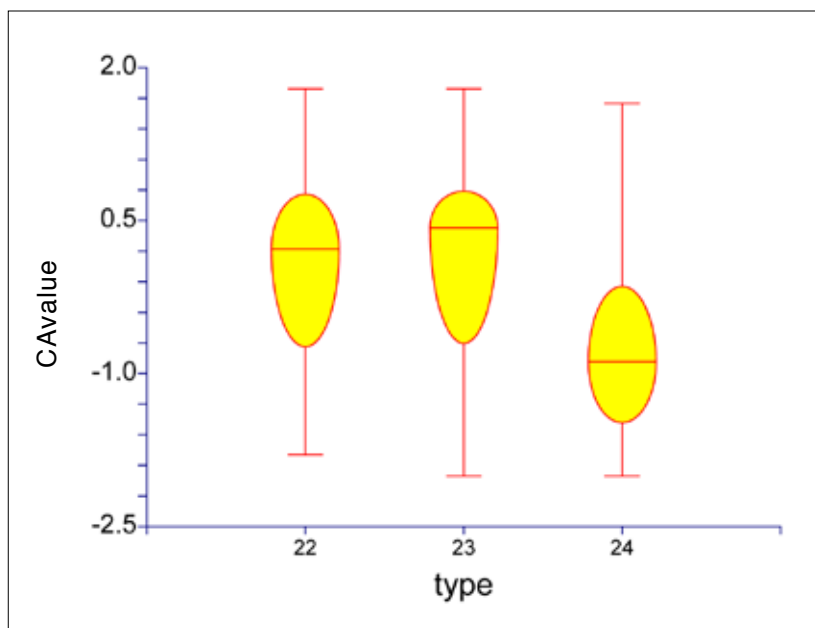


Figure 5-6. Red on Cream, CAvalue distributions by type.

Major Polychrome Types

One of these types, Incised Polychrome (15), was included in the CA; the other, Polychrome, White Dots (26), was not. Both types are early (table 5-11), but one, type 26, was significantly more so (table 5-12). This latter type for the most part occupied a limited time span but also had significant outliers extending late into the sequence (fig. 5-7). This result probably was due either to the physical mixing of sherds in the deposits or the vagaries of classification. Skewness and median values for these types vary inversely (see table 5-11).

Incised/Engraved Types

For the thirteen incised/engraved types, and in line with other findings, type skewness varies inversely with median CAvalue. Figure 5-8 depicts the relationship graphically ($r = -.863$, probability level = ***; figs. 5-9, 5-10). The alignment of types in figure 5-8 shows two clusters of incised/engraved types, one near each end of the plot. Table 5-13 compares the five types in the early cluster to the six types in the late by the mean

Table 5-9: Two Red on Cream types by CAvalue

CAvalue	Type 22	Type 23	Total
> .046	147 51.2%	224 61.4%	371 56.9%
< .046	140 48.8%	141 38.6%	281 43.1%
Total	287	365	652

Table chi-square = 6.75
Probability level = **

Table 5-10: Red on Cream types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
22	Red on Cream, Thin Lines	287	.876	-.032	.221	-.242
23	Red on Cream, Broad Lines	365	.946	.108	.429	-.522
24	Red on Cream, Complex	264	.941	-.663	-.884	.562

CAvalue for all eleven types. The probability is exceedingly low that differences are due to random distribution.

Early Incised/Engraved Types

Turning to examples of the early cluster, we find that type 51, Black Fine Engraved, has a low count coupled with a high standard deviation and appears to have little value as a chronological marker (table 5-14). By contrast, type 50, Red Fine Engraved, appears to be concentrated at the early end of the sequence. It seems possible that this type is from an earlier period of time than that encompassed by other types in our study.

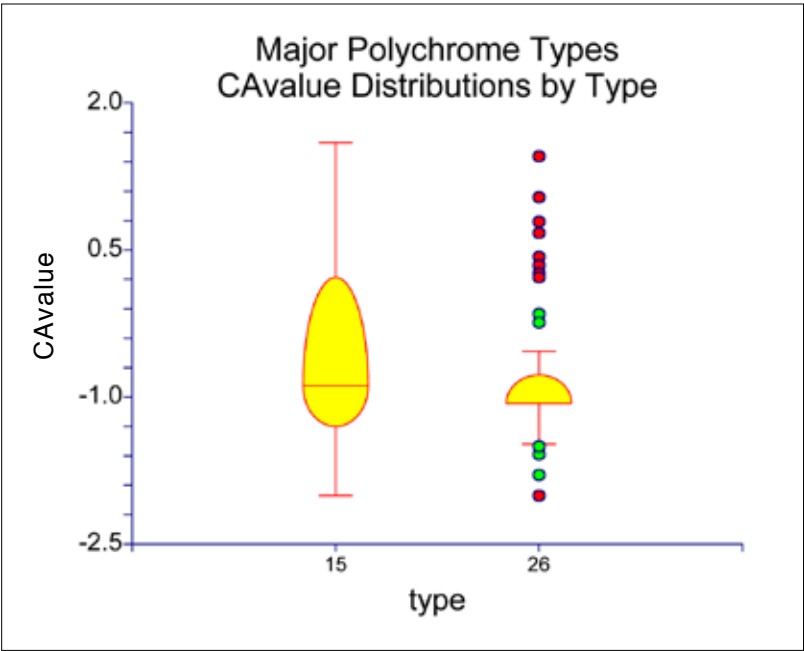


Figure 5-7. Major polychrome types, CAvalue distributions by type.

Table 5-11: Major polychrome types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
15	Incised Polychrome	269	.852	-.611	-.884	.475
26	Polychrome, White Dots	92	.697	-.816	-1.066	1.452

Of the remaining three types (40, 43, 47), type 43, Brushed or Combed Utility, is the earliest, but not significantly so. The distribution of early incised/engraved types can be seen graphically in figure 5-9, and figure 4-12 depicts a histogram of the distribution of type 47, Dark Red to Black Engraved, Complex.

Table 5-12: Major polychrome types by CAvalue

CAvalue	Type 15	Type 26	Total
> -.663	113 42.0%	20 21.7%	133 36.8%
< -.663	156 58.0%	72 78.3%	228 63.2%
Total	269	92	361

Table chi-square = 12.103

Probability level = ***

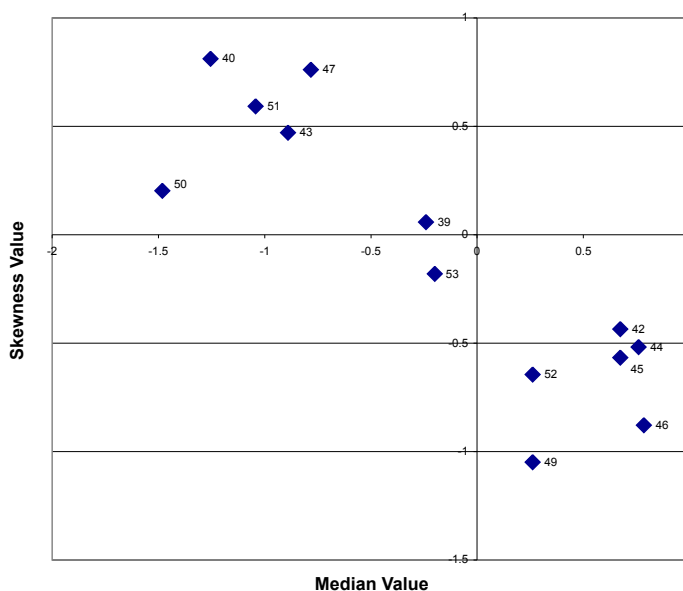


Figure 5-8. Median CAvalue by skewness, incised/engraved types.

Table 5-13: Incised/engraved types, early versus late by CAvalue

CAvalue	Early Types	Late Types	Total
> -.135	43 23.2%	159 78.7%	202 52.2%
< -.135	142 76.8%	43 21.3%	185 47.8%
Total	185	202	387

Table chi-square = 119.07

Probability level = ***

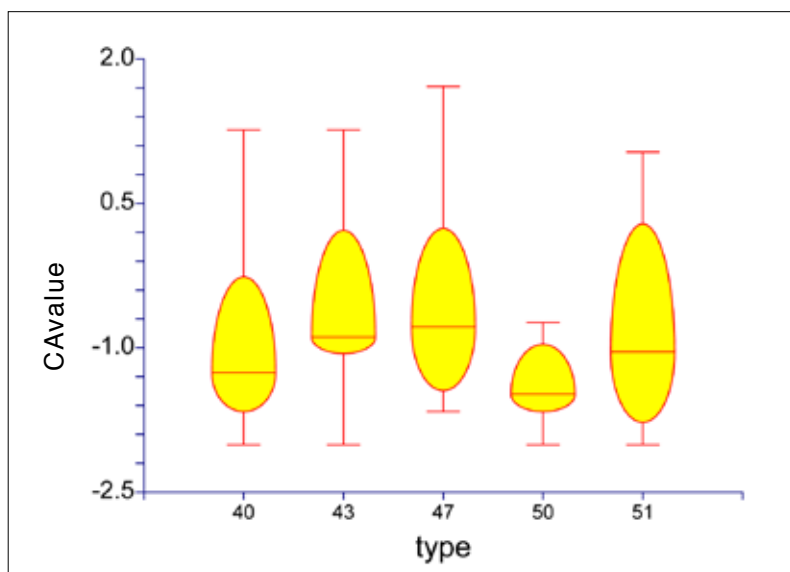


Figure 5-9. Early incised/engraved, CAvalue distributions by type.

Table 5-14: Early incised/engraved types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
40	Fine Engraved, Arcaded	58	.870	-.982	-1.255	.811
43	Brushed or Combed Utility	64	.836	-.567	-.890	.470
47	Red to Black Engraved, Complex	41	.920	-.633	-.782	.761
50	Red Fine Incised	13	.418	-1.352	-1.482	.203
51	Black Fine Engraved	9	1.132	-.825	-1.043	.593

Table 5-15: Late incised/engraved types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
42	Washboard Exterior	38	.511	.611	.674	-.435
44	Hatched from Rim Exterior, Incised	43	.515	.581	.759	-.518
45	Black Crude Engraved, Complex	37	.737	.309	.674	-.566
46	Black Crude Engraved, Suspended Triangles	14	.500	.785	.785	-.879
49	Exterior Incised, Deep Parallel Lines	21	.746	.189	.260	-1.049
52	Miscellaneous Broad-Line Incised	49	.609	.353	.262	-.645

Table 5-16: Types 49, 52 versus other late incised/engraved types by CAvalue

CAvalue	Types 49, 52	Other Late Types	Total
> .455	30 42.9%	85 64.4%	115 56.9%
< .455	40 57.1%	47 35.6%	87 43.1%
Total	70	132	202

Table chi-square = 8.653

Probability level = **

Late Incised/Engraved Types

Several of the late types listed in table 5-15 have relatively small standard deviations and are concentrated toward the late end of the sequence. This is especially true of type 46, Black Crude Engraved, Suspended Triangles. Except for the design motif, these sherds are identical to those of type 45, Black Crude Engraved, Complex, and this design motif may be very late in the sequence. The fourteen sherds of type 46 are scattered through nine different lots, so their distribution may not be the misleading product of one or two broken vessels.

Viewing the distribution of all late types in figure 5-10, we see that two of them, types 49 and 52, definitely appear earlier than the others. Figure 4-11 shows, as a histogram, the distribution of type 44, Hatched from Rim Exterior, Incised. The mean CAvalue for all late incised/engraved sherds is .455, and table 5-16 indicates that types 49 and 52 have significantly more sherds with low CAvalues when their combined distribution (above and below the mean) is compared to that for others.

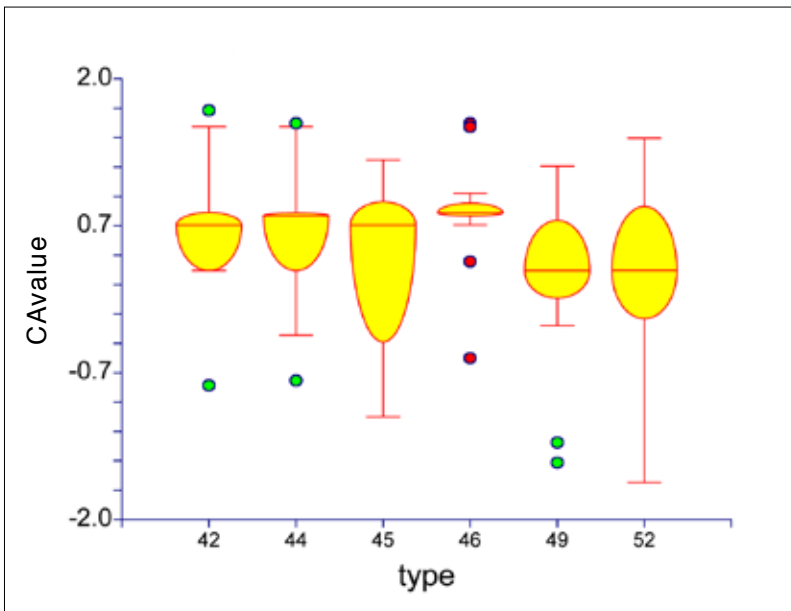


Figure 5-10. Late incised/engraved, CAvalue distributions by type.

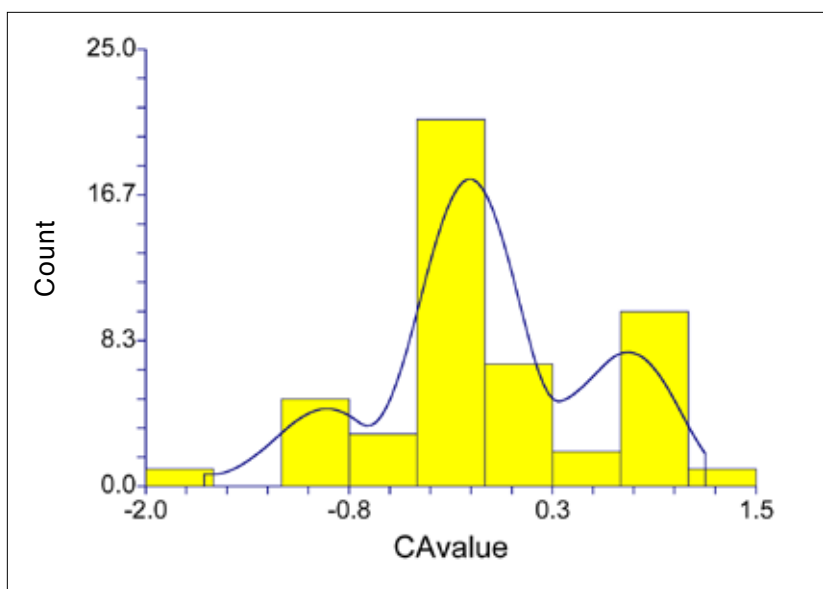


Figure 5-11. CAvalues for type 53, Complex Broad-Line Incised.

Intermediate Incised/Engraved Types

Two types, 39 and 53, fall between the two clusters in figure 5-8. Both their skewness values and median CAvalues are relatively close to 0. Type 39 is a residual type containing small well-finished sherds on which incising or engraving could be identified. The high standard deviation (table 5-17) indicates the presence of both early and late sherds. The distribution for type 53 (fig. 5-11) is more problematic, and with eighteen sherds from one lot, sampling error is a possibility.

Red Ware Types

While present in great abundance, red ware sherds were difficult to sort into potentially useful types. Sorting criteria did not readily suggest themselves, and many sherds seemed to be intergrades between the categories we did define. In some cases, type assignment seemed somewhat arbitrary. And for this reason, these types were not included in the CA. Yet the four type groupings do sort out chronologically to some extent

Table 5-17: Intermediate incised/engraved types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
39	Miscellaneous Fine Incised/Engraved	171	.977	-0.63	-.240	-.058
53	Complex, Broad-Line Incised	50	.618	-.048	-.199	-.180

when CAvalues are assigned to these sherds by lot and type designation. This is evident in their widely varying mean CAvalues (table 5-18) and in the graph of their distributions (fig. 5-12). Table 5-19 indicates that these differences in all probability are not the result of random fluctuation. Moreover, when the two early types, Red Fine Ware and Red Utility, are compared to one another, the difference is still significant, and the same is true for the two later types as well. Red ware type skewness values and medians show a significant negative correlation ($r = -.99$, probability level = **).

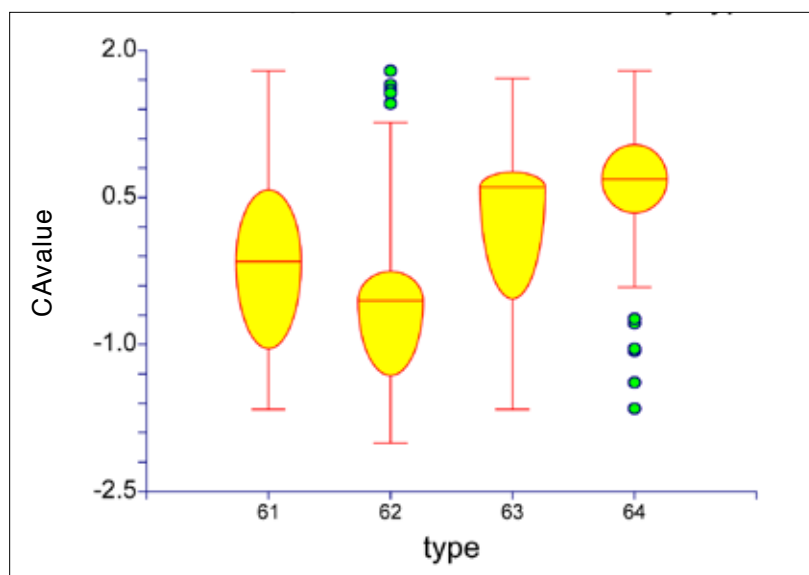


Figure 5-12. Red ware, CAvalue distributions by type.

Table 5-18: Red ware types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
61	Fine Red Ware	152	.868	-.179	-.152	.259
62	Red Utility	680	.945	-.595	-.553	.522
63	Red Streaky-Polish Jars	360	.856	.246	.605	-.529
64	Red Utility, Very Thick	208	.610	.650	.686	-.758

Table 5-19: Red ware types by CAvalue

CAvalue	Type 61	Type 62	Type 63	Type 64	Total
> -.149	72 47.4%	152 22.4%	251 69.7%	189 90.9%	664
< -.149	80 52.6%	528 77.6%	109 30.3%	19 9.1%	736
Total	152	680	360	208	1400

Table chi-square = 400.637

Probability level = ***

Other Major Plain Ware Types

In contradistinction to red ware types, other major types lacking decoration did figure in the CA. One of these encompassed sherds identified as comal fragments. These appear to have been present throughout the sequence but increased proportionately through time compared to red ware types. Table 5-20 compares Comales (type 2) proportions to red ware types combined (61, 62, 63, 64), on the one hand, and to grater bowl types combined (3, 7, 8, 9, 10, 11, 13), on the other. More than the others, grater bowl sherds tend to concentrate at the late end of the sequence. One can see differences among these major type groupings graphically by comparing figures 5-13, 5-14, and 5-15.

It must be mentioned that Long and Glassow discarded thousands of excavated potsherds prior to shipping the remainder of the collection to Los Angeles. Presumably, these were plain sherds, which they

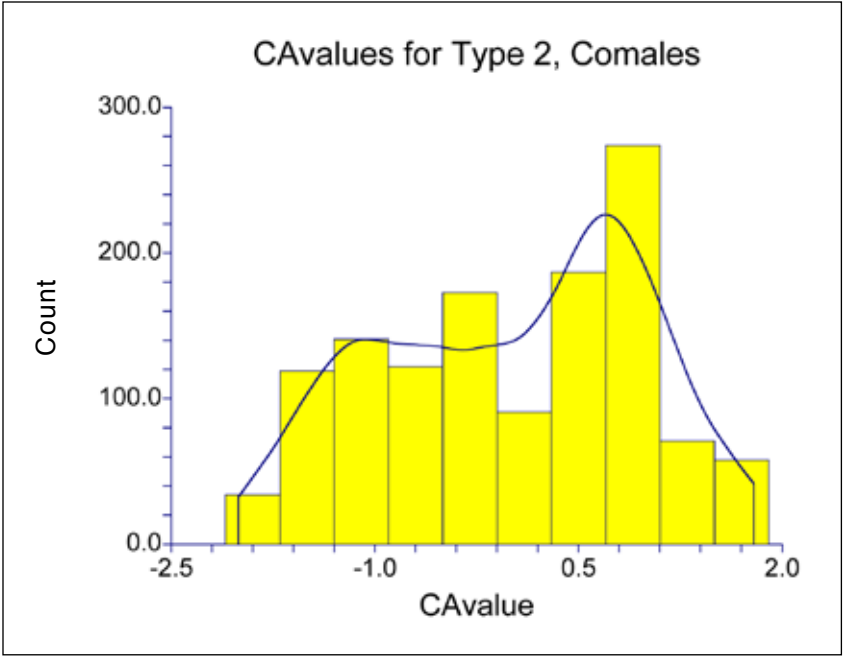


Figure 5-13. CAvalues for type 2, Comales.

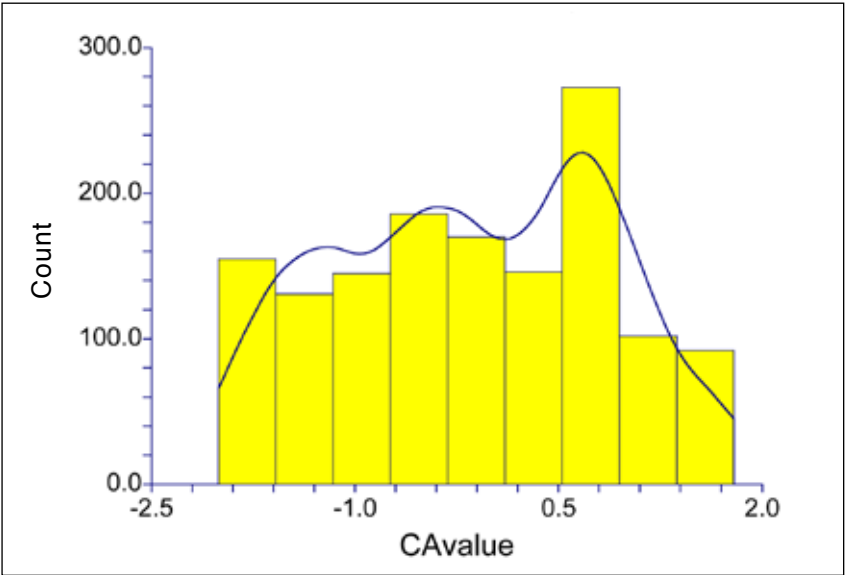


Figure 5-14. CAvalues for red ware types (61-64).

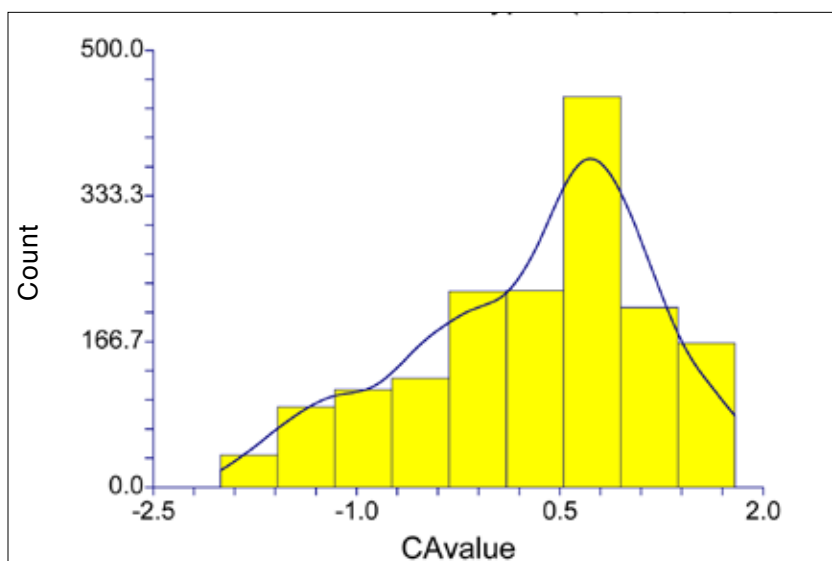


Figure 5-15. CAvalues for grater bowl types (3, 7, 8, 9, 10, 11, 13).

Table 5-20: Major type category by CAvalue

CAvalue	Comal	Red Ware	Grater Bowl	Total
> -.071	674 53.1%	630 45.0%	1075 65.8%	2379
< -.071	596 46.9%	770 55.0%	558 34.2%	1924
Total	1270	1400	1633	4303

Table chi-square = 135.878
Probability level = ***

tabulated by square and level and designated as “red” or “all other” (data archived at the UCLA Fowler Museum). Proportions of red ware versus other types reported here are based only on the extant collection.

Other major plain types are Gray/Buff Slipped and Polished (5) and Black/Brown Slipped and Polished (6). Their distributions along with that for Comales (2) are depicted in figure 5-16. Summary statistics for the same types are included in table 5-21. As indicated in several of the

CAs, Black/Brown Slipped and Polished sherds tend to be later than well-finished sherds with lighter surfaces (type 5), and the difference viewed through CAvalues for these sherds is statistically significant (table 5-22).

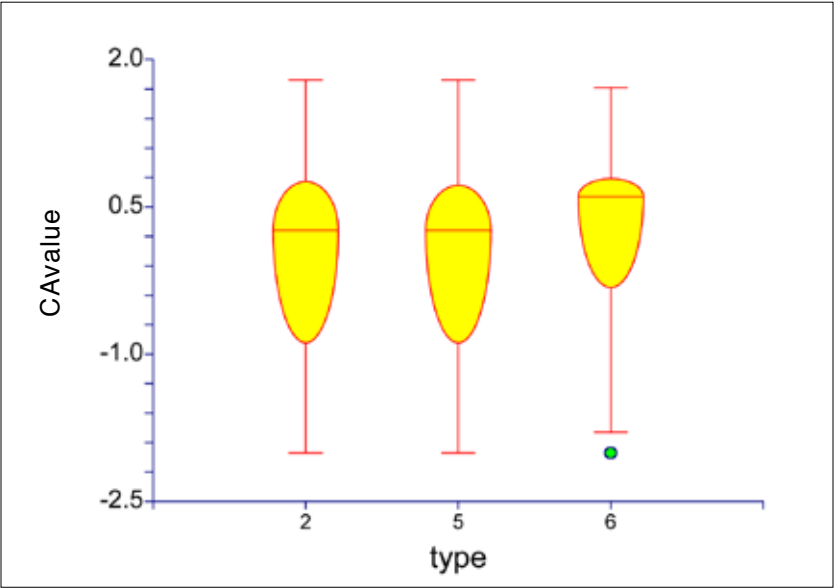


Figure 5-16. Other major plain ware types, CAvalue distributions by type.

Table 5-21: Other major plain types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
2	Comales	1270	.963	-.014	.261	-.225
5	Gray/Buf Slipped and Polished	314	.976	-.038	.261	-.330
6	Black/Brown Slipped and Polished	325	.870	.254	.606	-.635

Table 5-22: Plain slipped, polished types by CValue

CValue	Type 5	Type 6	Total
>.11	170 54.1%	215 66.2%	385
< .11	144 45.9%	110 33.8%	254
Total	314	325	639

Table chi-square = 9.624

Probability level = **

Other Low Frequency Types

Most of these types tend to occupy late positions in the sequence (table 5-23). Four types are either Black on Red or Red on Black bichrome. Since it was difficult at times to distinguish among the decorative techniques manifested on sherds of these types, we were not surprised to find them on the same time level (fig. 5-17).

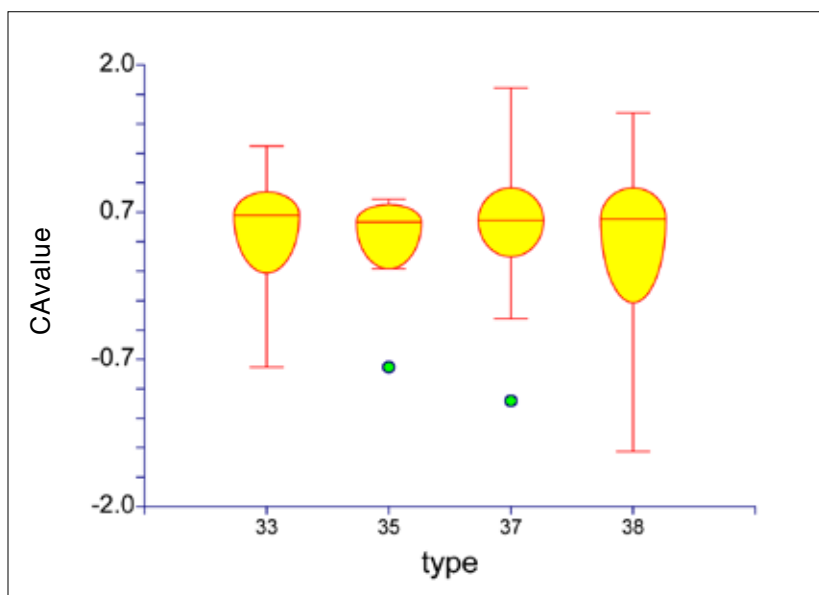


Figure 5-17. Red on Black and Black on Red types, CValue distributions by type.

Table 5-23: Other minority types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
27	Red on White	11	.995	.265	.343	-.039
28	Thick White Paint	12	.651	.607	.626	-1.234
29	Negative Painted	10	.769	.884	.900	-.627
30	Thick White Paint on Red	2	0	-.738	-.738	
31	White on Black	35	1.153	.033	.343	-.530
33	Red on Black	10	.606	.483	.640	-.872
34	Red on Black (?)	14	1.045	.132	.262	-.092
35	Red Wash/ Slip over Black	10	.593	.370	.579	-1.386
36	Black on Gray/White	33	.778	.466	.606	-.667
37	Black on Red	32	.571	.595	.592	-.327
38	Black on Red Striped	47	.792	.423	.606	-.422
41	Glazed Majolica	46	.692	.471	.451	-.248
48	Reed Punctate	25	.532	.661	.604	.278
60	Polished Utility	26	.867	-.748	-1.066	1.075
70	Black on Orange	5	1.113	-.889	-1.317	.675

Other minority type distributions are depicted in figure 5-18. Of these, one is the historic type Glazed Majolica ware (41), which is very distinctive and datable to the colonial era. Interestingly enough, in the collection at hand these historic sherds show up not as might be predicted at the late end of the sequence, but somewhat earlier (table 5-23). We will discuss this problem in a later chapter.

Residual Types

Two types are residual, and sherds we could not assign to other types were placed in one or the other. Included sherds either have eroded surfaces or are plain and lack distinctive features of decoration, surface finish, or vessel function. Distributions for the two types are summarized in table 5-24. In some ways, sherds of these two types tend to distribute through the sequence, as does the entire body of sherds where all types are combined (cf., figs. 5-19, 5-20, and 5-21), although for some reason sherds of the type Eroded Fine Ware (4) tend to be more plentiful late in the sequence compared to sherds of Gray/Buf Utility (1). The difference is statistically significant (table 5-25).

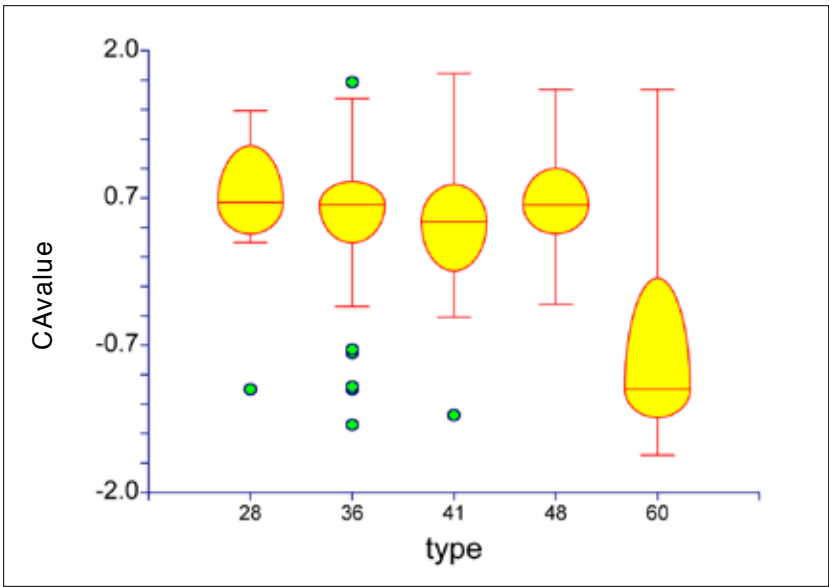


Figure 5-18. Other minority types, CAvalue distributions by type.

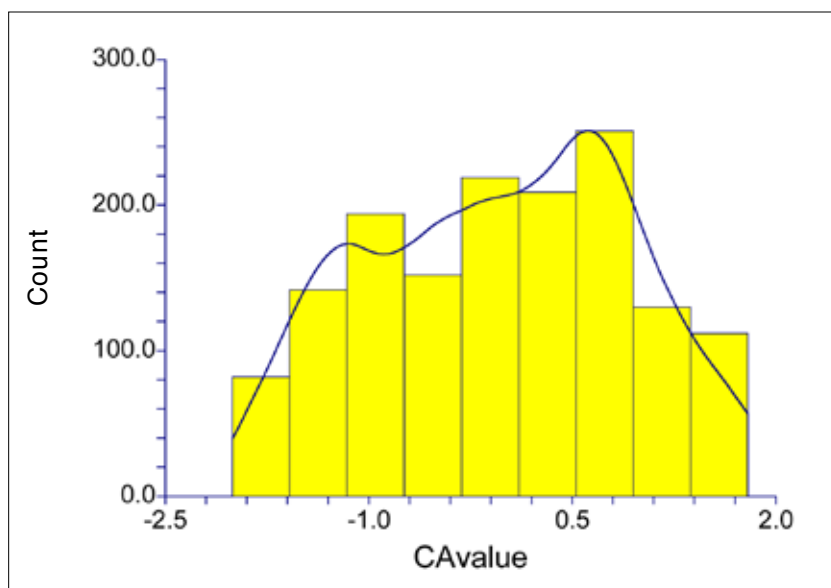


Figure 5-19. CAvalues for type 1, Gray/Buf Utility.

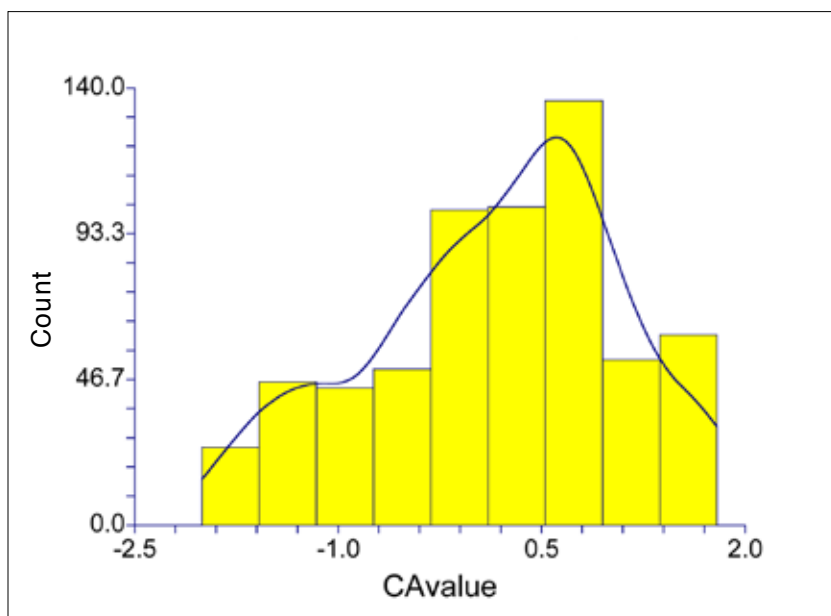


Figure 5-20. CAvalues for type 4, Eroded Fine Ware.

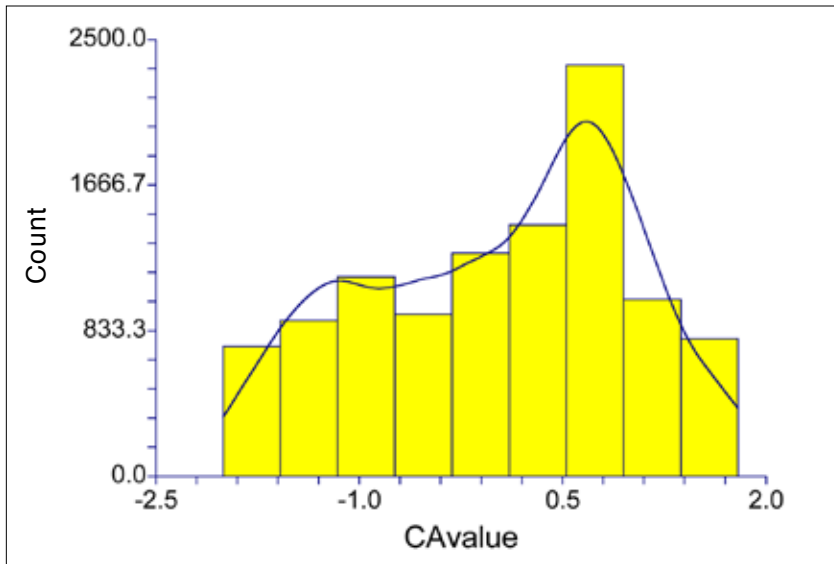


Figure 5-21. CAvalues for all types combined.

Table 5-24: Residual types, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CAvalue	Median CAvalue	Skewness
1	Gray/Buf Utility	1502	.952	-.029	.005	-.102
4	Eroded Fine Ware	618	.923	.158	.262	-.36

Table 5-25: Residual types by CAvalue

CAvalue	Type 1	Type 4	Total
> .026	741 49.3%	366 59.2%	1107
< .026	761 50.7%	252 40.8%	1013
Total	1502	618	2120

Table chi-square = 17.161
Probability level = ***

Table 5-26: CAvalues for unique sherds

Designation*	Type Code	Site	CAvalue
A	57	Las Cuevas	-.88405
B	58	Las Cuevas	1.03376
C	70	Anona	n/a†
D	71	Tiana	n/a
E	32	Las Cuevas	-1.66352
F	32	Las Cuevas	-.70465
G	32	Las Cuevas	1.02483
H	32	Las Cuevas	-.88405

*See chapter 3.

†Site not analyzed, that is, CAvalue not calculated.

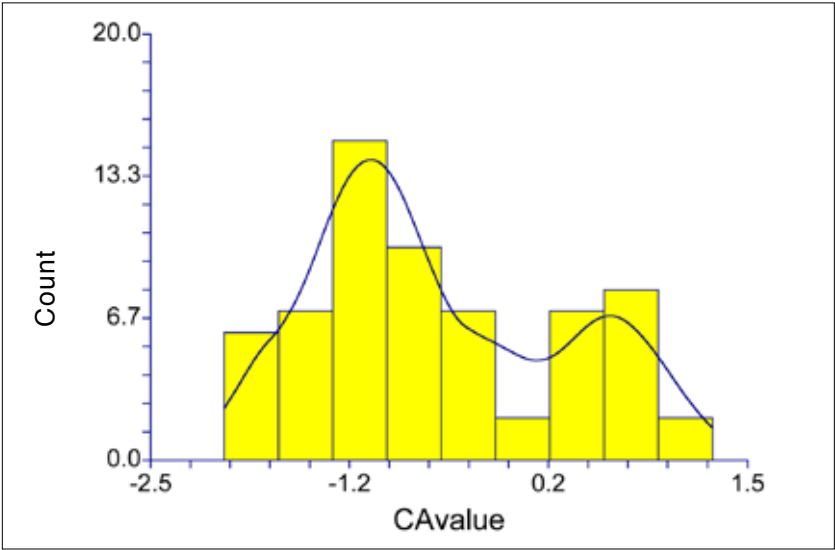


Figure 5-22. CAvalues for ceramic plaques.

Table 5-27: Other ceramic artifacts, summary statistics

Type Code	Description	Count	Standard Deviation	Mean CValue	Median CValue	Skewness
43	Brushed Plaques	64	.836	-.567	-.890	.470
81	Cookie-Cutter Figurines	9	1.028	-.071	-.298	.618
82	Spindle Whorls	15	.891	-.143	-.298	.664
83	Modeled Figurine Fragments	3	.099	-.997	-1.043	.665
85	Stamp	1	—	-1.653	-1.653	—
86	Molded Effigy Feet	13	.501	-.750	-.884	1.282
93	Whistle Fragment	1	—	.785	.785	—
94	Rattle Pellet	2	.001	.262	.262	—
95	Possible Pipe Stem	1	—	.262	.262	—
97	Historic Molded Figurine	1	—	.262	.262	—
98	Decorative Plaque Fragment	2	.412	.051	.051	—

Unique Sherds

A few sherds manifest unique designs and may be from low-frequency wares traded into the Etzatlán basin. Table 5-26 lists these sherds and the CA values of their square/level samples where they could be calculated, that is, for sherds falling within the three-site CA study sample.

Other Ceramic Artifacts

Included here are the ceramic Brushed Plaques (type 43) and the few ceramic artifacts not included in the ceramic data sets and described separately in chapter 3. Plaques were included in the ceramic data sets, and CA value distributions for the type are summarized in figure 5-22 and table 5-27. These plaques apparently were used through most of the Etzatlán sequence but most commonly in prehistoric times.

For the other artifacts, those that were not included in the ceramic data sets, CA values have been assigned where possible but only for the site of Las Cuevas. Results are summarized in table 5-27. Four categories are present in sufficient quantity to allow tentative chronological assessments. Mazatlán figurines and spindle whorls appear to be in the later portion of the prehistoric sequence, but standard deviations are high and the two samples relatively small. While only three modeled figurine fragments can be assessed, they are tightly clustered at a very early position in the sequence. The thirteen molded animal effigy vessel feet also fall into the early portion of the three-site sequence.

Alternative Analyses

By C. Roger Nance, Jan de Leeuw, Kathleen Prado, and David S. Verity

Introduction

†††In the previous chapter, we detailed type distributions through a proposed sequence based on a single analysis. At this point, we should address the following question: how would a different analyses or the use of different variables (types) affect the outcome? To what extent would different progressions of types or sherds materialize? We explored this issue through four additional analyses, three of them employing CA and the last the EDM method. Both of these statistical procedures are described in chapter 2.

Correspondence Analysis of Twenty-three Types: CA23

For this analysis, twenty-three types (listed in table 6-1) were selected based on their high frequencies, with type frequencies in the study sample varying between 78 and 1,270 sherds. A few types, those residual in nature, such as Gray/Buff Utility (1) and Eroded Fine Ware (4), were excluded. Figure 6-1 portrays the two-dimensional distribution of types, and inspection reveals a close parallel between this CA and the original three-site model (see fig. 4-7). In the original CA, Huistla Polychrome sherds were combined into two groups and occurred at the late end of the sequence. In figure 6-1, included Huistla Polychrome types (7, 8, 19, 20, 21) likewise cluster in the same relative position (although here the

time line is running in the opposite direction). Likewise, at the early end of both sequences occur Red on Buff, Early, and its equivalent, Red on Buff, Complex (24); three White on Red types (16, 17, 18) that were scattered in early positions in CA23 but which combined together in the original three-site CA; and type 15, Incised Polychrome. Near the center of both grids are Red on Buff, Late, combined in the original three-site analysis but occurring as types 22 and 23 for CA23; the Comales type (2); and Gray/Buff Slipped and Polished (5). The type Black/Brown Slipped and Polished (6) can be found in slightly later positions on both grids.

Table 6-1: Types in the additional analyses

Type Code	CA23	CA17	CA40	EDM23	Frequency	Name
2	+	+	+	+	1270	Comales
3	+	+	+	+	1254	Grater Bowls, General Form
5	+	+	+	+	314	Gray/Buff Slipped and Polished
6	+	+	+	+	325	Black/Brown Slipped and Polished
7	+		+	+	86	Grater Bowls, Huistla Polychrome, Complex
8	+		+	+	79	Grater Bowls, Huistla Polychrome, Parallel Lines
10			+		20	Grater Bowls, Black Band at Rim
11			+		32	Grater Bowls, Red on Buff
13	+		+	+	83	Grater Bowls, Fine Ware

14	+	+	+	+	249	Black and White on Red
15	+	+	+	+	269	Incised Polychrome
16	+	+	+	+	324	White on Red, Simple
17	+	+	+	+	157	White on Red, Complex
18	+		+	+	79	White on Red, Broad Stripe Outlined
19	+	+	+	+	148	Huistla Polychrome, Complex (zigzag)
20	+		+	+	78	Huistla Polychrome, Complex (unique)
21	+	+	+	+	245	Huistla Polychrome, Stripes
22	+	+	+	+	287	Red on Cream, Thin Parallel Lines
23	+	+	+	+	365	Red on Cream, Broad Parallel Lines
24	+	+	+	+	264	Red on Cream, Complex Design
26	+		+	+	92	Polychrome, White Dots
40			+		58	Fine Engraved, Arcaded
41			+		46	Historic, Glazed Majolica
42			+		38	Washboard Exterior
43			+		64	Brushed or Combed Utility

44			+		43	Hatched from Rim, Incised
45			+		37	Black Crude Engraved, Complex
46			+		14	Black Crude Engraved, Suspended Triangles
47			+		41	Dark Red to Black Engraved, Complex
48			+		25	Reed Punctate
49			+		21	Exterior Incised, Deep Parallel Lines
50			+		13	Red Fine Incised
52			+		49	Miscellaneous Broad-Line Incised
53			+		50	Complex Broad-Line Incised
61	+	+	+	+	152	Fine Red Ware
62	+	+	+	+	680	Red Utility Ware
63	+	+	+	+	360	Red-Painted Jars, Streaky Polish
64	+	+	+	+	208	Red Utility, Very Thick
90			+		99	Black and Red Types (33, 35, 37, 38)

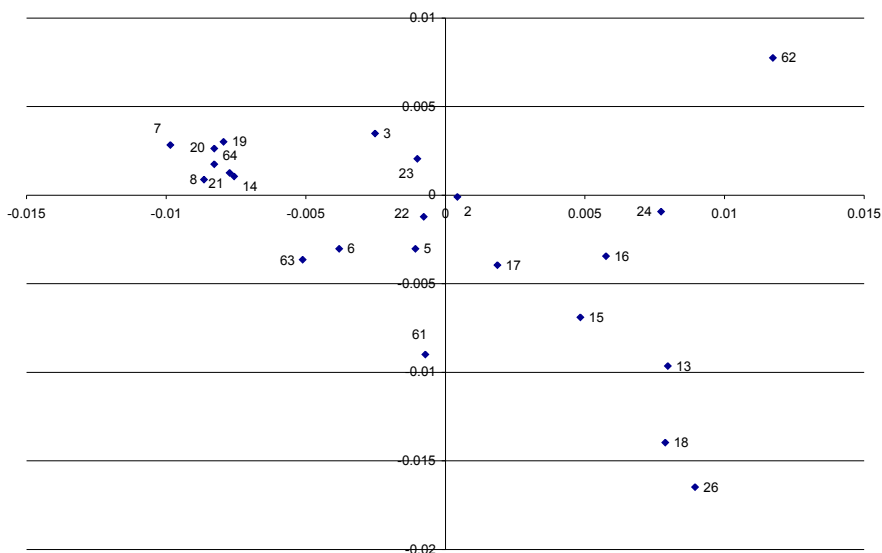


Figure 6-1. Type plot, CA23.

One interesting feature of figure 6-1 is the close proximity of points near the late end of the sequence and the fact that they become more dispersed the earlier their chronological positions. That is, in this analysis early types have more disparate distributions relative to one another than are found among their later counterparts. The same phenomenon can be seen operating in figure 6-2, which shows the plot for lots or samples of sherds in this same twenty-three-type analysis. The reason for this result is unknown, but it could be that early pottery on the site represents culturally distinct groups residing on the site at different periods of time and not continuous occupation by a single culture, as might have occurred later. Figure 6-2 identifies each lot by site, and, as in the three-site lot graph (see fig. 4-10), Tiana loci are all in early positions, in this case, all having positive x-axis values.

Another way to assess the degree of difference between the original CA and CA23 is to compare type median CA values from original CA lots, as presented in chapter 5, to type x-axis values of CA23, on a type-by-type basis. This comparison is portrayed graphically in figure 6-3, and one can see a strong negative correlation—negative, again, because

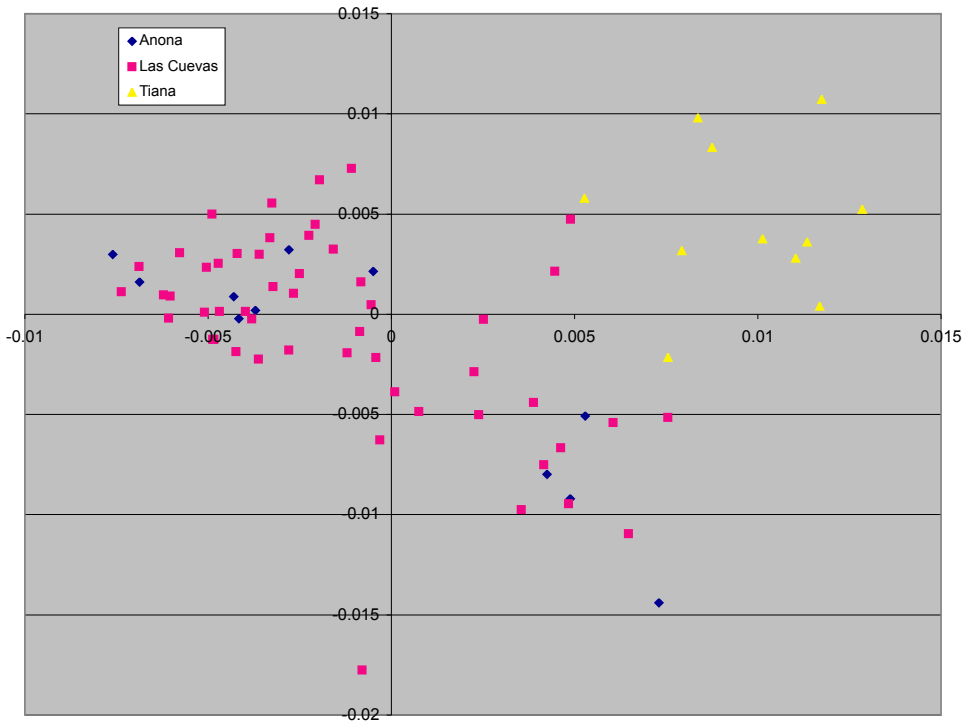


Figure 6-2. Sample plot, CA23.

the two CAs are aligned in opposite directions. The Pearson correlation coefficient of $r = -.9383$ has a significance level of ***; the two type progressions are strongly correlated.

The question remains, how well would CA23 serve in revealing the distribution of types in terms of its own lot (x-axis) CA values and presumably through time? Figure 6-4 shows these distributions for the six grater bowl types, although here the signs were reversed for CA values, which reverses the direction of the type distributions. This figure, then, is comparable to the similar grater bowl summary graph included for x-axis values from the original three-site CA (see fig. 5-1). For the CA23 graph, type distributions do have approximately the same relative positions, but separations are not as clear and distributions are not as consistent, that is, we see more outliers. Part of the problem is the smaller scale, -1.5 to 1.0 in figure 6-4 compared to -2.5 to 2.0 in figure 5-1.

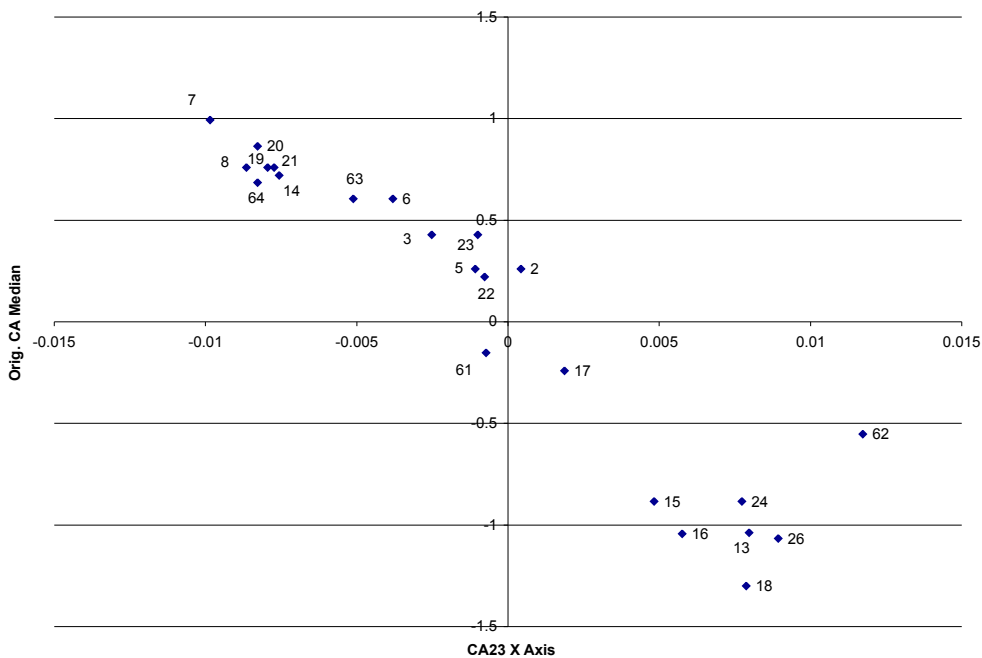


Figure 6-3. CA23 x-axis type value by type median from original CA lot values.

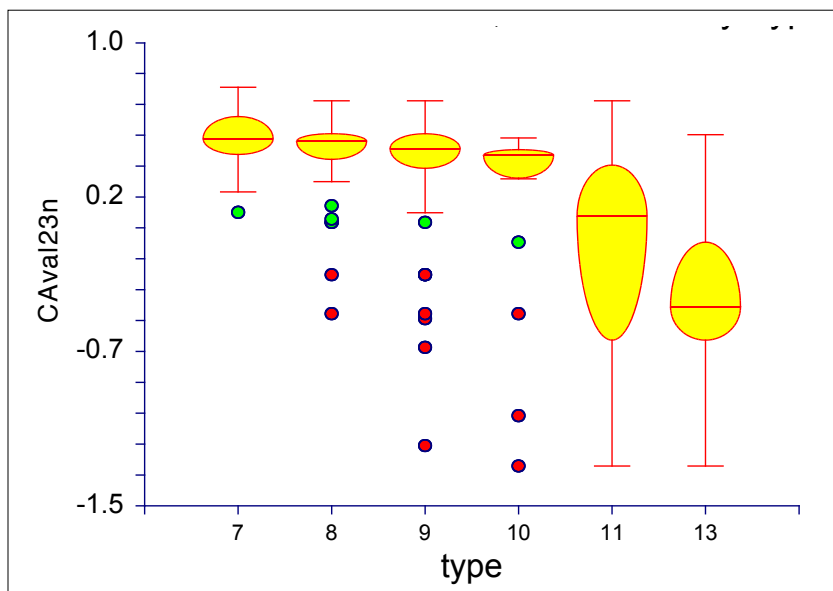


Figure 6-4. CA23 box plots, grater bowl lot CAvalue distributions by type.

In sum, CA23 does show very much the same sequence found in the original CA, but the former does not appear as useful in demonstrating the full range of type distributions. From the archaeologist's perspective, this lesser utility could be due to the inclusions of the red ware types 61–64. While high in frequency, sherds of these types proved difficult to classify, and these types might have had a destabilizing influence. One indication of this effect is that the type most out of alignment in figure 6-3 is type 62, Red Utility Ware.

Correspondence Analysis of Seventeen Types: CA17

To reduce the problem of sampling error, we also considered another CA in which included types were limited to those with at least 148 sherds. All seventeen of these types (listed in table 6-1) also figured in the twenty-three-type CA discussed previously.

Figure 6-5 portrays the two-dimensional grid for CA17 types. A comparison with the original three-site CA (see fig. 4-7) shows them to be close, albeit the two type progressions again run in opposite

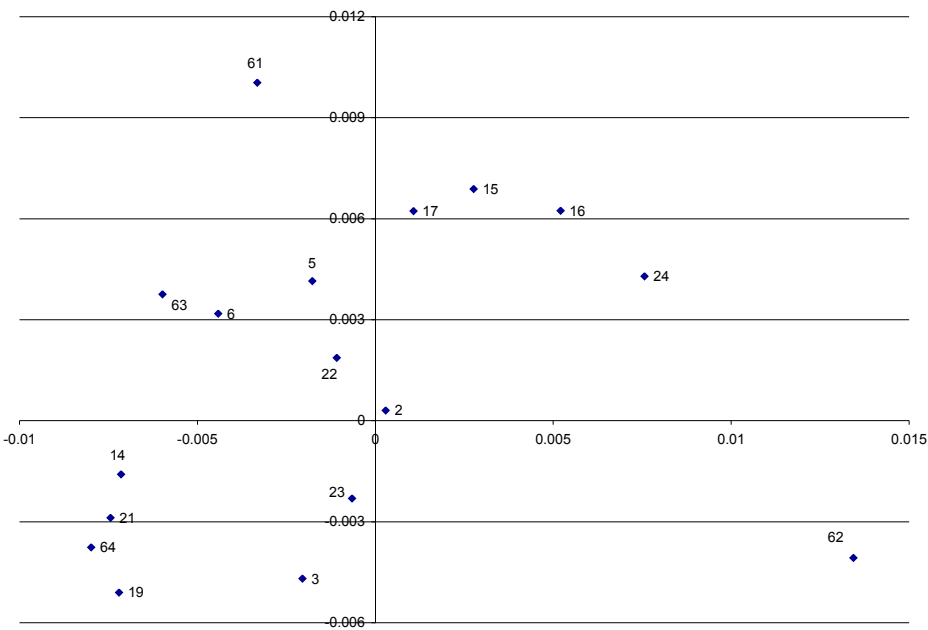


Figure 6-5. Type plot, CA17.

directions. In figure 6-5, two Huistla Polychrome types (14 and 21) anchor the late end of the sequence; Gray/BufF Slipped and Polished (5), Comales (2), and the two late Red on Buff types (22 and 23) occupy intermediate positions; and two White on Red types (16 and 17), Incised Polychrome (15), and the early Red on Buff type (24) are early. The type Black/Brown Slipped and Polished (6), is situated between the Huistla Polychrome and intermediate types. One can see essentially the same progression in the two figures.

Figure 6-6 contains the two-dimensional plot for lots in CA17. Here we see the same increasing density of points from early to late found in CA23 (see fig. 6-2), and to some extent the same holds for type plots (cf., figs. 6-5 and 6-1). Another comparison involves the median type values for the seventeen types taken from the original CA, in which lot x-axis values were applied to all sherds by type, compared to type x-axis values from CA17 (fig. 6-7). Again, we see a strong correlation ($r = -.8431$, significance level = ***) between the two type alignments. A final comparative view involves lot x-axis values for CA17, which were

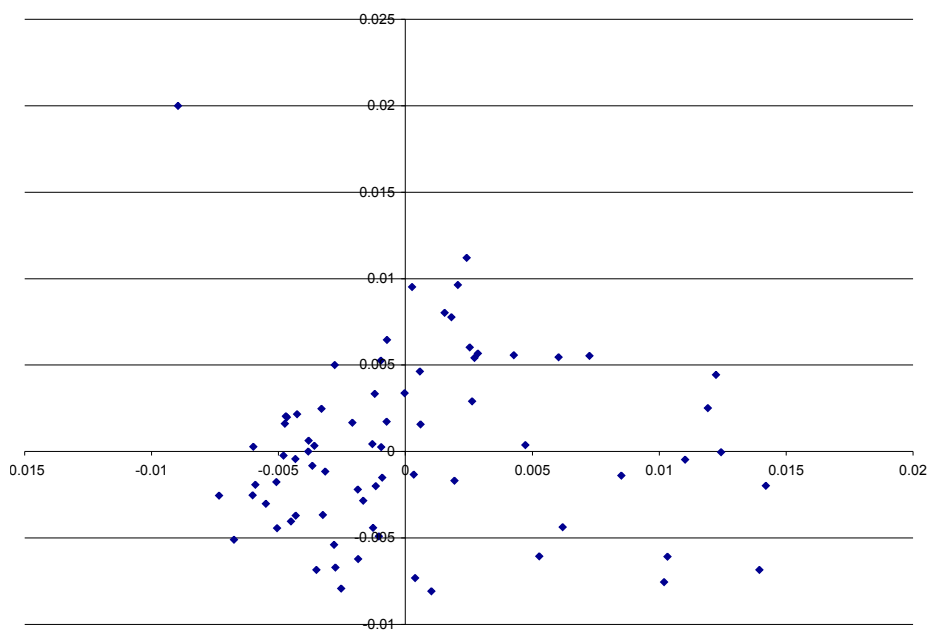


Figure 6-6. Sample plot, CA17.

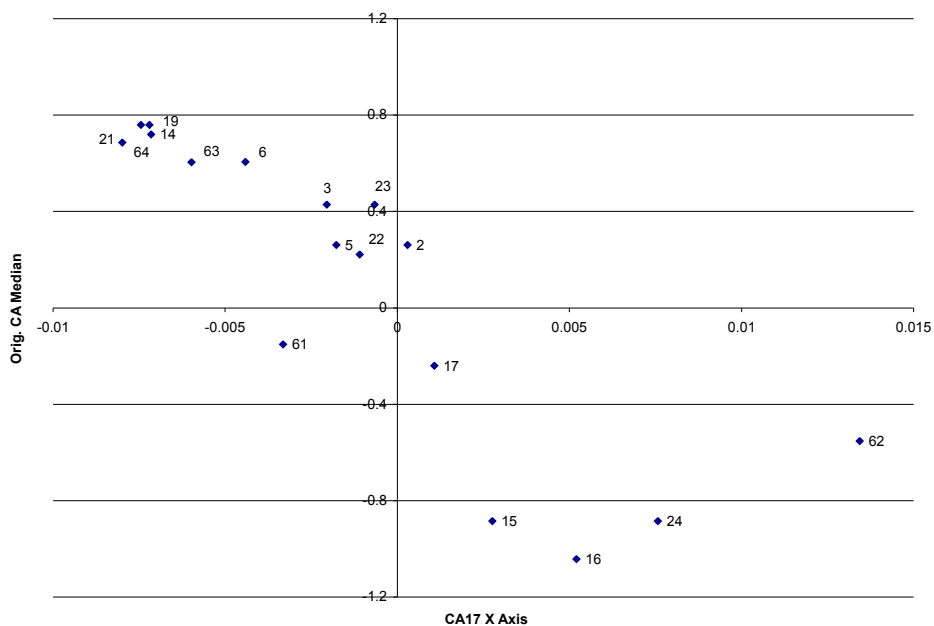


Figure 6-7. CA17 x-axis type value by type median from original CA lot values.

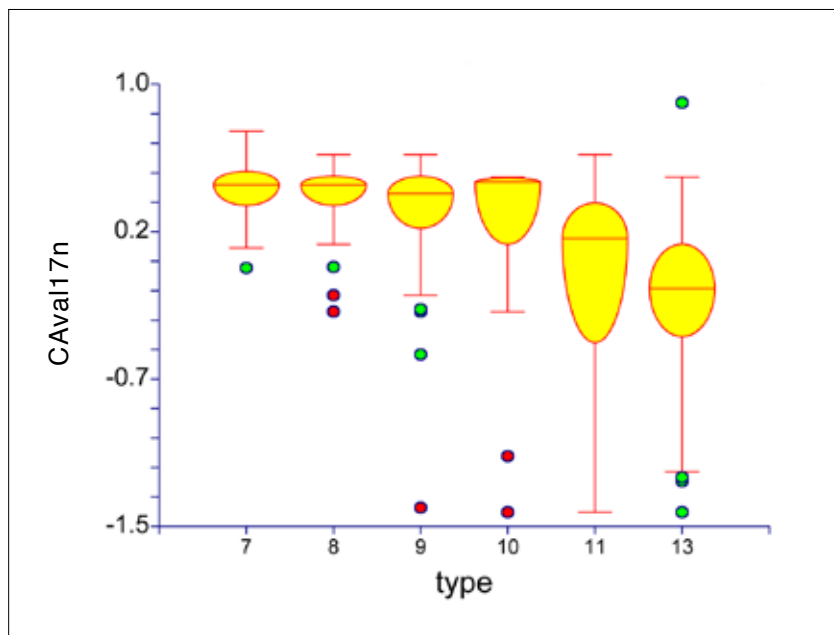


Figure 6-8. CA17 box plots, grater bowl lot CValue distributions by type.

summarized for sherds by type. Again, grater bowl types serve as an example (fig. 6-8) to let us evaluate how well this approach might depict a detailed picture of type distributions. Compared to figure 5-1, this rendition of grater bowl type distributions appears less satisfactory, similar to our finding for CA23.

Correspondence Analysis of Forty Types: CA40

In this third alternative analysis, we maximized the number of types included. Excluding residual types, we included all those with at least thirteen sherds, if they showed some tendency to cluster timewise in the original CA sequence. The forty included types are listed in table 6-1. One composite type is included, type 90, which contains data for four types (33, 35, 37, and 38). Because of the inclusion of types with low frequencies, we anticipated some volatility, and this volatility was revealed in the original two-dimensional type plot (fig. 6-9). Here, most type loci tend to be aligned roughly parallel to the x-axis, but five types can be identified as outliers: 46, 49, 52, 53, and 61. Most of these outlier

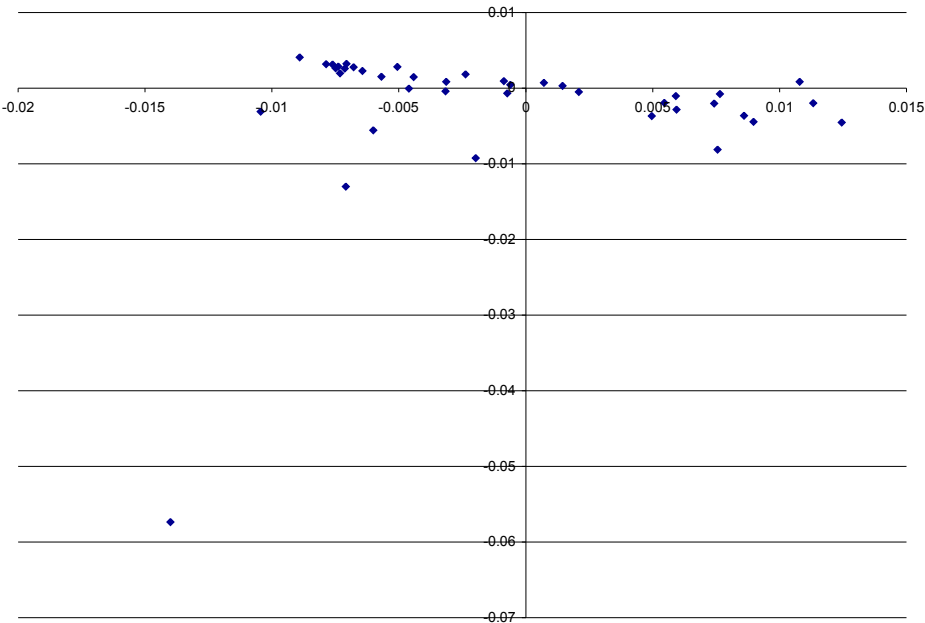


Figure 6-9. Type plot, CA40 (with outliers).

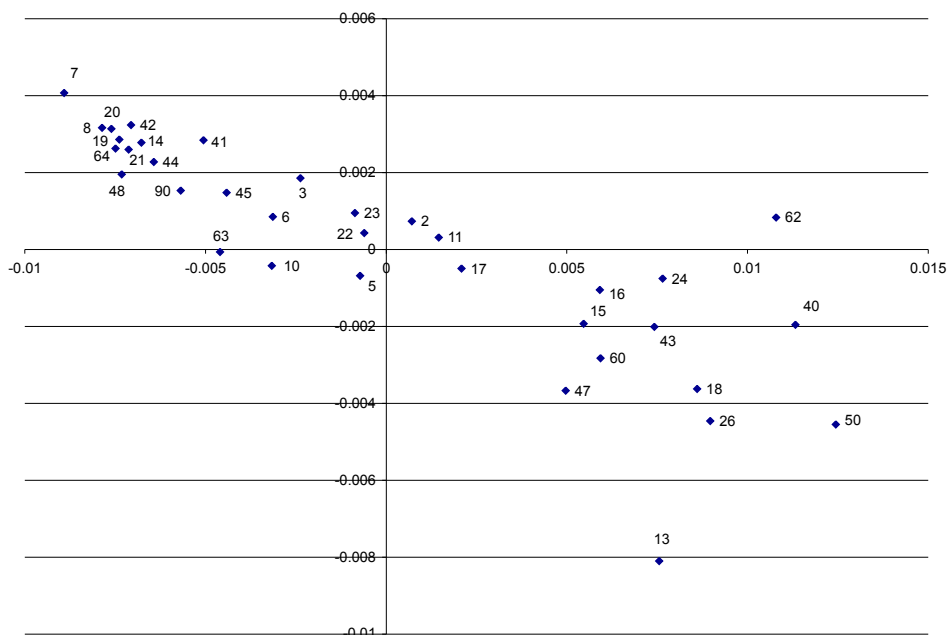


Figure 6-10. Type plot, CA40 (outliers removed).

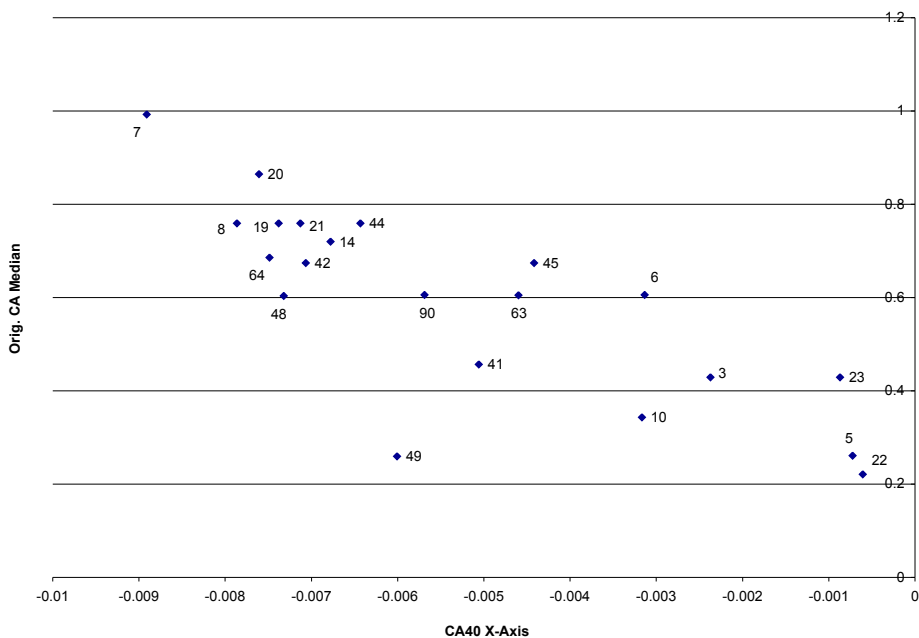


Figure 6-11. CA40 x-axis type value by type median from original CA lot values (late types).

types have low frequencies (fourteen to fifty sherds each in the data set). The exception, type 61, is a red ware type, and its reliability was suspect at the outset. We removed these types from further consideration and focused our attention on the remaining thirty-five types.

Figure 6-10, then, depicts the same plot but with the five outliers removed. Again, we see compatibility with the original three-site CA type plot (see fig. 4-7), and types line up consistently, as reported for CA23 and CA17; the type progression once more moves in the direction opposite that of the original. In figure 6-11, medians for type distributions, which we generated from the original CA lot x-axis values, are compared to type x-axis values for CA40 in a scattergram. Figure 6-11 depicts only late types, but when data for all forty points were compared, they were found to be highly correlated ($r = -.972$, significance level = ***).

It also is instructive to consider the lot distributions in the two-dimensional grid for CA40. As for types, we found the overall picture

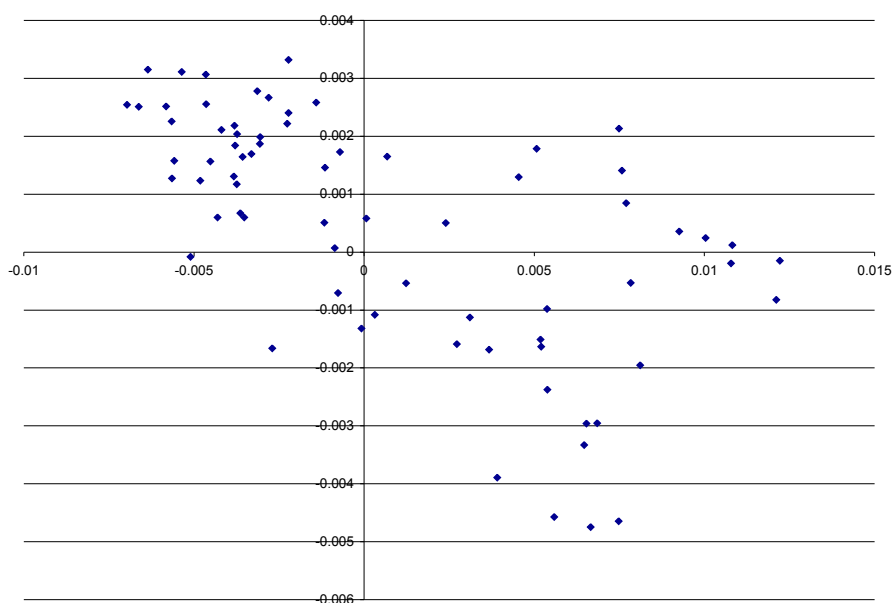


Figure 6-12. Sample plots, CA40 (outliers removed).

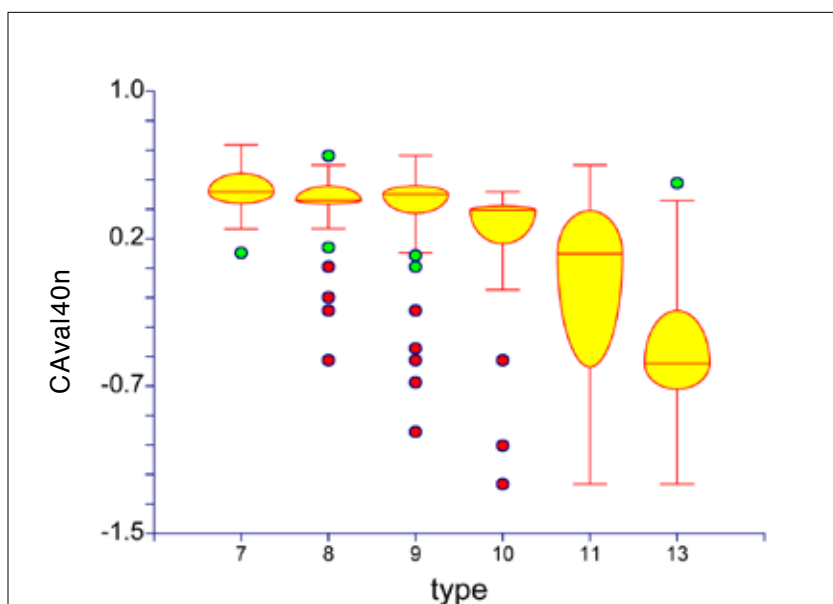


Figure 6-13. CA40 box plots, grater bowl lot CValue distributions by type.

somewhat distorted by the presence of outlier lots, but after the removal of three obvious outliers, the pattern of lots for CA40 (fig. 6-12) was similar to those for CA23 and CA17: closely spaced loci on the later portion of the grid and dispersed loci on the earlier portion.

Finally, we generated another box plot for the six grater bowl types, here using x-axis lot values from CA40 to summarize type distributions (fig. 6-13). As for CA23 and CA17, the results still seem inferior to those discovered in the original CA (see fig. 5-1).

Exponential Distance Model Analysis: EDM23

In this last alternative analysis, the EDM procedure was applied to the same twenty-three types included in CA23 (see table 6-1). This method produced a much more even array of types in the two-dimensional grid (fig. 6-14) without an increase in loci clustering through time. Also in this case, the type progression runs in the same direction as the original three-site CA: early types with negative x-axis values and late types with positive. Regarding the content of the two type progressions,

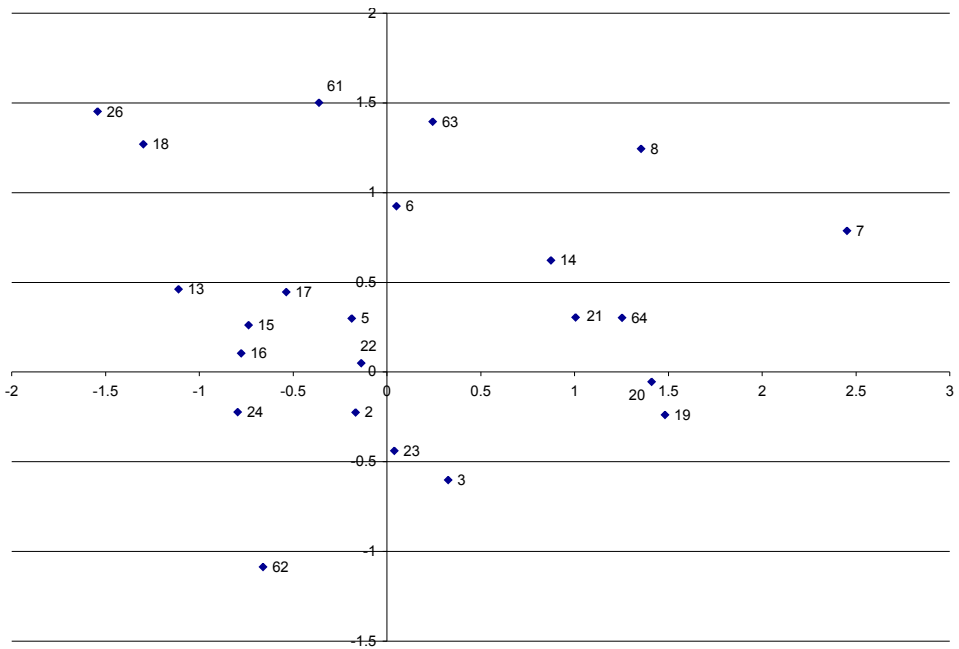


Figure 6-14. Type plot, EDM23.

parallels are consistent with those identified for the three previously described alternative CAs.

Figure 6-15 displays lot distributions for EDM23. As for types, we see no obvious evidence of an increased cluster of loci through the sequence. In this graph, loci are identified by site, and the early position of Tiana lots in the grid is consistent with findings from the original three-site CA (see fig. 4-10) as well as CA23 (see fig. 6-2).

We compared EDM23 type x-axis distributions to median values from the original CA lot summaries by type (fig. 6-16) and type x-axis distributions for CA23 (fig. 6-17). Correlations for both are high, negative in the case of CA23 and EDM23, where type orders are reversed. Relationships between these two CAs and the EDM analyses are curvilinear, and the order of types, one compared to the other, is remarkably similar. The EDM23 box plot (fig. 6-18) for grater bowl types is similar

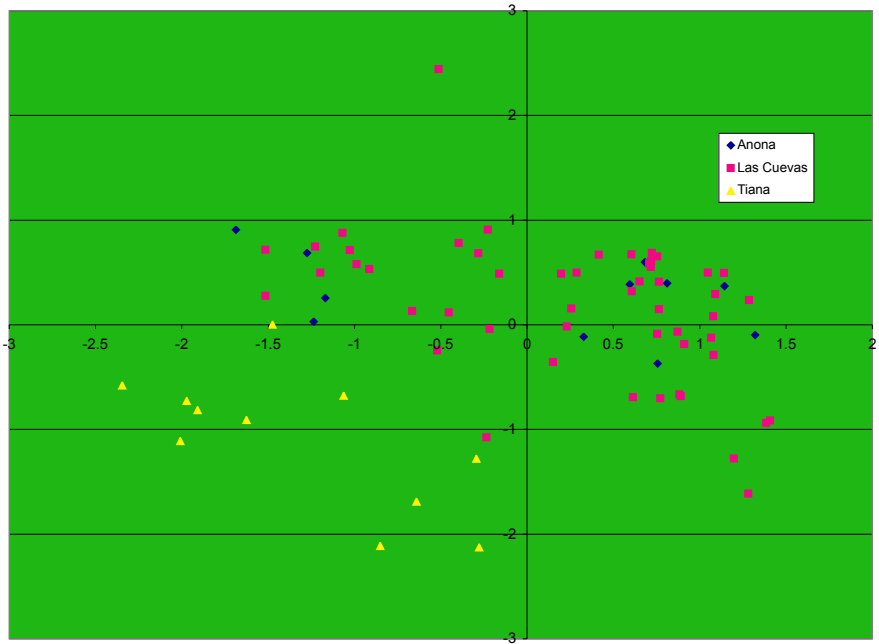


Figure 6-15. Sample plot, EDM23.

to but less satisfactory than that for the original CA; it displays more outliers and less spacing of types.

Summary

The aim of this brief chapter has been to explore the possibility that the original three-site CA and the site-specific CAs discussed earlier were somehow influenced by the selection of types for the analysis. In attempting to avoid this potential source of bias, we generated two additional CAs with type selection based only on type frequency: all types with more than 148 sherds (CA17) and all types with more than 78 sherds (CA23). In a third CA, we circumvented the selection process by including as many types as possible (CA40). The fourth analysis (EDM23) employed a different statistical procedure. Yet in all four of these alternative analyses, we found evidence for generally the same alignment of types. We can conclude, at least based on available findings, that this type alignment is embedded in the data analyzed.

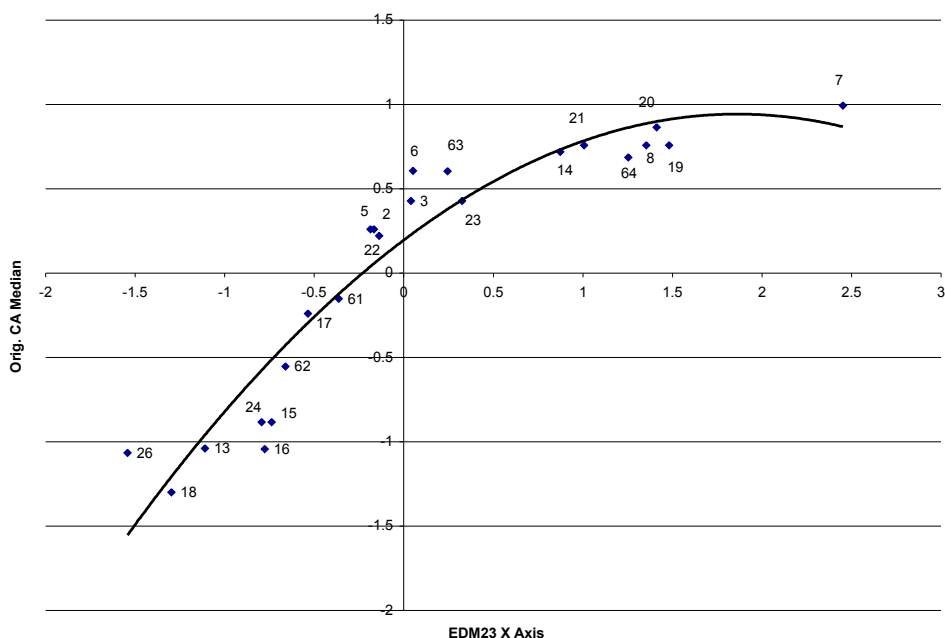


Figure 6-16. EDM23 x-axis type value by type median from original CA lot values.

A second result of this exercise was the discovery in analyses CA23, CA17, and CA40 of the clustering of lots near the later end of the sequence and their greater dispersal in earlier samples, showing more diverse distributions of early types. This finding may relate to an increase in settlement size or duration through time at these sites, but much more fieldwork and analysis will be required to explore this issue.

Finally, the four alternative analyses do say something about the actual distributions of types through the sequence, not just their relative positions on a CA grid. By applying lot CA values to all sherds in the study, not just those figuring in the CA, we could at least suggest the nature of the distribution for most individual types. Parallels across the different analyses can be seen in the four box plots included in this chapter.

Another way to glimpse similarities and differences in type distributions among the analyses has to do with skewness indices in type distributions. In chapter 5, we showed a positive correlation in the original

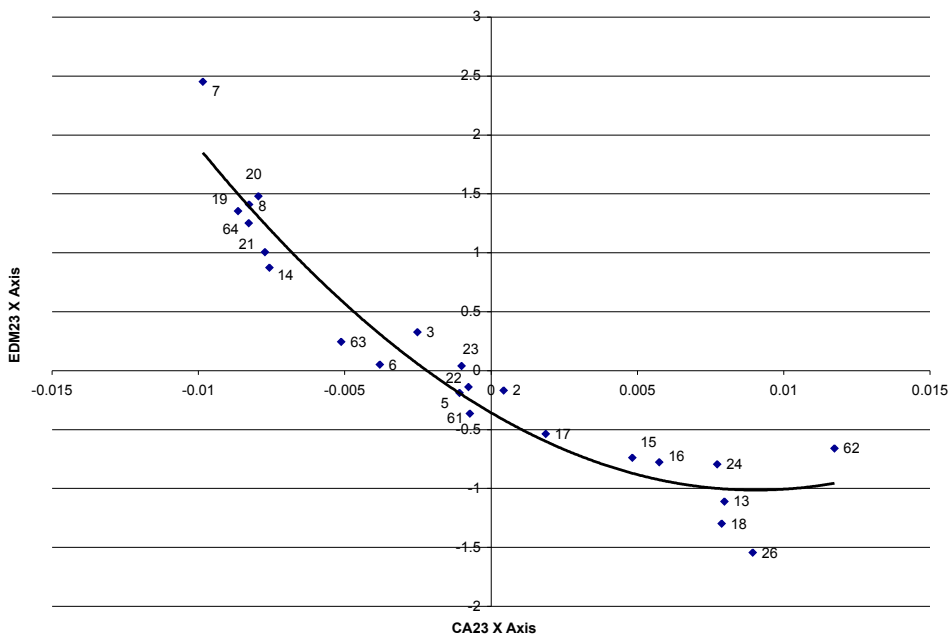


Figure 6-17. EDM23 x-axis type value by CA23 x-axis type value.

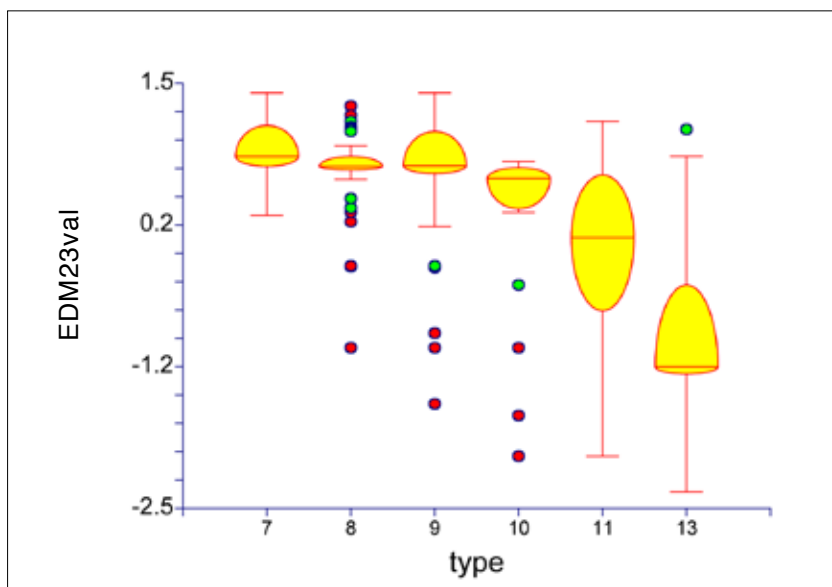


Figure 6-18. EDM23 box plots, grater bowl lot CAvalue distributions by type.

three-site CA between a type's median CA value and the skewness value of its distribution pattern. Early types tend to have positive skewness values and late types, negative values. Without delving too far into the intricacies of these alternative analyses, we included skewness indices for six of the incised/engraved types for each of the five analyses under consideration here (table 6-2). Keeping in mind that three of the type progressions are running in opposite directions to the original CA and that skewness value signs should be reversed in those cases, we can see evidence of correspondence for three of the four alternative analyses.

CHAPTER SEVEN

Table 6-2: Type skewness by alternative analysis

Age	Type	Original CA	CA17	CA23	CA40	EDM23
Late	44	-.518	2.677	2.416	2.379	-1.311
Late	45	-.566	1.387	1.237	1.343	-1.256
Late	46	-.879	.790	.865	.895	-1.004
Early	40	.811	.042	-.403	-.571	.574
Early	43	.470	.962	-.906	-.551	.351
Early	51	.593	.147	-.133	-.112	.122

Conclusions

By C. Roger Nance, Jan de Leeuw, and Phil C. Weigand

+++It remains to evaluate the ceramic sequence proposed herein through cross-site comparisons, which will involve comparing similar types of known age to their Etzatlán counterparts. A second topic of this chapter has to do with the presence of colonial Spanish pottery in the sequence and the fact that most of the sites studied here were occupied at the time of the conquest. What does our ceramic data set have to contribute, if anything, to an understanding of the Spanish conquest and its aftermath at Etzatlán? Finally, by way of summary and conclusion, we revisit the ceramic sequence and the statistical technique that produced it, correspondence analysis.

Beyond Etzatlán: Regional Ceramic Comparisons

Phil Weigand (personal communication 2004) obtained from two sites three radiocarbon assays that date Huistla Polychrome ceramics to the late fifteenth century. Two dates were processed by Beta Analytic and one by the University of Arizona. One of these sites is just south of the present-day community of Etzatlán, the Santa Clara Arroyo segment of the Etzatlán site (chapter 1); the other, Guachimontones de Teuchitlán, is 30 km to the east (Weigand 2007:111). These dates are certainly consistent with the finding that Huistla Polychrome is the latest indigenous pottery in our sequence. One ceramic assemblage probably

contemporary with Huistla Polychrome and associated types at Etzatlán is the Mylpa complex defined by Isabel Kelly (1945) for the Autlán-Tuxcacuesco area of Jalisco. There, the dominant decorated type is Autlán Polychrome, much of it from molcajetes with tripod feet. Sites of the Mylpa complex, Kelly (1945:5) believed, are the remains of settlements visited and described by Spanish observers in 1525.

Turning to the early portion of the sequence, we can begin with two Etzatlán types: White on Red, Complex (17) and Incised Polychrome (15). The relative chronological positions of these types, both in terms of median CA values (chapter 5) and one of the alternative analyses, CA23 (chapter 6), can be observed in figure 6-3. Incised Polychrome is clearly the earlier of the two. Much the same situation occurs in the stratified site of Amapa in Nayarit. Grosscup (1964) describes the ceramic sequence for Amapa in his dissertation and later published the work in Meighan, ed. 1976. Incised Polychrome is quite similar to Cerritos Polychrome, as is White on Red, Complex to Santiago White on Red. Based on stratigraphic distributions, Grosscup assigned Cerritos Polychrome to the Cerritos phase, which he estimated to date between AD 900 and 1100. Santiago White on Red is later, occurring in deposits of both the Ixcuintla and Santiago phases. Combined, these phases have estimated dates of AD 1100 to 1550. If Grosscup is correct, this evidence suggests that the Etzatlán sequence, at a minimum, began sometime before AD 1100.

How much earlier, however, is unclear. The Etzatlán type White on Red, Broad Strip Outlined (18) is another Amapa parallel and resembles Iago Polychrome, which in the Amapa sequence dates to the Cerritos phase. Another Cerritos phase type, Cerritos Engraved, is almost identical in its complex design motif to the Etzatlán type Fine Engraved Arcaded (40). Figure 6-10 shows these types, along with type 15, Incised Polychrome, clustered along with others at the early end of the sequence, and figure 6-3 provides much the same picture. It appears, then, given the close similarities between these three types (15, 18, and 40) and their Amapa counterparts, that the early portion of the Etzatlán sequence under investigation consists mainly of pottery contemporary with the Cerritos phase.

However, an even earlier occupation is probably represented in the Etzatlán pottery. Some design motifs on the sherds of the Etzatlán

type Red on Cream, Complex (24) are characteristic of Amapa Red on Orange, which is affiliated with the Amapa phase, estimated to date between AD 500 and 750. The Etzatlán type Polychrome, White Dots (26), with its aligned white dots on black stripes, has affinities to Gavilán Polychrome. This Amapa type is diagnostic of the Gavilán phase with bracketing dates estimated at AD 250–500, and in the relative sequence for Etzatlán, Polychrome, White Dots tends to be quite early (see figs. 6-3, 6-10).

This one type by itself may not seem like compelling evidence for Classic period occupation at Etzatlán, but several shaft tombs, described by Long (1966) and Weigand (1993), have been excavated or dug by looters at Las Cuevas (Long 1966; Weigand 1993), at other sites in the Laguna de Magdalena basin (Long 1966), and elsewhere in the vicinity (Corona Nuñez 1955). Tombs such as these are believed to date no later than AD 500 in north-central Jalisco. On this point, see Galván Villegas 1991:255–57, which reports shaft tombs from the vicinity of Guadalajara, and Beekman 1996:68–75. The early position of shaft burials and the elaborate hollow ceramic figurines that accompany them were observed by Gifford (1950) in his survey of the Ixtlán del Río area not far west of Etzatlán and by Mountjoy (1970) in a program of test excavations at sites near San Blas on the coast of Nayarit. Contemporary with the hollow figurines of Gifford's (1950:199, fig. 15b, e) Early Period is a form of polychrome pottery with the characteristic aligned white dots of the aforementioned Etzatlán early polychrome. More recently, Cabrero García and López Cruz (2007:241–43, table 1) have reported three shaft tombs from the site of El Piñon in northern Jalisco, which they radiocarbon-dated between AD 80 and 500. Beekman (2006:247) discusses radiocarbon dates from shell and bone collagen samples collected and submitted by Long from looted tombs at Las Cuevas.

Another site that produced useful data for our purposes is Tizapán El Alto. This large site is on the south shore of Lake Chapala and near the eastern border of Jalisco. Meighan and Foote (1968) excavated a small portion of a mound there that was 300 m long and contained midden to a depth of 2.4 m. At Tizapán El Alto, engraved polychrome occurs deep in the deposits, but white on red sherds become plentiful only above the depth of 1.2 m (Meighan and Foote 1968:table 7). As depicted in the publication, however, this Cojumatlán White on Red

pottery has painted designs dissimilar to both those of Santiago White on Red and the Etzatlán type, White on Red, Complex.

Schöndube (1994:110–12) conducted a large-scale survey and test-excavation program in southeastern Jalisco. He identified engraved polychrome sherds in his material that he saw as similar to and contemporaneous with early Tizapán El Alto counterparts discussed previously. Schöndube's (incised) Cojumatlán Polychrome, as illustrated, appears quite similar to the Incised Polychrome type from Etzatlán. Incised polychrome sherds also occur occasionally in components of the Amacueca phase, estimated to date between AD 1100 and 1532. (Ramírez Urrea 2005:323, 309). Sites of this phase are located in the Sayula basin, south of Lake Chapala, Jalisco.

Another pertinent finding from Tizapán El Alto has to do with comal sherds. Very few sherds from comales were found there, and those identified had upturned rims. By contrast, the large quantities of comal sherds from the four Etzatlán sites are from griddles that were essentially flat across their diameters. This evidence suggests that occupation at Tizapán El Alto ended before the adoption of the true comal and possibly before the manifestation of this vessel form at Etzatlán. Comal use appears to have been rare also at Amapa (Meighan 1976:140) as well as at the site of Cojumatlán (Lister 1949:46). The latter is not far from Tizapán El Alto on the shore of Lake Chapala, and occupation of the two nearby sites was in part synchronous (cf. Meighan and Foot 1968).

Meighan and Foote obtained four radiocarbon dates for Tizapán El Alto, three of which were internally consistent. All three of these dates were corrected through tree-ring calibration (Taylor and Berger 1968). From these, Meighan and Foote (1968:36–37, 120) estimated that the entire occupation of the site dated between AD 1000 and 1250 and that the later of their ceramic phases, when white on red pottery was in use, dated between AD 1100 and 1250.

All of this information suggests the following for the Etzatlán pottery under investigation: it represents occupations beginning at least by ca. AD 900 and possibly as early as AD 250. After AD 900, Incised Polychrome became an important decorated type. White on Red, Complex (type 17) pottery became popular in the vicinity after AD 1100, and the widespread use of comales did not begin until sometime after AD 1250. Following that development, Huistla Polychrome made its appearance.

At Las Cuevas and Anona, occupation may well have continued unbroken into the historic era, given the relatively early appearance of historic Majolica pottery (type 41) at these sites. Majolica and evidence for the continuation of indigenous types into historic times will be discussed in a later section of this chapter. Santiaguito, as indicated in chapter 4, shows the same range of major types and also yielded Majolica sherds, so Postclassic to historic occupation is indicated for that site as well. An even earlier Classic period occupation at Santiaguito is suggested by a few sherds of the type Polychrome, White Dots (seven sherds, .24 percent of the classified collection).

Finally, Glassow (1967) described much the same pottery through a comparable typology at Huistla, another site Long and Glassow excavated and located on the southwestern edge of Etzatlán (see fig 1-4). Huistla Polychrome (including bowls and molcajetes), red and buff pottery with banded and complex designs, incised polychrome, white on red sherds, and comales are all represented. We would ascribe the same time range to this material as well.

The Prehistoric/Historic Interface at Etzatlán

Abrupt Culture Change

Throughout the Etzatlán ceramic sequence, we have relatively little evidence that culture change was transitional, that early forms gradually gave way to later types through time. This process can be seen most clearly when type distributions are compared on a site-by-site basis. Table 7-1 deals with fourteen early pottery types and seventeen late types that were not included in the original three-site CA. As defined here, early types have median CAvalues of less than .5, and late types have median CAvalues greater than .5; Majolica-type sherds were excluded from the table. As can be seen, only 4.5 percent of sherds from the early site of Tiana were classified into late categories. There seems to have been no autochthonous development of these later types or the cultures they represent within the Etzatlán region, and since the Majolica type has median and mean CAvalues just below .5 (see table 5-23), the migration of new population(s) into the Etzatlán basin sometime around the onset of Spanish influence seems a reasonable possibility.

Majolica Pottery and Historic Occupation at Etzatlán

In our ceramic analyses, the type historic Glazed Majolica (41) apparently is not the latest in the sequence, nor does it show a typical late distribution, with a pronounced negative skewness value, which would be expected if Majolica sherds had been scattered over the site following termination of the indigenous occupation. In fact, Majolica sherds show a more or less normal distribution (fig. 7-1), and some indigenous types are clearly later in terms of (original three-site) CAvalue. Differences here are statistically significant (tables 7-2 and 7-3). The position of Majolica sherds compared with those of other late types is depicted in figure 6-10, which shows type distributions on the CA grid for the CA40 analysis. It also can be seen in figure 6-11, which includes late types plotted in terms of both original CA medians as well as CA40

Table 7-1: Indigenous types not in original CA by time and site

Age	Anona	Las Cuevas	Tiana	Total
Early	172 45.5%	817 46.3%	490 95.5%	1479 55.7%
Late	206 54.5%	949 53.7%	23 4.5%	1178 44.3%
Total	378	1766	513	2657

Table chi-square = 409.201
Probability level = ***

Table 7-2: Huistla Polychrome (7) and Glazed Majolica (41) by type and CAvalue

CAvalue	Type 7	Type 41	Total
> .804	51 59.3%	9 19.6%	60 45.5%
< .804	35 40.7%	37 80.4%	72 54.5%
Total	86	46	132

Table chi-square = 19.08706
Probability level = ***

Table 7-3: Huistla Polychrome (21) and Glazed Majolica (41) by type and CAvalue

CAvalue	Type 21	Type 41	Total
> .621	154 62.9%	19 41.3%	173
< .621	91 37.1%	27 58.7%	379
Total	245	46	291

Table chi-square = 7.4627
Probability level = **

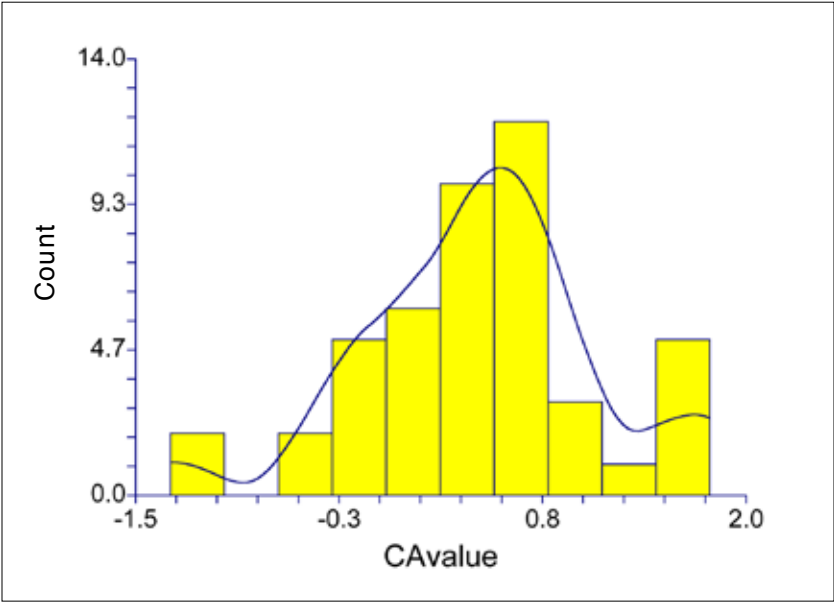


Figure 7-1. Historic Majolica ware by CAvalue.

x-axis values. Most types represented here have later chronological positions than does Glazed Majolica (type 41), including all five of the Huistla Polychrome types included in the CA40 analysis.

Before discussing the implications of these findings, we should consider the Majolica sherd distributions in more detail. With only forty-six Majolica sherds in the seventy-four lots studied, sampling error might seem a reasonable way to account for the type's early position. However, these sherds derive from twenty different lots from two sites—two from Anona and eighteen from Las Cuevas—so the relatively early peak does

not seem to result from a chance broken pot or two. The near-normal distribution also suggests that this chronological placement within the sequence is accurate.

Figure 5-21 summarizes the distribution of all classified potsherds from the three sites of Anona, Tiana, and Las Cuevas in terms of original CA value. The highest bar is just to the right of the 0.5 hatch mark. We can focus on the two sites with Majolica pottery, Anona and Las Cuevas (figs. 7-2 and 7-3). For both sites, the highest peak is in the same position, just above the 0.5 mark. Since the Majolica type has a mean of .47 and a median of .45, and assuming that potsherd frequencies can serve as a rough index for occupational intensity at these sites, our data suggest that the greatest amount of activity at these specific localities occurred after the beginning of the historic—not during the prehistoric—era at both Anona and Las Cuevas. Moreover, our data indicate that Huistla Polychrome types (7, 8, 19, 20, 21 in the CA40 analysis) all continued into historic times (after the Majolica peak) at Etzatlán, during which represented ceramic forms continued to evolve. For example, grater bowl type 7 is significantly later than type 8 (see table 5-3). As a matter of fact, these data do not clearly show that Huistla Polychrome was present in the vicinity of Etzatlán prior to the conquest. The box plots in figure 7-4 indicate that Huistla Polychrome types tend to extend into earlier lots than the Majolica, suggesting a prehistoric arrival for the former. But, as discussed previously, Majolica is earlier in the CA40 two-dimensional plot and tends to be significantly earlier in the sequence when compared with these types in chi-square tables, as in tables 7-2 and 7-3.

Other evidence that indigenous pottery was at least in part contemporary with Majolica comes from the Santiaguito statistical outlier discussed in chapter 4.

Also, and more subjectively, several of these types can be interpreted as representing a deteriorated form of the indigenous production system that developed under the pressures of acculturation. This interpretation is suggested by the Huistla Polychrome type Red and Black on Buff, Complex (Unique) (type 20), where instead of the carefully painted parallel black and red stripes found on the earlier type 21, designs are more free-form and executed in a casual or sloppy manner. Sherds of the Huistla Polychrome grater bowl type Red and Black on

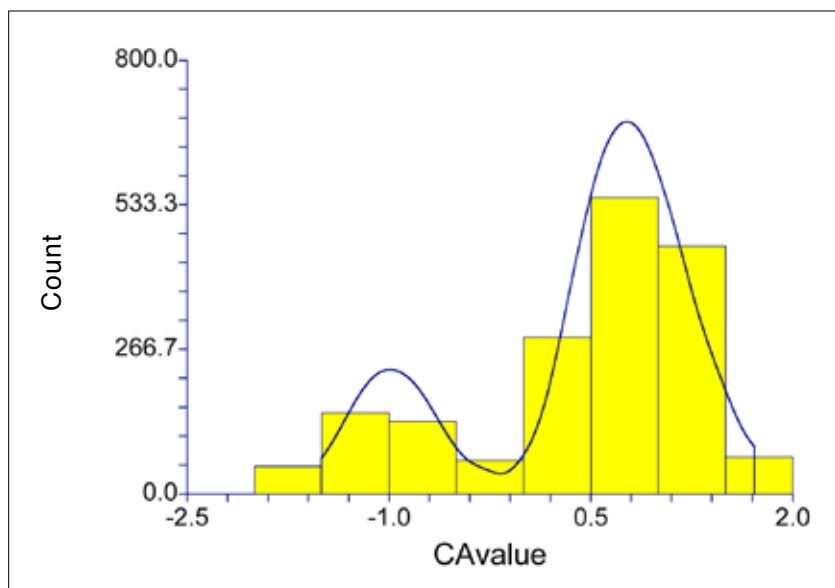


Figure 7-2. Distribution by CValue, all potsherds from Anona.

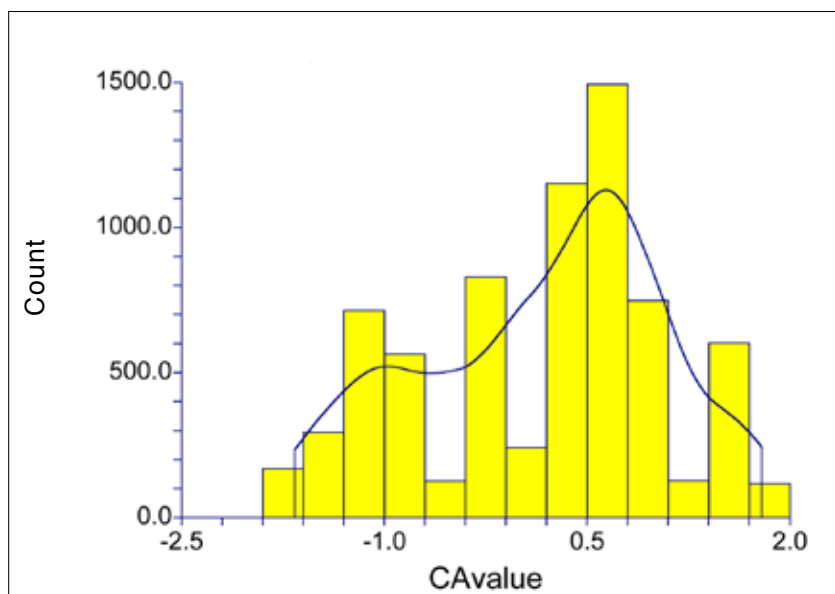


Figure 7-3. Distribution by CValue, all potsherds from Las Cuevas.

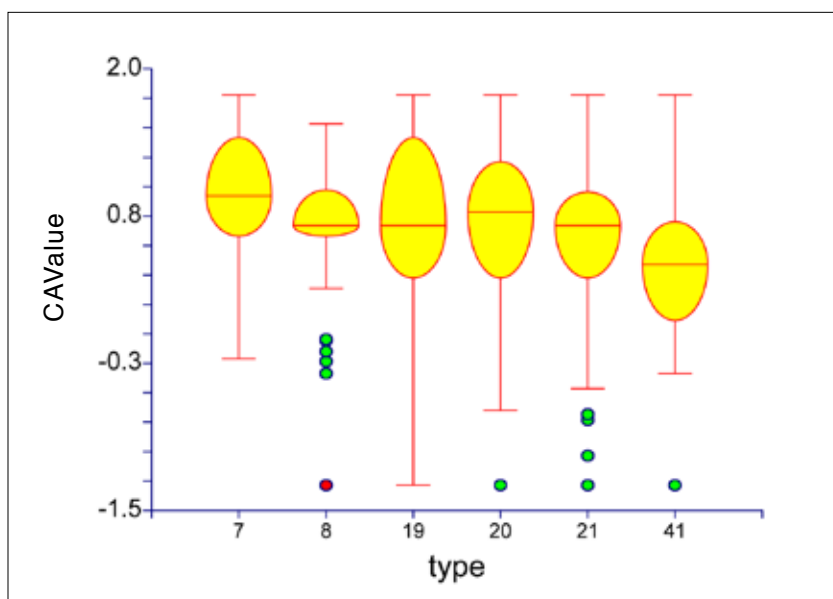


Figure 7-4. Huistla Polychrome and Glazed Majolica (41) types, CA values by type.

Buff, Complex (type 7), are marked by an additional black zigzag motif encircling the rim instead of the regular black and red parallel stripes of the earlier form (type 8). This motif does not seem to represent a normal development from the very simple and conservative design depicted uniformly on the earlier type. In these cases, one is reminded of the early historic type Ocmulgee Fields Incised, defined for Georgia in the southern United States, with its poorly executed incised and punctated decorations (Fairbanks 1956:48–49, plate 25).

Before turning to the ethnohistorical literature, we would like to point out the limitations of our archaeological data, as they pertain to the prehistoric/historic transition. First, we have no independent evidence that indigenous ceramics in fact persisted into the historic era, beyond the trends we have described. Second, it is difficult to say exactly what the increased quantity of “historic period” sherds along the CA value scale actually means. For example, it could simply mean more intense use of pottery by the early historic populace, compared with that of their late prehistoric progenitors. Alternatively, it could

indicate that the historic occupation represented by the Huistla Polychrome tradition persisted longer than its prehistoric counterpart. Finally, it could represent a population increase at the sites during the early historic period. A third general problem is that while this study deals with many potsherds, they came from relatively small areas of only two sites among the many in the Etzatlán vicinity. What happened locally might not be representative of regional trends. As a matter of fact, more compact settlements in early historic times, as suggested by higher densities of potsherds, do not necessarily indicate larger or more populous communities, which is especially true in this case, since we are unable to discuss the relative sizes of either of these two settlements as site boundaries expanded or contracted through time.

Zooarchaeology

Another source of information is the study of faunal remains from these same Etzatlán sites. Porcasi (2012) identified the bones of European-derived domesticated animals in her study collections, most of which came from excavation units that produced pottery we studied and most aligned chronologically through the CA for animal bones, as described in chapter 4.

Four identified bones from Anona were of cattle (*Bos taurus*) and one was of sheep (*Ovis aries*), together constituting 2.4 percent of the NISP (number of identified specimens) site sample (Porcasi 2012:table 4). The site of Santiaguito likewise produced twelve cattle bones and one sheep bone, 4 percent of the NISP site total (Porcasi 2012:table 7). As described previously, ceramics from Santiaguito were not included in the basic three-site analysis we have described. Nevertheless, we did classify much of the pottery excavated from the site, and this allowed us to include some lots from it in the CA for animal bones (chapter 4). As mentioned in chapter 4, the presence of historic occupation at Santiaguito is indicated not only by these domesticated animal remains but also by the recovery of Majolica type (41) sherds. Finally, Porcasi (2012:table 10) identified twenty-eight cattle bones from Las Cuevas, 2.7 percent of the site NISP total. By contrast, the site of Tiana yielded no bones of these domesticated species, no Majolica sherds, and very few potsherds of other late types at Etzatlán. These faunal data support the idea of historic-era occupation at the three sites in question. The same is true of the distribution of bones, which fell into lots included in

the CA for animal bones, bones to which Porcasi could assign relative chronological positions. Bones of non-native taxa tend to be relatively late in the CA for animal bone sequence compared to others (Porcasi 2012:table 13).

Ethnohistory

With all of this evidence in mind, we can search the ethnohistorical literature for information that might help explain these archaeological findings. The *Suma de visitas de pueblos por orden alfabético* (Paso y Troncoso 1905) is an anonymous summation of demographic and economic statistics probably compiled in 1546 or 1547 (Kubler 1942:618) for about nine hundred sixteenth-century towns in Mexico, including Nueva Galicia. In the *Suma*, Etzatlán (Yçatlan) is briefly described in two different entries. In one, Etzatlán and its satellite towns of Atitlán or Las Cuevas (Atitique), Tezontepeque (Teçontepeque), and one other (Atinque) are included (Paso y Troncoso 1905:135). These towns are described as having good land and abundant resources. Fishing is mentioned for the three satellite towns, including the island town of Atitlán. The populace for Etzatlán was differentiated into those who pay taxes (712 persons), those who do the work of the church (100 married men), and the 350 people who do not pay taxes. The overall Etzatlán population, according to this first *Suma* entry, was 1,262. Taxpayers (*tributarios*) were responsible for providing four blankets (*mantas*) and 2.5 pesos worth of gold dust every two months. In addition, each year they were taxed 320 fanegas of corn (possibly 500–512 bushels). The second entry is relatively brief and more general, describing the population of Etzatlán (both the town and surrounding settlements) as 1,310 (married) men and 376 bachelors. It mentions the lake with its quantity of fish and two islands (Paso y Troncoso 1905:126). We have, then, evidence for a continuing indigenous occupation of Etzatlán at least through one generation following the conquest.

In the earlier Cerezo/Coría account (chapter 1), we find a population estimate for Etzatlán of 600 men or, by extrapolation, 1,200 adults. Taking both accounts at face value, they in themselves suggest a somewhat stable population at Etzatlán between 1525 and 1546. For Atitlán or Las Cuevas, these estimates are 1,000 (in 1525) versus 650 (in 1546) and for Tezontepeque, 240 (1525) versus 238 (1546). Weigand (chapter

1) points out the difficulties in assessing the validity of such estimates, and the continuity suggested by these two early accounts may mask demographic volatility during decades following the conquest. Tello (1968:129) describes Tarascan mercenaries, brought to the province of Etzatlán by Nuño de Guzman in 1530, harassing, maltreating, and killing residents and burning towns over a lack of provisions (also, see chapter 1 on the Antonio de Mendoza occupation of Etzatlán in 1542).

Our archaeological findings suggest at least the possibility of a population increase in historic times at Anona and Las Cuevas, an idea these early written sources fail to support. However, we can say that for the Laguna de Magdalena basin, such a population rise at some communities in the early historic period is not beyond the realm of possibility. Kubler (1942) published population estimates for 156 sixteenth-century encomiendas in Mexico. For most, estimates were taken at three intervals, 1546–1547 (from the *Suma*), 1569–1571, and 1595–1597. Most of these communities, especially in the Archbishopric of Mexico and in Michoacán, showed population increases from 1546–1547 to 1569–1571 and then a marked decline from 1569–1571 to 1595–1597 (Kubler 1942:table I and fig. 2). Such may have occurred in Etzatlán and/or some surrounding communities as well.

Finally, we are left with the decline of Etzatlán and subsidiary towns, which seems to mark “the end of occupation” in our archaeological sequence. In the *Pintura del nuevo reino de Galicia*, dating to around 1542, and the *Ortelius* map of 1579 (see figs. 1-1, 1-2), the convento of Etzatlán, adjacent lake, and island towns are all featured in exaggerated scale. These maps indicate the importance of Etzatlán in colonial West Mexico prior to 1580. The situation, however, was soon to change. Cook and Borah (1971–1979:table 1, region IX, pt. A), employing tributary lists, estimate the 1568 Indian population of Etzatlán at 2,291, while a second estimate, based on a 1646 list for the same community, indicates a decline to around 626. In his 1621 *Descripción de la Nueva Galicia*, Domingo Lazaro de Arregui (Chevalier 1946:70) writes many town sketches but makes only a single mention of the pueblo of Etzatlán (Izatlan). The accompanying map includes “YZatlan,” but neither the large convento nor the lake with its islands and satellite towns are depicted. Evidence of widespread and devastating disease epidemics with concomitant population loss is ample in the *Relaciones geográficas*

del siglo XVI: Nueva Galicia, dating between 1579 and 1585 (Acuña 1988). Forced labor and the high tax load described previously must have taken their tolls as well. (See chapter 1 on the decline of Etzatlán.)

Contrasting Perspectives

Two co-authors of this work, Weigand (in chapter 1) and Nance (in this chapter), have included different narratives involving ethnohistorical accounts and the archaeology of Etzatlán. Weigand sees a large Postclassic population persisting to historic contact but then declining rapidly following the conquest. He bases this on his own archaeological surveys in Etzatlán and the vicinity as well as the ethnohistorical account of Tello and other sources. Nance sees more occupational activity in early historic times, compared with the Late Postclassic, when considering ceramic data from Las Cuevas and Anona. This view of changing settlement patterns around Etzatlán is admittedly very limited, as indicated previously, but it does at least point to the existence of early historic communities in the vicinity. Coupled with the ethnohistorical sources mentioned, the picture is one of substantial, mainly indigenous towns continuing a generation or more following the conquest.

Nance and Weigand agree that major population decline occurred following the conquest in West Mexico. The issue raised by the ceramic research reported here is how and where indigenous survivors redistributed themselves (or were redistributed by the Spanish) during early decades of the historic era. It seems likely that the matter can be explored effectively through future archaeological research. Certainly, excavation of early colonial structures should clarify whether or not they contain historic forms of Huistla Polychrome. Weigand spent decades examining approximately 41 km of trenches through the current town of Etzatlán that were dug during public works projects. Extending the typology developed here to the many thousands of collected potsherds from these trenches should provide more definitive information about the relative sizes of the late prehistoric and earliest colonial communities in Etzatlán.

The Ceramic Sequence at Etzatlán: A Final Comment

What, in summation, can be said of the ceramic sequence we have developed for Etzatlán? The proposed sequence has mainly to do with the Postclassic and historic eras. One clear finding involves the striking parallels we identified between Amapa and Etzatlán during the Cerritos

phase, defined for Amapa and dating between AD 900 and 1100. The site of Amapa is 16 km from the Pacific coast in central Nayarit and about 165 km northwest of Etzatlán as the crow flies. As we have indicated, the typological similarities are so detailed that they suggest close cultural ties between Etzatlán and communities in that direction during this period.

After the Cerritos phase, we identified only one clear tie with Amapa, White on Red, Complex (type 17), and its Amapa counterpart, Santiago White on Red. Otherwise, ceramic affinities in that direction become less distinct. At Etzatlán, comales become an important utility form, and the dominant form of decoration involves red-painted designs on cream or buff backgrounds. In the CAs for Las Cuevas, Anona, and, to some extent, Santiaguito, we find a cluster of types post-dating those of the Cerritos phase as well as predating another grouping of types, including those labeled Huistla Polychrome, at the late end of the sequence. These types include Red on Cream, Thin Parallel Lines (22); Red on Cream, Broad Parallel Lines (23); Grater Bowls, Red on Cream (11) and Comales (2); Gray/Buff Slipped and Polished (5); and White on Red, Complex (17). It seems likely that this cluster of types, in whole or part, will emerge as a ceramic phase at Etzatlán, as archaeology there continues to evolve. These types, as distributed, appear to represent a manifestation of the Aztatlán ceramic complex, defined loosely by Bell (1971:699–700) for West Mexico.

Finally, we should consider the cluster of types at the end of the sequence. As noted previously, it is difficult to determine from our data when Huistla Polychrome entered the sequence, whether before or around the same time as Glazed Majolica pottery (type 41); that is, before or synchronously with historic contact. What these data do indicate, however, is that indigenous pottery types continued to evolve after the beginning of Spanish influence. We have mentioned possible effects of acculturation in Etzatlán pottery, specifically calling attention to two Huistla Polychrome types (7 and 20), but this impact might extend to other types as well, for example, to the two Black Crude Engraved types (45 and 46). However, one can see through both the Amapa and Etzatlán sequences a general shift from well-executed incising and engraving on early types to much cruder design renditions later in the sequence. It is hard to say at this point if we are looking here at a general cultural trend or the sudden impact of European influence.

Correspondence Analysis at Etzatlán: An Archaeological Perspective

Based on this study, we can regard CA as an effective technique for constructing site-specific ceramic sequences in West Mexico, at least for sites resembling those encountered by Long and Glassow. Such determinations through traditional means have never been easy, especially for large sites with earthen mounds and platforms surrounded by relatively flat habitation areas. Deposits in intermound areas can be loaded with cultural debris but also can be relatively shallow and tend to be mixed. Archaeologists in the past have excavated deep pits through platforms or mounds in order to capture local sequences stratigraphically, but such projects are expensive and time-consuming. Also, one might be left with a large well-stratified sample, but one not entirely representative of the site as a whole.

The work of Long and Glassow and our study of ceramics from their excavations suggest that such intermound deposits are mixed, but only to a point, and that detailed ceramic sequences can be derived from them through use of the computer and CA. We should add, however, that there is no need to depend on samples from scattered test pits. If one were to excavate more intensively, the strategy would be to select samples from undisturbed contexts covering the full range of pottery at the site (cf. McCafferty 2001:14). Through CA, these samples might also generate information having to do with cultural dynamics other than or in addition to chronology (see Nance et al. 2003).

If this study contains a cautionary note, it is that archaeology is not pure science by any means and that CA, a logically closed system, can only work with the data fed into it via the computer. In other words, ceramic types have their limitations as constructs, the classification of large groups of potsherds is not an entirely consistent enterprise, and archaeologists might have a very limited understanding of the data they are intending to analyze, especially at a study's outset. These problems may be reflected in the distributional "static" of types through the site sequence (e.g., type outliers discussed in chapter 4), and such effects are hard to isolate from those of site mixing or random fluctuation. They also are behind the difficulties we had in selecting types to include in the various CAs as the study developed. Selecting types on the basis of their covariation (correlation coefficients) among samples from

Las Cuevas, based on type percentages by sample, cannot be deemed a good procedure, statistically speaking. Lumping types together to increase sample size and then adding types discovered to vary in their stratigraphic positions at Tiana added subjective qualities to the selection process. When we simplified the selection process and generated additional CAs based strictly on type frequencies, essentially the same sequence of types materialized (chapter 6). Interestingly enough, the original three-site CA seems to remain the best at summarizing type distributions along the x-axis time line. This is not to claim, however, that we discovered the optimal combination of types to be employed in CAs of our data sets. Love (2002) confronted this same problem when selecting variables for sequencing ceramics through statistical seriation. His research involved pottery from a Middle Preclassic site near the Pacific coast of Guatemala.

Our research suggests that CA potentially could attain broad-ranging applicability in West Mexican archaeology and that it should perform well in a variety of circumstances. Essentially the same type progressions were produced from the many and large samples of Las Cuevas as well as the relatively few but still large samples of Tiana and Anona. The more eroded collection from Santiaguito, with many samples eliminated, produced a close approximation of the sequence. The CA for animal bones, which involves both large and small potsherd samples from all four sites, also manifested the same progression of types.

Finally, we can point to one benefit of our approach in which we have emphasized the seriation of pottery types over the ordering of samples with the aim of constructing ceramic phases. The distribution of individual types actually does support the hypothesis that we have been dealing with chronology—that the seriation is in fact along a time line. At Etzatlán, most early types have distributions skewed positively, that is, tailed to the right, while most late types have distributions skewed negatively, that is, tailed to the left. These distributions occur for many types not included in any of the CAs. Altogether, they are representative of the truncated battleship curves Ford (1952) found for types at the beginnings and ends of his seriated ceramic sequences, with data derived from stratified sites in the Lower Mississippi Valley.

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